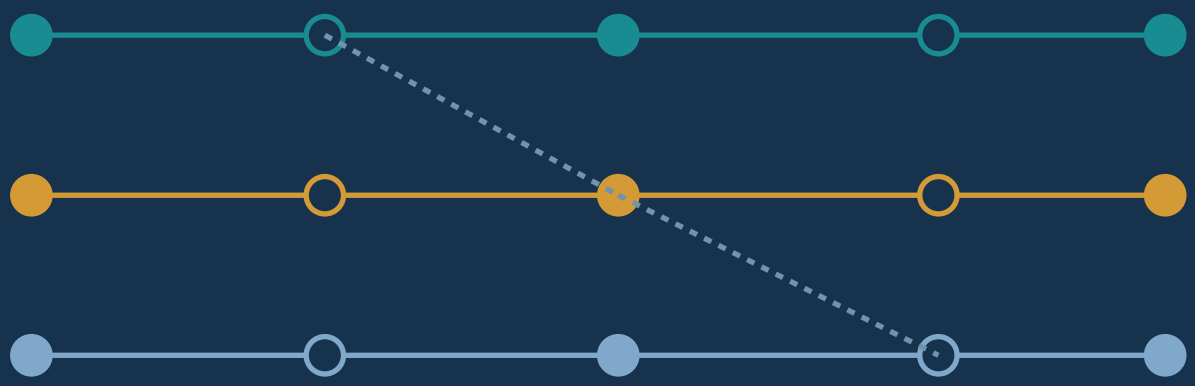
CAVE-EoT / RESEARCH HANDBOOK

From agreement to independent support

The papers, the four maps, the reasoning,
and a practical plan for your research project.



4 maps | 48 paper names | 37 source-identifiable entries

Plain-language synthesis, figures, paper guides and full evidence notes

13 October 2026 | Research synthesis and proposed design, not CAVE results

How to navigate this handbook

This is a reference book, not a linear reading assignment. Start with Chapters 1-4, then use Chapters 5-11 for the proposed method, experiments and thesis. The paper directory, exact maps, detailed notes and primary links follow. The PDF has a linked table of contents and document bookmarks.

Need	Go to
Understand the project	Chapters 1-2: agreement, independence, evidence and calibration
Learn what the maps imply	Chapter 3 and the complete-map appendix
Defend novelty	Chapter 4 and the closest paper profiles
Design an experiment	Chapters 5-9: operational target, reference, tests and cost
Write or discuss the thesis	Chapters 10-11: claims, questions and priorities
Find primary evidence	Paper directory, consolidated notes and bibliography

Evidence and scope

The four current maps are reproduced without changing marks. X means yes in the paper or a documented implemented variant; the negative mark means checked no; ? means unverified. Baselines and ablations can qualify, so a positive cell is not automatically a description of the proposed main method. This handbook makes the main-method resource boundary explicit.

Eleven intended identities remain unresolved. The last map was a rapid pass and leaves citation standing, complete-reading records and some release/status facts unresolved. Access to full text is not advertised as reading every line. The narrative is based on documented source reviews, available primary text and evidence notes; proposed CAVE choices are labeled as proposals.

Known archival disagreement: CoT, Map 3 row 45, has an unsupported X for extra inference LLM calls. It is retained in the map and not used as evidence for the one-pass CoT baseline. The original four map files have not been edited for this handbook.

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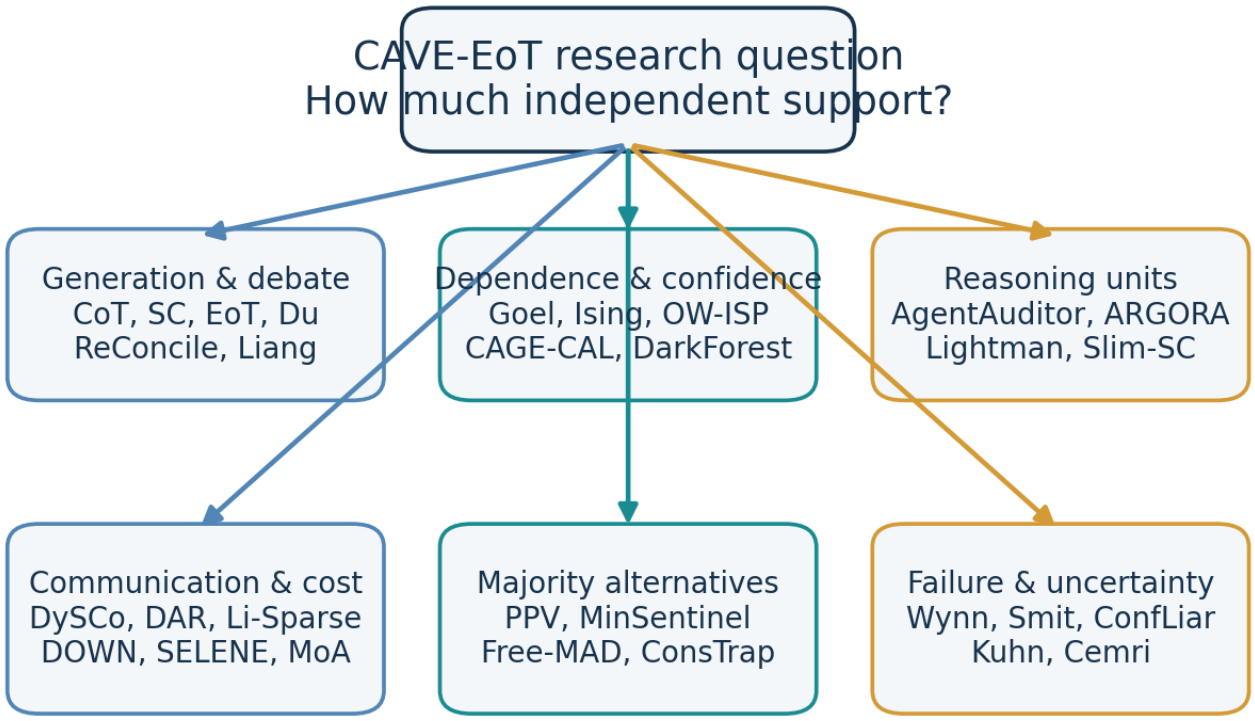
01 | Start here: what your project is really about

The research question in one sentence

CAVE-EoT asks whether a group of language models has several independent reasons to believe an answer, or several repetitions of essentially the same reason. It then asks whether the final decision can improve when repeated support is not counted as if it were new evidence.

This is a narrower and more useful question than asking whether multi-agent debate works. Debate changes the available answers, the arguments, the influence between agents and the computational cost at the same time. Your proposed layer should isolate one part of that process: the relationship between observable reasoning support and the final aggregation decision.

Imagine six agents. Four give answer A and repeat one flawed argument. Two give answer B through genuinely different supporting paths. Majority vote chooses A. An evidence-aware rule might choose B. But the interesting scientific work is not the example itself. It is defining what counts as the same argument, identifying when overlap is more than honest solvers would normally share, and showing that the resulting decision is better under fair comparisons.



Families are an editorial reading aid; membership and method scope overlap.

Figure 1. An editorial map of the reviewed families. Arrows connect research needs, not claims of historical derivation.

What the four maps give you

The maps organize the literature along four different axes. Map 1 describes the setting, agents and communication. Map 2 describes decision rules and signals. Map 3 describes dependence, reasoning representation, training and verification. Map 4 describes evaluation, theoretical background and the relationship to CAVE-EoT.

Together they contain 215 criteria applied to 48 paper names: 10,320 cells. All cells have been processed. This is not the same as every cell being verified. There are 37 source-identifiable entries and 11 aliases whose intended Paper Key citation has not been recovered. The unresolved aliases account for 2,365 question marks across the four maps. A further 120 question marks concern individual properties or bibliographic/reading records. The marked tables contain 3,370 X, 4,465 negative marks and 2,485 question marks.

The main literature synthesis uses the 37 source-identifiable entries. The 11 unresolved names remain visible in the directory and full maps, because silently choosing a plausible paper would make the comparison misleading. Publication year follows the reviewed version or published venue, so the chronology is not always the first arXiv date.

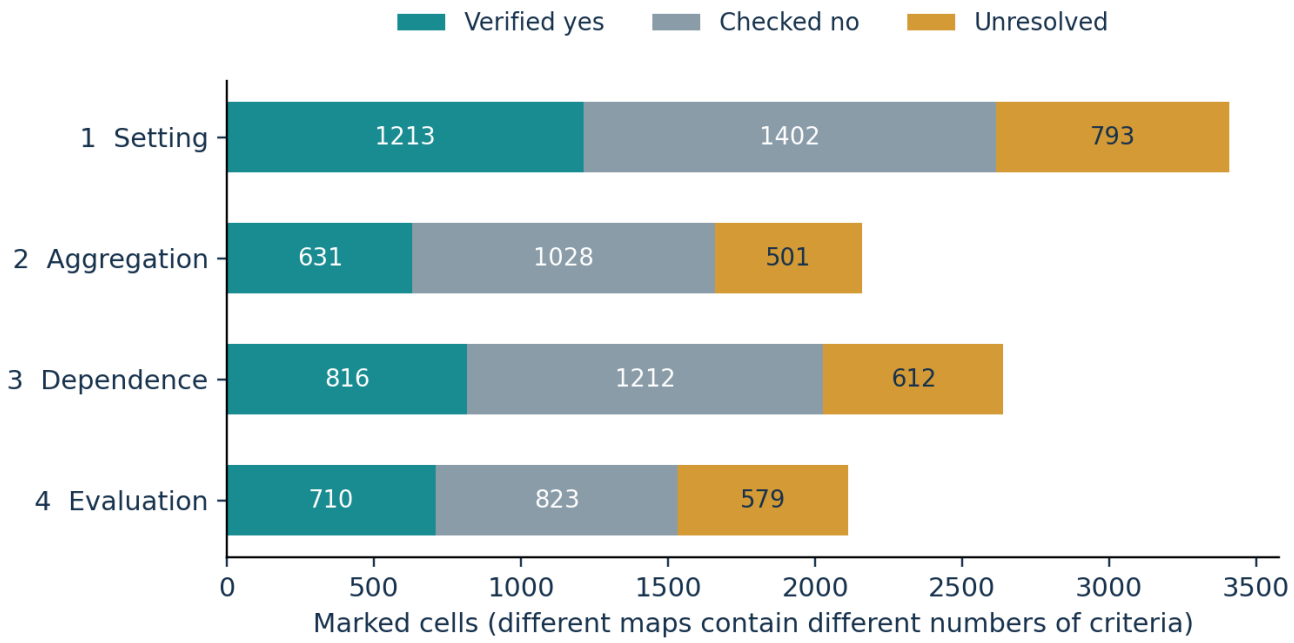


Figure 2. Exact mark counts in the current four maps. Different criterion totals make raw bar lengths unsuitable as paper-quality scores.

What you should remember before everything else

Agreement is a statement about outputs. Independence is a statement about how the support was generated or how the errors depend on each other. Correctness is a statement about the answer or inference. These three properties can move separately.

A unanimous group can be wrong. A diverse group can contain many irrelevant guesses. Two independent arguments can make the same factual error. Two agents can honestly derive the same elementary calculation. A confident answer can be unsupported. A correct answer can arrive through an invalid rationale. A structurally different rationale can be a paraphrase of the same hidden premise.

Therefore, a credible CAVE result needs at least two successes: it handles dependence or redundant support sensibly, and its final answers improve or remain reliable under an appropriate budget. Final accuracy alone cannot prove that the dependence estimator works. Conversely, detecting duplicated

support does not automatically make a decision correct.

How to use this handbook

Read the concept and synthesis chapters first. Then read the closest-neighbor comparison and the proposed design. Use the paper directory when a short name appears. Use the full-map appendix for the exact archival marks. Use the consolidated evidence notes when you need a section, page, scope qualification or unresolved detail.

Three kinds of statements are kept separate. Paper findings describe what a reviewed source reports. Map summaries describe marks in the supplied comparison artifacts, including implemented baselines and ablations. Proposed design choices are recommendations for a future CAVE implementation and evaluation. No CAVE benchmark results are claimed: this handbook does not report a built or experimentally validated CAVE system.

02 | The concepts that make the literature readable

Votes, claims, steps and chains

A vote is an agent's final answer. A claim is a proposition, such as a numerical result or a statement about the world. A step links premises to a conclusion through an inference. A chain is an ordered or partially ordered set of steps supporting an answer. A graph can represent shared premises, branches, support and contradiction between these steps.

An answer-level rule may see that five agents choose option C. A reasoning-level rule may see that three of those agents rely on the same unsupported assumption. A step-level rule may localize that assumption instead of comparing two long responses as indivisible objects. A chain-level rule asks which complete supporting paths remain after shared support has been accounted for.

These are design levels, not automatic improvements. Smaller units give more detail, but they introduce extraction errors, alignment errors and computational work. A sentence is not always an inference; one sentence can contain several inferences. A long chain can be verbose without containing more evidence.

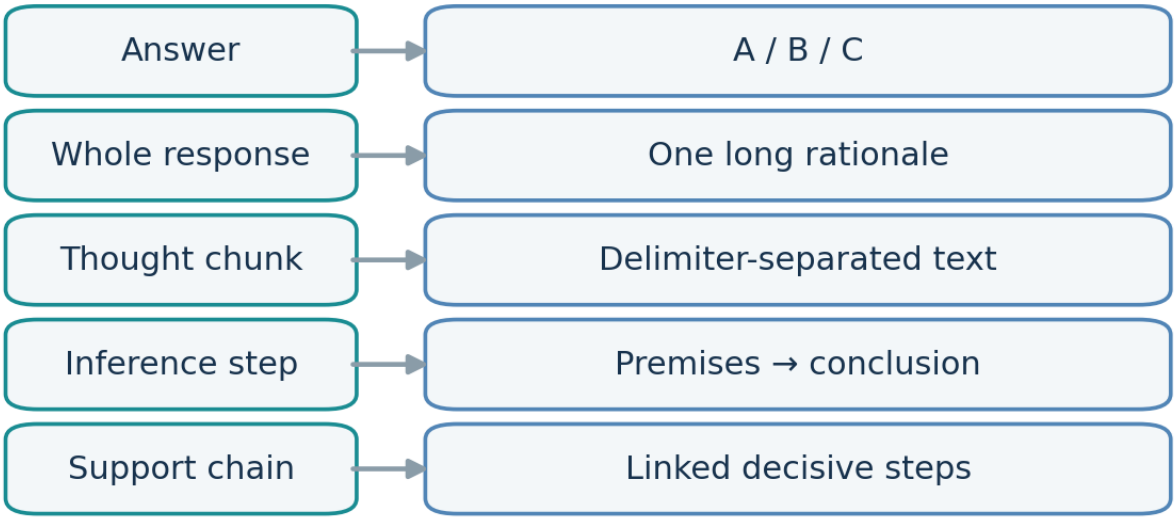


Figure 3. The unit being represented changes the question an aggregator can answer.

Four different meanings of independence

Generation independence means agents did not see one another's current outputs when generating a response. Error independence means their correctness events do not depend on one another, under a specified conditioning set. Evidence-source independence means their supporting information does not come from the same source or upstream assumption. Derivation independence concerns how much their actual support paths depend on or repeat each other.

Independent generation does not guarantee independent errors. Agents may share model families, pretraining, prompts, retrieved sources and familiar mistakes. Distinct model names do not guarantee distinct knowledge. Different wording does not guarantee different reasoning. Even different reasoning

graphs can share the one premise that determines the answer.

Your no-communication reference should therefore be named precisely. It estimates overlap expected from solvers operating independently under the chosen task and generation conditions. It does not certify that those solvers have statistically independent knowledge or errors in an absolute sense.

Why correlation reduces the value of a crowd

For equally variable measurements with a common pairwise correlation rho, the variance of their average is sigma-squared times [1 + (N - 1)rho] divided by N. Matching that variance to an independent average gives an effective sample-size illustration:

```
N_eff = N / [1 + (N - 1)rho]
```

For N = 10, independent measurements have N_eff = 10. At rho = 0.5, the variance-equivalent count is about 1.82. At rho = 1, the ten perfectly correlated measurements provide the averaging benefit of one independent measurement. This is a derivation under a simple exchangeable model, not a measured CAVE result and not a universal formula for reasoning chains. DAR provides related effective-sample-size motivation; Ising and OW-ISP use more explicit label-aggregation models.

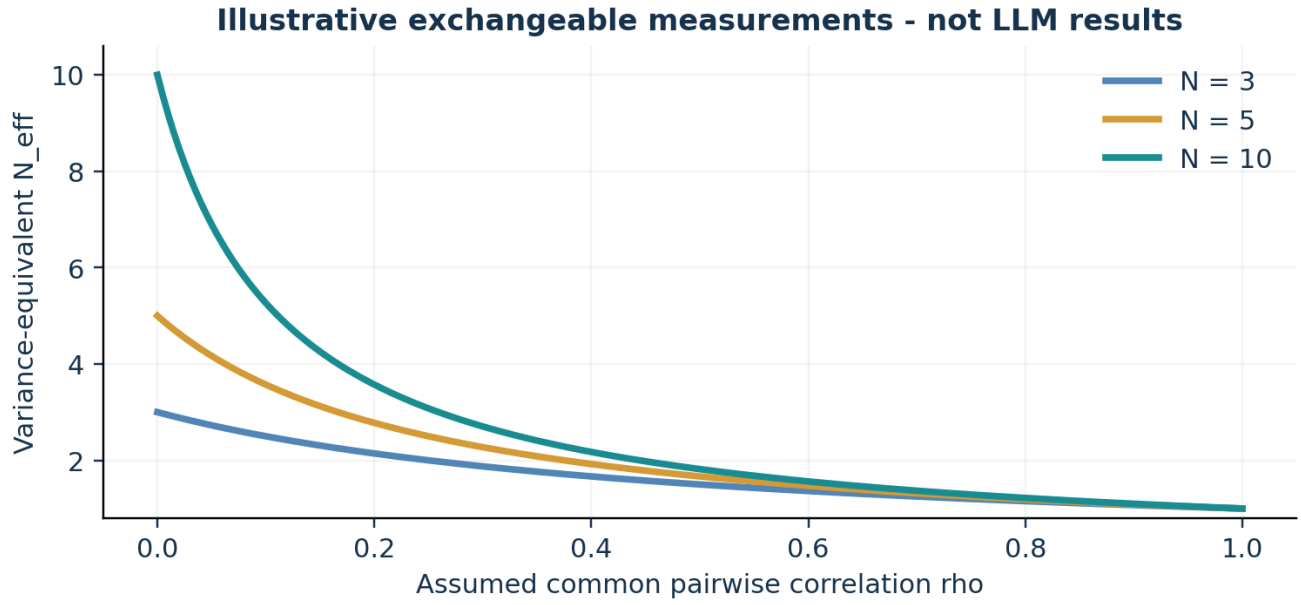


Figure 4. A simple exchangeable-correlation illustration. The curves are mathematical examples, not measured language-model results.

The lesson is not to estimate one correlation number and declare the project complete. The lesson is to specify the variables, conditioning assumptions and purpose of a dependence measure. Correlation of answer correctness across a dataset is not the same object as overlap between two supporting derivations for one question.

Honest overlap and suspicious overlap

Suppose every solver must calculate the area of a rectangle. Repeating the area formula is normal. Suppose all solvers copy an unusual unsupported claim from one persuasive peer. That overlap has a different interpretation. The task, shared premises and available information influence how much overlap is expected even before communication.

An overlap score without a reference can penalize legitimate standard reasoning. A reference that is too permissive can hide copied support. A reference formed from unrelated questions can underestimate legitimate overlap on the actual question. This is why question matching is central, not a cosmetic detail

in the calibration procedure.

The evidence you can observe may also be insufficient for a perfect distinction. Two identical traces could have been generated independently or copied. Without provenance, interventions or a reliable statistical reference, the text alone cannot reveal the true history with certainty. A good system should report an operational dependence proxy, not pretend to recover invisible causality perfectly.

Diversity is useful, but it is not the target

Answer diversity counts distinct final predictions. Lexical diversity counts differences in words. Semantic diversity measures differences in meaning or embeddings. Reasoning diversity concerns different inference paths. Information diversity concerns different evidence sources. Model diversity concerns backbones, providers or capabilities.

These quantities can disagree. Two lexical variants can mean the same thing. Two semantically distinct answers can both be wrong. Two different derivations can share a false premise. Several correct agents can converge on one canonical proof because the task admits very little useful variation.

ReConcile, Zhu, DAR, MoA and Smit show why diversity deserves attention. CAVE should test whether derivation-level information adds something beyond simpler diversity statistics. If an answer-diversity feature or whole-response embedding explains the same gain, the stronger chain-independence story has not yet earned its place.

Confidence, calibration and discrimination

Confidence is a reported or computed belief in an answer. Calibration asks whether stated probabilities correspond to observed frequencies. Discrimination asks whether a score ranks correct answers above incorrect answers. AUROC measures discrimination; ECE is a calibration summary. A score can rank well and still be poorly calibrated.

Verbal confidence and token likelihood are also different objects. High probability for a sequence of familiar words does not guarantee the truth of its conclusion. Confidence may be recomputed during discussion, while historical reliability may be estimated over many labeled questions. Neither tells you directly whether the current support was copied.

The maps expose these distinctions across ReConcile, Zhu, CAGE-CAL, Kuhn, SELENE and ConfLiar. Do not compare their scores as if every score were a calibrated probability of correctness, and never reinterpret such a score as a calibrated probability of independence without a separate validation.

Training-free needs a clear resource statement

A method can freeze the debating LLMs while training a classifier, fitting an EM model, learning a graph encoder, building a labeled memory bank or tuning thresholds on outcome labels. Those are different resource settings.

For this project, a strong statement would be: no fine-tuning of solvers, no newly trained dependence encoder or classifier, and no outcome-labeled fitting of the main chain estimator. A fixed pretrained encoder or extractor and an unlabeled empirical no-communication reference can still introduce substantial inference cost. State those dependencies instead of hiding them inside the phrase training-free.

The existing maps sometimes mark training-free because a paper implements a frozen baseline or ablation, even when its full proposed method trains an auxiliary component. The paper-directory resource line and the main-method comparison in this handbook prevent that broad marking convention from becoming a false description of the primary system.

03 | What each map tells you

Map 1: communication is a separate design axis

The first map shows that the literature includes independent sampling, iterative discussion, label/judge aggregation and hybrids that independently initialize before exchanging messages. A source can implement more than one setting through baselines or ablations. Topology entries are similarly nonexclusive.

For CAVE, the most useful implication is to isolate the aggregation layer from the debate engine. If you simultaneously change the model mix, prompts, number of rounds, topology, message budget and decision rule, an improved answer cannot be attributed cleanly to independence-aware evidence accounting.

Keep at least one simple, reproducible host protocol. Record all initial answers, messages, revisions, tools and final predictions. Then replay the identical record through majority, confidence weighting, a judge, semantic deduplication and CAVE. A second experiment can investigate how the layer transfers across protocols; it should not replace the first controlled comparison.

Useful protocol families include full broadcast, fixed sparse neighbors, sequential discussion, dynamic selected communication and token-level continuation. DySCo changes selected edges; Li-Sparse changes neighborhood structure; DAR changes retained messages; DOWN and SELENE change when or how long to deliberate; ConsTrap changes the collaboration unit to a token prefix. These interventions affect exposure and therefore the conditions under which overlap develops.

Map 2: final decisions use different evidence

Majority uses answer counts. Confidence-weighted rules attach belief to those answers. Reliability-aware rules incorporate estimated observer quality. Judges synthesize or adjudicate. Trajectory rules use changes over time. Fitted probabilistic rules use joint label patterns. Step/argument methods use local evidence or structural strength.

These rules should not be ranked by sophistication alone. They answer different questions and pay different costs. A trajectory rule can recover a minority without checking an inference. A trained auditor can evaluate a branch without a no-communication null. A likelihood-weighted rule can favor a fluent wrong path. A reliable observer can still share a decisive mistake with another reliable observer.

The full maps retain every mark exactly as delivered, including an explicitly disputed CoT cell in Map 3 row 45. Its X for extra inference LLM calls is not supported by the reviewed one-pass-plus-calculator description. This handbook does not use that mark as evidence for an extra LLM call. The original map is not silently corrected.

An especially useful comparison for CAVE is between answer-only rules and rules that use reasoning information. Answer-only aggregation establishes whether extra derivation processing is necessary. Whole-response embedding aggregation establishes whether step extraction is necessary. Step-level semantic deduplication establishes whether null calibration is necessary. Each comparison removes one simpler explanation for an apparent gain.

Map 3: the most important gap and its limits

Among the 37 source-identifiable entries, the marked map records step/derivation extraction for AgentAuditor, Slim-SC, ARGORA and Lightman. These entries represent very different units and purposes: atomic reasoning alignment, delimiter-separated thoughts, argument nodes and supervised process steps. Treating them as one identical technique would hide the most important distinctions.

The marked null-calibration row has one X, Goel. That X concerns chance-adjusted model-output agreement, not a complete per-question derivation-null pipeline. CAGE-CAL has matched no-communication controls and centered correctness correlations, but the map distinguishes these from a calibrated distribution of expected reasoning overlap. The marked honest-shared-versus-copied separation row has no X among identified entries.

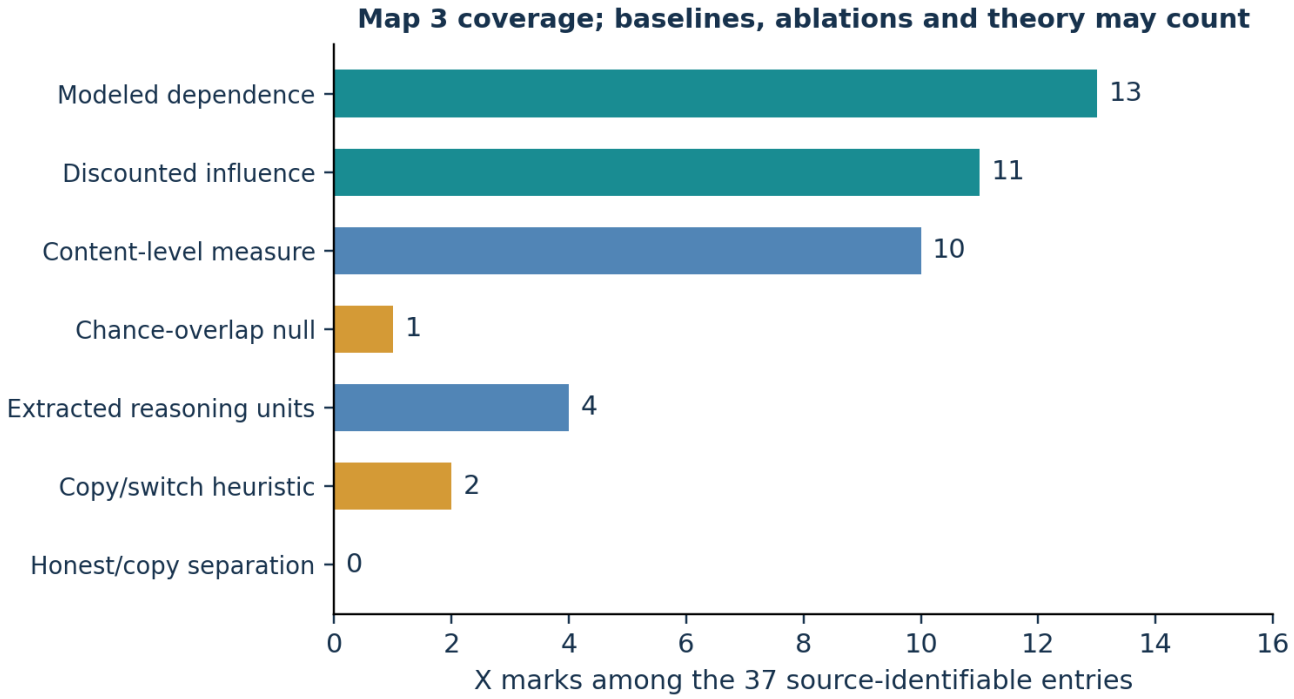


Figure 5. Positive archival marks in Map 3. A theory result, baseline or ablation may qualify; this is not a main-method-only census of all literature.

This supports a corpus-limited research opportunity: combine operational reasoning units, a task-matched honest-solver reference and effective support accounting. It does not prove global novelty. The unresolved aliases include authors and topics potentially important for correlated voting, ensemble diversity and source copying. Broader literature can also contain a relevant combination under a different name.

There is a second limit. The map's modeled-dependence row combines analytical models, learned models and implemented variants. The discounted-dependence row combines different mechanisms and levels. A row total is descriptive of this marked corpus, not a count of interchangeable operational systems.

Map 4: evaluation must test the mechanism

Math provides objective answers and some executable checking. Knowledge and commonsense QA reveal shared misconceptions and persuasive mistakes. Code allows execution-based outcome checks, with the risk that test suites are incomplete. Forecasting emphasizes shared evidence and temporal leakage. Human-rated evaluation exposes judge preferences and positioning effects.

Known-dependence controls are especially important because normal accuracy benchmarks do not reveal the true copying or shared-information mechanism. The map includes controlled information, corruption or reference interventions as well as explicitly generative statistical simulations. These are not all the same strength of ground truth. A topology intervention knows who could communicate; it does not necessarily know which reasoning step was copied.

The last map was deliberately completed as a rapid pass. Its code, citation and complete-reading cells include many question marks. These are bibliographic or process gaps, not findings that the methods lack reasoning or evaluation. Citation standing is not used to assign importance in this report. The phrase read in full is not inferred from access to a complete PDF or from a detailed source note.

What the combined maps say

The literature already provides many ways to improve a group: better initialization, model diversity, selective messages, trusted peers, calibrated confidence, stronger adjudicators, graph representations and different collaboration units. The useful claim for CAVE cannot simply be that votes can be correlated or that reasoning should be examined.

The more specific target is excess overlap at a meaningful derivation level, relative to what comparable solvers would share without communication, followed by a transparent support rule. The decisive experiment is whether this extra structure improves dependence handling and decision quality beyond answer-only weighting, whole-trace geometry and uncalibrated step deduplication.

04 | The closest comparisons and the novelty boundary

Start with the neighbors that can invalidate a weak claim

AgentAuditor already atomizes traces, aligns equivalent reasoning and audits branches. CAGE-CAL already uses a matched no-communication reference for dependence-sensitive answer calibration. DarkForest already discounts correlated answer support and restricts coordinator disclosure. Ising already explains why independent-vote models miss useful joint structure. PPV already uses reasoning geometry for aggregation. Slim-SC already removes thought-level semantic redundancy. ARGORA already represents and weights argument structure.

A literature review that mentions only vanilla debate would miss these comparisons. They should appear explicitly in the thesis problem statement, related work and experimental rationale. Their proximity is useful: it makes the project's distinction precise and supplies meaningful baselines.

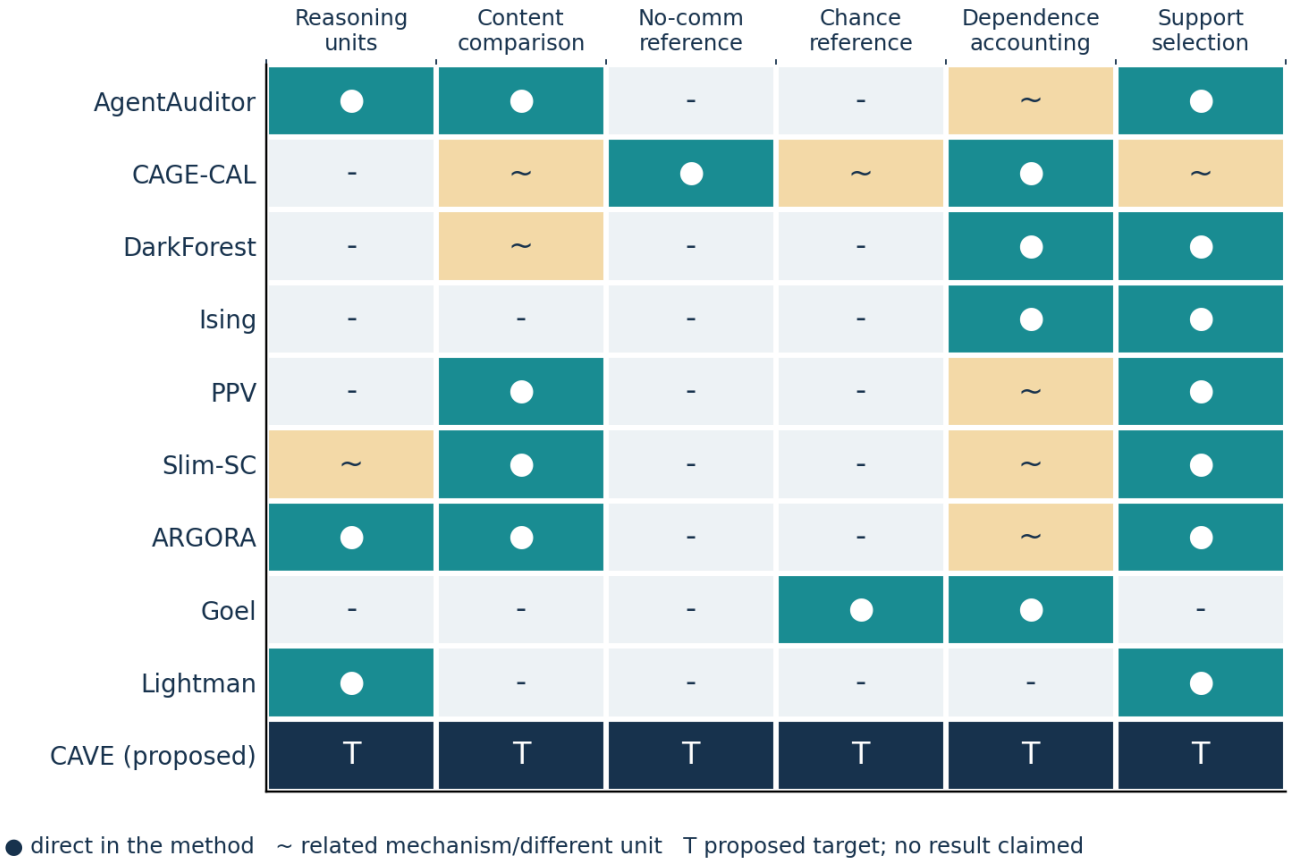


Figure 6. Qualitative main-method comparison from the source notes. A tilde denotes a related mechanism or different unit/target; CAVE entries are proposed goals.

A comparison needs four questions

First, what is the unit: label, full response, chunk, inference step, argument or complete chain? Second, what is the target: correctness, confidence, agreement, redundancy, copying proxy or conditional error dependence? Third, what supervision or fitting is required? Fourth, what does the decision rule actually count or score?

For example, a semantic step tree can improve adjudication while leaving independent-support estimation undefined. A dependence graph can use correctness labels across nearby questions without reading derivations. A process reward model can verify mathematical steps while requiring supervised training. A response-pruning method can remove duplicate thoughts while lacking an honest-overlap reference.

These are substantive distinctions. They should replace broad statements such as other methods ignore reasoning or nobody considers dependence. The maps themselves contradict those broad statements.

A defensible preliminary contribution statement

A careful statement is: we study a training-free, per-question aggregation layer that models overlap between observable supporting derivations relative to a matched no-communication reference, then evaluates whether dependence-adjusted support improves decisions under controlled redundant-evidence conditions and matched compute budgets.

This is a proposal, not a completed claim of novelty or empirical success. It becomes a research contribution only after the representation, statistic, decision rule, baselines and evaluation are specified and the experiments support the stated improvement.

Claims that need to be softened

Do not claim to detect true copying from text alone. Say excess-overlap or dependence proxy unless provenance ground truth is available. Do not claim all common reasoning is dependent in the harmful sense. Do not claim confidence is useless: some papers improve it through calibration or use it productively in restricted settings. Do not claim debate never helps: observed results vary by task, capability mix, prompts and protocol.

Do not call every frozen-model system training-free in the same sense. Do not call every sparse communication graph a reasoning graph. Do not call answer entropy a chain-independence measure. Do not label simple centering or a confidence threshold a null calibration without identifying its null population and target statistic.

What a reviewer would reasonably ask

What information does CAVE use that a strong dedup baseline lacks? What does the no-communication reference preserve? What does it remove? How many independent generations are needed? Does the method protect honest shared derivations? Can it recognize the same flawed premise after paraphrasing? Does it improve the same recorded answer pool, or did an extra call generate a new answer? Does performance transfer to another extractor, model family and topology? Can the apparent gain be reproduced by adding the same budget to self-consistency?

The design and evaluation chapters below answer these as experimental requirements. The full-map and evidence appendices let you locate the literature that motivates each requirement.

05 | A proposed CAVE-EoT specification

Define the output before the algorithm

A useful output record should contain the selected answer, its visible supporting chains, the raw answer counts, a dependence-adjusted support score, the overlap reference used, verification results and diagnostic warnings. The score should be described as an operational support score unless it has been separately calibrated as a probability.

Do not report a precise number of true independent reasons merely because the estimator returns a real number. A structural cluster count, a variance-equivalent count and a reference-relative support credit have different meanings. Name the quantity your implementation actually produces.

The proposal here has two views. The structural view describes unique visible supporting paths and shared decisive steps. The statistical view describes how unusual their overlap is relative to comparable no-communication solvers. They are complementary descriptions; a single scalar should not erase the difference between them.

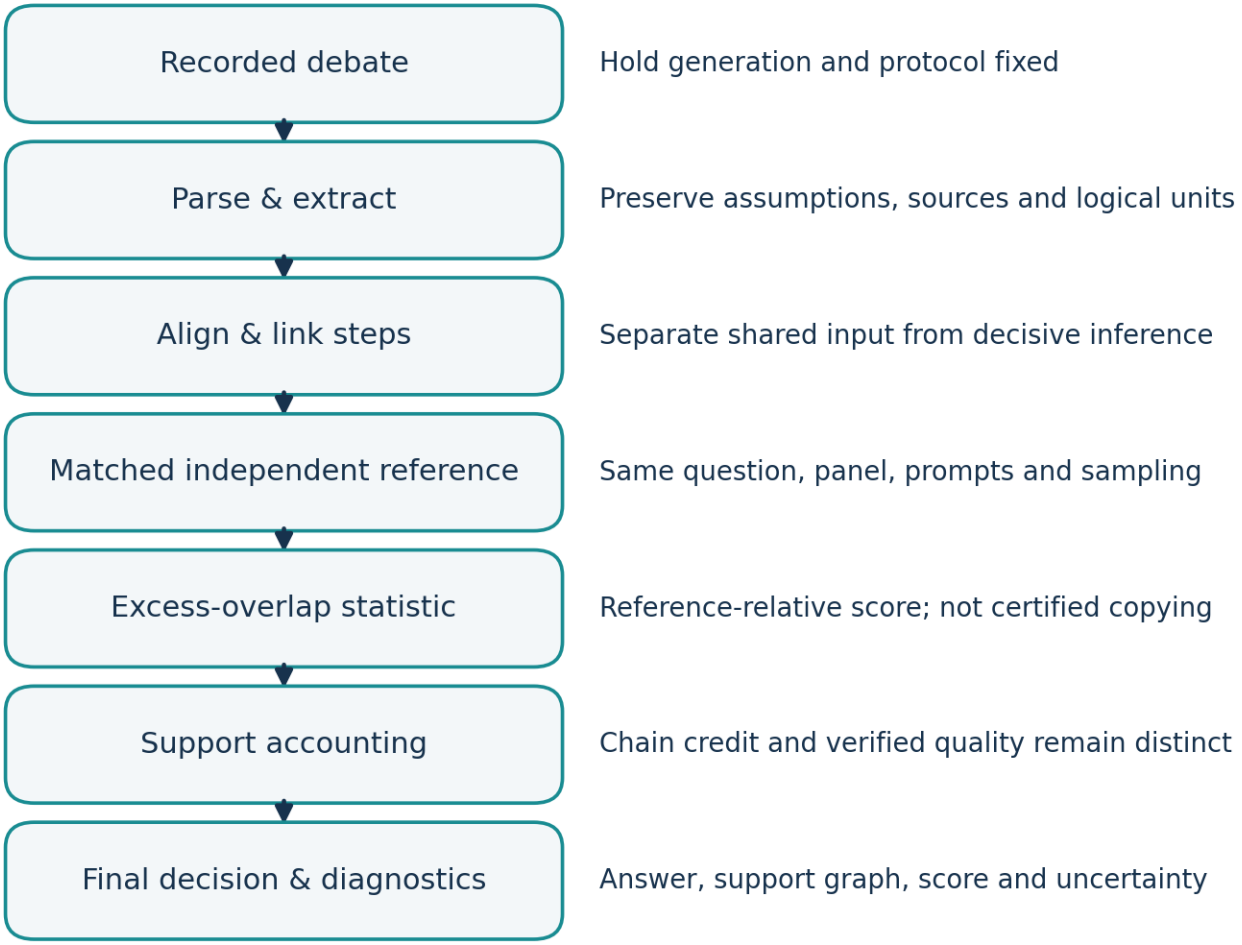


Figure 7. Proposed CAVE analysis pipeline. Every stage needs an inspectable record; no validated CAVE implementation is claimed.

Stage A: record the debate

Store the complete question, task constraints, model identity, role, prompt version, generation settings, round number, parent messages, tool outputs, token counts and final answer. Give every message and agent a stable identifier. Preserve the pre-discussion responses separately.

This makes exposure observable: an agent cannot have copied a peer message it did not receive through the recorded protocol, although both agents might still share an external source or prior knowledge. Exposure is useful causal information, not proof that copying happened. A received message may be ignored or critically rejected.

If answers are extracted from long messages, keep both the raw output and parsed value. A parser failure should be visible. Otherwise an invalid format can masquerade as a disagreement, a vote change or a failed estimator.

Stage B: extract meaningful support units

An extraction unit should specify its claim, premises, conclusion, local inference type, assumptions, relevant source IDs and supporting text span. It should say whether the unit is a problem restatement, a factual premise, a derived inference, a contradiction or a final answer. The original span is essential for auditing the extractor.

A pragmatic first version can use a frozen extractor with a strict schema, plus deterministic parsing for equations and answer fields. A second version can reduce extractor calls through delimiter or sentence heuristics, but it should not claim equal logical fidelity without testing that assumption.

Do not force every sentence into an independent step. Merge continuations that jointly express one inference. Split compound sentences when they contain distinct premises and conclusions. Preserve negation, units, domain restrictions and quantifiers. The difference between $x = 3$ and $x =$ plus-or-minus 3 is not a stylistic difference.

Stage C: represent and compare support

Construct a local graph for each response. Nodes are support units; edges indicate that one premise or inference supports another. A response can have several valid paths to the same answer. If the extractor proposes a graph, retain uncertainty about its edges instead of treating them as ground truth.

Use embeddings to retrieve candidate matches, then apply a stricter comparison to important pairs. Similar vocabulary is weak evidence. Matching should consider the proposition, assumptions, conclusion and logical role. When both units contain comparable symbolic expressions, normalized equations or executable checks can help. For open-ended factual reasoning, source IDs and explicit entailment checks may matter more.

Separate common task input from decisive support. Two agents repeating the question do not add two reasons. Two agents relying on the same decisive source claim may share support even when the surrounding language differs. A generic introductory sentence can dominate an embedding if the representation is poorly controlled.

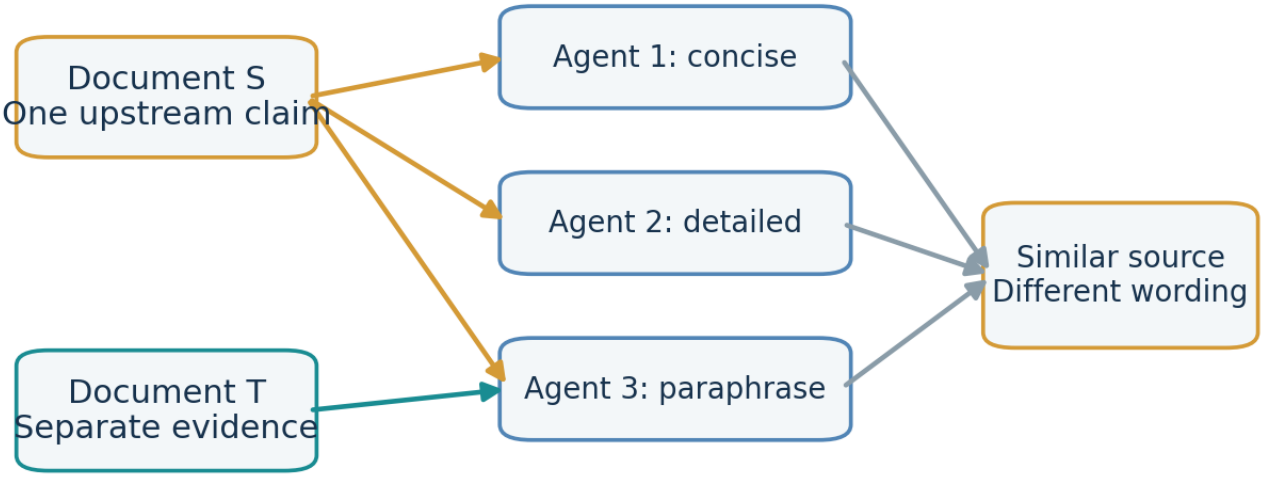


Figure 8. Different wording can share an upstream source. The second source illustrates a partially shared support path.

Stage D: build the honest-solver reference

For question q, independently generate comparable solver panels without peer communication. Match the models, roles, question, permitted sources, sampling settings, output format and approximate reasoning budget. Compute the same overlap statistic on these panels as on the observed debate.

The reference answers: how much overlap would occur under this particular independent-generation protocol? It does not answer how much overlap any perfectly independent mind would have. Models can share priors and sources even in the no-communication condition. That shared baseline belongs in the interpretation.

One initial independent panel is useful but usually insufficient to estimate a distribution reliably. Repeated panels provide variation. They also cost real tokens. Reusing cached independent starts helps, provided the cache is matched to the question and generation conditions. A reference from another domain, another model mixture or a different rationale length can distort the comparison.

Stage E: quantify excess overlap

Let S_obs be an observed similarity or overlap statistic, and let S_1 through S_B be its matched no-communication values. Possible summaries include S_obs minus the null mean, an empirical upper-tail rank or a bounded excess-overlap score. Each has different sensitivity and interpretation.

A simple descriptive bounded score for a statistic in [0,1] is:

```
excess = max(0, (S_obs - mean(S_null)) / (1 - mean(S_null) + epsilon))
```

This is a proposed heuristic normalization, not a validated probability of copying. It becomes unstable or uninformative when the reference already lies near one, and epsilon should not silently decide the result. Show the raw values and reference spread alongside the normalized score.

For an empirical one-sided test, a familiar conservative Monte Carlo form is (1 + b)/(B + 1), where b is the number of null values at least as extreme as the observed statistic. Its testing interpretation requires the correct null and appropriate exchangeability. Zero exceedances do not justify a p-value of zero. Phipson and Smyth provide methodological background on randomly drawn permutation/Monte Carlo p-values [Phipson-Smyth].

A z-score can be a descriptive feature, but should not automatically be interpreted through a Gaussian tail. If the null variance is near zero, report that condition or use an empirical/range rule; do not manufacture certainty by dividing by an arbitrary tiny constant. The reference design matters more than decorating a raw cosine with statistical notation.

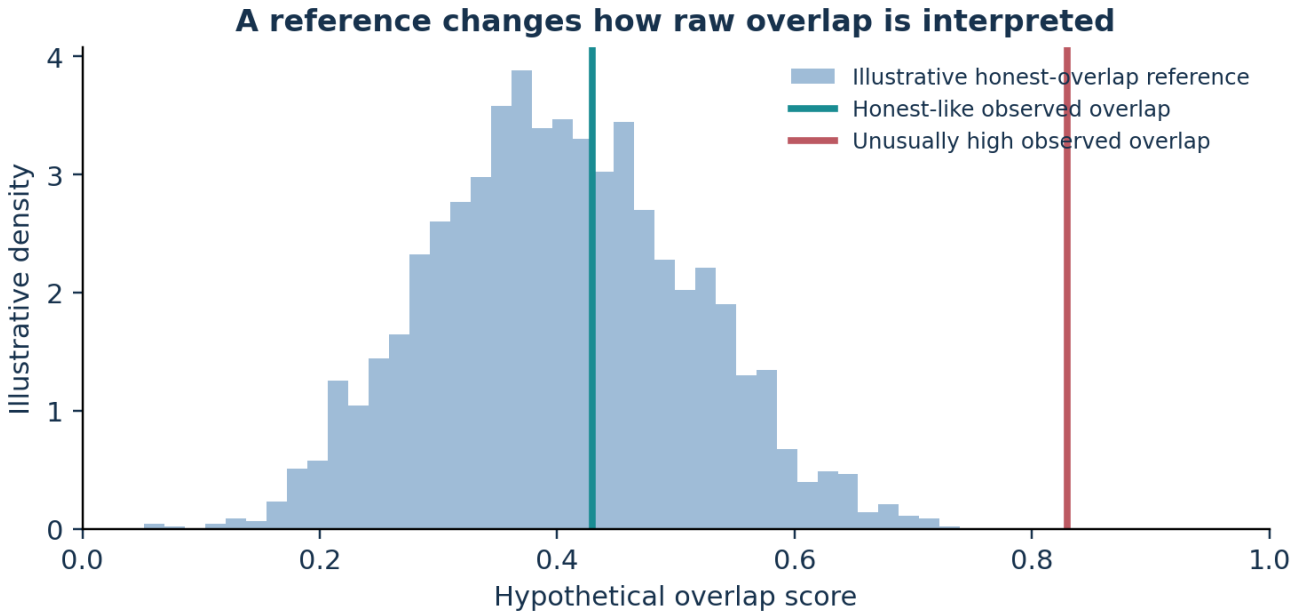


Figure 9. Hypothetical scores and a simulated illustrative reference. This is an explanation of interpretation, not an empirical test result.

Stage F: turn overlap into support credit

The aggregation rule should preserve the distinction between truth quality and dependence. One possible design assigns a quality term to each supporting path and discounts its contribution when it shares unusually strong decisive support with already-accounted paths. Another groups structurally redundant paths before scoring their verified support. Both need clear rules for partial sharing.

An illustrative score for answer a is a sum over its support groups: group credit times evidence quality. The exact credit function is part of the research problem. It should be bounded, auditable and accompanied by sensitivity experiments. A repeated claim should not obtain unlimited influence merely because another agent restates it at length.

Be careful with transitive similarity. If A resembles B, and B resembles C, connected components can merge A and C even when they express incompatible inferences. A graph component is not automatically a semantic equivalence class. Consider stricter complete-link criteria, explicit incompatibility checks or local graph alignment, and report the resulting tradeoff.

Likewise, counting every unique step can reward verbosity. Count coherent answer-supporting paths or carefully normalized support groups; distinguish a path's decisive information from decorative intermediate text. Several unique but irrelevant steps must not overpower one valid derivation.

Stage G: decide and expose the evidence

Select the answer with the strongest defined support score, or abstain when evidence is insufficient under an explicitly specified policy. Expose why the score differs from the raw vote: which paths share support, what the reference expected, which steps were checked and where the extraction remained ambiguous.

For a research prototype, returning a traceable explanation matters more than returning a visually persuasive confidence number. The analysis should let a reviewer inspect a failure at the level of the raw message, extracted step, match, null value, group credit and final decision.

A crucial limitation: the baseline can contain shared mistakes

If independent no-communication solvers all use the same familiar misconception, high overlap may be normal under the reference. An excess-overlap detector can then correctly report little communication-induced excess while the whole group remains wrong. This is not necessarily an estimator bug; it exposes the difference between baseline shared knowledge errors and added dependence from the current interaction.

To handle that case, CAVE may need a separate quality/verification channel, an upstream-source analysis or a broader model-error diagnostic. Do not force the same null statistic to solve all three problems. Goel, Ising, DivEvid and Lightman illuminate different parts of this separation.

06 | Worked examples and counterexamples

Example 1: the wrong majority repeats one flaw

Consider x-squared = 9. Four agents conclude x = 3 only, sharing the same missing-sign argument. Two others conclude x = plus-or-minus 3: one factorizes the polynomial and one reasons through absolute value. In a controlled test with known derivation origins, the four wrong responses trace back to one flawed supporting path.

Raw votes favor the wrong answer four to two. Structurally distinct supporting paths can favor the correct answer two to one. A symbolic checker may also expose the missing solution. These are separate explanations for a better decision; the experiment must include a verification-only baseline so a simple calculator or solver is not mistaken for an independence contribution.

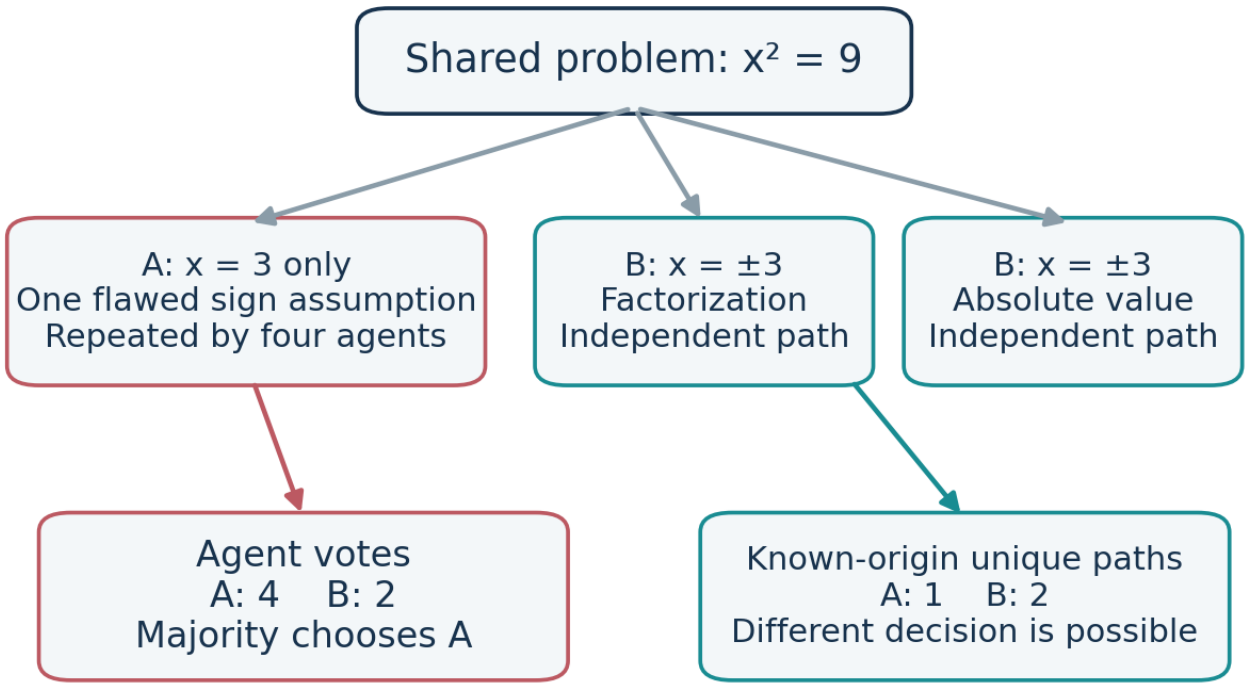


Figure 10. A controlled, known-origin toy case. Symbolic verification is a separate possible explanation for improvement and needs its own baseline.

The example also tests semantic precision. The wrong and correct square-root statements may be very close in an embedding space. The decisive sign/domain condition must survive extraction and matching. A method that merges them because they look similar can create a new error while trying to remove redundancy.

Example 2: every honest solver uses the same formula

A rectangle has sides 6 and 5. Each solver independently calculates area = 6 times 5 = 30. The reasoning overlap is high because the task is simple and the standard calculation is unavoidable.

An unconditional similarity penalty can incorrectly treat all but one solver as useless. A matched reference should reveal that this amount of overlap is unsurprising. However, the visible structural proof is still largely the same. This is why independent validation credit and structurally distinct proof count should not be conflated. Decide which quantity the experiment intends to recover.

The desirable behavior is not a universally fixed count. It is that the system does not invent suspicious copying, does not create extra distinct derivations from repeated wording, and does not let the choice of language style determine the answer.

Example 3: paraphrase hides the same source

Agent 1 says a study establishes claim P. Agent 2 writes a long explanation about P. Agent 3 uses a different sentence structure but relies on the same study and inference. Lexical diversity is high, but the decisive information source is shared.

A source-aware representation can reveal that shared root. An embedding-only method may or may not detect it. A no-communication baseline must match the sources available to the solvers; if every honest solver receives that same study, the overlap may be expected even without interaction.

For controlled validation, preserve source IDs and author the copy mechanism deliberately. Do not infer a real-world plagiarism or copying accusation merely from semantic resemblance. The purpose is evidence accounting under a known experimental mechanism.

Example 4: a copied path adds one useful fact

A peer repeats another agent's argument but adds a genuinely new constraint that repairs a missing assumption. Treating the entire response as a zero-value copy loses useful information. Treating it as a fully independent new chain double-counts the inherited support.

A step graph should represent the shared subpath and the additional fact separately. The decision rule then needs a partial-credit policy. Test it with known shared and novel components, rather than choosing a credit formula because it looks plausible on one attractive example.

Example 5: diverse independent errors

Five solvers produce five different wrong answers using different arguments. Independence or diversity alone cannot recover the truth. An evidence-aware rule may still choose a wrong answer because the candidate pool lacks a correct supported path.

Report candidate coverage: whether a correct answer or valid derivation existed anywhere in the pool. Separate failures of generation from failures of selection. If no valid candidate exists, an aggregator's failure cannot demonstrate that dependence adjustment was useless; equally, an aggregator must not claim it generated a correct answer unless an extra generation step actually did so.

Example 6: agreement increases while evidence decreases

During debate, a correct minority abandons its independent derivation after hearing a confident but unsupported majority. The final answers become more consistent, while the variety of defensible support shrinks. A simple terminal vote may hide the lost evidence.

Record intermediate states and provenance. Compare terminal-only aggregation with trajectory-aware aggregation such as Free-MAD and with CAVE's actual chosen history scope. If CAVE uses the whole history, a terminal-only baseline is not the only fair comparator.

Example 7: a judge changes the problem

A strong judge can produce a new answer that was not supported by any original candidate. This may improve accuracy, but it changes the task from selecting support to generating a new solution. The cost and capability of that judge can dominate the experiment.

Create two conditions. In selection mode, the judge must choose among existing answer clusters and cite available support. In synthesis mode, it may propose a new answer, and the experiment logs that extra generation separately. Label these conditions so a reviewer can interpret the source of improvement.

07 | An evaluation plan that can defend the mechanism

Use two complementary experimental tracks

Track A uses controlled support origins, sharing and corruption. Its purpose is to test the dependence/redundancy mechanism. Track B uses ordinary reasoning benchmarks with objective or human-verified outcomes. Its purpose is to test useful decision quality and cost. Neither track replaces the other.

In Track A, construct original derivations and deliberately duplicate or paraphrase some of them. Keep the source links so the underlying structure is known. Vary the number of repeated agents, fraction of shared decisive steps, amount of new evidence, correctness of the inherited premise and communication exposure.

In Track B, use a small task suite that includes objective mathematical answers and at least one less canonical reasoning/QA setting. Add code or forecasting only when their outcome checks and evidence provenance can be managed well. Expanding domains before the estimator is understood often increases noise rather than convincing evidence.

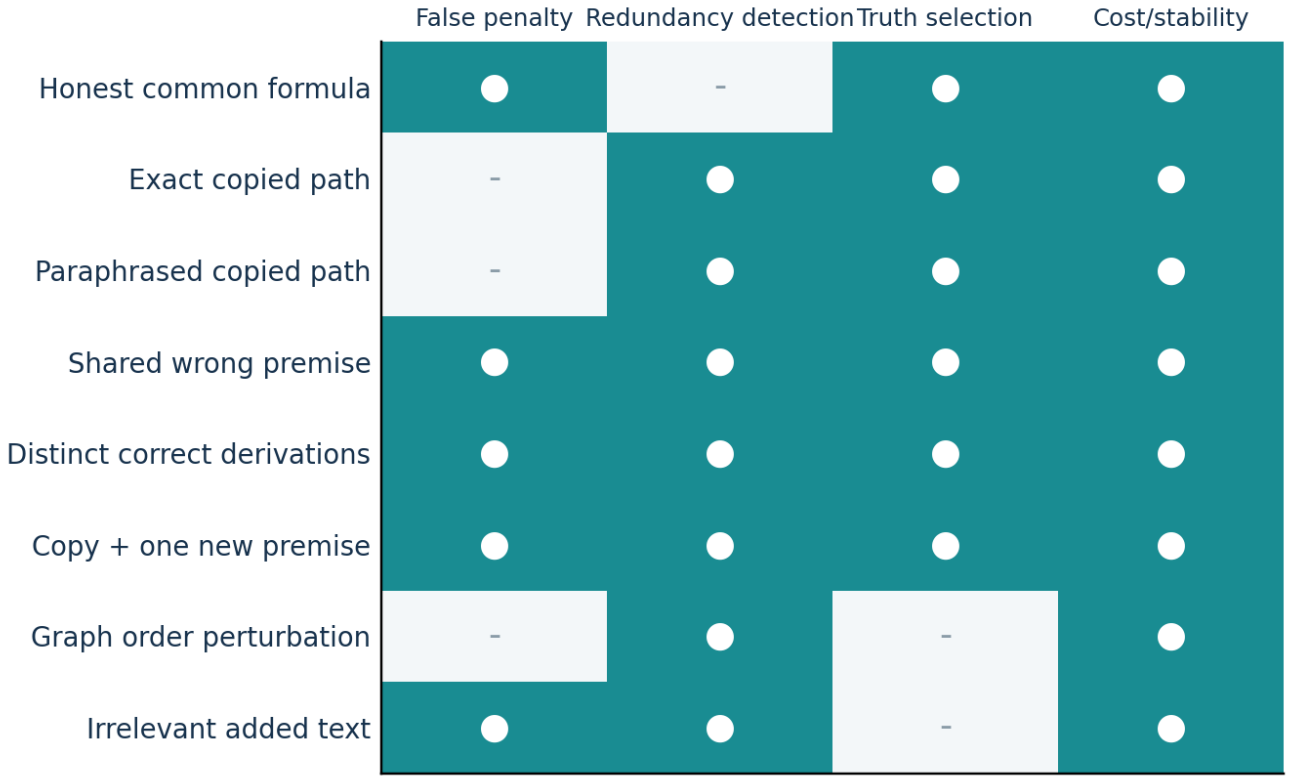


Figure 11. Proposed controlled tests and the kinds of failure they can diagnose.

The controls that should exist early

A no-sharing independent condition tests false penalties on honest solvers. An exact-copy condition tests obvious redundancy. A paraphrase-copy condition tests surface invariance. A legitimate-common-step condition tests task-specific expected overlap. A shared-wrong-premise condition tests whether the baseline itself can contain correlated misconceptions. A partially shared condition tests graded credit. An exposure control tests whether a suspected copied step could have arrived through the recorded protocol.

Also perturb graph order, response length, agent names and paraphrase style while preserving the argument. The answer and dependence description should not change radically merely because an irrelevant introduction became longer. A wrong negation or domain change, in contrast, should change the logical match.

The controls need not be enormous. A carefully labeled suite of interpretable cases can reveal a broken matching or credit rule before expensive large-model experiments. Keep synthetic mechanisms explicit so they are not advertised as ordinary real-world prevalence estimates.

Baselines in layers

Start with single CoT and equal-budget self-consistency. Add independent majority, vanilla debate majority, confidence-weighted aggregation and a frozen LLM judge. Add whole-response embedding similarity/delegation and uncalibrated step deduplication. Add a simple answer-level reliability or higher-order label rule where it is practical. Include a trajectory rule if CAVE uses history.

For the closest structural comparison, a frozen AgentAuditor-like branch audit or ARGORA-like argument representation can be more realistic for a final-year project than retraining a published large auxiliary model. Make clear that a simplified implementation is not a complete reproduction of the published method. Provide a faithful description of what was retained and changed.

Run ablations that remove one CAVE ingredient: no extraction, answer-only overlap, whole-trace embeddings, no null, unmatched null, no quality check, no chain grouping, terminal-only history and no dependence discount. These ablations answer why the new machinery is needed.

Separate generation from aggregation

Use one recorded debate as a shared input to all post-hoc rules. This is a paired comparison: each rule faces the same question, answers and messages. It isolates selection differences. Then run an end-to-end comparison where protocols can generate different candidates, with total cost and candidate coverage measured.

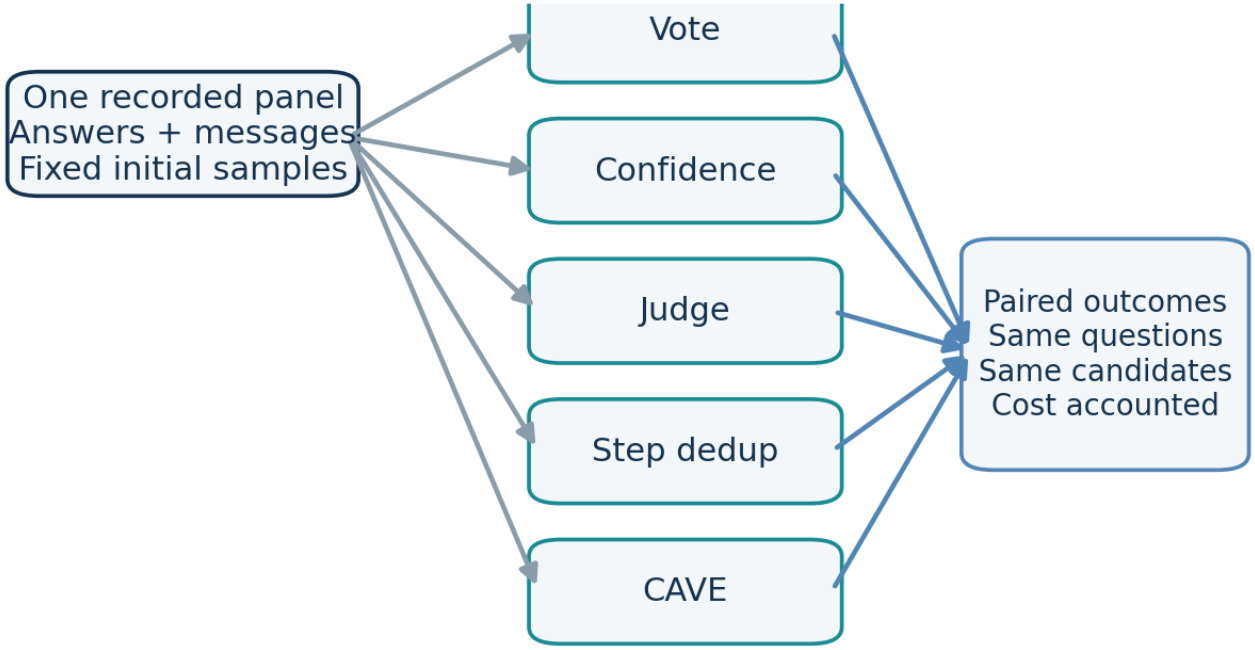


Figure 12. A paired replay experiment compares aggregators on the same candidate pool before changing the debate protocol.

Do not let CAVE silently inspect ground truth to choose a path, fit a threshold on the test set, or use a stronger model than the baselines without labeling that condition. Do not compare a three-sample baseline with a CAVE pipeline that secretly generates dozens of null samples and call the result budget-matched.

Metrics for decision quality

Report final accuracy or the task's proper outcome metric. Report candidate coverage and accuracy conditional on a correct candidate being available. Report correct-majority preservation and correct-minority recovery. For changed decisions, report flip precision: the fraction of flips that improve the answer under the chosen definition. Report helpful and harmful flips separately and their net effect.

If the method can abstain, report both coverage and accuracy/risk on accepted questions. If it emits confidence, report calibration and discrimination separately. Do not call an arbitrary support score a probability just because it falls between zero and one.

Metrics for the dependence target

When the sharing mechanism is known, measure whether the estimator detects redundant/shared support, how often it penalizes legitimate independent/common reasoning, and whether its support credit tracks the known controlled structure. Evaluate exact copies, semantic copies, partial sharing and independently repeated canonical derivations separately.

A scalar area-under-curve can summarize discrimination, but the failure categories are more informative. Show false positives on honest common steps and false negatives on paraphrased shared assumptions. Compare the quality of a recovered step graph with a small human-inspected subset. Independence estimation errors can originate in extraction rather than in the null statistic itself.

Metrics for cost and stability

Count generated and consumed tokens for solvers, debate messages, extractors, judges, null-reference generation and any verification calls. Report embedding and graph-processing time, memory use where important, and total wall-clock latency under the actual hardware and parallelism. Call counts alone do not imply equal token or time costs.

Repeat across seeds and report uncertainty on paired outcomes. Use question-level resampling when the scientific unit is the question; multiple traces from one question should not masquerade as independent test questions. Predefine primary comparisons. Treat large collections of exploratory feature tests as exploratory, with their testing choices visible.

Do not rank papers by unrelated headline scores

An accuracy on one dataset, a win rate on an open-ended preference benchmark and an AUROC on confidence discrimination are different metrics. A reported token reduction can use a different input/output accounting boundary. A stronger model panel can dominate an apparent method advantage. The maps help locate these differences; they do not make the numbers interchangeable.

For the thesis, compare your own reproduced methods under a common setup. Use published numbers as context with their task and comparator stated, not as a leaderboard declaring one paper universally best.

08 | Build a research framework that can explain its failures

Modules with a stable boundary

Use separate modules for agent management, the debate protocol, answer parsing, step extraction, graph construction, overlap/reference estimation, support accounting, answer selection and evaluation. Store the output of each stage. A failure should be traceable to a module rather than hidden inside one long script.

The most valuable engineering decision is replay. A recorded panel can be evaluated many times without generating the discussion again. This allows new match thresholds, a different support rule or a corrected parser to be tested against the same raw evidence. Keep generation and aggregation experiment IDs separate.

Python, a model-serving layer such as vLLM, Hugging Face Transformers, a sentence-embedding toolkit, SymPy and pandas fit these roles. vLLM documents serving/inference; Transformers documents pretrained-model use; Sentence Transformers documents embeddings; SymPy documents symbolic mathematics. These tools do not supply validated independence estimation by themselves [Toolkit].

A minimal record

The schema below is illustrative. It is a proposed research interface, not a report of an existing CAVE implementation.

```
{
  "question_id": "q017",
  "agent_id": "agent2",
  "model_id": "pinned-model-revision",
  "round": 1,
  "parent_message_ids": ["m03", "m04"],
  "raw_text": "...",
  "parsed_answer": "B",
  "steps": [
    {
      "step_id": "s2",
      "kind": "inference",
      "premise_ids": ["s1"],
      "conclusion": "...",
      "assumptions": ["..."],
      "source_ids": ["docA"],
      "text_span": [48, 119]
    }
  ],
  "token_usage": {"input": 0, "output": 0}
}
```

A graph-match record should additionally preserve the compared step IDs, raw similarity features, logical compatibility result, reference statistic, threshold/rule version and the final decision about the pair. Use schema validation. Invalid extraction should be counted and visible, not automatically filled with plausible premises.

A high-level algorithm

The pseudocode separates candidate generation from the proposed analysis. Each function needs its own documented assumptions and checks.

```
record = load_recorded_panel(question_id)
```

```
answers = normalize_answers(record)
steps = extract_observable_steps(record)
graphs = link_local_support(steps)
matches = propose_and_validate_matches(graphs)
null_panels = load_or_generate_matched_independent_panels(
    question_id, models, roles, prompts, sampling, budget
)
null_stats = measure_same_overlap_statistic(null_panels)
dependence = compare_with_reference(matches, null_stats)
quality = verify_supported_units(graphs, allowed_tools)
credits = assign_support_credit(graphs, dependence, quality)
result = select_answer(answers, credits)
save_all_intermediate_records(result)
```

This is intentionally not a claim that one specific credit formula is already correct. The scientific work includes testing whether step extraction, the comparison statistic and the reference actually recover the intended distinction.

Keep the null cost visible

With three agents and two exchange rounds, a simple synchronous debate uses three initial calls plus six revision calls: nine generation calls. Three final-trace extraction calls bring that illustrative count to twelve. Nineteen fresh independent null panels add fifty-seven solver calls, bringing the count to sixty-nine before any additional judge verification.

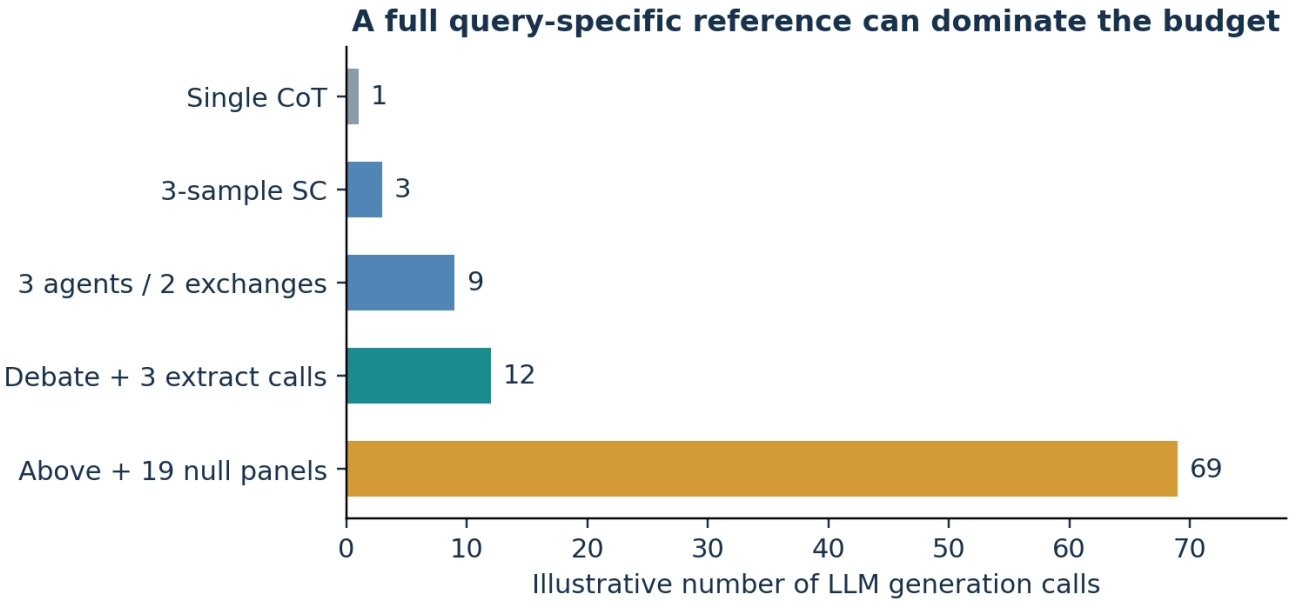


Figure 13. Illustrative generation-call accounting. Tokens, wall time and parallelism must still be measured.

These are illustrative call counts, not actual token or latency benchmarks. Calls may have very different lengths and run in parallel. Nevertheless, the example reveals a real design pressure: query-specific null generation can dominate the budget. A sophisticated statistic with an unaffordable reference is not automatically a useful inference-time layer.

Study cost tiers. A small reference can provide a descriptive excess score but has limited statistical resolution. A larger reference can support stronger empirical-tail interpretation but costs more. An approximate reusable/task-conditioned reference is cheaper but changes the validity question and must be evaluated as an approximation. Report both the full reference and practical approximation, rather than letting a cheap proxy inherit the full method's claim.

If B = 19 and the empirical-tail count b = 0, (1 + b)/(B + 1) is 0.05. With B = 7, the smallest such value is 0.125. Small reference sets cannot express arbitrarily strong tail evidence. More digits in the output

do not improve that resolution.

Reproducibility records

Pin model and tokenizer revisions, prompt templates, sampling settings, dataset versions and source cutoffs. Save random seeds and document which operations remain nondeterministic. Include the extractor and embedding model identity, even if both are frozen. Report cached versus fresh calls and how parallelism affects latency.

Store outcomes in a tidy table keyed by question, panel, protocol and aggregation rule. Keep raw messages and graphs in structured files linked by IDs. The evaluation script should reproduce tables from those records, not from manually copied notebook cells.

09 | Failure modes your system should expect

Extraction and alignment failures

An extractor may omit a decisive premise, invent an implicit assumption, split one inference into several apparent votes, merge incompatible claims, lose a negation or attach a source to the wrong step. An embedding may match two similar-looking but contradictory expressions or miss a paraphrased shared premise.

Test extraction separately on a small audited subset. Measure how errors propagate to overlap and final decisions. A good final answer on an easy task can hide a poor representation; a poor final answer can result from generation even when extraction is accurate. Inspect both successful and failed cases.

The graph can create false confidence

A clean-looking graph may be an extractor's interpretation rather than a faithful structure generated by the solver. Edge direction can be uncertain. A path can omit the unsupported leap on which it depends. A large connected component can be caused by transitive similarity, not a single meaningful argument.

Return graph uncertainty and original spans where possible. Provide an audit view showing the graph beside the raw text. Do not advertise a precise reasoning DAG as a recovered internal computation if only natural-language outputs were observed.

Quality and independence can conflict

A repeated valid proof can be better than several independent invalid ones. A weak but unique claim should not receive a bonus merely because nobody else said it. A false source can be independently cited by many agents. A copied argument can be correct.

Separate the quality rule from the dependence rule. This lets you explain whether a failure came from false verification, wrong overlap interpretation or a poor combination of the two. Lightman illustrates quality verification; Goel/Ising illustrate dependence; neither automatically solves the other target.

Observable reasoning is not guaranteed internal reasoning

Turpin and colleagues report that CoT explanations can rationalize biased answers without acknowledging the biasing input. Lanham and colleagues use interventions on reasoning text and find that models vary in how much their final predictions depend on it. These additional sources are not columns in the four maps; they are methodological background [Faithfulness].

For CAVE, define the object as observable supporting derivations. Testing intervention sensitivity and verification can strengthen the interpretation, but similarity between explanations alone does not reveal internal causal reasoning. This limit belongs in the thesis definition and discussion, not only in a footnote after a strong causal claim.

The no-communication reference can be wrong

A mismatched task, source pool, model mix, role prompt, temperature or reasoning length can change the overlap distribution. A reference containing leaked peer messages is not a no-communication reference. A bank selected using test outcomes creates an additional information advantage.

Stress-test reference mismatch deliberately. Compare exact question-matched panels, nearby-question references and cross-domain references. If performance changes dramatically, the result tells you which conditioning variables are scientifically important. Report that dependence instead of claiming a universal threshold.

The aggregation score can reward artifacts

More steps can create more apparent evidence. Longer explanations can create more matches. A particular agent can dominate the reference if it generates many samples. One answer can gain credit through diverse but irrelevant subclaims. A graph threshold can turn small representation changes into large discrete grouping changes.

Normalize for the scientific unit being counted. Cap repeated contribution from the same path or source where the chosen definition requires it. Test length/style perturbations and the order in which paths are credited. If sequential crediting depends strongly on order, the score needs a symmetric formulation or an explicit order-sensitivity analysis.

Majority override is a precision problem

A minority can be correct, but most minorities are not automatically correct. Aggressive overturning can erase correct majorities. Measure the safety of the rule: beneficial flips, harmful flips and preservation of originally correct decisions. MinSentinel is especially relevant for that evaluation discipline.

Do not optimize the override threshold on the test set. If a labeled development set is used to tune a practical rule, state that resource explicitly. A more conservative method can be preferable if its small gain is stable and its false overrides are understood.

10 | Turn the research into a defensible final-year thesis

Research questions that have separate answers

RQ1: Does the proposed reference-relative overlap statistic distinguish controlled redundant support from legitimate common reasoning better than uncalibrated similarity? RQ2: Does dependence-adjusted support improve decisions on the same recorded candidate pool? RQ3: What extra cost and representation errors does the layer introduce? RQ4: Does the result transfer across model families, task types, reasoning lengths and debate protocols?

These questions make a useful result possible even when not every benchmark improves. For example, RQ1 can reveal that the estimator fails on shared canonical proofs. RQ2 can show that a strong verification-only baseline explains the gain. RQ3 can show that the full null is too expensive and an approximation changes reliability. Those findings are scientifically valuable if presented clearly.

A thesis structure

Introduce the disagreement between agreement and evidence, then define the scope of observable reasoning and operational independence. Review baseline generation/debate, confidence/reliability aggregation, statistical dependence, reasoning structure and verification as separate families. Position CAVE against the closest neighbors rather than against only majority vote.

Describe the representation, reference distribution, score and resource requirements. Present controlled mechanism validation before broad benchmark results. Show paired replay comparisons, matched budgets and ablations. Use qualitative cases to explain typical failures, not only to display a favorable example. Discuss unresolved assumptions and the conditions under which the method should or should not be used.

A useful contribution ladder

The first deliverable is a reproducible trace corpus and replay framework. The second is an auditable extraction/overlap baseline. The third is a properly specified matched reference. The fourth is an evaluated support-credit rule. The fifth is an experimental account of when the combination succeeds and fails.

Do not promise a large learned system if the project is explicitly training-free. A simple validated contribution can be stronger than a complex unvalidated architecture. The complexity should earn its place through ablations that remove it.

Questions to discuss with the supervisor

What exact kind of independence is the main target? Are the units inference steps, full paths or evidence sources? Is the no-communication reference per question, per task family or reusable? Which resource boundary defines training-free? Which controlled cases count as ground truth? Which strongest baseline must be reproduced? Is the main claim about minority rescue, redundant-evidence discounting or improved accuracy at a fixed budget?

Bring the closest-neighbor matrix and one complete worked case. They make the conversation concrete: the issue is not simply that many papers exist, but that each design choice already has a nearby alternative whose assumptions and outputs differ.

Writing rules that avoid overclaiming

Prefer under the evaluated protocol to in general. Prefer source-identifiable reviewed corpus to all literature. Prefer observable derivation overlap to internal thought dependence. Prefer reference-relative support score to probability of independence unless the latter has been validated. Prefer full-system resource statement to training-free without qualification.

When a paper contributes a baseline, theory and proposed method, attribute each property to the right component. If a source describes future work, do not write it as a completed experiment. If a benchmark score is estimated, proxy-based, human-rated or oracle-assisted, keep that boundary visible.

11 | Reading priorities and the next concrete steps

The smallest set that explains the problem

Begin with CoT and SC to understand the generation/vote baseline. Read EoT and Du for the host debate. Read Wynn and Smit for persuasion, answer loss and baseline strength. Read Goel for chance-adjusted output dependence. This gives the motivation and the reason a simple majority comparison is insufficient.

Then inspect AgentAuditor and CAGE-CAL most carefully. They constrain the novelty claim about step structure and counterfactual dependence. Add DarkForest, Ising, OW-ISP, PPV, Slim-SC and ARGORA to understand calibrated support, statistical dependence, geometry, pruning and argument structure.

Use ReConcile, Zhu, SELENE, DAR and MinSentinel to refine confidence, credibility, diversity and minority evaluation. Use Lightman for verification, DivEvid for source asymmetry, and Cemri for failure analysis. The remaining identified papers provide communication, trajectory and evaluator alternatives. Priority here is an editorial relevance judgment, not a citation ranking.

Resolve the missing key before a final novelty claim

The unresolved columns are ReasCons, Estornell, Kim, CanDebate, Cortex, MARGIN, Huang, Ladha, Kuncheva, Prelec and Dong. Their names should not be replaced with whichever paper seems familiar. The unresolved-entry directory preserves the candidate distinctions previously found, where available.

Some of those missing entries may concern exactly the statistical voting and source-copying theory relevant to CAVE. The missing identities therefore limit a final novelty claim even though the four tables have no blank cells. Obtaining the actual title and identifier is a bibliography task with direct research value.

A practical first implementation cycle

Build a replayable three-agent protocol and a parser. Produce a small inspected set of step records. Implement answer voting, whole-response similarity and uncalibrated step dedup. Construct a small known-origin sharing suite. Add a matched no-communication reference and inspect its distributions. Only then add a support-credit rule and run paired decision comparisons.

After the first cycle, choose one unresolved weakness to improve: extraction fidelity, logical matching, null cost, partial sharing or quality verification. Change one mechanism at a time so the next experiment has an interpretable result.

What success would look like

A successful final-year study would leave a reader able to reproduce the setup, understand the independence target, inspect how support is discounted, verify that honest overlap is not punished arbitrarily, and see when the final decision improves relative to strong matched baselines. The result could be positive, conditional or partly negative. What matters is that the proposed mechanism and the evidence for it are clear.

The next parts of this handbook supply the reference material for that work: a guide for each source-identifiable paper, unresolved identities, the complete four comparison maps, consolidated evidence notes and linked primary references.

12 | Paper directory

Each guide distinguishes the main mechanism, required fitting/supervision, CAVE relevance and a useful experiment. Priority labels are editorial judgments about this project. The counts summarize all archival marks for that paper, not paper quality or main-method-only features. Primary links and deeper section/page notes are reproduced later.

DySCo | Communication and credibility

PROJECT RELEVANCE: HIGH

Wanshuang Gou and Zihan Liu, Dynamic Trust-Aware Sparse Communication Topology for LLM-Based Multi-Agent Consensus, arXiv:2606.01828v1 (2026).

Main method and resources

Prompt-only; online trust heuristics

Independent answers are followed by selected peer exchanges. Trust, divergence, relevance and a communication budget choose useful edges. A trust-aware terminal vote and stopping rule complete the system. The study measures accuracy, tokens, latency and agent scaling.

What matters for CAVE-EoT

The useful comparison is whether CAVE improves evidence accounting when the debate protocol is held fixed. DySCo improves who talks; CAVE aims to change how support is counted. Avoid treating a trust score as a probability of independence.

A concrete comparison or lesson

Use its sparse communication idea as an optional protocol, then replay identical recorded outputs under different aggregators.

Archival coverage	X	No	?
Map 1	30	41	0
Map 2	26	19	0
Map 3	21	30	4
Map 4	20	21	3

Primary source 1

DarkForest | Calibrated aggregation

PROJECT RELEVANCE: HIGH

Yi Li et al., DarkForest: Less Talk, Higher Accuracy for Multi-Agent LLMs, arXiv:2605.25188v1 (2026).

Main method and resources

Frozen LLMs; fitted held-out reliability parameters

Agents first solve independently. Outputs are parsed into candidate records and answer clusters. Reliability, confidence, format quality, support-pattern reliability and dependence discounts construct a belief state. A coordinator receives restricted evidence; a guardrail can preserve the preferred candidate.

What matters for CAVE-EoT

A very close aggregation neighbor. Your distinction must be at derivation level and in the honest-solver overlap reference, not merely in replacing majority vote or withholding full traces. It also makes disclosure policy an experimental variable.

A concrete comparison or lesson

Compare answer-level discounting with step/chain-level accounting, using the same candidate answers and an equal budget.

Archival coverage	X	No	?
Map 1	35	36	0
Map 2	28	17	0
Map 3	26	29	0
Map 4	21	21	2

Primary source 1

AgentAuditor | Reasoning structure and verification

PROJECT RELEVANCE: HIGH

Wei Yang, Shixuan Li, Gengshuo Liu, Heng Ping, Peiyu Zhang, Paul Bogdan and Jesse Thomason, Auditing Multi-Agent LLM Reasoning Trees Outperforms Majority Vote and LLM-as-Judge, arXiv:2602.09341v2 (2026). The PDF author line includes Gengshuo Liu, whom the abstract-page metadata omits.

Main method and resources

Trained ACPO auditor; frozen variants also reported

Reasoning traces are atomized and aligned into a tree. The auditor inspects competing evidence at important branch divergences. Anti-Consensus Preference Optimization trains adjudication against misleading support frequency. Experiments compare majority vote and flat LLM judging across reasoning tasks and multi-agent frameworks.

What matters for CAVE-EoT

The strongest direct neighbor for step extraction and semantic deduplication. Step parsing and a reasoning tree alone cannot carry your novelty claim. The defensible distinction is the dependence target, no-communication null, effective support rule and controlled validation.

A concrete comparison or lesson

Reproduce a frozen branch-audit or semantic-dedup baseline before comparing the trained version, whose supervision changes the resource setting.

Archival coverage	X	No	?
Map 1	35	36	0
Map 2	21	24	0
Map 3	31	24	0
Map 4	22	20	2

Primary source 1

CAGE-CAL | Counterfactual calibration

PROJECT RELEVANCE: HIGH

Jiatan Huang, Mingchen Li, Ziming Li, Sunjae Kwon, Hong Yu and Chuxu Zhang, Counterfactual Graph for Multi-Agent LLM Calibration, arXiv:2605.30653v1 (2026).

Main method and resources

Trained graph encoder and correctness head

Matched independent and post-communication agent graphs represent local correctness correlations, model/role groups, answers and topology. A learned calibrator predicts confidence in the panel answer; calibrated scores can select among completed topology panels. Experiments report ECE, AUROC and correctness.

What matters for CAVE-EoT

Counterfactual comparison already exists in nearby work. CAVE must explain why an honest-solver distribution over derivation overlap is a different target from learned answer/agent calibration. Frozen base LLMs do not make its full estimator training-free.

A concrete comparison or lesson

Compare the target, required labels, representation and output before comparing benchmark scores.

Archival coverage	X	No	?
Map 1	39	32	0
Map 2	22	23	0
Map 3	30	25	0
Map 4	21	20	3

Primary source 1

Zhu | Confidence and diversity

PROJECT RELEVANCE: HIGH

Xiaochen Zhu, Caiqi Zhang, Yizhou Chi, Tom Stafford, Nigel Collier and Andreas Vlachos, Demystifying Multi-Agent Debate: The Role of Confidence and Diversity, Findings ACL 2026, Anthology ID 2026.findings-acl.1694.

Main method and resources

Training-free initialization; trained confidence policy

The paper separates diverse initialization from confidence-sensitive debate updates. Greedy starting-answer diversity has a training-free component; calibrated confidence expression/perception uses supervised and reinforcement learning. Dirichlet-based belief analysis provides conditions for beneficial updates.

What matters for CAVE-EoT

Useful for exposing the difference between a diverse answer pool, calibrated confidence and independent supporting derivations. Do not call its complete confidence system training-free merely because one ablation is.

A concrete comparison or lesson

Use diversity-only initialization as a clean ablation and keep the trained confidence intervention in a separate resource category.

Archival coverage	X	No	?
Map 1	31	40	0
Map 2	19	26	0
Map 3	17	38	0
Map 4	23	19	2

Primary source 1

Ising | Statistical dependence aggregation

PROJECT RELEVANCE: HIGH

Balasubramanian, Podkopaev and Kasiviswanathan, Dependence-Aware Label Aggregation for LLM-as-a-Judge via Ising Models, ICML/PMLR 306 (2026). The author-hosted January 29 preprint prints the expanded author names Krishnakumar Balasubramanian and Shiva Prasad Kasiviswanathan.

Main method and resources

Unsupervised generalized EM fit

Binary judge labels are aggregated with class-conditioned Ising dependence. Learned fields and pairwise couplings alter Bayes decisions relative to independence-based voting. Theory and synthetic examples show that reliability weights alone can miss information in correlated agreement.

What matters for CAVE-EoT

This is a strong statistical counterexample to treating each vote as an independent unit. It operates on labels, not supporting derivations; it also requires model fitting. Its dependence law is more explicit than an embedding heuristic.

A concrete comparison or lesson

Borrow its demand for a clear dependence target and controlled simulations, rather than presenting semantic cosine as an equivalent probabilistic law.

Archival coverage	X	No	?
Map 1	25	46	0
Map 2	20	25	0
Map 3	23	32	0
Map 4	21	20	3

Primary source 1

Primary source 2

SELENE | Selective debate and evidence weighting

PROJECT RELEVANCE: HIGH

Akshay Verma, Swapnil Gupta, Deepak Gupta, Prateek Sircar and Siddharth Pillai, SELENE: Selective and Evidence-Weighted LLM Debating for Efficient and Reliable Reasoning, EACL Industry 2026, Anthology ID 2026.eacl-industry.7.

Main method and resources

Prompt-only LLMs; validation-tuned gates

Semantic disagreement and a confidence-likelihood gap determine when debate is worthwhile. Evidence-weighted, paraphrase-based judgments and stability stopping reduce repeated work. The reported tasks are QA-style datasets with accuracy, cost and judgment stability analyses.

What matters for CAVE-EoT

A close credibility/efficiency comparator. Confidence agreement, judge stability and evidence-quality weighting do not themselves establish independent evidence chains. The exact weighting descriptions have an unresolved map cell.

A concrete comparison or lesson

Separate gate, weight and stopping effects; keep an always-debate CAVE condition to test the aggregator alone.

Archival coverage	X	No	?
Map 1	35	36	0
Map 2	24	20	1
Map 3	26	29	0
Map 4	17	24	3

Primary source 1

DAR | Diversity-aware communication

PROJECT RELEVANCE: HIGH

Manh Nguyen, Anh Nguyen, Dung Nguyen, Svetha Venkatesh and Hung Le, Hear Both Sides: Efficient Multi-Agent Debate via Diversity-Aware Message Retention, arXiv:2603.20640v2, April 14, 2026.

Main method and resources

Prompt-only retention; auxiliary embedding analysis

A selector retains original disagreeing peer messages while preserving a fixed budget; selection does not rewrite the retained arguments. The paper analyzes redundancy, correlation and effective sample size, and studies accuracy, tokens and message diversity.

What matters for CAVE-EoT

Its effective-sample-size discussion is important background. A theoretical variance-equivalent sample count is different from question-specific, structurally justified independent chain credit. Disagreement may preserve a wrong outlier as well as a correct minority.

A concrete comparison or lesson

Compare random retention, diversity retention, semantic deduplication and CAVE on identical available messages.

Archival coverage	X	No	?
Map 1	40	31	0
Map 2	17	28	0
Map 3	28	27	0
Map 4	23	19	2

Primary source 1

Goel | Error dependence and oversight

PROJECT RELEVANCE: HIGH

Shashwat Goel et al., Great Models Think Alike and this Undermines AI Oversight, ICML 2025, PMLR 267:19621-19678; arXiv:2502.04313v2, June 12, 2025.

Main method and resources

Diagnostic metric; gold outcomes needed for error analysis

CAPA adjusts probabilistic model agreement for agreement expected under an independent-response reference. Broad model comparisons link similarity to shared errors, judge affinity and weak-to-strong oversight. The object is model-output dependence over tasks.

What matters for CAVE-EoT

The clearest motivation for chance adjustment, but its null and signal level differ from CAVE. Do not claim the project invents chance-adjusted agreement. Explain the move from dataset-level outputs to per-question derivations.

A concrete comparison or lesson

Use CAPA-inspired diagnostics across datasets while keeping your derivation statistic separately defined.

Archival coverage	X	No	?
Map 1	21	50	0
Map 2	14	31	0
Map 3	25	30	0
Map 4	18	24	2

Primary source 1

Primary source 2

Wynn | Debate failure diagnosis

PROJECT RELEVANCE: HIGH

Andrea Wynn, Harsh Satija and Gillian K. Hadfield, Talk Isn't Always Cheap: Understanding Failure Modes in Multi-Agent Debate, ICML MAS Workshop 2025; arXiv:2509.05396v2, October 13, 2025.

Main method and resources

Frozen homogeneous and heterogeneous panels

The study compares independent starts with multi-agent discussion and measures correct-to-wrong and wrong-to-correct changes. It examines peer pressure and capability imbalance, showing that debate can corrupt previously correct positions in particular settings.

What matters for CAVE-EoT

Direct motivation for protecting evidence from persuasive repetition. Capability differences can change who influences whom; a majority is not a quality guarantee. These findings do not imply debate always hurts.

A concrete comparison or lesson

Track answer changes, minority preservation and incorrect persuasion, not only final average accuracy.

Archival coverage	X	No	?
Map 1	32	39	0
Map 2	10	35	0
Map 3	17	38	0
Map 4	18	24	2

Primary source 1

EoT | Cross-model exchange

PROJECT RELEVANCE: HIGH

Zhangyue Yin, Qiushi Sun, Cheng Chang, Qipeng Guo, Junqi Dai, Xuanjing Huang and Xipeng Qiu, Exchange-of-Thought: Enhancing Large Language Model Capabilities through Cross-Model Communication, EMNLP 2023, ACL Anthology 2023.emnlp-main.936.

Main method and resources

Prompt-only; confidence/stability heuristics

Exchange-of-Thought compares memory, report, relay and debate communication patterns. Agents exchange rationale/answer information, with confidence and adaptive termination heuristics. The paper analyzes communication complexity and tests reasoning performance with different model combinations.

What matters for CAVE-EoT

The natural host/baseline for CAVE-EoT. Treat the proposed layer as a replayable aggregator rather than attributing a new debate topology to it. Whole rationale exchange is not extraction of independent proof units.

A concrete comparison or lesson

Hold its protocol constant, record all rounds, and compare vote, confidence, judge, dedup and CAVE decisions offline.

Archival coverage	X	No	?
Map 1	38	33	0
Map 2	16	29	0
Map 3	18	37	0
Map 4	18	23	3

Primary source 1

Primary source 2

PPV | Geometry and delegation

PROJECT RELEVANCE: HIGH

** Yasushi Sakai, Allen Song, Kent Larson, When Does Delegation Beat Majority? A Delegation-Based Aggregator for Multi-Sample LLM Inference, arXiv:2606.08098v3 (July 2026). Full text: https://arxiv.org/html/2606.08098 . “PPV” names the Propagational Proxy Voting mechanism applied to multi-sample LLM inference here; it is not a literal multi-agent discussion protocol. Separate Paper Key absent, so identity remains subject to Paper Key cross-check.

Main method and resources

Training-free delegation over many samples

Groups of sampled reasoning outputs supply answer entropy and centered reasoning embeddings. A delegation matrix routes voting influence and its stationary behavior determines the answer. A two-block analysis characterizes when a coherent minority can overturn a majority.

What matters for CAVE-EoT

A close content-geometry aggregator. Centering embeddings is not automatically a chance-calibrated honest-solver null. Geometry can reflect shared style, brevity or a coherent wrong argument without provenance.

A concrete comparison or lesson

Include a whole-reasoning embedding/delegation baseline, then test whether step-level information adds value beyond it.

Archival coverage	X	No	?
Map 1	19	52	0
Map 2	20	23	2
Map 3	21	33	1
Map 4	17	22	5

Primary source 1

Primary source 2

MinSentinel | Learned minority override

PROJECT RELEVANCE: HIGH

** Chuan He, Zebin Chen, Zhengyi Yang, Shaobo Qiao, Mingchen Ju, Jiate Liu, Dong Wen, Guanfeng Liu, Minority Sentinel: When to Overturn Majority Voting in Multi-Agent LLM Debates, AgentSearch Workshop at SIGIR 2026, arXiv:2606.29270v1. Full text: https://arxiv.org/html/2606.29270 .

Main method and resources

LightGBM classifier plus LLM semantic features

Debate dynamics, vote metadata and semantic audit features form a debate fingerprint. A trained classifier decides when a 2:1 majority should be overturned. The study reports flip precision, net gain, preservation constraints and repeated validation.

What matters for CAVE-EoT

The project must compete with a concrete minority-rescue decision, not only with majority vote. This comparator uses outcome-labeled training and an external semantic audit, so budget and supervision need separate accounting.

A concrete comparison or lesson

Adopt its flip precision/net gain measures and include a simple conservative override baseline.

Archival coverage	X	No	?
Map 1	31	40	0
Map 2	24	21	0
Map 3	26	29	0
Map 4	18	23	3

Primary source 1

Primary source 2

OW-ISP | Higher-order label aggregation

PROJECT RELEVANCE: HIGH

Rui Ai, Yuqi Pan, David Simchi-Levi, Milind Tambe and Haifeng Xu, Beyond Majority Voting: LLM Aggregation by Leveraging Higher-Order Information, ICML 2026, PMLR 306:1311-1340; primary PDF arXiv:2510.01499v2, May 19, 2026.

Main method and resources

Unsupervised reliability estimation; no debate required

Optimal weighting uses first-order model accuracy; informed selection uses conditional relationships among model answers. The paper analyzes Bayes-optimal aggregation under assumptions, estimates useful quantities without labels in practical variants, and studies correlated ensembles and capability imbalance.

What matters for CAVE-EoT

A central answer-level statistical baseline. It demonstrates that higher-order information can improve on counting labels. Its assumptions and dataset-level estimation must not be confused with per-question reasoning independence.

A concrete comparison or lesson

Compare answer-only higher-order information with derivation-aware information under the same initial predictions.

Archival coverage	X	No	?
Map 1	25	46	0
Map 2	16	29	0
Map 3	19	36	0
Map 4	22	19	3

Primary source 1

Primary source 2

DivEvid | Information asymmetry and forecasting

PROJECT RELEVANCE: HIGH

** Yuante Li, Yicheng Tao, Kate Zhang, Taozhi Wang, Gefei Gu, Yaxin Zhou, Diverse Evidence, Better Forecasts: Multi-Agent Deliberation Under Information Asymmetry, arXiv:2607.01661v1 (July 2026), method named **InfoDelphi**. Primary: https://arxiv.org/html/2607.01661 . The abbreviation “DivEvid” uniquely and directly matches **Diverse Evidence**, and method/topic match this comparison map; still request the Paper Key entry if strict key-level identity certification is necessary.

Main method and resources

Prompt-only; disjoint private evidence by design

InfoDelphi routes retrieved documents into public and private subsets, shares forecasts and rationale summaries, and pools final forecasts with confidence weights. The paper models common/private error covariance and evaluates forecasting with controlled evidence routing.

What matters for CAVE-EoT

Independent-looking reasoning can share an upstream information source. Record evidence provenance, not just text similarity. Information asymmetry is a design intervention, different from measuring dependence after the fact.

A concrete comparison or lesson

Use controlled public/private evidence as one validation condition and guard against future-information leakage.

Archival coverage	X	No	?
Map 1	42	29	0
Map 2	20	25	0
Map 3	22	33	0
Map 4	22	18	4

Primary source 1

Primary source 2

R²-MAD | Memory and reliability

PROJECT RELEVANCE: HIGH

Xuanfa Jin, Zhijian Ma, Yongcheng Zeng, Xinyu Cui, Haifeng Zhang and Jun Wang, Remember and Reweight: Enhancing Multi-Agent Debate with Experience Memory and Confidence Estimation, arXiv:2609.03619v1, September 3, 2026.

Main method and resources

Frozen LLMs; outcome-labeled experience memory

Experience retrieval and computed peer reliability guide debate under shared misconceptions. Recursive summaries preserve state; memory selection balances relevance, success and redundancy. The terminal decision still uses majority voting, with cost and memory/confidence ablations.

What matters for CAVE-EoT

A useful credibility/memory comparator, but a frozen language model can still depend on labeled historical data. Retrieved case diversity is not the same as current-panel derivation independence.

A concrete comparison or lesson

Report the provenance of memory labels and compare CAVE with and without shared retrieved memories.

Archival coverage	X	No	?
Map 1	41	30	0
Map 2	23	22	0
Map 3	31	24	0
Map 4	21	19	4

Primary source 1

Choi | Voting versus debate theory

PROJECT RELEVANCE: HIGH

Choi, Zhu and Li, Debate or Vote: Which Yields Better Decisions in Multi-Agent Large Language Models?, NeurIPS 2025; arXiv:2508.17536v2.

Main method and resources

Frozen panels; practical and oracle interventions

The paper separates independent majority voting from subsequent debate. A Dirichlet-based framework analyzes belief trajectories as a martingale under stated assumptions. Oracle locks and practical majority-guided interventions explore when updates can improve outcomes.

What matters for CAVE-EoT

The essential skeptical baseline: extra interaction must justify its cost. A martingale model is conditional on its update assumptions, not a universal theorem that debate never helps.

A concrete comparison or lesson

Compare matched independent vote and debate, then show whether CAVE improves the same recorded candidate pool.

Archival coverage	X	No	?
Map 1	38	33	0
Map 2	10	35	0
Map 3	20	35	0
Map 4	24	18	2

Primary source 1

Free-MAD | Trajectory-aware aggregation

PROJECT RELEVANCE: SUPPORTING

Cui et al., Free-MAD: Consensus-Free Multi-Agent Debate, Findings ACL 2026; arXiv:2509.11035.

Main method and resources

Prompt-only; fixed trajectory-scoring coefficients

The method favors consensus-free discussion and scores answer labels over the whole debate history. Initial support, abandoned answers, new answers, retained answers and round decay contribute to a final trajectory score. Minority terminal answers can win.

What matters for CAVE-EoT

A direct test of whether history alone explains any apparent CAVE gain. Answer switching supplies a signal, but neither switching nor stubbornness is a proof-quality certificate.

A concrete comparison or lesson

Include its trajectory rule or a faithful simplified counterpart while keeping reasoning-overlap extraction separate.

Archival coverage	X	No	?
Map 1	36	35	0
Map 2	19	26	0
Map 3	23	32	0
Map 4	19	23	2

Primary source 1

ReConcile | Diverse-model confidence debate

PROJECT RELEVANCE: HIGH

Justin Chih-Yao Chen, Swarnadeep Saha and Mohit Bansal, ReConcile: Round-Table Conference Improves Reasoning via Consensus among Diverse LLMs, ACL 2024, pp.7066-7085, ACL Anthology 2024.acl-long.381; arXiv:2309.13007.

Main method and resources

Frozen models; confidence rescaling and demonstrations

Diverse LLMs exchange explanations and confidence in a round-table setting, with weighted decisions and consensus-based termination. Demonstration, model-mixture and confidence analyses examine when heterogeneous collaboration helps.

What matters for CAVE-EoT

An important strong baseline for credibility. Verbal confidence may be useful after rescaling, yet confidence and independence measure different things. Different model names alone do not certify independent errors.

A concrete comparison or lesson

Implement uniform and confidence-weighted versions and preserve the same initial panel for comparison.

Archival coverage	X	No	?
Map 1	46	24	1
Map 2	21	24	0
Map 3	24	31	0
Map 4	21	21	2

Primary source 1

Primary source 2

DOWN | Selective collaboration

PROJECT RELEVANCE: SUPPORTING

Eo et al., Debate Only When Necessary: Adaptive Multiagent Collaboration for Efficient LLM Reasoning, arXiv:2504.05047v2.

Main method and resources

Prompt-only; validation-selected threshold

Initial confidence determines whether to accept one response or invoke peer debate. Internal token probabilities or verbal confidence support the gate; the active branch shares confidence and reasoning and ends with vote or synthesis judge.

What matters for CAVE-EoT

Useful for computing the cost of doing nothing on easy questions. Keep confidence gating separate from dependence estimation; a correct confidence threshold does not solve copied support.

A concrete comparison or lesson

Measure accuracy and full cost with an always-on aggregator and a gated version separately.

Archival coverage	X	No	?
Map 1	39	32	0
Map 2	18	27	0
Map 3	20	35	0
Map 4	18	23	3

Primary source 1

ConfLiar | Reasoning-quality diagnosis

PROJECT RELEVANCE: SUPPORTING

Keramati et al., The Confident Liar: Diagnosing Multi-Agent Debate with Log-Probabilities and LLM-as-Judge, ACL 2026 SRW; arXiv:2606.10296.

Main method and resources

Frozen role agents and external meta-evaluator

Constructor, Auditor and Synthesizer roles produce a structured exchange. Token log-probability windows and independent rubric judgments are compared with downstream performance. The reported experiment is rubric scoring; other domains in the protocol are not established experiments.

What matters for CAVE-EoT

Strong evidence that internal confidence features and reasoning quality can diverge. Token windows are not automatically logical steps, and a separate judge introduces its own reliability question.

A concrete comparison or lesson

Use confidence-quality alignment as a diagnostic and keep actual experiments distinct from future protocol descriptions.

Archival coverage	X	No	?
Map 1	19	51	1
Map 2	14	31	0
Map 3	15	40	0
Map 4	12	29	3

Primary source 1

Slim-SC | Thought-level redundancy pruning

PROJECT RELEVANCE: HIGH

Hong et al., Slim-SC: Thought Pruning for Efficient Scaling with Self-Consistency, EMNLP 2025; arXiv:2509.13990.

Main method and resources

Training-free sampling and embedding pruning

Parallel reasoning chains are pruned when delimiter-separated thoughts become sufficiently similar. A warm-up protects common initial restatements; random or internal-diversity rules choose which chain survives. The surviving answers are voted.

What matters for CAVE-EoT

A close redundancy baseline. Thought chunks are weaker units than explicit premises/inferences, and a warm-up delay does not statistically distinguish honest overlap from copying. Its token/latency results matter for your budget.

A concrete comparison or lesson

Compare threshold-only pruning and warm-up pruning with a query-matched null before claiming better redundancy handling.

Archival coverage	X	No	?
Map 1	21	49	1
Map 2	13	31	1
Map 3	25	30	0
Map 4	20	22	2

Primary source 1

ConsTrap | Adversarial majority and token collaboration

PROJECT RELEVANCE: HIGH

Jiayuan Liu, Shiyi Du, Weihua Du, Mingyu Guo, Vincent Conitzer. The Consensus Trap: Rescuing Multi-Agent LLMs from Adversarial Majorities via Token-Level Collaboration, arXiv:2604.17139v1. Primary: https://arxiv.org/html/2604.17139v1 .

Main method and resources

Frozen continuation agents; formal stability assumptions

Coordinated false beliefs can defeat response-level majority voting. Token-level round-robin continuation and voting over resulting trajectories are analyzed with impossibility and stability results, then tested under controlled corruption and capability mixtures.

What matters for CAVE-EoT

An alternative way to resist wrong majorities, not simply another vote weight. Its results are conditional on the corruption mechanism and formal dynamics; natural shared misconceptions are a separate test.

A concrete comparison or lesson

Use known coordinated corruption as a stress test and distinguish token collaboration from post-hoc CAVE replay.

Archival coverage	X	No	?
Map 1	36	34	1
Map 2	11	33	1
Map 3	24	31	0
Map 4	21	20	3

Primary source 1

Primary source 2

ARGORA | Argument graphs and causal strength

PROJECT RELEVANCE: HIGH

Youngjin Jin, Hanna Kim, Kwanwoo Kim, Chanhee Lee, Seungwon Shin. ARGORA: Orchestrated Argumentation for Causally Grounded LLM Reasoning and Decision Making, arXiv:2601.21533v1. Primary: https://arxiv.org/html/2601.21533v1 .

Main method and resources

Prompt-only experts; formal argument-strength machinery

Experts contribute claims and supporting/attacking arguments. Semantic novelty pruning and quantitative argumentation build rooted argument structures; strength and intervention-based influence help choose a decision. Reversal metrics and structural ablations characterize behavior.

What matters for CAVE-EoT

A close structural neighbor. Argument nodes, causal influence scores and proof-chain independence are different targets; a communication workflow graph is also different from a reasoning graph.

A concrete comparison or lesson

Compare simple argument-tree dedup/strength scoring with CAVE and explain where a shared decisive premise affects credit.

Archival coverage	X	No	?
Map 1	47	24	0
Map 2	28	17	0
Map 3	29	26	0
Map 4	25	17	2

Primary source 1

Primary source 2

Li-Sparse | Sparse debate topology

PROJECT RELEVANCE: SUPPORTING

Li et al., Improving Multi-Agent Debate with Sparse Communication Topology, Findings EMNLP 2024; arXiv:2406.11776.

Main method and resources

Prompt-only fixed/probabilistic communication graphs

Agents independently start, consult selected neighbors for fixed rounds and then vote. The study varies sparsity, capabilities and incorrect reference information, showing how reduced exposure can alter both accuracy and input-token cost.

What matters for CAVE-EoT

A useful low-cost debate host. Sparsity changes who can copy whom, so a different graph can confound an apparent aggregation improvement. Keep topology fixed in the principal comparison.

A concrete comparison or lesson

Use dense and sparse protocols as separate conditions, reusing cached initial answers.

Archival coverage	X	No	?
Map 1	45	26	0
Map 2	9	35	1
Map 3	21	34	0
Map 4	21	20	3

Primary source 1

MoA | Layered response synthesis

PROJECT RELEVANCE: SUPPORTING

Wang et al., Mixture-of-Agents Enhances Large Language Model Capabilities, ICLR 2025; arXiv:2406.04692.

Main method and resources

Frozen proposers and aggregators; searched architectures

Independent proposers feed successive aggregation layers. Model quality and output diversity influence architecture choices. Open-ended response benchmarks compare synthesis quality, token-priced budgets and adapted debate methods.

What matters for CAVE-EoT

A synthesis baseline, not an independent-evidence estimator. A polished aggregate can combine wrong premises; quality preference scores do not directly reveal the number of independent support chains.

A concrete comparison or lesson

Use it for open-ended tasks with explicit grounding evaluation, rather than forcing its chat objective into short-answer vote metrics.

Archival coverage	X	No	?
Map 1	38	33	0
Map 2	10	35	0
Map 3	23	32	0
Map 4	20	22	2

Primary source 1

CoT | Single-solver foundation

PROJECT RELEVANCE: FOUNDATION

Wei et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS 2022; arXiv:2201.11903.

Main method and resources

Frozen few-shot prompting

Chain-of-Thought prompting supplies worked reasoning examples before a new question. Arithmetic, commonsense and symbolic tasks test model scale and prompt design. An external calculator variant repairs arithmetic expressions; generated rationales are not a multi-agent dependence estimator.

What matters for CAVE-EoT

The basic single-agent control. A readable explanation is an observable trace, not certified internal computation. Preserve this baseline so extra sampling, discussion and extraction each justify their cost.

A concrete comparison or lesson

Compare one CoT pass, equal-budget sampled CoT and multi-agent debate before adding CAVE.

Archival coverage	X	No	?
Map 1	10	61	0
Map 2	1	44	0
Map 3	13	42	0
Map 4	15	26	3

Primary source 1

Primary source 2

SC | Sampling and majority foundation

PROJECT RELEVANCE: FOUNDATION

Wang et al., Self-Consistency Improves Chain of Thought Reasoning in Language Models, ICLR 2023; arXiv:2203.11171.

Main method and resources

Frozen stochastic decoding; likelihood variants

Self-consistency samples multiple reasoning paths, parses their answers and chooses consistent support, with unweighted and likelihood-weighted aggregation comparisons. Sampling, prompt and model-mixture analyses examine robustness.

What matters for CAVE-EoT

The key zero-communication ensemble baseline. Alternative paths and answer agreement can help even without debate; an improvement over one CoT answer alone would not establish the value of your new layer.

A concrete comparison or lesson

Match samples and total token budget, not merely the number of named agents.

Archival coverage	X	No	?
Map 1	24	47	0
Map 2	17	28	0
Map 3	16	39	0
Map 4	16	25	3

Primary source 1

Du | Vanilla multi-agent debate

PROJECT RELEVANCE: HIGH

Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum and Igor Mordatch, Improving Factuality and Reasoning in Language Models through Multiagent Debate, ICML 2024, PMLR 235:11733-11763; arXiv:2305.14325.

Main method and resources

Frozen peers; prompt and moderator variants

Independent agents exchange rationales, revise through debate and aggregate terminal responses. Reasoning, factuality and chess-related tasks, prompt/model/round ablations and answer-loss examples characterize performance and limitations.

What matters for CAVE-EoT

The cleanest traditional debate host. The report should distinguish benefits of new candidate generation from benefits of selecting among the same candidates.

A concrete comparison or lesson

Freeze its recorded conversations and evaluate several aggregators on them.

Archival coverage	X	No	?
Map 1	40	30	1
Map 2	15	30	0
Map 3	25	30	0
Map 4	20	21	3

Primary source 1

Primary source 2

Liang | Dissent and judge-led debate

PROJECT RELEVANCE: HIGH

Tian Liang et al., Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate, EMNLP 2024:17889-17904; arXiv:2305.19118.

Main method and resources

Frozen debaters, judge and evaluated variants

Opposing debaters share a conversation history and a judge decides or stops. Counterintuitive tasks, capability mixes and disagreement analyses explore whether adversarial perspectives improve reasoning and where extra rounds can hurt.

What matters for CAVE-EoT

Dissent is a mechanism, not a guarantee of independent evidence. Judge agreement may reflect persuasion or same-backbone preference, so evidence/provenance diagnostics remain useful.

A concrete comparison or lesson

Include a frozen judge baseline and measure whether independent support survives conflict.

Archival coverage	X	No	?
Map 1	41	29	1
Map 2	16	29	0
Map 3	23	32	0
Map 4	21	21	2

Primary source 1

ChatEval | Multi-agent evaluation

PROJECT RELEVANCE: SUPPORTING

Chan et al., ChatEval: Towards Better LLM-based Evaluators through Multi-Agent Debate, ICLR 2024; arXiv:2308.07201.

Main method and resources

Frozen persona evaluators

A referee panel evaluates candidate texts through sequential or simultaneous discussion, including a summarizer variant. Comparison labels are voted and numeric ratings are averaged. Human-aligned response-evaluation benchmarks test roles and communication.

What matters for CAVE-EoT

Do not transfer evaluator-human kappa into a claim about debater independence. A quality judge is itself a source whose errors and preferences can be correlated.

A concrete comparison or lesson

Use it as an evaluator baseline where human-rated text quality is the objective, with position/role controls.

Archival coverage	X	No	?
Map 1	33	38	0
Map 2	14	31	0
Map 3	20	35	0
Map 4	17	24	3

Primary source 1

Smit | Debate strategy benchmark

PROJECT RELEVANCE: HIGH

Andries Smit, Nathan Grinsztajn, Paul Duckworth, Thomas D. Barrett and Arnu Pretorius, Should we be going MAD? A Look at Multi-Agent Debate Strategies for LLMs, ICML 2024, PMLR:smit24a.

Main method and resources

Frozen models; configuration tuning

A broad comparison of self-consistency, ensemble refinement and debate strategies studies accuracy, cost, diversity, answer changes and agreement-prompt tuning. It reports cases where untuned debate trails simpler alternatives.

What matters for CAVE-EoT

Useful protection against weak baselines and configuration cherry-picking. A tuned baseline may erase an apparent improvement over a poorly configured debate.

A concrete comparison or lesson

Tune all competing methods fairly on separate development questions and report transfer and token cost.

Archival coverage	X	No	?
Map 1	44	27	0
Map 2	16	29	0
Map 3	24	30	1
Map 4	21	21	2

Primary source 1

Xiong | Inter-consistency and capability imbalance

PROJECT RELEVANCE: SUPPORTING

Xiong et al., Examining Inter-Consistency of Large Language Models Collaboration: An In-depth Analysis via Debate, Findings EMNLP 2023; arXiv:2305.11595.

Main method and resources

Frozen mixed-capability models and judge

FORD obtains independent stances and arguments, debates inconsistent cases, and uses voting or a stronger judge. Commonsense tasks measure disagreement, compromise and domination in comparable, mismatched and round-table panels.

What matters for CAVE-EoT

Model-strength imbalance can change both disagreement and influence. A weaker stubborn agent can create diversity without valuable evidence; a stronger judge can dominate without increasing independent support.

A concrete comparison or lesson

Evaluate balanced and mismatched panels and count correct-to-wrong changes by agent role.

Archival coverage	X	No	?
Map 1	42	28	1
Map 2	15	30	0
Map 3	18	37	0
Map 4	17	25	2

Primary source 1

Kuhn | Semantic-answer uncertainty

PROJECT RELEVANCE: HIGH

Lorenz Kuhn, Yarin Gal and Sebastian Farquhar, Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation, ICLR 2023; arXiv:2302.09664v3, April 15, 2023.

Main method and resources

Frozen generator; pretrained entailment classifier

Meaning-equivalent answers are grouped and their probability mass supports semantic entropy. QA experiments compare uncertainty discrimination with predictive/lexical baselines and p(True) judging. Semantic equivalence acts at answer level.

What matters for CAVE-EoT

Answer clustering and reasoning clustering are different operations. Entropy can estimate uncertainty in an answer set without revealing whether its supporting derivations share a decisive assumption.

A concrete comparison or lesson

Use semantic answer normalization when exact strings differ, while keeping derivation overlap separate.

Archival coverage	X	No	?
Map 1	19	52	0
Map 2	15	30	0
Map 3	18	37	0
Map 4	16	26	2

Primary source 1

Primary source 2

Lightman | Process verification

PROJECT RELEVANCE: HIGH

Hunter Lightman et al., Let's Verify Step by Step, ICLR 2024; arXiv:2305.20050.

Main method and resources

Trained outcome/process reward models; human supervision

Human judgments on intermediate mathematical steps support process reward modeling. Sampled solutions are ranked or voted with verifier information, and experiments compare process versus outcome supervision, active learning and out-of-distribution performance.

What matters for CAVE-EoT

Correctness checking and independence estimation are complementary. Two independent chains can both be wrong; a verified step can still be repeated many times. The trained verifier is not a training-free CAVE component.

A concrete comparison or lesson

Start with executable arithmetic checks; put PRM-assisted variants in a separate supervision/resource category.

Archival coverage	X	No	?
Map 1	26	43	2
Map 2	22	23	0
Map 3	22	32	1
Map 4	15	27	2

Primary source 1

Cemri | System failure taxonomy

PROJECT RELEVANCE: SUPPORTING

Cemri et al., Why Do Multi-Agent LLM Systems Fail?, NeurIPS 2025 Datasets and Benchmarks; arXiv:2503.13657.

Main method and resources

Human study; prompt-based trace annotator

The work develops and validates a taxonomy of multi-agent failures from execution traces, including communication, task, verification and termination failures. A prompt-based annotator scales labeling; workflow interventions are tested in math/software cases.

What matters for CAVE-EoT

A practical error-analysis template. A final accuracy gain does not reveal whether extraction failed, information was lost, the verifier was wrong or the termination rule stopped too soon.

A concrete comparison or lesson

Annotate a sample of CAVE failures by mechanism and keep raw messages, tools and final decisions inspectable.

Archival coverage	X	No	?
Map 1	35	33	3
Map 2	16	29	0
Map 3	21	34	0
Map 4	16	26	2

Primary source 1

Dawid-Skene | Observer reliability background

PROJECT RELEVANCE: FOUNDATION

A. P. Dawid and A. M. Skene, Maximum Likelihood Estimation of Observer Error-Rates Using the EM Algorithm, Applied Statistics 28(1):20-28 (1979); DOI 10.2307/2346806.

Main method and resources

Unsupervised latent-class EM fit

Categorical human observer judgments infer latent class consensus and observer confusion matrices without known true labels. Conditional independence assumptions and practical limitations are explicit. The printed clinical example demonstrates heterogeneous reliability estimation.

What matters for CAVE-EoT

The historical statistical ancestor of reliability-aware aggregation. It is not a multi-LLM debate method, and its conditional independence assumption is exactly the kind of assumption your dependence analysis must examine.

A concrete comparison or lesson

Use it as a conceptual baseline or label-aggregation comparison, with the assumptions stated rather than silently inherited.

Archival coverage	X	No	?
Map 1	15	56	0
Map 2	11	34	0
Map 3	11	44	0
Map 4	13	30	1

Primary source 1

Primary source 2

13 | Unresolved identities and what to obtain

Every one of these columns has 215 question marks across the four maps. A familiar surname, plausible acronym or related title is insufficient to establish the intended citation. The source-identifiable corpus is therefore incomplete for a final novelty claim, especially around correlated voting and source dependence.

ReasCons

No verified full title or identifier matches the alias. Obtain the original Paper Key entry; do not replace it with a similarly named reasoning-consensus method.

Needed: full title, author initials/list, year, venue/status and arXiv/DOI/primary URL. Candidate properties are not assigned to the column.

Estornell

Prior records found both Multi-LLM Debate: Framework, Principals, and Interventions (NeurIPS 2024) and ACC-Collab/actor-critic collaboration work (ICLR 2025). The key has not selected one.

Needed: full title, author initials/list, year, venue/status and arXiv/DOI/primary URL. Candidate properties are not assigned to the column.

Kim

The surname does not identify author initials, title, year or ID. Several reasoning/dependence works are possible.

Needed: full title, author initials/list, year, venue/status and arXiv/DOI/primary URL. Candidate properties are not assigned to the column.

CanDebate

A controlled-debate candidate appeared in earlier searches, but no evidence securely maps that title to this short name.

Needed: full title, author initials/list, year, venue/status and arXiv/DOI/primary URL. Candidate properties are not assigned to the column.

Cortex

CortexDebate: Debating Sparsely and Equally and CORTEX: Collaborative LLM Agents for High-Stakes Alert Triage are different candidates. No substitution is justified.

Needed: full title, author initials/list, year, venue/status and arXiv/DOI/primary URL. Candidate properties are not assigned to the column.

MARGIN

Exact alias searches did not establish an unambiguous paper. The word also appears as an ordinary method/metric term.

Needed: full title, author initials/list, year, venue/status and arXiv/DOI/primary URL. Candidate properties are not assigned to the column.

Huang

Large Language Models Cannot Self-Correct Reasoning Yet and Large Language Models Can Self-Improve are different Huang-authored works. The surname alone does not choose one.

Needed: full title, author initials/list, year, venue/status and arXiv/DOI/primary URL. Candidate properties are not assigned to the column.

Ladha

Multiple jury-theory works are possible, including correlated-vote analyses. Obtain the exact title/DOI before transferring theoretical properties.

Needed: full title, author initials/list, year, venue/status and arXiv/DOI/primary URL. Candidate properties are not assigned to the column.

Kuncheva

Several ensemble-diversity/error-dependence publications share this author. Exact citation is required.

Needed: full title, author initials/list, year, venue/status and arXiv/DOI/primary URL. Candidate properties are not assigned to the column.

Prelec

A Bayesian truth serum for subjective data (2004) and A solution to the single-question crowd wisdom problem (2017) are different relevant works.

Needed: full title, author initials/list, year, venue/status and arXiv/DOI/primary URL. Candidate properties are not assigned to the column.

Dong

Integrating Conflicting Data: The Role of Source Dependence and Truth Discovery and Copying Detection in a Dynamic World are distinct source-copying papers from the same author team. The key must choose the intended paper.

Needed: full title, author initials/list, year, venue/status and arXiv/DOI/primary URL. Candidate properties are not assigned to the column.

14 | Complete four-map appendix

The wide original maps are split into two paper panels and short row panels for legibility. Every original label and mark is retained. Read broad cells with the scope notes: an implemented baseline, ablation or theoretical analysis can qualify. These matrices are an archive of the completed maps, not silent corrections or a global literature census.

Map 1: setting, agents and communication. Map 2: aggregation and weighting signals. Map 3: dependence, reasoning content, training and verification. Map 4: evaluation, theory and links. The per-paper counts in the directory and cell totals in Chapter 1 are computed directly from these same artifacts.

Map 1 | papers 1-24, rows 1-24

Criterion / original row number	DySCo	DarkForest	AgentAuditor	CAGE-CAL	Zhu	Ising	SELENE	DAR	Goel	Wynn	EoT	ReasCons	PPV	MinSentinel	OW-ISP	DivEvid	R²-MAD	Estornell	Kim	Choi	CanDebate	Free-MAD	Cortex	ReConcile
01 Setting: independent sampling (no communication)	–	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
02 Setting: multi-round debate	X	–	X	X	X	–	X	X	–	X	X	?	–	X	–	X	X	?	?	X	?	X	?	X
03 Setting: judge / label aggregation	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
04 Setting: hybrid (independent first, then communication)	X	X	–	X	X	–	X	X	–	X	X	?	–	X	–	X	X	?	?	X	?	X	?	X
05 Goal: accuracy	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
06 Goal: token / communication cost	X	X	X	–	–	–	X	X	–	–	X	?	–	–	–	–	–	?	?	–	?	X	?	X
07 Goal: calibration	–	X	–	X	X	X	X	–	–	–	–	?	–	–	–	X	–	?	?	–	?	–	?	X
08 Goal: diagnosis / theory only	–	–	–	–	–	–	–	–	X	X	–	?	–	–	–	–	–	?	?	–	?	–	?	–
09 Goal: robustness	X	X	X	X	X	X	X	X	–	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
10 Failure targeted: false consensus / correlated errors	–	X	X	X	–	X	–	X	X	X	–	?	X	X	X	X	X	?	?	–	?	X	?	X
11 Failure targeted: conformity / sycophancy	–	X	X	X	X	–	X	X	–	X	–	?	–	X	–	X	X	?	?	X	?	X	?	X
12 Failure targeted: confident-wrong peers	X	X	X	X	X	X	X	X	–	–	X	?	–	X	–	X	X	?	?	–	?	–	?	X
13 Failure targeted: shared misconception	–	X	X	–	–	X	–	–	–	–	–	?	–	X	–	X	X	?	?	–	?	–	?	X
14 Failure targeted: redundant messages	X	X	X	–	–	–	X	X	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
15 Failure targeted: quadratic communication cost	X	–	–	–	–	–	–	–	–	–	X	?	–	–	–	–	–	?	?	–	?	–	?	–
16 Failure targeted: lost correct minority	–	X	X	–	X	–	X	X	–	X	X	?	X	X	–	–	X	?	?	X	?	X	?	X
17 Pipeline position: before debate	–	X	–	–	X	–	X	–	–	X	–	?	–	X	–	X	–	?	?	–	?	–	?	X
18 Pipeline position: during debate	X	–	–	–	X	–	X	X	–	X	X	?	–	X	–	X	X	?	?	X	?	X	?	X
19 Pipeline position: after debate (aggregation)	X	X	X	X	X	X	X	X	–	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
20 Pipeline position: controller of when to debate	–	–	–	–	–	–	X	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	X
21 Stance: debate helps	X	–	–	X	X	–	X	X	–	X	X	?	–	–	–	X	X	?	?	X	?	X	?	X
22 Stance: debate hurts	–	X	–	X	X	–	X	X	–	X	–	?	–	–	–	X	X	?	?	X	?	X	?	X
23 Stance: no gain beyond voting	–	–	–	–	X	–	–	–	–	X	–	?	–	–	–	–	–	?	?	X	?	–	?	–
24 Independence of agents: assumed	–	–	–	–	–	X	–	–	X	–	–	?	X	–	X	X	X	?	?	X	?	–	?	–

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 1 | papers 1-24, rows 25-48

Criterion / original row number	DySCo	DarkForest	AgentAuditor	CAGE-CAL	Zhu	Ising	SELENE	DAR	Goel	Wynn	EoT	ReasCons	PPV	MinSentinel	OW-ISP	DivEvid	R²-MAD	Estornell	Kim	Choi	CanDebate	Free-MAD	Cortex	ReConcile
25 Independence of agents: challenged	–	X	X	X	–	X	–	X	X	X	–	?	X	X	X	X	X	?	?	–	?	X	?	X
26 Independence of agents: modeled	–	X	X	X	–	X	–	X	X	–	–	?	–	–	X	X	X	?	?	X	?	X	?	–
27 Contribution: method	X	X	X	X	X	X	X	X	X	–	X	?	X	X	X	X	X	?	?	X	?	X	?	X
28 Contribution: analysis	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
29 Contribution: theory	X	X	X	–	X	X	–	X	X	–	X	?	X	–	X	X	X	?	?	X	?	X	?	–
30 Contribution: benchmark / resource	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	X	–	?	?	–	?	–	?	–
31 Models: one model sampled many times	X	–	X	–	X	–	X	X	–	X	X	?	X	–	–	X	X	?	?	X	?	–	?	X
32 Models: one model with personas / roles	–	–	–	X	–	–	–	–	–	–	X	?	–	–	–	–	X	?	?	X	?	–	?	–
33 Models: same family, different sizes	–	–	X	–	–	–	–	–	X	–	–	?	–	–	X	–	–	?	?	–	?	X	?	?
34 Models: different families	–	X	X	X	–	X	–	–	X	X	X	?	–	X	X	X	–	?	?	–	?	X	?	X
35 Models: strong + weak mix	–	–	–	–	–	X	–	–	X	X	X	?	–	–	X	–	–	?	?	–	?	X	?	X
36 Models: open-weight	–	X	X	X	X	X	–	X	X	X	–	?	X	–	X	X	X	?	?	X	?	X	?	X
37 Models: closed API	X	–	–	–	–	X	X	–	–	X	X	?	–	X	X	X	X	?	?	X	?	–	?	X
38 Diversity source: temperature	–	–	–	–	X	–	–	X	–	–	X	?	X	–	–	–	X	?	?	X	?	–	?	X
39 Diversity source: roles / personas	–	–	–	X	–	–	–	–	–	–	X	?	–	X	–	–	X	?	?	X	?	–	?	–
40 Diversity source: different models	–	X	X	X	–	X	–	–	X	X	X	?	–	X	X	–	–	?	?	–	?	X	?	X
41 Diversity source: diverse initial answers	X	X	X	X	X	–	–	–	–	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
42 Diversity source: information asymmetry	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	X	X	?	?	–	?	–	?	X
43 Diversity source: seeded hints	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
44 Diversity: measured	X	X	X	X	X	–	X	X	X	X	–	?	X	X	X	X	X	?	?	–	?	–	?	X
45 Diversity: enforced	X	–	–	X	X	–	–	X	–	–	–	?	–	X	–	X	X	?	?	X	?	X	?	X
46 Agent roles: asymmetric (proposer / critic / judge)	–	X	X	X	–	–	X	X	X	–	X	?	–	X	–	–	X	?	?	X	?	–	?	–
47 Capability gap considered	–	–	X	–	–	X	–	–	X	X	X	?	–	–	X	X	–	?	?	–	?	X	?	X
48 Shared pretraining / same-provider correlation considered	–	X	X	X	–	X	–	X	X	–	–	?	–	X	X	–	–	?	?	–	?	–	?	X

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 1 | papers 1-24, rows 49-71

Criterion / original row number	DySCo	DarkForest	AgentAuditor	CAGE-CAL	Zhu	Ising	SELENE	DAR	Goel	Wynn	EoT	ReasCons	PPV	MinSentinel	OW-ISP	DivEvid	R²-MAD	Estornell	Kim	Choi	CanDebate	Free-MAD	Cortex	ReConcile
49 Communication: none	–	–	–	X	–	X	–	–	X	X	–	?	X	X	X	X	–	?	?	X	?	–	?	X
50 Communication: limited	X	X	X	X	X	–	X	X	–	–	X	?	–	–	–	X	–	?	?	X	?	X	?	–
51 Communication: full broadcast	–	–	–	X	X	–	X	X	–	X	X	?	–	X	–	X	X	?	?	X	?	X	?	X
52 Topology: fully connected	–	–	–	X	X	–	X	X	–	X	X	?	–	X	–	X	X	?	?	X	?	X	?	X
53 Topology: fixed sparse	–	–	–	X	–	–	–	X	–	–	X	?	–	–	–	–	–	?	?	X	?	–	?	–
54 Topology: dynamic sparse	X	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
55 Topology: learned / chosen per query	X	–	–	X	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
56 Topology: ring / star / tree	–	–	–	X	–	–	–	–	–	–	X	?	–	–	–	–	–	?	?	X	?	–	?	–
57 Topology: sequential relay	–	–	–	X	–	–	–	–	–	–	X	?	–	–	–	–	–	?	?	–	?	–	?	–
58 Passes: full reasoning	–	–	X	–	X	–	X	X	–	X	X	?	–	X	–	–	X	?	?	X	?	X	?	X
59 Passes: answer only	–	–	–	X	–	X	X	X	–	–	–	?	–	–	X	–	–	?	?	X	?	X	?	X
60 Passes: answer + confidence	–	X	–	–	X	–	X	X	–	–	X	?	–	–	–	X	X	?	?	–	?	–	?	X
61 Passes: selected subset	X	X	X	X	–	–	–	X	–	–	X	?	–	–	–	X	X	?	?	X	?	–	?	X
62 Passes: structured evidence	X	X	X	–	–	–	X	–	–	–	–	?	–	–	–	X	X	?	?	–	?	X	?	X
63 Rounds: single round	–	X	–	X	–	X	–	X	–	–	–	?	X	–	X	X	–	?	?	X	?	X	?	X
64 Rounds: multiple fixed rounds	X	–	–	X	X	–	X	X	–	X	–	?	–	X	–	X	X	?	?	X	?	X	?	–
65 Stopping: early stop on consensus	X	–	–	–	–	–	X	–	–	–	X	?	–	–	–	–	X	?	?	–	?	–	?	X
66 Stopping: confidence threshold	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
67 Debate trigger: selective (only when needed)	–	–	–	–	–	–	X	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	X
68 Message filtering: by disagreement / diversity	X	–	X	–	–	–	–	X	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
69 Message filtering: by trust / relevance	X	X	X	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
70 Message filtering: by confidence threshold	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
71 Communication cost reported	X	X	X	–	–	–	X	X	–	–	X	?	–	–	–	X	X	?	?	–	?	X	?	–

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 1 | papers 25-48, rows 1-24

Criterion / original row number	DOWN	MARGIN	ConfLiar	Slim-SC	ConsTrap	ARGORA	Li-Sparse	MoA	CoT	SC	Du	Liang	ChatEval	Smit	Huang	Xiong	Kuhn	Lightman	Cemri	Ladha	Kuncheva	Dawid-Skene	Prelec	Dong
01 Setting: independent sampling (no communication)	X	?	–	X	X	X	X	X	–	X	X	X	X	X	?	X	X	X	–	?	?	X	?	?
02 Setting: multi-round debate	X	?	–	–	–	X	X	X	–	–	X	X	X	X	?	X	–	–	X	?	?	–	?	?
03 Setting: judge / label aggregation	X	?	X	X	X	X	X	X	–	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
04 Setting: hybrid (independent first, then communication)	X	?	–	–	–	X	X	X	–	–	X	–	X	X	?	X	–	–	X	?	?	–	?	?
05 Goal: accuracy	X	?	–	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
06 Goal: token / communication cost	X	?	–	X	X	X	X	X	–	–	X	X	X	X	?	–	X	–	–	?	?	–	?	?
07 Goal: calibration	–	?	–	–	–	X	–	–	–	X	–	–	X	–	?	–	–	X	–	?	?	–	?	?
08 Goal: diagnosis / theory only	–	?	X	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	X	?	?	X	?	?
09 Goal: robustness	X	?	–	X	X	X	X	X	X	X	X	–	–	X	?	X	X	X	X	?	?	X	?	?
10 Failure targeted: false consensus / correlated errors	X	?	–	X	X	X	X	–	–	–	X	X	–	X	?	X	–	–	–	?	?	X	?	?
11 Failure targeted: conformity / sycophancy	X	?	X	–	X	X	X	–	–	–	X	X	–	X	?	X	–	–	–	?	?	–	?	?
12 Failure targeted: confident-wrong peers	–	?	X	–	X	X	–	–	–	–	X	X	–	–	?	X	–	–	X	?	?	–	?	?
13 Failure targeted: shared misconception	–	?	–	X	X	X	–	–	–	–	–	X	–	–	?	–	–	–	–	?	?	–	?	?
14 Failure targeted: redundant messages	X	?	–	–	X	X	–	–	–	–	–	–	–	–	?	–	–	–	X	?	?	–	?	?
15 Failure targeted: quadratic communication cost	–	?	–	–	–	–	X	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
16 Failure targeted: lost correct minority	X	?	–	X	X	–	–	–	–	–	–	–	–	X	?	X	–	X	X	?	?	–	?	?
17 Pipeline position: before debate	X	?	–	–	–	X	X	X	–	–	–	–	–	–	?	X	–	–	X	?	?	–	?	?
18 Pipeline position: during debate	X	?	X	–	X	X	X	X	–	–	X	X	X	X	?	X	–	–	X	?	?	–	?	?
19 Pipeline position: after debate (aggregation)	X	?	X	X	X	X	X	X	–	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
20 Pipeline position: controller of when to debate	X	?	–	–	–	–	–	–	–	–	–	X	–	X	?	X	–	–	X	?	?	–	?	?
21 Stance: debate helps	X	?	–	–	–	X	X	X	–	–	X	X	X	X	?	X	–	–	X	?	?	–	?	?
22 Stance: debate hurts	X	?	–	–	–	X	X	X	–	–	X	X	X	X	?	X	–	–	–	?	?	–	?	?
23 Stance: no gain beyond voting	–	?	–	–	–	X	X	–	–	–	–	–	–	X	?	–	–	–	–	?	?	–	?	?
24 Independence of agents: assumed	–	?	–	X	–	–	–	–	–	X	X	–	–	X	?	X	X	X	–	?	?	X	?	?

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 1 | papers 25-48, rows 25-48

Criterion / original row number	DOWN	MARGIN	ConfLiar	Slim-SC	ConsTrap	ARGORA	Li-Sparse	MoA	CoT	SC	Du	Liang	ChatEval	Smit	Huang	Xiong	Kuhn	Lightman	Cemri	Ladha	Kuncheva	Dawid-Skene	Prelec	Dong
25 Independence of agents: challenged	X	?	–	–	X	X	X	–	–	–	X	X	–	X	?	X	–	–	–	?	?	X	?	?
26 Independence of agents: modeled	–	?	–	–	X	–	X	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
27 Contribution: method	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
28 Contribution: analysis	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
29 Contribution: theory	–	?	–	X	X	X	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
30 Contribution: benchmark / resource	–	?	–	–	–	–	–	–	X	–	X	X	–	X	?	–	–	X	X	?	?	–	?	?
31 Models: one model sampled many times	X	?	–	X	X	X	X	X	–	X	X	X	X	X	?	–	X	X	X	?	?	–	?	?
32 Models: one model with personas / roles	X	?	X	–	–	X	–	X	–	–	X	X	X	X	?	–	–	–	X	?	?	–	?	?
33 Models: same family, different sizes	–	?	?	–	X	–	–	X	–	–	–	?	–	–	?	?	–	?	–	?	?	–	?	?
34 Models: different families	X	?	–	–	X	–	X	X	–	X	X	X	–	–	?	X	–	–	–	?	?	–	?	?
35 Models: strong + weak mix	–	?	–	–	X	–	X	X	–	X	X	X	–	–	?	X	–	X	–	?	?	–	?	?
36 Models: open-weight	X	?	–	X	X	X	X	X	X	X	X	X	X	X	?	X	X	–	X	?	?	–	?	?
37 Models: closed API	X	?	X	–	–	X	X	X	X	X	X	X	X	X	?	X	–	?	X	?	?	–	?	?
38 Diversity source: temperature	–	?	–	?	?	–	X	X	–	X	?	–	–	X	?	–	X	X	?	?	?	–	?	?
39 Diversity source: roles / personas	–	?	X	–	–	X	–	–	–	–	X	X	X	X	?	–	–	–	X	?	?	–	?	?
40 Diversity source: different models	X	?	–	–	X	–	X	X	–	X	X	X	–	–	?	X	–	–	–	?	?	–	?	?
41 Diversity source: diverse initial answers	X	?	–	X	X	X	X	X	–	X	X	X	X	X	?	X	X	X	X	?	?	–	?	?
42 Diversity source: information asymmetry	–	?	–	–	X	X	–	–	–	X	–	–	–	–	?	X	–	–	X	?	?	–	?	?
43 Diversity source: seeded hints	–	?	–	–	X	–	X	–	–	–	–	–	–	–	?	–	–	X	–	?	?	–	?	?
44 Diversity: measured	–	?	–	X	–	X	X	X	–	X	X	X	–	X	?	X	X	–	–	?	?	–	?	?
45 Diversity: enforced	X	?	X	X	X	X	X	X	–	–	–	X	X	X	?	X	–	–	X	?	?	–	?	?
46 Agent roles: asymmetric (proposer / critic / judge)	X	?	X	–	–	X	–	X	–	–	X	X	X	X	?	X	–	X	X	?	?	–	?	?
47 Capability gap considered	–	?	–	–	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
48 Shared pretraining / same-provider correlation considered	–	?	–	–	–	–	–	–	–	–	–	X	–	–	?	X	–	X	–	?	?	–	?	?

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 1 | papers 25-48, rows 49-71

Criterion / original row number	DOWN	MARGIN	ConfLiar	Slim-SC	ConsTrap	ARGORA	Li-Sparse	MoA	CoT	SC	Du	Liang	ChatEval	Smit	Huang	Xiong	Kuhn	Lightman	Cemri	Ladha	Kuncheva	Dawid-Skene	Prelec	Dong
49 Communication: none	X	?	–	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	–	?	?	X	?	?
50 Communication: limited	–	?	X	–	X	X	X	–	–	–	X	–	X	X	?	X	–	–	X	?	?	–	?	?
51 Communication: full broadcast	X	?	–	–	–	X	X	X	–	–	X	X	X	X	?	X	–	–	?	?	?	–	?	?
52 Topology: fully connected	X	?	–	–	–	X	X	X	–	–	X	X	X	X	?	X	–	–	?	?	?	–	?	?
53 Topology: fixed sparse	–	?	X	–	X	X	X	–	–	–	–	–	–	–	?	–	–	–	X	?	?	–	?	?
54 Topology: dynamic sparse	–	?	–	–	–	–	X	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
55 Topology: learned / chosen per query	–	?	–	–	–	–	–	X	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
56 Topology: ring / star / tree	–	?	–	–	X	X	X	X	–	–	–	–	X	X	?	–	–	–	X	?	?	–	?	?
57 Topology: sequential relay	–	?	X	–	X	–	–	X	–	–	–	X	X	X	?	X	–	–	X	?	?	–	?	?
58 Passes: full reasoning	X	?	X	–	X	X	X	X	–	–	X	X	X	X	?	X	–	X	X	?	?	–	?	?
59 Passes: answer only	X	?	–	X	X	X	X	–	–	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
60 Passes: answer + confidence	X	?	–	–	–	–	–	–	–	X	–	–	–	–	?	–	X	X	X	?	?	–	?	?
61 Passes: selected subset	–	?	–	–	–	X	X	–	–	–	–	–	–	X	?	–	–	X	–	?	?	–	?	?
62 Passes: structured evidence	–	?	X	–	–	X	–	–	–	–	–	–	–	–	?	–	–	X	–	?	?	–	?	?
63 Rounds: single round	X	?	X	X	–	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
64 Rounds: multiple fixed rounds	X	?	–	–	–	X	X	X	–	–	X	X	X	X	?	–	–	–	–	?	?	–	?	?
65 Stopping: early stop on consensus	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	X	–	–	–	?	?	–	?	?
66 Stopping: confidence threshold	X	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
67 Debate trigger: selective (only when needed)	X	?	–	–	–	–	–	–	–	–	–	–	–	–	?	X	–	–	–	?	?	–	?	?
68 Message filtering: by disagreement / diversity	–	?	–	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
69 Message filtering: by trust / relevance	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
70 Message filtering: by confidence threshold	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
71 Communication cost reported	–	?	–	–	–	–	X	X	–	–	X	X	X	X	?	–	–	–	–	?	?	–	?	?

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 2 | papers 1-24, rows 1-24

Criterion / original row number	DySCo	DarkForest	AgentAuditor	CAGE-CAL	Zhu	Ising	SELENE	DAR	Goel	Wynn	EoT	ReasCons	PPV	MinSentinel	OW-ISP	DivEvid	R²-MAD	Estornell	Kim	Choi	CanDebate	Free-MAD	Cortex	ReConcile
01 Final rule: plain majority vote	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	–	X	?	?	X	?	X	?	X
02 Final rule: last-round consensus	X	X	–	X	X	–	X	X	–	X	X	?	–	X	–	X	X	?	?	X	?	X	?	X
03 Final rule: confidence-weighted vote	X	X	–	–	–	–	–	–	–	–	–	?	X	–	–	X	–	?	?	–	?	X	?	X
04 Final rule: trust-weighted vote	X	–	–	–	–	–	–	–	–	–	–	?	X	–	–	–	–	?	?	–	?	–	?	–
05 Final rule: reliability / accuracy-weighted vote	X	X	–	–	–	X	?	–	–	–	–	?	–	–	X	–	–	?	?	–	?	–	?	–
06 Final rule: LLM judge	–	X	X	–	–	–	X	–	X	–	–	?	–	X	–	X	–	?	?	–	?	X	?	X
07 Final rule: trained auditor / adjudicator	–	–	X	–	–	–	–	–	–	–	–	?	–	X	–	–	–	?	?	–	?	–	?	–
08 Final rule: meta-classifier that overturns the majority	–	–	–	–	–	X	–	–	–	–	–	?	–	X	–	–	–	?	?	–	?	–	?	–
09 Final rule: fitted probabilistic dependence model	–	–	–	–	–	X	–	–	–	–	–	?	–	–	X	–	–	?	?	–	?	–	?	–
10 Final rule: calibrated belief over answer clusters	–	X	–	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
11 Final rule: delegation / vote routing	–	–	–	–	–	–	–	–	–	–	–	?	X	–	–	–	–	?	?	–	?	–	?	–
12 Final rule: trajectory score	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	X	?	–
13 Final rule: step / DAG weights	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
14 Answer equivalence: exact match	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
15 Answer equivalence: semantic clustering	X	X	–	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
16 Answer equivalence: structured parsing	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
17 Decision uses the whole debate trajectory	X	–	–	–	–	–	–	–	–	–	X	?	–	X	–	–	X	?	?	–	?	X	?	–
18 Can overturn the majority	X	X	X	–	–	X	X	–	–	–	X	?	X	X	X	X	–	?	?	–	?	X	?	X
19 Output: answer only	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	–	X	?	?	X	?	X	?	X
20 Output: answer + confidence	–	X	–	X	X	X	X	X	–	–	X	?	–	X	–	X	X	?	?	–	?	X	?	X
21 Output: belief distribution	–	X	–	–	–	X	–	–	–	–	–	?	–	–	–	X	–	?	?	–	?	–	?	–
22 Fuses a single signal	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
23 Fuses several signals	X	X	X	X	X	X	X	X	X	–	X	?	X	X	X	X	X	?	?	–	?	X	?	X
24 Aggregation level: answer	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 2 | papers 1-24, rows 25-45

Criterion / original row number	DySCo	DarkForest	AgentAuditor	CAGE-CAL	Zhu	Ising	SELENE	DAR	Goel	Wynn	EoT	ReasCons	PPV	MinSentinel	OW-ISP	DivEvid	R²-MAD	Estornell	Kim	Choi	CanDebate	Free-MAD	Cortex	ReConcile
25 Aggregation level: reasoning path	–	–	X	–	–	–	X	X	–	–	–	?	X	X	–	–	X	?	?	–	?	–	?	–
26 Aggregation level: step	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
27 Aggregation level: token	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
28 Signal: raw vote count	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	–	X	?	?	X	?	X	?	X
29 Signal: verbalized confidence	X	X	–	X	X	–	X	–	–	–	–	?	?	–	–	X	–	?	?	–	?	X	?	X
30 Signal: token log-prob / likelihood confidence	–	–	–	X	–	–	X	X	X	–	–	?	?	–	–	–	X	?	?	–	?	–	?	–
31 Signal: answer variability / entropy / self-consistency	X	X	X	X	X	–	X	X	–	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
32 Signal: calibrated confidence	–	X	–	X	X	X	X	–	–	–	–	?	–	–	–	X	–	?	?	–	?	–	?	X
33 Signal: trust / credibility	X	–	–	–	–	–	–	–	–	–	X	?	X	–	–	X	X	?	?	–	?	–	?	–
34 Signal: reliability / accuracy prior	X	X	–	–	–	X	–	–	X	–	–	?	X	–	X	–	X	?	?	–	?	–	?	–
35 Signal: disagreement / divergence	X	X	X	X	X	X	X	X	X	–	–	?	X	X	X	X	X	?	?	–	?	–	?	X
36 Signal: diversity of starting answers	X	X	X	X	X	–	X	X	–	–	–	?	X	X	–	–	X	?	?	–	?	–	?	–
37 Signal: task relevance	X	–	X	X	–	–	X	–	–	–	–	?	–	–	–	X	X	?	?	–	?	–	?	–
38 Signal: parse / format quality	–	X	X	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
39 Signal: evidence quality (judge rubric / paraphrase variance)	X	X	X	–	–	–	X	–	X	–	–	?	–	X	–	X	X	?	?	–	?	–	?	X
40 Signal: debate trajectory / dynamics	X	–	–	X	–	–	X	–	–	–	X	?	–	X	–	–	X	?	?	–	?	X	?	–
41 Weights: fixed	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
42 Weights: learned	–	X	X	X	X	X	–	–	–	–	–	?	–	X	X	–	–	?	?	–	?	–	?	–
43 Weights: from held-out calibration	–	X	–	X	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
44 Weights: updated online	X	–	–	–	X	–	–	–	–	–	–	?	–	–	–	–	X	?	?	–	?	–	?	X
45 Shows that confidence is unreliable	–	–	–	X	X	X	X	X	–	–	–	?	X	X	–	X	–	?	?	–	?	–	?	X

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 2 | papers 25-48, rows 1-24

Criterion / original row number	DOWN	MARGIN	ConfLiar	Slim-SC	ConsTrap	ARGORA	Li-Sparse	MoA	CoT	SC	Du	Liang	ChatEval	Smit	Huang	Xiong	Kuhn	Lightman	Cemri	Ladha	Kuncheva	Dawid-Skene	Prelec	Dong
01 Final rule: plain majority vote	X	?	–	X	X	X	X	–	–	X	X	X	X	X	?	X	–	X	–	?	?	–	?	?
02 Final rule: last-round consensus	X	?	–	–	–	X	X	–	–	–	X	–	X	X	?	X	–	–	–	?	?	–	?	?
03 Final rule: confidence-weighted vote	–	?	–	–	–	–	–	–	–	X	–	–	–	–	?	–	–	X	–	?	?	–	?	?
04 Final rule: trust-weighted vote	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
05 Final rule: reliability / accuracy-weighted vote	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	X	?	?
06 Final rule: LLM judge	X	?	X	–	–	X	–	X	–	–	X	X	X	X	?	X	–	–	X	?	?	–	?	?
07 Final rule: trained auditor / adjudicator	–	?	–	–	–	–	–	–	–	–	–	X	–	–	?	–	–	X	–	?	?	–	?	?
08 Final rule: meta-classifier that overturns the majority	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
09 Final rule: fitted probabilistic dependence model	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
10 Final rule: calibrated belief over answer clusters	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
11 Final rule: delegation / vote routing	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
12 Final rule: trajectory score	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	X	–	?	?	–	?	?
13 Final rule: step / DAG weights	–	?	–	–	–	X	–	–	–	–	–	–	–	–	?	–	–	X	–	?	?	–	?	?
14 Answer equivalence: exact match	X	?	–	X	X	X	X	–	–	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
15 Answer equivalence: semantic clustering	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	X	–	–	?	?	–	?	?
16 Answer equivalence: structured parsing	X	?	–	?	?	X	?	–	–	X	X	X	X	X	?	X	X	X	X	?	?	–	?	?
17 Decision uses the whole debate trajectory	–	?	X	–	X	X	–	X	–	–	X	X	X	X	?	X	–	–	X	?	?	–	?	?
18 Can overturn the majority	X	?	X	X	–	X	–	X	–	X	–	X	–	X	?	X	–	X	–	?	?	X	?	?
19 Output: answer only	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
20 Output: answer + confidence	X	?	X	–	–	X	–	–	–	X	–	–	–	–	?	–	X	X	X	?	?	X	?	?
21 Output: belief distribution	–	?	–	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	X	?	?
22 Fuses a single signal	X	?	X	X	X	X	X	X	–	X	X	X	X	X	?	X	X	X	X	?	?	–	?	?
23 Fuses several signals	X	?	X	X	–	X	–	X	–	X	–	–	X	–	?	X	X	X	X	?	?	X	?	?
24 Aggregation level: answer	X	?	X	X	X	X	X	X	–	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 2 | papers 25-48, rows 25-45

Criterion / original row number	DOWN	MARGIN	ConfLiar	Slim-SC	ConsTrap	ARGORA	Li-Sparse	MoA	CoT	SC	Du	Liang	ChatEval	Smit	Huang	Xiong	Kuhn	Lightman	Cemri	Ladha	Kuncheva	Dawid-Skene	Prelec	Dong
25 Aggregation level: reasoning path	–	?	X	X	–	X	–	–	–	X	–	–	–	–	?	–	–	X	X	?	?	–	?	?
26 Aggregation level: step	–	?	–	X	–	X	–	–	–	–	–	–	–	–	?	–	–	X	–	?	?	–	?	?
27 Aggregation level: token	–	?	X	–	X	–	–	–	–	X	–	–	–	–	?	–	–	–	–	?	?	–	?	?
28 Signal: raw vote count	X	?	–	X	X	X	X	–	–	X	X	X	X	X	?	X	–	X	–	?	?	X	?	?
29 Signal: verbalized confidence	X	?	–	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	X	?	?	–	?	?
30 Signal: token log-prob / likelihood confidence	X	?	X	–	–	–	–	–	–	X	–	–	–	–	?	–	X	X	–	?	?	–	?	?
31 Signal: answer variability / entropy / self-consistency	X	?	–	X	X	X	X	–	–	X	X	X	–	X	?	X	X	X	–	?	?	–	?	?
32 Signal: calibrated confidence	–	?	–	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
33 Signal: trust / credibility	–	?	–	–	–	–	–	–	–	–	X	–	–	–	?	–	–	–	–	?	?	–	?	?
34 Signal: reliability / accuracy prior	–	?	–	–	–	–	–	X	–	–	–	–	–	–	?	–	–	–	–	?	?	X	?	?
35 Signal: disagreement / divergence	–	?	–	X	–	X	–	–	–	–	X	–	–	X	?	X	X	–	–	?	?	–	?	?
36 Signal: diversity of starting answers	–	?	–	–	–	–	–	X	–	–	–	–	–	–	?	–	X	–	–	?	?	–	?	?
37 Signal: task relevance	–	?	–	–	–	X	–	–	–	–	–	–	X	–	?	–	–	–	X	?	?	–	?	?
38 Signal: parse / format quality	–	?	–	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	X	?	?	–	?	?
39 Signal: evidence quality (judge rubric / paraphrase variance)	X	?	X	–	–	X	–	–	–	–	–	X	X	X	?	–	X	X	X	?	?	–	?	?
40 Signal: debate trajectory / dynamics	–	?	–	–	X	X	–	–	–	–	–	X	–	X	?	–	–	–	–	?	?	–	?	?
41 Weights: fixed	X	?	X	X	X	X	X	X	–	X	X	X	X	X	?	X	X	X	X	?	?	–	?	?
42 Weights: learned	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	X	–	?	?	X	?	?
43 Weights: from held-out calibration	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
44 Weights: updated online	–	?	–	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
45 Shows that confidence is unreliable	X	?	X	–	–	X	–	–	–	X	X	X	–	–	?	–	X	X	X	?	?	–	?	?

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 3 | papers 1-24, rows 1-24

Criterion / original row number	DySCo	DarkForest	AgentAuditor	CAGE-CAL	Zhu	Ising	SELENE	DAR	Goel	Wynn	EoT	ReasCons	PPV	MinSentinel	OW-ISP	DivEvid	R²-MAD	Estornell	Kim	Choi	CanDebate	Free-MAD	Cortex	ReConcile
01 Dependence: ignored	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
02 Dependence: mentioned as motivation only	–	–	–	–	X	–	X	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
03 Dependence: modeled	–	X	X	X	–	X	–	X	X	–	–	?	–	–	X	X	X	?	?	X	?	X	?	–
04 Dependence: discounted	–	X	X	X	–	X	–	X	–	–	–	?	–	X	X	–	X	?	?	–	?	X	?	–
05 Dependence: measured at content level	–	–	X	X	–	–	–	X	–	–	–	?	X	X	–	–	–	?	?	–	?	–	?	X
06 Measure: pairwise couplings	–	–	–	X	–	X	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
07 Measure: counterfactual no-communication baseline	–	–	–	X	–	–	–	–	–	X	–	?	–	X	–	X	–	?	?	X	?	X	?	X
08 Measure: error overlap beyond chance	–	–	–	X	–	–	–	–	X	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
09 Measure: second-order answer correlation	–	–	X	X	–	X	–	X	X	–	–	?	–	–	X	X	–	?	?	–	?	–	?	–
10 Measure: agent-level discount	–	X	–	–	–	X	–	X	–	–	–	?	–	–	–	–	X	?	?	–	?	–	?	–
11 Measure: information asymmetry by design	–	X	–	–	–	–	X	–	–	–	–	?	–	–	–	X	X	?	?	–	?	–	?	X
12 Signal level: model	–	X	–	X	–	X	–	–	X	X	X	?	–	X	X	–	–	?	?	X	?	X	?	X
13 Signal level: agent	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
14 Signal level: answer	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
15 Signal level: message	X	X	X	–	X	–	X	X	–	X	X	?	X	X	–	X	X	?	?	X	?	X	?	X
16 Signal level: whole-reasoning embedding	?	–	–	–	–	–	X	X	–	–	–	?	X	–	–	–	X	?	?	–	?	–	?	X
17 Signal level: step / derivation	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
18 Signal level: token	–	–	–	X	–	–	X	X	X	–	–	?	?	–	–	–	X	?	?	–	?	–	?	–
19 Scope: per question	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
20 Scope: per agent	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
21 Scope: per dataset	–	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
22 Null calibration (corrects for overlap expected by chance)	–	–	–	–	–	–	–	–	X	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
23 Extracts steps / derivations	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
24 Reasoning structure: flat	X	X	X	X	X	X	X	X	X	X	X	?	X	X	–	X	X	?	?	X	?	X	?	X

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 3 | papers 1-24, rows 25-48

Criterion / original row number	DySCo	DarkForest	AgentAuditor	CAGE-CAL	Zhu	Ising	SELENE	DAR	Goel	Wynn	EoT	ReasCons	PPV	MinSentinel	OW-ISP	DivEvid	R²-MAD	Estornell	Kim	Choi	CanDebate	Free-MAD	Cortex	ReConcile
25 Reasoning structure: tree	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
26 Reasoning structure: DAG	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
27 Reasoning structure: graph	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
28 Cross-trace overlap read as support	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
29 Cross-trace overlap read as redundancy	X	X	X	X	X	X	X	X	X	X	–	?	X	X	X	X	X	?	?	X	?	X	?	X
30 Semantic clustering of reasoning	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
31 Detects copying / switching without own derivation	–	–	–	–	–	–	–	–	–	–	–	?	–	X	–	–	X	?	?	–	?	–	?	–
32 Separates honest shared reasoning from copied reasoning	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
33 Similarity: embedding	?	–	X	X	–	–	X	X	–	–	–	?	X	–	–	–	X	?	?	–	?	–	?	X
34 Similarity: LLM judge	X	–	–	–	–	–	–	X	X	–	–	?	–	X	–	X	X	?	?	–	?	–	?	–
35 Similarity: lexical	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
36 Similarity: structural	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
37 Needs thresholds / hyperparameters	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
38 Training: training-free	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
39 Training: small calibration split	–	X	–	X	–	–	X	–	–	–	–	?	–	X	–	–	–	?	?	–	?	–	?	–
40 Training: trained classifier	–	–	–	X	–	X	–	–	X	–	–	?	–	X	–	–	–	?	?	–	?	–	?	–
41 Training: trained auditor	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
42 Training: learned graph encoder	–	–	–	X	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
43 Training: EM-fitted model	–	–	–	–	–	X	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
44 Needs gold labels at scale	–	–	X	X	X	–	–	–	X	–	–	?	X	X	X	–	X	?	?	X	?	–	?	–
45 Extra LLM calls at inference	X	X	X	X	X	–	X	X	X	X	X	?	X	X	–	X	X	?	?	X	?	X	?	X
46 Verification: LLM judge	X	X	X	X	–	X	X	–	X	–	–	?	X	X	–	–	X	?	?	–	?	–	?	X
47 Verification: executable / symbolic check	?	X	–	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
48 Verification: process reward model	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 3 | papers 1-24, rows 49-55

Criterion / original row number	DySCo	DarkForest	AgentAuditor	CAGE-CAL	Zhu	Ising	SELENE	DAR	Goel	Wynn	EoT	ReasCons	PPV	MinSentinel	OW-ISP	DivEvid	R²-MAD	Estornell	Kim	Choi	CanDebate	Free-MAD	Cortex	ReConcile
49 Verification: tool / retrieval	?	–	–	–	–	–	–	–	–	–	–	?	–	–	–	X	X	?	?	–	?	–	?	–
50 Verifies intermediate steps	X	–	X	–	–	–	X	X	–	–	–	?	–	X	–	–	–	?	?	–	?	–	?	X
51 Reports token cost vs a baseline	X	X	X	–	–	–	X	–	–	–	X	?	–	–	–	–	X	?	?	–	?	X	?	–
52 Efficiency claim: tokens	X	X	X	–	–	–	X	X	–	–	X	?	–	–	–	–	X	?	?	–	?	X	?	–
53 Efficiency claim: latency	X	–	–	–	–	–	X	X	–	–	–	?	–	–	X	–	X	?	?	–	?	X	?	–
54 Inference-only (no fine-tuning)	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
55 Scales to many agents / samples	X	X	–	X	–	X	–	X	X	–	X	?	X	–	X	X	X	?	?	X	?	X	?	X

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 3 | papers 25-48, rows 1-24

Criterion / original row number	DOWN	MARGIN	ConfLiar	Slim-SC	ConsTrap	ARGORA	Li-Sparse	MoA	CoT	SC	Du	Liang	ChatEval	Smit	Huang	Xiong	Kuhn	Lightman	Cemri	Ladha	Kuncheva	Dawid-Skene	Prelec	Dong
01 Dependence: ignored	X	?	X	X	X	X	X	X	–	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
02 Dependence: mentioned as motivation only	–	?	X	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	X	?	?
03 Dependence: modeled	–	?	–	–	X	–	X	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
04 Dependence: discounted	–	?	–	X	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
05 Dependence: measured at content level	–	?	–	X	–	X	–	X	–	–	–	X	–	–	?	–	–	–	–	?	?	–	?	?
06 Measure: pairwise couplings	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
07 Measure: counterfactual no-communication baseline	X	?	–	–	X	–	X	X	–	–	X	–	X	X	?	X	–	–	–	?	?	–	?	?
08 Measure: error overlap beyond chance	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
09 Measure: second-order answer correlation	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
10 Measure: agent-level discount	–	?	–	X	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
11 Measure: information asymmetry by design	–	?	–	–	X	–	–	–	–	–	–	–	–	–	?	–	–	–	X	?	?	–	?	?
12 Signal level: model	X	?	–	X	X	–	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	–	?	?
13 Signal level: agent	X	?	X	X	X	X	X	X	–	X	X	X	X	X	?	X	–	–	X	?	?	X	?	?
14 Signal level: answer	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
15 Signal level: message	X	?	X	–	X	X	X	X	–	–	X	X	X	X	?	X	X	X	X	?	?	–	?	?
16 Signal level: whole-reasoning embedding	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
17 Signal level: step / derivation	–	?	–	X	–	X	–	–	X	–	–	–	–	–	?	–	–	X	X	?	?	–	?	?
18 Signal level: token	X	?	X	–	X	–	–	–	–	X	X	–	–	X	?	–	X	X	–	?	?	–	?	?
19 Scope: per question	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
20 Scope: per agent	X	?	X	X	X	X	X	X	–	X	X	X	X	X	?	X	–	–	X	?	?	X	?	?
21 Scope: per dataset	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
22 Null calibration (corrects for overlap expected by chance)	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
23 Extracts steps / derivations	–	?	–	X	–	X	–	–	–	–	–	–	–	–	?	–	–	X	–	?	?	–	?	?
24 Reasoning structure: flat	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	–	X	X	?	?	–	?	?

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 3 | papers 25-48, rows 25-48

Criterion / original row number	DOWN	MARGIN	ConfLiar	Slim-SC	ConsTrap	ARGORA	Li-Sparse	MoA	CoT	SC	Du	Liang	ChatEval	Smit	Huang	Xiong	Kuhn	Lightman	Cemri	Ladha	Kuncheva	Dawid-Skene	Prelec	Dong
25 Reasoning structure: tree	–	?	–	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
26 Reasoning structure: DAG	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
27 Reasoning structure: graph	–	?	–	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
28 Cross-trace overlap read as support	X	?	–	–	X	X	X	X	–	X	X	X	X	X	?	X	X	X	X	?	?	–	?	?
29 Cross-trace overlap read as redundancy	X	?	–	X	X	X	X	X	–	–	X	X	–	X	?	X	X	–	–	?	?	–	?	?
30 Semantic clustering of reasoning	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
31 Detects copying / switching without own derivation	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
32 Separates honest shared reasoning from copied reasoning	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
33 Similarity: embedding	–	?	–	X	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
34 Similarity: LLM judge	–	?	–	–	–	X	–	–	–	–	X	–	X	–	?	–	–	–	–	?	?	–	?	?
35 Similarity: lexical	X	?	–	X	X	X	X	X	–	X	X	X	X	X	?	X	X	X	–	?	?	X	?	?
36 Similarity: structural	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
37 Needs thresholds / hyperparameters	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
38 Training: training-free	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	–	?	?
39 Training: small calibration split	X	?	–	–	–	–	–	X	–	–	–	–	–	–	?	–	–	–	X	?	?	–	?	?
40 Training: trained classifier	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	X	X	–	?	?	–	?	?
41 Training: trained auditor	–	?	–	–	–	–	–	–	–	–	–	X	–	–	?	–	–	X	–	?	?	–	?	?
42 Training: learned graph encoder	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
43 Training: EM-fitted model	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	X	?	?
44 Needs gold labels at scale	X	?	–	X	–	X	–	–	–	–	–	–	–	X	?	–	–	X	–	?	?	–	?	?
45 Extra LLM calls at inference	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	–	?	?
46 Verification: LLM judge	–	?	X	–	–	X	–	–	–	–	X	X	X	X	?	–	X	–	X	?	?	–	?	?
47 Verification: executable / symbolic check	–	?	–	–	–	–	–	–	X	–	X	–	–	–	?	–	–	?	X	?	?	–	?	?
48 Verification: process reward model	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	X	–	?	?	–	?	?

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 3 | papers 25-48, rows 49-55

Criterion / original row number	DOWN	MARGIN	ConfLiar	Slim-SC	ConsTrap	ARGORA	Li-Sparse	MoA	CoT	SC	Du	Liang	ChatEval	Smit	Huang	Xiong	Kuhn	Lightman	Cemri	Ladha	Kuncheva	Dawid-Skene	Prelec	Dong
49 Verification: tool / retrieval	–	?	–	–	–	–	–	–	X	–	–	–	–	–	?	–	–	–	X	?	?	–	?	?
50 Verifies intermediate steps	–	?	–	–	–	X	–	–	X	–	X	X	–	?	?	–	–	X	X	?	?	–	?	?
51 Reports token cost vs a baseline	–	?	–	X	X	–	X	X	–	–	X	X	–	X	?	–	–	–	–	?	?	–	?	?
52 Efficiency claim: tokens	–	?	–	X	X	X	X	X	–	–	X	X	–	X	?	–	–	–	–	?	?	–	?	?
53 Efficiency claim: latency	–	?	–	X	X	X	–	X	–	–	–	–	X	X	?	–	X	–	–	?	?	–	?	?
54 Inference-only (no fine-tuning)	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	–	?	?
55 Scales to many agents / samples	–	?	–	X	X	X	X	X	–	X	X	X	X	X	?	X	X	X	–	?	?	X	?	?

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 4 | papers 1-24, rows 1-24

Criterion / original row number	DySCo	DarkForest	AgentAuditor	CAGE-CAL	Zhu	Ising	SELENE	DAR	Goel	Wynn	EoT	ReasCons	PPV	MinSentinel	OW-ISP	DivEvid	R²-MAD	Estornell	Kim	Choi	CanDebate	Free-MAD	Cortex	ReConcile
01 Benchmark: math	X	X	X	X	X	–	–	X	–	X	X	?	–	X	–	–	X	?	?	X	?	X	?	X
02 Benchmark: commonsense / knowledge QA	X	X	X	X	X	–	X	X	X	X	X	?	X	X	X	–	X	?	?	X	?	X	?	X
03 Benchmark: code	–	X	–	–	–	–	–	–	–	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
04 Benchmark: judging / evaluation sets	–	–	–	–	X	X	–	X	X	–	–	?	–	–	X	–	–	?	?	X	?	–	?	–
05 Benchmark: forecasting	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	X	–	?	?	–	?	–	?	–
06 Baseline: single CoT	X	X	X	X	X	–	X	X	–	X	X	?	–	–	–	–	X	?	?	X	?	X	?	X
07 Baseline: self-consistency / majority	X	X	X	X	X	X	X	X	–	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
08 Baseline: vanilla debate	X	X	X	X	X	–	X	X	–	X	X	?	–	–	–	X	X	?	?	X	?	X	?	X
09 Baseline: ReConcile	–	X	–	–	–	–	–	–	–	–	–	?	–	–	–	X	–	?	?	–	?	X	?	–
10 Baseline: Mixture-of-Agents	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	X	–	?	?	–	?	–	?	–
11 Baseline: LLM judge	–	X	X	X	–	–	X	–	X	–	–	?	–	X	–	X	–	?	?	–	?	X	?	–
12 Baseline: trust-weighted aggregation	X	–	–	–	–	X	–	–	–	–	–	?	–	–	X	–	–	?	?	–	?	–	?	X
13 Metric: accuracy	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
14 Metric: calibration (ECE / AUROC)	–	–	–	X	X	–	–	–	–	–	–	?	–	X	–	X	–	?	?	–	?	–	?	X
15 Metric: token cost	X	X	X	–	–	–	X	X	–	–	X	?	–	–	–	–	X	?	?	–	?	X	?	–
16 Metric: error correlation / overlap	X	–	–	X	–	X	–	X	X	–	–	?	X	–	X	X	–	?	?	–	?	–	?	X
17 Metric: flip precision / net gain	–	–	–	–	–	–	–	–	–	X	–	?	X	X	–	–	–	?	?	X	?	–	?	X
18 Controlled known-dependence validation	–	–	–	X	–	X	–	–	–	–	–	?	–	–	X	X	–	?	?	–	?	–	?	–
19 Human-labelled evaluation	–	–	–	–	X	–	–	X	–	–	–	?	–	–	–	–	–	?	?	X	?	–	?	–
20 Seeds / confidence intervals / significance tests	X	–	X	X	X	X	–	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
21 Ablations	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
22 Theory: none	–	X	–	X	–	–	X	–	–	X	–	?	–	X	–	–	–	?	?	–	?	–	?	X
23 Theory: Bayesian	–	–	–	–	X	X	–	–	–	–	–	?	–	–	X	–	X	?	?	X	?	–	?	–
24 Theory: martingale	–	–	–	–	X	–	–	–	–	–	–	?	–	–	–	–	–	?	?	X	?	–	?	–

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 4 | papers 1-24, rows 25-44

Criterion / original row number	DySCo	DarkForest	AgentAuditor	CAGE-CAL	Zhu	Ising	SELENE	DAR	Goel	Wynn	EoT	ReasCons	PPV	MinSentinel	OW-ISP	DivEvid	R²-MAD	Estornell	Kim	Choi	CanDebate	Free-MAD	Cortex	ReConcile
25 Theory: Condorcet / jury	–	–	X	–	–	X	–	–	–	–	–	?	–	–	X	–	–	?	?	X	?	–	?	–
26 Theory: Ising / latent factor	–	–	–	–	–	X	–	–	–	–	–	?	–	–	–	X	X	?	?	–	?	–	?	–
27 Theory: complexity / stability	X	–	–	–	–	–	–	X	X	–	X	?	X	–	X	–	X	?	?	–	?	–	?	–
28 Theory: closed-form condition	X	–	X	–	X	X	–	X	X	–	–	?	X	–	X	X	X	?	?	X	?	–	?	–
29 Failure analysis / negative results	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
30 Evaluates 10 or more models	–	–	–	–	–	–	–	–	X	–	–	?	–	–	–	–	–	?	?	–	?	–	?	–
31 Code released	?	X	X	?	X	?	?	X	X	X	?	?	?	?	?	X	X	?	?	X	?	X	?	X
32 Open models / data / prompts shared	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
33 Limitations stated by the authors	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
34 Peer-reviewed venue	–	–	–	–	X	X	X	–	X	X	X	?	?	X	X	?	?	?	?	X	?	X	?	X
35 Preprint only	X	X	X	X	–	–	–	X	–	–	–	?	?	–	–	?	?	?	?	–	?	–	?	–
36 Year: 2026	X	X	X	X	X	X	X	X	–	–	–	?	X	X	X	X	X	?	?	–	?	X	?	–
37 Year: 2025	–	–	–	–	–	–	–	–	X	X	–	?	–	–	–	–	–	?	?	X	?	–	?	–
38 Year: before 2025	–	–	–	–	–	–	–	–	–	–	X	?	–	–	–	–	–	?	?	–	?	–	?	X
39 Highly cited foundation	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?
40 Relation to CAVE-EoT: threat to novelty	–	X	X	X	–	X	X	–	–	–	–	?	X	X	X	–	–	?	?	–	?	–	?	–
41 Relation to CAVE-EoT: supports motivation	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
42 Relation to CAVE-EoT: baseline candidate	X	X	X	X	X	X	X	X	–	X	X	?	X	X	X	X	X	?	?	X	?	X	?	X
43 Relation to CAVE-EoT: background theory	–	–	X	–	X	X	–	X	X	–	X	?	X	–	X	X	X	?	?	X	?	–	?	–
44 Read in full	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 4 | papers 25-48, rows 1-24

Criterion / original row number	DOWN	MARGIN	ConfLiar	Slim-SC	ConsTrap	ARGORA	Li-Sparse	MoA	CoT	SC	Du	Liang	ChatEval	Smit	Huang	Xiong	Kuhn	Lightman	Cemri	Ladha	Kuncheva	Dawid-Skene	Prelec	Dong
01 Benchmark: math	X	?	–	X	X	X	X	–	X	X	X	X	–	X	?	–	–	X	X	?	?	–	?	?
02 Benchmark: commonsense / knowledge QA	X	?	–	X	X	X	X	–	X	X	X	X	–	X	?	X	X	–	X	?	?	–	?	?
03 Benchmark: code	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	X	?	?	–	?	?
04 Benchmark: judging / evaluation sets	–	?	X	–	–	X	X	X	–	–	–	–	X	–	?	–	–	–	X	?	?	–	?	?
05 Benchmark: forecasting	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
06 Baseline: single CoT	X	?	–	X	X	X	X	–	X	X	X	X	–	X	?	–	–	–	–	?	?	–	?	?
07 Baseline: self-consistency / majority	X	?	–	X	X	X	X	–	–	X	X	X	X	X	?	X	X	X	–	?	?	–	?	?
08 Baseline: vanilla debate	X	?	–	–	–	X	X	X	–	–	X	X	X	X	?	X	–	–	–	?	?	–	?	?
09 Baseline: ReConcile	–	?	–	–	–	X	–	X	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
10 Baseline: Mixture-of-Agents	–	?	–	–	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
11 Baseline: LLM judge	X	?	–	–	–	X	–	X	–	–	X	X	X	X	?	X	X	–	–	?	?	–	?	?
12 Baseline: trust-weighted aggregation	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
13 Metric: accuracy	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
14 Metric: calibration (ECE / AUROC)	–	?	X	–	–	–	–	–	–	X	–	–	–	–	?	–	X	–	–	?	?	–	?	?
15 Metric: token cost	–	?	–	X	X	–	X	X	–	–	X	X	–	X	?	–	–	–	–	?	?	–	?	?
16 Metric: error correlation / overlap	–	?	–	X	–	–	–	X	–	–	–	X	X	X	?	X	–	–	–	?	?	–	?	?
17 Metric: flip precision / net gain	X	?	–	–	X	X	–	–	–	–	X	X	–	X	?	X	–	–	–	?	?	–	?	?
18 Controlled known-dependence validation	–	?	–	–	X	X	X	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
19 Human-labelled evaluation	–	?	X	–	–	–	–	X	X	–	X	–	X	–	?	–	–	X	X	?	?	X	?	?
20 Seeds / confidence intervals / significance tests	X	?	–	X	X	–	X	X	X	X	X	X	X	X	?	–	X	X	X	?	?	–	?	?
21 Ablations	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	–	?	?
22 Theory: none	X	?	X	–	–	–	–	X	X	X	X	X	X	X	?	X	X	X	X	?	?	–	?	?
23 Theory: Bayesian	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	X	?	?
24 Theory: martingale	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

Map 4 | papers 25-48, rows 25-44

Criterion / original row number	DOWN	MARGIN	ConfLiar	Slim-SC	ConsTrap	ARGORA	Li-Sparse	MoA	CoT	SC	Du	Liang	ChatEval	Smit	Huang	Xiong	Kuhn	Lightman	Cemri	Ladha	Kuncheva	Dawid-Skene	Prelec	Dong
25 Theory: Condorcet / jury	–	?	–	X	X	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
26 Theory: Ising / latent factor	–	?	–	–	–	–	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	X	?	?
27 Theory: complexity / stability	–	?	–	–	X	X	X	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
28 Theory: closed-form condition	–	?	–	X	X	X	X	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
29 Failure analysis / negative results	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
30 Evaluates 10 or more models	–	?	–	–	–	–	–	X	X	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
31 Code released	?	?	?	X	?	X	?	X	?	?	?	X	?	X	?	X	X	X	X	?	?	–	?	?
32 Open models / data / prompts shared	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
33 Limitations stated by the authors	X	?	X	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
34 Peer-reviewed venue	–	?	X	X	–	–	X	X	X	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
35 Preprint only	X	?	–	–	X	X	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
36 Year: 2026	–	?	X	–	X	X	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
37 Year: 2025	X	?	–	X	–	–	–	X	–	–	–	–	–	–	?	–	–	–	X	?	?	–	?	?
38 Year: before 2025	–	?	–	–	–	–	X	–	X	X	X	X	X	X	?	X	X	X	–	?	?	X	?	?
39 Highly cited foundation	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?
40 Relation to CAVE-EoT: threat to novelty	–	?	–	X	–	X	–	–	–	–	–	–	–	–	?	–	–	–	–	?	?	–	?	?
41 Relation to CAVE-EoT: supports motivation	X	?	X	X	X	X	X	X	–	X	X	X	X	X	?	X	X	X	X	?	?	X	?	?
42 Relation to CAVE-EoT: baseline candidate	X	?	–	X	X	X	X	X	X	X	X	X	X	X	?	X	X	X	–	?	?	X	?	?
43 Relation to CAVE-EoT: background theory	–	?	–	–	X	X	X	–	–	–	–	–	–	–	?	–	–	–	–	?	?	X	?	?
44 Read in full	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	?	X	?	?

X = yes in reported scope – = checked no ? = unverified. Original cells are reproduced; row/column order is unchanged.

15 | Consolidated evidence notes

These are the short source-review notes produced while completing the maps, grouped by paper. They retain section/page references and scope qualifications. Historical notes can differ in marking conventions or be more tentative than a later clarification; the main narrative explains the consequential distinctions. Row references belong to the named map or map pair, not to PDF page numbers.

The notes are paraphrases rather than reproductions of the papers. Repeated source links are retained for traceability. Priority queues, routine upload details and whole-session handoffs are omitted because the paper-specific notes and final map archive carry the useful information.

Evidence notes | DySCo

Criteria Map 1 Batch 1 Evidence Log

Source: Gou & Liu (2026), arXiv:2606.01828v1; https://arxiv.org/html/2606.01828 (Method §4; Experiments §5; Discussion §6)

• Setting/position (rows 2-4, 17-19): Method §4 initial independent solve, sparse multi-round discussion, final trust-weighted vote; independent first then communication (Eq. 3, 10).

• Trust/diversity (rows 12, 38, 41, 44-45, 68-69): §4.2 Eq. 4 measures divergence and includes historical trust, confidence, task relevance and token cost to choose Top-k neighbors; §5 ablates diversity score.

• Messages/topology (rows 50, 54-55, 61-62): §4.2-4.3 selects per-round sparse directed edges and sends structured short critiques containing answer, critical reason and potential counterexample; not full reasoning broadcast.

• Stopping/cost (rows 15, 65, 71): §4.5 consensus-entropy or stability early stop; §5 reports tokens/latency and ablates compression and stopping.

• Models (rows 31, 37): §5.1 six agents use same gpt-3.5-turbo backbone; system prompts unified; no open-weight, different-family or strong/weak mix is tested.

• Scope/negative judgments: §4-5 supplies no statistical independence estimator, calibration objective, controlled seeded-hint experiments, or confidence-threshold message filter; related work is not counted as own design.

Criteria Maps 2 3 Batch 1 Evidence Log

Identity: Wanshuang Gou and Zihan Liu, Dynamic Trust-Aware Sparse Communication Topology for LLM-Based Multi-Agent Consensus, arXiv:2606.01828v1 (2026). Source: paper; method equations 3-10, Experiments, Discussion and prompt appendix.

• M2 1-5/17-18/22-24/28-29/31/33-37/39-41/44: final last-round voting multiplies trust and verbal confidence; recursive trust combines prior trust, verifier quality and agreement. A nonuniform vote can overturn the raw majority. Plain voting/single-signal marks include the implemented SC baseline. Fixed scoring coefficients coexist with online trust updates.

• M2 14-16/19: equality and optional semantic normalization/structured fields; final return is an answer. No trained/dependence-based final adjudicator or calibrated posterior. Discussion warns about confidence, without demonstrating its unreliability (45: -).

• M3 1/13-15/19-20/24/28-29: agent/answer/message heuristics, flat rationales; agreement reinforces trust, divergence discourages redundant exchange. No statistical-dependence estimator, null calibration, copied-versus-honest separation or atomic step extraction. Critique compression is not a derivation graph.

• M3 16/33/47/49 (?): embedding divergence and executable/retrieval verifier alternatives are mentioned, but the reported configuration does not identify their use. Judge critiques and normalization are specified (34/46); critiques inspect reasoning errors (50).

• M3 37-38/45/51-55: tunable budgets/thresholds, no fine-tuning, extra critique/revision calls, token/latency comparisons and scaling to 12 agents. No calibration split, trained auditor or graph encoder.

Counts: M2 26 X, 19 -, 0 ?; M3 21 X, 30 -, 4 ?.

Criteria Map 4 Evidence Log

Identity: Wanshuang Gou and Zihan Liu, Dynamic Trust-Aware Sparse Communication Topology for LLM-Based Multi-Agent Consensus, arXiv:2606.01828v1 (2026). Sources: https://arxiv.org/pdf/2606.01828v1

• §§4-6: GSM8K/LogiQA/StrategyQA, CoT/vote/dense-debate and trust-only comparisons, accuracy/tokens/latency, three seeds and component ablations. Proposition 1 analyzes consensus stability/its condition; this is not known-dependence simulation or Bayesian/null-calibration validation. Appendix prompts are supplied. Relevance: motivation/baseline; no atomic derivation-independence novelty claim. Current arXiv record has no venue registration. Citation standing, code release and a literal complete-reading record remain unverified.

Evidence notes | DarkForest

Criteria Map 1 Batch 1 Evidence Log

Source: Li et al. (2026), arXiv:2605.25188v1; https://arxiv.org/html/2605.25188 (§§3-4, App. A, B, D)

• Setting/communication (rows 1, 3-4, 17, 19, 50, 63): §3.1 independent candidate agents; coordinator receives policy-restricted compact beliefs in one call. Not iterative multi-round debate; independent stage is followed by centralized coordination.

• Independence (rows 10, 24-26, 48): §3.2 explicitly disclaims full statistical independence because models may share data/architecture; §3.4 discounts known correlated support and calibrates empirical support-pattern reliability; Appendix A discusses dependence as latent agent type.

• Calibration (rows 7, 26, 44): §3.4 estimates agent and joint-support reliability from held-out examples, bounds confidence influence, computes posterior-like belief; Appendix D.4 calibrations ablated. No content-step dependence estimator.

• Models/roles (rows 34, 36, 40, 46): §4.1 and Appendix B.1 use Qwen, Mathstral, finance-Llama and Saul open-weight 7B-8B agents with domain differences plus centralized coordinating model.

• Messages (rows 59-62, 69): §§3.5-3.6 coordinator receives canonical candidate labels, confidence, support-pattern, posterior and uncertainty; exposure controlled and full traces withheld by default. Appendix D.3 compares belief summary, reasoning summary and full traces.

• Validation/cost (rows 5-6, 71): §4 six benchmarks, token-cost measurements; Appendix D ablations for independent evidence, calibration, disclosure, guardrail. Errors after communication motivate approach (§1).

• Boundaries: The paper's coordinator+guardrail preserves minority possibilities but does not run selective debate, formal dynamic topology, seeded-hint tests or step-by-step content-dependence extraction; baseline debate results are not counted as DarkForest features.

Criteria Maps 2 3 Batch 1 Evidence Log

Identity: Yi Li et al., DarkForest: Less Talk, Higher Accuracy for Multi-Agent LLMs, arXiv:2605.25188v1 (2026). Source: paper, §§3.2-3.6, A/B and D.1-D.6.

• M2 5-6/10/14-16/18-24/28-29/31-32/34-36/38-39/41-43: candidate clusters receive frozen reliability, joint-support, parse-quality, confidence and dependence factors; normalized belief is disclosed to a coordinator, with deterministic override. Belief-stage output includes candidate confidences/distribution; the complete pipeline returns an answer. M2 1-3/22 and M3 1/38 include implemented majority, ReConcile and uniform-weight baselines/ablations. No full generative dependence model (M2 9: -; A.3 explicitly says posterior-like).

• M3 3-4/10-15/19-21/28-29/39: empirical support-pattern reliability and agent discounts model shared support at answer/agent level. Calibration uses 50-114 held-out examples per benchmark (B, Table 5), then freezes parameters. No chance-overlap correction, matched dependence counterfactual, atomic derivation extraction or honest-versus-copied separation.

• M3 24/35/37/45-47/51-52/54-55: canonical equality, flat traces in D.3, thresholds, one coordinator call, token comparisons and 3→5-agent scaling. Unit-test/program execution is benchmark verification (B.2), not an inference-time evidence checker. LLM weights remain unchanged; full orchestration uses fitted calibration parameters.

• M2 45/M3 53: -; bounded confidence and latency motivation are not an empirical confidence-reliability or latency result. Answer clusters are not reasoning clusters (M3 30: -).

Counts: M2 28 X, 17 -, 0 ?; M3 26 X, 29 -, 0 ?.

Criteria Map 4 Evidence Log

Identity: Yi Li et al., DarkForest: Less Talk, Higher Accuracy for Multi-Agent LLMs, arXiv:2605.25188v1 (2026). Sources: https://arxiv.org/pdf/2605.25188v1

• §4/B: MATH, HumanEval, MMLU-Pro, finance/law reasoning; six baseline families and token/quality comparisons. D ablates reliability/calibration/independence/disclosure/guardrail; calibrated internal beliefs do not establish ECE/AUROC reporting or a controlled known-dependence test. Paper explicitly releases code and prompt/configuration details. Relevance: answer-level dependence discounts/calibrated beliefs overlap the proposed aggregation layer (40), although

there is no chance-calibrated step-chain count. Literal full-reading/citation status is unverified.

Evidence notes | AgentAuditor

Criteria Map 1 Batch 1 Evidence Log

Source: Yang et al. (2026), arXiv:2602.09341v2; https://arxiv.org/html/2602.09341 (§§3-6; Appendices E-G)

• Version/identity: arXiv:2602.09341v2 (3 Sep 2026) includes Gengshuo Liu in author line in addition to six names in the earlier project description; paper title and identifier remain the same. Paper Key itself was not supplied.

• Step extraction/overlap (rows 14, 44, 61-62, 68): §4.1.1 segments complete agent traces into atomic reasoning steps; §4.1.2 cosine similarity clusters equivalent steps into a Reasoning Tree with agent-support sets; §4.2 selects critical divergence points and compact packets for the Auditor. This is alignment/deduplication, NOT a null-calibrated independent evidence-chain count.

• Independence/correlation (rows 10, 25-26, 48): §1 notes shared pretraining, alignment and prompting errors; Appendix E and F formalize why correlated errors break majority voting using pairwise correctness correlation. Theory is not an empirical question-level error-dependence estimator.

• Sycophancy/training (rows 11-13, 16): §5 Anti-Consensus Preference Optimization trains an Auditor using evidence-verified majority/minority branch preferences; §6.2 evaluates majority-wrong minority-correct subsets.

• Settings/roles (rows 1-3, 19, 46): §6 integrates the final aggregation plug-in into LLM-Debate, DyLan, GPTSwarm, AgentPrune; App. G additionally uses three independently generated candidate traces. Auditor is an asymmetric adjudication role, not a redesigned debate communication topology.

• Models/capability (rows 31, 33-34, 36, 47): §6.4 tests Qwen2.5 3B and 7B vs LLaMA 3B backbones and weaker-to-stronger trends; do not interpret backbone comparisons as a simultaneous strong+weak agent mixture.

• Cost (rows 6, 71): §6.3 reports total audit input/output tokens vs LLM-as-Judge and LLM-as-Solver (973 vs 1762, 2046).

• Boundaries: related-work topology comparisons not counted; selective audited evidence is sent to post-debate Auditor and does not require initial agents to communicate under new topology.

Criteria Maps 2 3 Batch 1 Evidence Log

Identity: Wei Yang, Shixuan Li, Gengshuo Liu, Heng Ping, Peiyu Zhang, Paul Bogdan and Jesse Thomason, Auditing Multi-Agent LLM Reasoning Trees Outperforms Majority Vote and LLM-as-Judge, arXiv:2602.09341v2 (2026). The PDF author line includes Gengshuo Liu, whom the abstract-page metadata omits. Source: paper, §§4-6 and E-G.

• M2 1/6-7/14/16/18-19/22-26/28/31/35-39/41-42: majority fallback/baselines, flat and trained judges, branch/step adjudication, exact/alias answers and structured packets. Support counts are auxiliary; contextual rubric checks facts, logic, constraints, assumptions and format/hygiene. Learned Auditor parameters coexist with fixed vote baselines. Branch selection is not numerical step/DAG weighting (13: -). Final reasoning traces are not a debate-time trajectory score (12/17/40: -).

• M3 3-5/9/13-15/17/19-21/23/25/29-30/33/36: atomization, cosine/centroid step alignment conditioned on a shared prefix, tree branches and semantic deduplication. Population correctness correlation is a theoretical model; it is not an empirical per-question dependence estimate. Shared steps retain auxiliary support (28); multiplicity is also treated as redundancy (29). No DAG/general reasoning graph, honest-versus-copied separation or null calibration.

• M3 1/24/35/38/54: implemented voting/flat-judge and prompt-only variants. M3 37/41/44-46/50-52: trained ACPO Auditor uses benchmark gold-derived preferences, thresholds, extra local calls and intermediate-step rubric checks; token comparison in §6.3. No process-reward model, trained graph encoder or executable/retrieval verifier. Native three-agent experiments do not demonstrate many-agent scaling (55: -).

Counts: M2 21 X, 24 -, 0 ?; M3 31 X, 24 -, 0 ?.

Criteria Map 4 Evidence Log

Identity: Wei Yang, Shixuan Li, Gengshuo Liu, Heng Ping, Peiyu Zhang, Paul Bogdan and Jesse Thomason, Auditing Multi-Agent LLM Reasoning Trees Outperforms Majority Vote and LLM-as-Judge, arXiv:2602.09341v2 (2026). The PDF author line includes Gengshuo Liu, whom the abstract-page metadata omits. Sources: https://arxiv.org/pdf/2602.09341v2

• §§4-6/E-G: six math/knowledge/QA tasks; CoT/SC/vote/flat-judge comparisons and four MAS integrations; reports accuracy, audit token budget, seeds and component/capability ablations. E-F derives correlated-majority failure

conditions; a theoretical correlation parameter is not an empirical error-correlation metric or known-copying validation. Code is linked. Relevance: atomic-step alignment, semantic deduplication and reasoning-tree audit are the strongest direct design overlap; no honest-independent overlap null is supplied.

Evidence notes | CAGE-CAL

Criteria Map 1 Batch 1 Evidence Log

Source: Huang et al. (2026), Counterfactual Graph for Multi-Agent LLM Calibration, arXiv:2605.30653v1, https://arxiv.org/html/2605.30653

• §§3, 5.2-5.4 (rows 7,10,25,26): compares matched iid and post-communication graphs and estimates instance-neighborhood correctness correlation and model/role/answer/topology group hyperedges; a TRAINED graph encoder calibrates plurality-answer probability, not a training-free reasoning-step independence estimator.

• §4 and Appendix B (rows 10-12,21-22): analyses diversity-induced under-confidence and communication-induced over-confidence/false consensus; peer exchange can correct or corrupt answers depending on topology.

• Appendix D.2 (rows 32,34,36,39-41,44-46,48): four distinct open-weight backbones paired with five prompting roles, reusing the same panel across topologies; correlated family behavior explicitly modeled.

• Appendix D.3 (rows 1-4,49-53,56-57,59,61,63-64): explicitly implements iid, two-round all-to-all answer broadcast, sequential chain, hub-spoke and tree; shared messages comprise prior agent ANSWERS rather than full reasoning derivations.

• §6.5 and Appendix F.1 (row 55): CAGE-Select chooses one of five whole-topology completed panels per query using learned calibrated confidence; it DOES NOT dynamically prune communication edges during one ongoing debate, nor pre-trigger debate selectively.

• Appendix D.5 (rows 6,71 marked negative): reports GPU training/rollout computation cost, not measured inter-agent communication-token cost; no direct reduction objective.

• §§5.3-5.4 and Appendix C (rows 58,62 marked negative): sentence embeddings are of final answers; graph edges are local correctness correlations and topology, NOT extracted reasoning steps or transmitted structured derivations.

• §9 (rows 54,67-70 marked negative): online adaptive communication and dynamic agent selection are described as future work; CAGE-Cal calibrates completed panel outcomes rather than filtering messages.

Criteria Maps 2 3 Batch 1 Evidence Log

Identity: Jiatan Huang, Mingchen Li, Ziming Li, Sunjae Kwon, Hong Yu and Chuxu Zhang, Counterfactual Graph for Multi-Agent LLM Calibration, arXiv:2605.30653v1 (2026). Source: paper, §§3-6 and B-D/F.

• M2 1-2/14/16/19-20/22-24/28-32/35-37/40-43/45: panel plurality is scored by a learned correctness calibrator; CAGE-Select chooses among completed topology panels. It does not select a minority within a panel (18: -), fit a full generative dependence law (9: -), or emit a distribution over competing answers (10/21: -). Signals include vote/entropy, mean answer-token likelihood, query-neighborhood relevance, IID-versus-communicated state and trained calibration; elicited confidence and single-signal marks include implemented baselines.

• M3 3-9/12-14/18-21/28-29/33/35: agent/answer dependency graphs use locally estimated Pearson correctness correlations, family/role/answer hyperedges and matched cached IID panels. Answer embeddings and identity groups are content inputs (5), not reasoning overlap. Pearson centering measures excess co-error over independent marginals (8); no overlap statistic is calibrated against a chance-null distribution (22: -). This distinction preserves the explicit counterfactual-baseline mark (7).

• M3 24/37-40/42/44-46/54-55: raw CoT roles, thresholds, baseline inference-only variants, learned graph encoder/classifier, gold-derived training targets and 100-query validation calibration per benchmark (D.5). Extra judge calls are confidence baselines; the calibrator uses cached panels. Base LLMs are frozen; full CAGE-CAL still trains its encoder/head. F.3 compares N=10 and N=20.

• M3 23/25-27/30-32/47-53: -; no extracted reasoning tree/DAG, reasoning clustering, copied-versus-honest discriminator, step/tool verifier or token/latency-efficiency comparison. GPU training cost is distinct from communication-token cost.

Counts: M2 22 X, 23 -, 0 ?; M3 30 X, 25 -, 0 ?.

Criteria Map 4 Evidence Log

Identity: Jiatan Huang, Mingchen Li, Ziming Li, Sunjae Kwon, Hong Yu and Chuxu Zhang, Counterfactual Graph for Multi-Agent LLM Calibration, arXiv:2605.30653v1 (2026). Sources: https://arxiv.org/pdf/2605.30653v1

• §§5-6/C-F: TriviaQA/TruthfulQA/MMLU-Pro/GSM8K/BBH, calibration/likelihood/judge and vote baselines, ECE/AUROC/accuracy and local correctness correlations. Matched cached IID/post-communication panels and controlled topologies support row 18; this is communication-induced dependence control, not ground-truth copied-step labels. Three rollout seeds, Wilcoxon significance and ablations are reported. Code release unverified; prompts/configurations are published. Relevance: strong counterfactual-calibration overlap, but a trained answer/agent graph differs from training-free derivation counting.

Evidence notes | Zhu

Criteria Map 1 Batch 2 Evidence Log

Primary source: Zhu et al., Demystifying Multi-Agent Debate: The Role of Confidence and Diversity, Findings ACL 2026, https://aclanthology.org/2026.findings-acl.1694.pdf (22 pp).

• §3 p.4: five homogeneous agents sampled independently from one model, initially give answers, run five fixed multi-turn fully connected debate rounds, with majority vote after final round (rows 1-4,31,51-52,64).

• §4.1 pp.4-5: generate 10 candidates for 5 slots, calculate unique-answer diversity and greedily choose a maximally diverse initial set; explicitly TRAINING-FREE selection only; protects correct minority, does not filter during-round messages (rows 16-17,41,44-45).

• §4.2 p.5 and App. B pp.13-14: confidence expression/calibration and confidence perception are trained using SFT/GRPO/LoRA; distinction: full system is NOT training-free (rows 7,12,18,27).

• §5 pp.5-6: Dirichlet-categorical theoretical model; positive correlation of confidence and correctness gives submartingale; does NOT estimate statistical dependence of derivations/errors (row 29 yes; independence-modeled row 26 no).

• §6 pp.6-7: Qwen2.5-7B and Llama3.1-8B tested as separate homogeneous panels, not mixed families simultaneously; open-weight, sampled with T=1.0 top-p=.9; tests effect of temperature in App. C p.14 (rows 31,36,38; 34/35/40 no).

• §6 and App. A pp.12-13: each peer receives full earlier reasoning/answer/confidence via all-to-all exchange; numerical confidence changes influence rather than deleting messages (rows 50-52,58,60).

• §7 pp.7-8: measures correlation of diversity with improvement, evaluates worse-than-vote vanilla trajectories and better interventions; no direct message-token-cost reporting found despite reporting computational GPU-hours in App. B (rows 21-23,44,71 no).

• Scope: independence of sampled initial answers is not statistical independence of correctness events; no dependence/null overlap estimator, asymmetric judge, dynamic topology, debate trigger, early stopping or message filtering.

Criteria Maps 2 3 Batch 2 Evidence Log

Identity: Xiaochen Zhu, Caiqi Zhang, Yizhou Chi, Tom Stafford, Nigel Collier and Andreas Vlachos, Demystifying Multi-Agent Debate: The Role of Confidence and Diversity, Findings ACL 2026, Anthology ID 2026.findings-acl.1694. Source: publisher PDF, 22 PDF pages; §§3-7, Appendices A-D. Page numbers below are PDF page positions.

• M2 1-2/14/16/19-20/24/28: §3 p.4 ends every variant with majority voting on terminal answers; Appendix A p.13 specifies answer/confidence tags. Terminal consensus here means last-round vote, not required unanimity. Confidence accompanies intermediate agent outputs; the final pooled vote returns an answer. No separate final judge, learned adjudicator, minority override or confidence-weighted terminal vote (3-13/18: -). Learn2Agg is a trained debate-update ablation, not a separate final-rule classifier (§7 p.8).

• M2 22-23/29/31-32/35-36/41-42/44: diversity-aware greedy initialization selects distinct answer IDs (§4.1 p.4); confidence-modulated revisions use learned calibrated verbal scores (§4.2 p.5). Uniform vote is a single-signal/fixed-weight variant; the combined process uses diversity, votes and confidence. Confidence is re-expressed each round and enters online weighted updates in the theoretical model (D.3 p.17), rather than a fitted error-dependence law. There is no held-out post-hoc weight calibration (43: -).

• M2 17/40: -; each actual update sees the previous round only, explicitly Markovian in B.2 pp.13-14. The theoretical belief process and plotted correctness transitions are not a whole-trajectory decision feature. Format penalties in GRPO are training rewards, not an inference aggregation feature (38: -). No token likelihood, reliability prior, evidence rubric or step/DAG weighting is used.

• M2 45: X; Table 6 p.13 measures vanilla confidence miscalibration (ECE 0.257/0.217) and improves it through training. This is the paper's own evidence, beyond related-work warnings.

• M3 1-2/13-15/19-21/24/28-29/35: vote/update rules do not estimate statistical dependence; conformity/diversity collapse motivates intervention (§§1-2 pp.2-3). Exact answer multiplicity is support for voting and duplicate initial answers are redundant for coverage. Confidence is question/agent specific; calibration and diversity correlations are also evaluated over datasets (§7 pp.7-8, B.1 p.13). Rationales remain flat text; prompting stepwise reasoning does not extract, align or cluster atomic derivations.

• M3 3-11/16-18/22-23/25-27/30-34/36: -; the DCM analysis models belief updates, without fitted inter-agent error dependence, chance-overlap correction, matched no-communication calibration, semantic reasoning embeddings, a copying detector, or honest-versus-copied separation. It does not construct a reasoning graph.

• M3 37-38/44-45/54: X; diversity-only and vanilla variants are training-free/inference-only, while the full confidence method uses SFT plus GRPO/LoRA, including 5k confidence-target and 10k difficult training examples (B.1-B.2 pp.13-14). This is substantial gold-supervised policy training, not a small calibration split, EM fit, trained auditor or graph/classifier head (39-43: -). Extra candidate sampling and debate turns require additional LLM calls. No dedicated verification judge, executable check, retrieval or process reward model (46-50: -); no measured token/latency-efficiency or large-panel scaling claim (51-53/55: -; larger populations are future work, Limitations p.9).

Criteria Map 4 Evidence Log

Identity: Xiaochen Zhu, Caiqi Zhang, Yizhou Chi, Tom Stafford, Nigel Collier and Andreas Vlachos, Demystifying Multi-Agent Debate: The Role of Confidence and Diversity, Findings ACL 2026, Anthology ID 2026.findings-acl.1694. Sources: https://aclanthology.org/2026.findings-acl.1694.pdf

• §§5-7/A-D: seven math/commonsense/knowledge/preference tasks, CoT/SC/vanilla debate and confidence/diversity ablations. Reports accuracy, confidence ECE and diversity/outcome correlations; these are not error-overlap metrics (16: -). DCM/Bayesian belief updates and submartingale conditions are its own analysis. Own code linked; Findings ACL 2026. Motivation/baseline/background; no derivation-null mechanism.

Evidence notes | Ising

Criteria Map 1 Batch 2 Evidence Log

Primary source: Balasubramanian, Podkopaev & Kasiviswanathan, Dependence-Aware Label Aggregation for LLM-as-a-Judge via Ising Models, ICML/PMLR 306 (2026), https://www.shivakasiviswanathan.com/LLMaaJ.pdf (46 pp), proceedings.mlr.press/v306/balasubramanian26a.html.

• §§1-2 pp.1-6: independently obtained per-item binary judge votes are input to a post-hoc classifier; its own pipeline has NO judge-to-judge communication, debate, or per-query topology (rows 1,3,19,49,59).

• §2 pp.3-6 and Appendix B pp.17-18: conditional independence is a baseline assumption that the paper CHALLENGES; pairwise couplings W^(0), W^(1) encode class-conditional voting correlation, including redundant agreement from shared failures. Thus rows 24-26 all apply across compared variants; NOT semantic overlap of reasoning steps.

• §2 pp.4-6 and §3 pp.6-9: class-dependent Ising yields quadratic Bayes log odds; class-independent Ising yields correlation-adjusted linear weighted voting; formal finite-K and asymptotic error separation (rows 10,12,13,27-29).

• §4.1 pp.9-10: actual six-model judge panel combines Claude Sonnet/Haiku 4.5 (closed), gpt-oss-120B, Llama 4 Maverick/Scout and DeepSeek-R1 (open); varying model strengths/sizes and different families are covered, but no agents debate or exchange their traces (rows 34-37,40,47,59).

• §4 and Table 1 pp.9-11: class-independent/dependent Ising parameters fitted by generalized EM on unlabeled item votes, outperform CI weighted/uniform vote on relevance, toxicity, summarization; work targets test accuracy and posterior calibration but paper's main real-world table reports accuracy, not ECE.

• Appendix G.3 pp.42-46: LLM judge prompt may elicit reasoning internally, but Ising aggregator consumes binary labels ONLY, not full reasoning or semantic evidence. No cost/topology/rounds/message-selection rows beyond no-communication.

• Caveat row 24: applies to explicitly modeled/compared CI BASELINE; paper's proposed Ising method does not itself assume judge independence. If map intends rows exclusively for proposed method, this is a potential disagreement to revisit.

• No null-calibrated step overlap, seeded-hint validation, early-stopping, token-cost goal, or sent structured derivations.

Criteria Maps 2 3 Batch 2 Evidence Log

Identity: Balasubramanian, Podkopaev and Kasiviswanathan, Dependence-Aware Label Aggregation for LLM-as-a-Judge via Ising Models, ICML/PMLR 306 (2026). The author-hosted January 29 preprint prints the expanded author names Krishnakumar Balasubramanian and Shiva Prasad Kasiviswanathan. Sources: proceedings identity; author-hosted full text reviewed, 46 PDF pages. Section/page references concern that full text.

• M2 1/5/8-9/14/16/18-24/28/32/34-35/41-42: §§2.1-2.2 pp.4-5 derive a quadratic Bayes classifier and correlation-adjusted linear weighted vote; §4/Table 1 pp.9-10 implements uniform and reliability-weighted CI baselines. Appendix D pp.30-32 fits fields/couplings/class priors with generalized EM and outputs both posterior probabilities and thresholded labels. The fitted statistical classifier can select a minority label (Appendix B pp.17-19). These are binary class posteriors, not semantic answer clusters (10/15: -); output confidence/distribution marks refer to the explicitly emitted posterior record.

• M2 22-23: the uniform baseline uses only vote count; the dependence-aware classifier combines learned per-judge strengths, class prior and pair interactions. Input labels are XML-parsed and compared exactly (G.3 pp.44-46). The aggregator consumes those labels, not the judges' prose, relevance/quality rubrics, verbal confidence or token probabilities (25-27/29-31/33/36-40: -). No debate, trajectory, final LLM adjudicator, routing or step weighting occurs (2-4/6-7/11-13/17: -).

• M2 43-44: -; weights come from offline unlabeled training samples, not a held-out confidence-calibration split or online per-question updates (§4 p.10, D pp.30-32). M2 45: X; Appendix B p.18 exhibits a CI posterior of 0.968 for the opposite label from the dependence-aware posterior, despite using correct judge marginals. This demonstrates dependence-induced confidence failure, not verbal-confidence miscalibration.

• M3 1/3-4/6/9-10/12-14/19-21/28-29: the CI baseline ignores dependence; the proposed model fits class-conditional pair couplings and discounts redundant agreement (§§1-2 pp.1-5). Parameters are model/judge specific and learned over each task's dataset, with a posterior for each item. Shared vote labels can provide support or redundant evidence. Dependence is operational, rather than motivation only (2: -).

• M3 5/7-8/11/16-18/22-23/25-27/30-34/36: -; there is no semantic reasoning-overlap estimator, matched no-communication counterfactual, explicitly chance-adjusted error-overlap statistic, copied-versus-honest discriminator, extraction of derivations, or reasoning tree/DAG/graph. Ising graphical models encode judge-vote dependency, not reasoning structure. Quadratic vote interactions justify second-order answer-correlation (9), without implying null calibration (22). Hypothesis tests suggested in the conclusion pp.11-12 are future guidance, not implemented overlap calibration.

• M3 24/35/37-38/40/43/46/54-55: prompts elicit flat reasoning and binary XML labels (G.3); exact label matching, manually tuned EM tolerances/regularization, an EM-fitted statistical classifier and frozen LLM judges are used (D pp.30-32). Training-free applies to the implemented uniform-vote baseline, not fitted Ising. No gold labels are needed for fitting (44: -); benchmark gold is evaluation-only. A small labeled validation split is suggested, not implemented (39: -). Scaling is supported by large-K theory and simulations up to 300 judges (F.2 pp.40-42), beyond the six-LLM real panel.

• M3 15/41-42/45/47-53: -; aggregation uses binary votes without message-level reasoning features, a trained auditor/graph encoder, extra aggregator LLM calls, intermediate-step verification, executable/retrieval/PRM checks or token/latency comparisons. M3 46 is the initial LLM annotation/summary-quality judging stage, not a final LLM adjudicator.

Criteria Map 4 Evidence Log

Identity: Balasubramanian, Podkopaev and Kasiviswanathan, Dependence-Aware Label Aggregation for LLM-as-a-Judge via Ising Models, ICML/PMLR 306 (2026). The author-hosted January 29 preprint prints the expanded author names Krishnakumar Balasubramanian and Shiva Prasad Kasiviswanathan. Sources: https://proceedings.mlr.press/v306/balasubramanian26a.html, https://www.shivakasiviswanathan.com/LLMaaJ.pdf

• §§2-4/B-F: binary judge relevance/toxicity/summarization tasks, uniform/CI reliability-weighted voting, accuracy and pairwise dependence. Synthetic Curie-Weiss/latent-correlation simulations supply known-dependence validation. Own Bayes classifier, Ising/latent-factor separation and decision conditions support theory rows. Unlabeled EM fitting is not calibrated-confidence ECE evaluation; six actual judges are fewer than ten models. Code release unclear; prompts supplied. Relevance: statistical dependence-aware aggregation/theory benchmark, without content-level chains.

Evidence notes | SELENE

Criteria Map 1 Batch 2 Evidence Log

Primary source: Verma et al., SELENE: Selective and Evidence-Weighted LLM Debating for Efficient and Reliable Reasoning, EACL Industry 2026, https://aclanthology.org/2026.eacl-industry.7.pdf (10 pp).

• §§3.1-3.3 pp.3-4 and Algorithm 1 p.10: independent initial outputs followed by conditional bounded multi-round all-peer debate; disagreement D and confidence-likelihood gap M gate debate ON/OFF; early stop based on semantic-stability ΔD < epsilon, not a dynamic message-filtering topology (rows 1-4,17-20,50-52,64-65,67).

• §3.2 pp.3-4: D is cosine distance among query+rationale+answer embeddings, M is difference between self-confidence and answer log-likelihood; NOT null-corrected step-level evidence dependence (rows 7,12,44). Appendix Algorithm 1 uses answer-only embeddings as shorthand whereas §3.2 explicitly specifies rationale-inclusive embedding.

• §3.4 pp.4-5: after debate, EWSC perturbs/paraphrases external evidence and performs three parallel judge evaluations, aggregates scores weighted by cross-evaluation variance. This is evidence-robustness calibration, not independent evidence-chain counting (rows 3,9,19,46,62).

• §4.3 p.5 and Appendix A pp.8: primary GPT-4 Turbo closed API backbone shared by all agents; separate generalization tests with GPT-4o-mini and Claude Haiku, NOT mixed-model simultaneous panels (rows 31,37; 34/35/40 no).

• Appendix B pp.8-9: initial agent messages contain answer+confidence and structured intermediate reasoning; agents see all prior messages and compare rationales; final evidence judge sees candidate answer plus perturbed/subsampled evidence (rows 58-60,62). Multiple pass types can apply at distinct stages.

• Tables 1-5 pp.5-6: reports debate skip rates, normalized token costs 1.9× vs 3.7× and judge variance stability; SDI reduces repeated exchanges but does not eliminate quadratic all-to-all fanout when active (rows 6,14,71).

• Appendix D p.10: the debate loop stops on semantic stability, while the BEFORE-debate trigger uses confidence/disagreement thresholds; neither filter removes individual transmitted peer messages (rows 65,67 yes; rows 66,68-70 no).

• No tested independence-of-errors model, no shared-hint/step extraction, no seeded validation, no agent-level information asymmetry in the debate.

Criteria Maps 2 3 Batch 2 Evidence Log

Identity: Akshay Verma, Swapnil Gupta, Deepak Gupta, Prateek Sircar and Siddharth Pillai, SELENE: Selective and Evidence-Weighted LLM Debating for Efficient and Reliable Reasoning, EACL Industry 2026, Anthology ID 2026.eacl-industry.7. Source: publisher PDF, 10 PDF pages; §§3-5 and Appendices A-D.

• M2 1-2/6/14/16/18-20/22-25/28: the implemented self-consistency baseline uses ordinary voting (§4.1 p.5); the skip branch adopts consensus then applies a single factual judge, and the debate branch selects the highest judged candidate score (§3.4 p.4, Algorithm 1 p.10). Judge selection can favor a minority answer. Initial records contain binary answer, reasoning and confidence; judge records emit correctness scores, with the complete algorithm returning an answer. No normalized belief distribution/semantic answer clusters, dependence-law fit, trained final auditor, routing or step/DAG weighting (7-13/15/21/26-27: -). Whole-path embedding aggregation refers to SDI's rationale-inclusive pairwise pooling, not extracted step support (25: X).

• M2 5: ?; §3.4 p.4 and Algorithm 1 p.10 describe variance-weighted judging, but exp(-Var_k[s_i]) is constant across the K evaluations of candidate i and cancels from the printed normalized score. Thus the formula yields the mean judged score. Reliability-weighted final voting cannot be verified from the conflicting description/pseudocode. The PDF equation was visually checked; this is not a text-extraction error.

• M2 29-32/35-37/39-41/45: SDI uses verbal confidence, answer-token likelihood, initial semantic diversity/disagreement and confidence alignment (§3.2 p.3). EWSC judges factual support/contradiction against evidence and repeats checks on evidence variants (§3.4 p.4, B p.9). The paper calls the fixed sigmoid(log-likelihood) feature calibrated probability (C p.9); this X identifies the stated feature, without endorsing empirical calibration or a learned calibration fit. Stability of successive debate states controls stopping (40), while the judge receives terminal candidates, not the whole round history (17: -). Calibration gap/qualitative hidden-overconfidence cases and measured judge instability support confidence unreliability (C.1 p.9, Table 5 p.6).

• M2 3-4/33-34/38/42-44: -; no explicit confidence/trust/accuracy-weighted final vote, parse-quality score, learned aggregation parameters, calibration-split-derived weights or iterative online weight updates. SDI thresholds are tuned on validation data, which is distinct from fitting EWSC weights (§4.3 p.5). Uniform baseline and fixed score transforms

account for 41.

• M3 1-2/11/13-16/18-21/24/28-29/33/35: redundant exchanges motivate the approach, but neither module estimates statistical dependence. Shared semantic answers justify skipping repetitive dialogue; repeated final judgments provide support. SDI embeds the complete query/rationale/answer (§3.2 p.3), whereas Algorithm 1 p.10 abbreviates it to query/answer. EWSC deliberately gives parallel judge calls paraphrased or subset-sampled evidence, so information asymmetry applies at that judging stage (11), not the debating agents. Exact binary equality includes the implemented self-consistency baseline (35).

• M3 3-10/17/22-23/25-27/30-32/34/36: -; cosine disagreement is a content-diversity heuristic, not an operational statistical-dependence measurement/discount or chance-overlap null. No couplings, counterfactual independence baseline, reasoning clusters, extracted atomic steps, DAG/tree/graph, copying detector or honest-versus-copied separation is provided. Factual evidence judgment is verification (46), not cross-trace semantic-equivalence judging (34).

• M3 37-39/45-46/50-54: training-free, frozen models with a small validation set for SDI thresholds (§4.3 p.5); extra debate and three parallel judgment calls, peer checks for factual errors/unsupported reasoning (B pp.8-9), token comparisons in Tables 1/3 pp.5-6 and an explicit negligible-latency-overhead claim (Table 1 p.5). No raw wall-clock latency measurement is given. External evidence is supplied, not retrieved; retrieval feedback is future work (§§6-7 pp.6-7; 47-49: -). No trained classifier/auditor/graph encoder, EM fit or gold-label training at scale (40-44: -). Larger agent/sample panels are not demonstrated (55: -).

Criteria Map 4 Evidence Log

Identity: Akshay Verma, Swapnil Gupta, Deepak Gupta, Prateek Sircar and Siddharth Pillai, SELENE: Selective and Evidence-Weighted LLM Debating for Efficient and Reliable Reasoning, EACL Industry 2026, Anthology ID 2026.eacl-industry.7. Sources: https://aclanthology.org/2026.eacl-industry.7.pdf

• §§3-5/A-D: BoolQ/CosmosQA/Internal-QnA, single CoT, self-consistency, MAD/CFMAD and judge comparisons; accuracy, normalized token cost and judge stability, gating/weighting ablations. Factual-calibration language does not establish a reported ECE/AUROC metric; no known-dependence or atomized-step test. Published EACL Industry 2026, appendix prompts provided; code not verified. Relevance: credibility/evidence weighting and selective exchange are nearby methods, but not independent-chain estimation.

Evidence notes | DAR

Criteria Map 1 Batch 2 Evidence Log

Primary source: Nguyen et al., Hear Both Sides: Efficient Multi-Agent Debate via Diversity-Aware Message Retention (DAR), arXiv:2603.20640 (2026), https://arxiv.org/pdf/2603.20640 (24 pp).

• §§3.1-3.4 pp.3-5, Appendix A.2 Algorithm 1 p.14: per debate round a separate LLM filter F takes previous responses and previous majority vote, returns IDs of messages to retain without rewriting them; all other agents reason from selected authentic context. Final answer remains majority vote (rows 2-4,18-19,46,61,68).

• §3.4 pp.4-5: simplified exchangeable correlated estimators model with constant pairwise ρ derives variance $\sigma^2/N(1+(N-1)\rho)$ and effective sample size $N_{eff}=N/(1+(N-1)\rho)$. This models statistical redundancy THEORETICALLY; it DOES NOT estimate individual evidence-chain dependence for actual problems (rows 25-26,29,48).

• §3.4 p.5 and qualitative Fig.3 p.8: retaining disagreeing answers amplifies minority-correct response, suppressing duplicate incorrect-majority voices; filters peer messages rather than step-level evidence or null-calibrated overlap (rows 10-11,14,16,44-45).

• §4.1 pp.5-6 and Appendices A.4-A.5 pp.15-18: separate homogeneous-agent panels for Qwen2.5-1.5B/3B, Falcon3-7B, Llama3.1-8B; experiments with 4 agents, also N=2/8. Although multiple model families/sizes are evaluated, they are not mixed in one debate (rows 31,36; 33-35/40 no).

• §4.1 pp.6-7: actual DAR is tested over both all-to-all Decentralized MAD and FIXED Sparse MAD protocols; §3.4 stresses no change to topology, so dynamic sparse topology not claimed (rows 50-53 yes, 54-55 no).

• §3.2-3.3 pp.3-4, Appendix A.2-A.3 pp.14-15: agents retain full original peer messages, append ANLL uncertainty cues and prior majority-vote answer to prompts. Multiple pass-types occur: full peer text, answer-only majority anchor, and answer+uncertainty score (rows 58-60).

• §4.3 p.9: reports average retained messages per round with up to 32% reduction, modest latency overhead (3-14%). R=1 and R=2 results reported in Appendix A.5; default two fixed rounds, no adaptive consensus stopping (rows 6,63-64,71).

• §4.3 p.9 and Table 3 p.8: diversity measured as pairwise MiniLM cosine distance; confidence-based message filtering exists as tested comparison (Certain Answers), but DAR core retains by disagreement; not by confidence threshold (row 44/68 yes, 70 no).

• No trained semantic independence estimator; selector is prompt-only (training-free) but uses LLM for index choice; no seeded-hint experiments, alternative-topology creation, null-calibrated derivation overlap, selective per-query debate start or confidence-based early stopping.

Criteria Maps 2 3 Batch 2 Evidence Log

Identity: Manh Nguyen, Anh Nguyen, Dung Nguyen, Svetha Venkatesh and Hung Le, Hear Both Sides: Efficient Multi-Agent Debate via Diversity-Aware Message Retention, arXiv:2603.20640v2, April 14, 2026. Source: primary PDF, 24 PDF pages; §§3-4, Appendices A.1-A.7.

• M2 1-2/14/16/19-20/22-25/28/30-31/35-36/41: terminal mode over all agents is unchanged (§3.3 p.4, Algorithm 1 p.14); majority/single-signal and prompt ablations are implemented (§4.1 pp.5-6). Peer records communicate answer plus token-likelihood uncertainty. Full-message retention compares differing reasoning paths, rather than extracted steps (§4.3 p.8). Multiple signals are uncertainty, previous vote and diversity; the final answer is still a plain vote. No final confidence/trust/reliability weighting, judge, fitted dependence-law classifier, calibrated cluster posterior, routing or step weighting (3-13/15/21/26-27: -).

• M2 17-18/40: -; the selector sees previous-round responses and a majority anchor, not the whole debate trajectory. Figure 3 p.8 recovers an initially minority-correct answer by changing later responses, not by overriding the raw majority of terminal votes. No historical trajectory feature is computed. Weights remain uniform; changing retained message IDs is context selection, not learned/calibration-derived/online numerical vote weights (42-44: -).

• M2 29/32-34/37-39: -; actual uncertainty is computed ANLL, not verbal self-confidence, calibrated probabilities, an accuracy prior, trust, task relevance or a quality rubric. Figure 1's numerical-confidence illustration does not establish a verbal-confidence input. Index-only filtering preserves message format; parse quality is not an aggregation score. M2 45: X; uncertainty-only filtering underperforms across conditions (§4.2 p.7), and Certain Answers ablation loses to diversity retention (Table 3 p.8), supporting the paper's own claim that uncertainty alone is misaligned with correctness. This is filtering evidence, not an ECE analysis.

• M3 1/3-5/9-10/13-16/18-21/24/28-29/33-35: §3.4 pp.4-5 models exchangeable correlated estimators analytically, deriving effective sample size from a fixed average correlation rho. Runtime index selection suppresses redundant agent messages using semantic disagreements; this is a heuristic agent/message discount, not an estimated per-question rho or error model. Ordinary-vote/no-filter baselines ignore dependence. Whole-response MiniLM cosine-distance measurements in §4.3 p.9/Figures 11-12 pp.16/18 are validation-only; the inference selector is an LLM. Signals include per-agent ANLL, answers and messages, with query-level selection and dataset-level diversity analysis. Exact distinct-answer filtering is a tested ablation (§4.3 p.8). The prior-vote anchor treats repeated answers as support, while the retention module treats repetition as redundancy.

• M3 2/6-8/11-12/17/22-23/25-27/30-32/36: -; dependence goes beyond motivation through theory and retention, without fitted pair couplings, error-overlap beyond chance, matched no-communication null, model-family dependence features, atomic derivation extraction, reasoning clusters/graphs, copying detection or honest-versus-copied separation. Index selection does not create information-asymmetric debate agents or a structural reasoning-similarity measure. The correlation model supports 9, not null-calibrated overlap (22).

• M3 37-38/45/50/52-55: prompt-only filter, frozen LLMs, fixed/tunable N/R/sampling hyperparameters and extra filter calls (Algorithm 1 p.14, Table 6 p.17). Qualitative generated traces recheck intermediate arithmetic (A.7/Figure 15 p.24); this is fallible peer re-reasoning, not a dedicated verification judge, executable/symbolic checker or process reward model (46-49: -). No classifier/auditor/graph encoder/EM fit, calibration split or large-scale gold training (39-44: -).

• M3 51: -; Table 4 p.9 compares retained message counts (up to 32% fewer), not token totals. M3 52: X for the stated token/context-efficiency rationale (§1 p.1 and §4.3 p.9), with that measurement limit. M3 53: X for the bounded-latency-overhead claim and measured 3-14% increase (Table 11 p.21), not a latency speedup. N=2/4/8 experiments establish agent scaling (Figure 2 p.7).

Criteria Map 4 Evidence Log

Identity: Manh Nguyen, Anh Nguyen, Dung Nguyen, Svetha Venkatesh and Hung Le, Hear Both Sides: Efficient Multi-Agent Debate via Diversity-Aware Message Retention, arXiv:2603.20640v2, April 14, 2026. Sources: https://arxiv.org/pdf/2603.20640v2

• §§3-4/A: arithmetic/GSM8K, MMLU/CSQA/HellaSwag and HH-RLHF; CoT/SC/dense/sparse debate and retention baselines. Accuracy, tokens and MiniLM pairwise diversity, three-seed mean±SD and retention/capacity ablations. Proposition 3.1 gives a correlation/effective-sample-size variance bound (28), with message-scaling analysis (27). Similarity ablation is not known-copying ground truth. Code/prompts released. Relevance: motivation, communication baseline and correlation background; no calibrated independent derivations.

Evidence notes | Goel

Criteria Map 1 Batch 3 Evidence Log

Identity confirmed: Shashwat Goel et al., Great Models Think Alike and this Undermines AI Oversight, ICML 2025 (PMLR 267, pp. 19621-19678); arXiv:2502.04313. Sources: https://proceedings.mlr.press/v267/goel25b.html and full-text https://arxiv.org/html/2502.04313 (v2, 12 June 2025).

• §2.1-2.2: CAPA computes probabilistic agreement minus agreement expected under an **independent-response null** based on accuracy/confidence. It measures statistical output-error dependence, **not content-level chain independence**, and does not transmit reasoning between agents (rows 1,24-29,49).

• §3.1-3.2: 9 judge models score 39 other model outputs independently, demonstrating affinity bias of judges toward similar models (rows 3,8,10,46,47; no inter-agent debate/communication).

• §4.1-4.2: 12 weak-supervisor/strong-student pairs and 180 training runs quantify gains from complementary model knowledge; training does not mean a multi-agent debate aggregation procedure (rows 5,35,47).

• §5.1-5.2: 130 independently scored models from multiple developers on MMLU-Pro and BBH; CAPA increases with capability, highlighting shared mistakes and error correlations (rows 10,34,40,44,48).

• Appendix B.10 & D.5: experiments with open-weight judges from Qwen, Llama, Gemma and Ministral families, several sizes; no closed-source judge/model was established as part of study (rows 33,34,36,37-).

• Appendix A.1-A.6: chance-null derivation, multiparty extension and theoretical bounds on CAPA (row 29).

• **Scope**: no debate rounds, topology/message filtering, role/persona prompting, stopping rule, measured communication cost or independent derivation-step extraction. The paper's goal is diagnostic oversight analysis rather than a proposed multi-agent deliberation protocol (row 8 X).

Criteria Maps 2 3 Batch 3 Evidence Log

Identity: Shashwat Goel et al., Great Models Think Alike and this Undermines AI Oversight, ICML 2025, PMLR 267:19621-19678; arXiv:2502.04313v2, June 12, 2025. Sources: proceedings record, full PDF, 58 pages.

• M2 1/6/14/16/19/22/24/28/39/41: reference-based evaluation uses five frozen LLM judges, each returning a binary correctness label, then uniform majority voting (B.3.2 pp.28-29, B.11 pp.39-42). Free-form final answers are regex-extracted (B.8 p.37); the reference judge compares them with correct/incorrect options. These are evaluation rules, not a proposed debate aggregator. No final minority override, weighted/dependence-based decision, answer-confidence/distribution aggregation output or trajectory/step rule (2-13 except 6, 17-18/20-21: -).

• M2 23/30/34-35: CAPA combines observed option-probability agreement with model-specific mean correct-option probabilities and an independent chance expectation (§2.2 pp.3-4, A.1 pp.18-19); output log probabilities are taken from evaluation logs (B.9 p.37). These X marks describe diagnostic metric inputs, not weights applied to the reference jury. Weak-to-strong confidence-weighted loss is training, not confidence-weighted final voting or learned ensemble weights (3/32/42-44: -; C.1 pp.45-46). Judge affinity bias does not demonstrate numerical confidence miscalibration (45: -).

• M3 3/8-9/12-14/18-22/29: CAPA models/quantifies probabilistic answer dependence beyond an accuracy-conditioned independent-response null, including differing wrong labels. It has dataset-level similarity and model-specific marginals; per-question scope also includes the implemented reference jury. Chance subtraction/normalization supports 22, but no agents/votes are discounted (4/10: -), and no matched no-communication intervention is performed (7: -). Binary label/probability association is not semantic reasoning-content dependence (5: -).

• M3 1/24/28/34-35: the reference jury counts repeated correctness labels as support without correcting dependence; generator CoT is flat prose. Reference grading explicitly judges semantic match to answer options (B.11.2 p.39), not cross-trace step similarity. No derivation extraction, reasoning clustering/graph, copying detector or honest-versus-copied separation (16-17/23/25-27/30-33/36: -).

• M3 37-38/40/44-46/54-55: CAPA and frozen-judge diagnosis are training-free/inference-only; the separate weak-to-strong study trains binary classifiers with LoRA, ground-truth-supervised weak models and strong elicitation controls (10k training / 1k validation / 5k test split; C.1 pp.45-46). No small calibration fit, trained auditor/graph encoder or EM model (39/41-43: -). The chance marginals use benchmark gold (A.1 p.18). Extra judge calls perform final-answer verification; no executable/tool/PRM/intermediate-step checker or token/latency-efficiency comparison (47-53: -). Multi-model CAPA is derived in A.2 p.19 and applied to large model sets (§5 p.8).

Criteria Map 4 Evidence Log

Identity: Shashwat Goel et al., Great Models Think Alike and this Undermines AI Oversight, ICML 2025, PMLR 267:19621-19678; arXiv:2502.04313v2, June 12, 2025. Sources: https://proceedings.mlr.press/v267/goel25b.html, https://arxiv.org/pdf/2502.04313v2

• §§2-5/A-D: CAPA chance-adjusted output agreement/error overlap, LM judging and weak-to-strong oversight, MMLU-Pro/BBH and 130-model comparisons. Own probability bounds/closed expressions, uncertainty reporting and sensitivity analyses; natural model similarities are not known-copying validation. Project page/code and sample predictions are shared. ICML 2025. Motivation/background rather than a debate aggregation baseline; output-error dependence is distinct from derivation dependence.

Evidence notes | Wynn

Criteria Map 1 Batch 3 Evidence Log

Identity confirmed: Andrea Wynn, Harsh Satija and Gillian K. Hadfield, Talk Isn't Always Cheap: Understanding Failure Modes in Multi-Agent Debate, arXiv:2509.05396v2 (13 Oct 2025), https://arxiv.org/pdf/2509.05396 (13 pp).

• §3 pp.3-4: independent starting answers + reasoning, then two all-peer message rounds, ending with majority vote. No learned weighting or semantic provenance aggregation (rows 1-4,17-19,51-52,58,64).

• §4.2 p.4, Table 1 p.6: GPT-4o-mini (closed API), Llama-3.1-8B and Mistral-7B (open-weight) tested both in homogeneous groups and genuinely heterogeneous, different-capability panels (rows 31,34-37,40,47).

• §5.1-5.2 pp.5-6: compares majority-vote-without-debate to post-debate majority vote, finds decline on CSQA and MMLU in many combinations, but gains on some tasks/configurations (rows 9-11,21-23,49).

• §6.1-6.2 pp.7-8: counts correction and deterioration transitions (correct→wrong / wrong→correct), conditions answer flips on number of initially agreeing peers. Isolated initially correct minorities are especially vulnerable. **Measures social influence; does not explicitly fit a pairwise statistical dependence estimator of models/derivations** (rows 11,16,25,26-,44).

• §6.3 pp.8-9: controlled correctness-payoff prompting intervention tests sycophancy mechanism, finds no consistent mitigation; not a new communication selection, trained aggregation, or judge model (rows 8,27-).

• Scope: no topology selection, per-message filtering, confidence-weighted voting, null-corrected step overlap, or reported cost optimization.

Criteria Maps 2 3 Batch 3 Evidence Log

Identity: Andrea Wynn, Harsh Satija and Gillian K. Hadfield, Talk Isn't Always Cheap: Understanding Failure Modes in Multi-Agent Debate, ICML MAS Workshop 2025; arXiv:2509.05396v2, October 13, 2025. Source: primary PDF, 13 pages; §§3-7.

• M2 1-2/14/16/19/22/24/28/31/41: independent starts, two peer-revision rounds and uniform majority over terminal answer labels (§3 pp.3-4). The implemented no-debate majority baseline includes same-model independent samples (Table 1 p.6). Parenthesized option labels/numeric answer fields support structured extraction and exact answer identity (§4.3 p.5). All decision weighting remains equal; final answer only. No weighted vote, judge, classifier, fitted dependence law, semantic clustering, belief distribution or minority override is used.

• M2 17/23/25-27/29-40/42-45: -; full peer rationales are read, but no explicitly scored path/step/token fusion is constructed. Only previous-round context enters the update. Transition/diversity/social-influence statistics in §6 pp.7-10 diagnose behavior; they are not features used by an aggregation rule or a whole-trajectory score. Confidence estimates/critical-evaluation mechanisms are proposed in Discussion, rather than implemented. The related-work example of a confident false argument is not the paper's own confidence-reliability result.

• M3 1/7/12-15/19-21/24/28-29/35: the voting protocol does not correct dependence. Table 1 p.6 explicitly contrasts initial no-communication votes with post-debate votes on the same fixed query/panel setting: 7 is a performance counterfactual control, not a null-calibrated overlap estimator. Model/agent/question/task effects and correct↔wrong transition rates are measured (§6.1-6.4 pp.7-10); agreement supports voting and social reinforcement, while convergence on shared errors is treated as harmful redundancy. Raw reasoning is flat bullet-point prose.

• M3 2-6/8-11/16-18/22-23/25-27/30-34/36: -; social dependence is empirically investigated, not mentioned only as motivation, but no pairwise probabilistic error/derivation model, coupling, second-order correlation estimator, content-overlap metric, chance null, semantic clustering or reasoning graph is fitted. Counting answer switches does not establish switching without an original derivation, copying, or honest-versus-copied separation (31-32: -).

• M3 37-38/45/54: fixed/tunable agent/round/context settings, prompt-only correctness-payoff intervention, frozen models and extra revision calls (§§4/6.3 pp.4-5/9-10); optional overflow summarization is specified in §3 p.4. No calibration split, trained classifier/auditor/encoder/EM fit or gold-supervised runtime rule (39-44: -); gold labels are used for evaluation/diagnosis. No dedicated judge, intermediate-step verifier, executable/PRM/tool checker, measured token-cost/latency-efficiency or larger-panel scaling experiment (46-53/55: -). Agents are three per group, though their model composition varies.

Criteria Map 4 Evidence Log

Identity: Andrea Wynn, Harsh Satija and Gillian K. Hadfield, Talk Isn't Always Cheap: Understanding Failure Modes in Multi-Agent Debate, ICML MAS Workshop 2025; arXiv:2509.05396v2, October 13, 2025. Sources: https://arxiv.org/pdf/2509.05396v2

• §§3-7: homogeneous/heterogeneous three-model panels, math/commonsense/knowledge tasks, independent voting versus vanilla debate, correct→wrong/wrong→correct counts and peer-pressure intervention. Five seeds and negative results; no fitted/error-overlap metric or known-dependence labels. Source code is explicitly released. Workshop 2025 status follows the previously confirmed ICML MAS record. Relevance: motivation and debate baseline; confident persuasion does not establish independent evidence.

Evidence notes | EoT

Criteria Map 1 Batch 3 Evidence Log

Identity confirmed: Zhangyue Yin et al., Exchange-of-Thought: Enhancing Large Language Model Capabilities through Cross-Model Communication, EMNLP 2023, pp.15135-15153, ACL Anthology ID 2023.emnlp-main.936. Primary: https://aclanthology.org/2023.emnlp-main.936.pdf (19 pp).

• §4.1 pp.4-5: first round independently elicits rationales and answers (r,a), subsequent rounds share prior traces. Four concrete configurations: Memory (fully connected/bus), Report (centralized star), Relay (directed ring), Debate (binary tree); both all-to-all and fixed sparse message paths apply. These are **fixed topology alternatives**, not a learned query-specific topology (rows 1-4,18,31,50-53,56-58,61; rows 54-55 -).

• §4.2 pp.5-6: derives $O(n^2)$ messages for Memory and linear message volumes for star/ring/tree (e.g., $3n-2$ and $2n$). Specifically targets quadratic communication cost (rows 6,15,29,71).

• §4.3 p.6, Appendix C p.14: adaptive termination on one model's repeated answer or all-model/majority consensus. Maximum five rounds used; **not** fixed multiple rounds for every example (row 65 X, 64 -).

• §4.4 p.6: confidence Ci estimated as frequency of most common answer within an agent's earlier rounds; low confidence makes reasoning less credible. Confidence is included in prompts, **not fitted probability calibration**, and no transmission blocked by confidence threshold (rows 12,60 X; 7,66,70 -).

• §5.1 p.6, §5.3 pp.8-9 and Fig.11: primary GPT-3.5 panel sampled multiple times, mixed GPT-3.5/GPT-4/Claude-2 in heterogeneous panels; larger GPT-4 contribution improves performance. Temperature 1 used to elicit varied reasoning. Commercial API models only (rows 31,34-35,37-41,47).

• Appendix D p.15: specific personae Kitty, Ben, Peter; distinct roles tied to graph center/leaf nodes (rows 32,39,46).

• §5.2-5.3 pp.7-9, Fig.9: compares accuracy with CoT and self-consistency; cost in dollars calculated using API token prices, lower than several inference-sampling baselines (rows 5,6,71).

• Scope: no chance-corrected model-error dependence or null-calibrated semantic step overlap; confidence-stability heuristic is not evidence-chain independence. EoT passes whole rationales/answers rather than extracted independent step chains.

Criteria Maps 2 3 Batch 4 Evidence Log

Identity: Zhangyue Yin, Qiushi Sun, Cheng Chang, Qipeng Guo, Junqi Dai, Xuanjing Huang and Xipeng Qiu, Exchange-of-Thought: Enhancing Large Language Model Capabilities through Cross-Model Communication, EMNLP 2023, ACL Anthology 2023.emnlp-main.936. Sources: publisher record, primary PDF, 19 PDF pages. Page references here are physical PDF pages.

• M2 1-2/14/16/19/22/24/28/41: main EoT takes the terminal majority; consensus stops early if all three agree within the initial five rounds, otherwise the round-five majority is used (§4.3-5.1 p.6, C p.14). SC baselines also use uniform majority. Numeric/option fields are regex-extracted and compared exactly (B p.14). No weighted terminal vote, fitted dependence law, classifier, calibrated cluster posterior, routing or step-weighted final rule (3-13: -). Parent nodes in the Debate topology generate/revise solutions; this does not create a separately specified final LLM judge (6: -).

• M2 17-18/20/23/31/33/40: per-agent confidence is the frequency of its most common answer across all preceding rounds, exposed with peer solutions and recomputed each round (§4.4 p.6, C-D pp.14-15). This cumulative history steers revision and serves as prompt-level credibility, not verbal self-confidence or calibrated probability (29-30/32: -). The consistent-output variant terminates when one agent repeats its answer and returns that agent's answer (C p.14); 18 follows because that rule can select an answer without terminal majority support. Main majority EoT does not itself override its terminal majority. Numerical vote weights do not update (42-44: -); the changing confidence tag is an input to generation.

• M2 15/21/25-27/34-39/45: -; no semantic answer clustering, emitted belief distribution, scored reasoning path/step/token fusion, accuracy prior, runtime diversity/disagreement/relevance/parse-quality/evidence rubric or empirical confidence-unreliability result. Warning in the prompt that high confidence can be wrong is not such a result (D p.15). Full rationales are exchanged without step extraction.

• M3 1/12-15/19-21/24/28/35: dependence is not corrected in voting. The method uses heterogeneous models/agents, answer histories and full messages; exact agreement is support (§§4-5 pp.4-9). Raw reasoning is flat prose. No statistical dependence model/content-overlap measure, pairwise coupling, excess-error chance statistic, matched no-communication dependence control or explicit agent dependence discount (2-11: -). Ordinary SC benchmarks differ

in sampling budget and are not a matched dependence null.

• M3 16-18/22-23/25-27/29-34/36: -; no reasoning embedding, atomic derivation extraction, tree/DAG/graph of reasoning, cross-trace redundancy score, reasoning clustering, copying detector or honest-versus-copied distinction. The four graph-shaped communication topologies in §4.1 pp.4-5 connect agents, not extracted reasoning steps. Cumulative answer stability is not chance-overlap calibration.

• M3 37-38/45/54-55: frozen prompt-only models, tunable rounds/temperature/prompt threshold, extra peer revision calls, and implemented SC sample counts up to ten (§5 p.6, Figures 9-10 p.9, C-D pp.14-15). No fitted calibration/classifier/auditor/graph encoder/EM model or gold training at scale (39-44: -). No dedicated judge, executable/symbolic checker, PRM, tool/retrieval grounding or implemented intermediate-step verifier (46-50: -); generic peer re-examination is not a specified step-checking mechanism.

• M3 51-52: X; Figure 9 p.9 compares token-priced computational cost against CoT/SC, using explicit input/output-token pricing (B p.14), and claims a cost advantage. M3 53: -; theoretical communication-volume/propagation-hop comparisons (§4.2 p.5) do not report inference latency or a wall-clock latency-efficiency claim.

Criteria Map 4 Evidence Log

Identity: Zhangyue Yin, Qiushi Sun, Cheng Chang, Qipeng Guo, Junqi Dai, Xuanjing Huang and Xipeng Qiu, Exchange-of-Thought: Enhancing Large Language Model Capabilities through Cross-Model Communication, EMNLP 2023, ACL Anthology 2023.emnlp-main.936. Sources: https://aclanthology.org/2023.emnlp-main.936/, https://aclanthology.org/2023.emnlp-main.936.pdf

• §§4-5/C-D: math/commonsense/symbolic tasks, CoT/SC and graph-protocol comparisons, accuracy and token-priced inference cost, prompt/graph/model-capability studies. Own $O(n^2)$/linear communication-count analysis supports complexity, not a closed-form correctness condition. EMNLP 2023; published prompts. Code release not verified. Relevance: named system baseline, motivation and communication-complexity background.

Evidence notes | ReasCons

Criteria Map 1 Batch 3 Evidence Log

The short name ReasCons appears as a comparison-map column, but **the separate Paper Key is not supplied** and no full title, author, year, identifier, or primary source for this key could be identified unambiguously. Targeted searches for "ReasCons" LLM, "ReasCons" reasoning, and project documents gave no verified matching source. Do not substitute a similarly titled 'reasoning consensus' paper.

• **All rows 1-71:** ?, not -, because its identity and source cannot be verified.

• To resolve: supply the Paper Key entry containing the ReasCons full title and arXiv/DOI/paper URL. Future session can then reassess this column only with user permission to overwrite ? marks, because current task explicitly forbids changing non-blank cells.

Criteria Maps 2 3 Final Evidence Log

M2 rows 1-45 and M3 rows 1-55: **?**. The intended title/authors/identifier cannot be verified from this alias; the Paper Key remains unavailable. Retained Map 1 logs already document the identity gap. Candidate papers, surname matches and remembered methods were not substituted. These are identity uncertainties, not negative findings about an identified paper. No source can be attributed securely to this column until its Paper Key entry is supplied.

Criteria Map 4 Evidence Log

Identity unresolved; intended Paper Key entry remains unavailable. Sources:

• Rows 1-44: ?. No verified title/author/ID maps this alias to a source; all substantive and tracking properties remain unresolved.

Evidence notes | PPV

Criteria Map 1 Batch 4 Evidence Log

Identity used: Yasushi Sakai, Allen Song, Kent Larson, When Does Delegation Beat Majority? A Delegation-Based Aggregator for Multi-Sample LLM Inference, arXiv:2606.08098v3 (July 2026). Full text: https://arxiv.org/html/2606.08098 . “PPV” names the Propagational Proxy Voting mechanism applied to multi-sample LLM inference here; it is not a literal multi-agent discussion protocol. Separate Paper Key absent, so identity remains subject to Paper Key cross-check.

• **§4.1-4.2:** one **Qwen3-1.7B** solver samples **128** chains-of-thought per question at temperature **1.5**, partitioned into **16 groups of eight**. Per-group answer letter/entropy and centered reasoning embeddings are extracted (rows **1,3,19,31,36,38,41,44,49,63**, X); Qwen3-Embedding-8B is an embedding encoder, **not** a second collaborating solver (rows **33-35**, -).

• **§4.2-4.4:** confidence lever `When` is group letter entropy, and routing lever `Whom` is positive, centered reasoning-embedding cosine. These feed an absorbing-Markov delegation **matrix**; no agents receive each other's messages or exchange rationales. Do **not** treat voting edges as communication topology (rows **50-62**, -; row **49**, X).

• **§5, §6.1, §7.1-7.3:** gains over majority vote on MMLU-Pro, 10-6 **wrong majority overturned by a correct reasoning-coherent minority**, and an exact two-block flip condition. Supports goal accuracy, robustness to erroneous majority, correct-minority recovery, and method/theory (rows **5,9,10,16,27-29**, X).

• **§2.1, §4.2, §7:** samples are independent solver draws, but superficial agreement can hide divergent reasoning structures. The method quantifies geometry, **not** provenance-dependent reasoning steps or a null-corrected estimator of evidence independence (rows **24-25**, X; **26**, -). It does **not** assume a debate peer's vote is a separately trained model's vote.

• **§2.4, §8:** studies confidence modes and unreliable `P(True)`, but does **not** construct calibrated probabilities as the output target (row **7**, -), nor does it stop/filter a debate based on confidence (rows **65-70**, -).

• **§5:** no reported communication-cost optimization, because there is no peer exchange (rows **6,71**, -). The 128-sample inference budget is described, not a communication-cost comparison.

Criteria Maps 2 3 Batch 4 Evidence Log

Identity checked from the PDF: Yasushi Sakai, Allen Song and Kent Larson, When Does Delegation Beat Majority? A Delegation-Based Aggregator for Multi-Sample LLM Inference, arXiv:2606.08098v3, July 12, 2026. Source: primary PDF, 13 pages. The abstract lookup failed; the full PDF was accessible. Method §§3-4 p.4, experiments §§5-8 pp.5-9.

• M2 1/3-4/11/14/16/18-19/22-25/28/31/33/35-36/41: 128 independent CoT samples form 16 fixed groups of eight. Group answers are majority letters, When retains entropy-based confidence mass on the group's answer, and Whom routes remaining mass to peers by centered whole-reasoning cosine (§4 p.4). Terminal answer maximizes absorbed policy mass (§3 Eq.2 p.4). Uniform majority and single-signal modes are implemented. Confidence and peer trust affect the final routed vote; geometric scoring aggregates whole paths, not extracted steps. A wrong 10-6 majority is actually overturned (§6.1 pp.5-6). Exact MCQ letters are not semantic clustering. Entropy/geometry concern starting independent samples, without a debate trajectory (2/12/17/40: -).

• M2 5-10/13/15/20-21/26-27/32/37-39/42-44: -; no trained final adjudicator, fitted probabilistic dependence law, calibrated cluster posterior, emitted answer-confidence/belief-distribution decision, step/token aggregation, calibrated confidence, relevance/parse-quality/evidence rubric or learned/held-out/online vote-weight update. Absorbed mass is internal vote allocation, not a calibrated posterior. Per-question matrix construction followed by powers of a fixed matrix is not online evidence-based updating.

• M2 34: X only for the implemented best-dictator baseline: choose the voter index with highest dataset accuracy (§5 p.5). This is a gold-aware reference selection, not a reliability-weighted main PPV rule. M3 44: X for that selection's gold-label requirement; main PPV requires no labels. Oracle/pass@16 and correctness diagnostics otherwise use gold for evaluation.

• M2 45: X; own P(True) experiment has AUROC 0.472, below chance, and the confidently wrong entropy example is explicit (§§6.2/8 pp.6/9). M2 29-30 and M3 18: ?; the paper says it computed a same-model P(True) verifier signal, but does not specify whether confidence was verbalized, read from token probabilities or extracted another way. Its token-level implementation cannot be verified from the accessible paper. Main entropy confidence does not need token likelihood.

• M3 1/5/13-16/19-21/24/28-29/33/35: the ordinary-majority baseline ignores dependence. Main PPV measures whole-trace cosine and removes the common question component by per-question centering (§4.2 p.4, §6.4 p.7). Repeated coherent reasoning is rewarded as support in vote routing; 29 applies narrowly to removing the uninformative common question component. This is not a discount for copied or correlated evidence chains (4/10: -). The routing matrix models delegation, not a joint error/dependence distribution (3/6/8-9: -). Agent scope is synthetic groups, not communicating solvers; dataset scope includes best-dictator selection and diagnostics. No model-family signal, message-exchange intervention or asymmetric private evidence is used (7/11-12: -).

• M3 17/22-23/25-27/30-32/34/36: -; neither centering nor the Miller-Madow small-sample entropy correction (§4.2 p.4) corrects overlap expected under an independent chance null. No atomic step extraction, reasoning tree/DAG/graph, semantic reasoning-cluster algorithm, copying detection, honest-versus-copied separation or structural similarity. The Markov graph's nodes are voters/answer policies, not derivations; plotted clusters are grouped by answer label.

• M3 37-38/45-46/54-55: fixed sampling/group settings and inference-only aggregation, with a pretrained embedding model and no fit to this task; scales to 128 traces/16 voter groups (§§4-5 pp.4-5). Extra same-model self-verification was actually tested for P(True) (§8.1 p.9), accounting for 45-46, though not used by the best main mode. No small calibration split, trained classifier/auditor/graph encoder, EM, executable/tool/PRM or intermediate-step verification (39-43/47-50: -). Linear-boundary grid search validates a theoretical vote-mass prediction (§7.3 pp.8-9); it is not a trained correctness classifier deployed in PPV. A supervised polarity classifier is future work.

• M3 51-53: -; the fixed sample budget is specified, but no token-cost versus baseline or token/latency efficiency comparison is reported. Controlled geometry/entropy ablations and numeric two-block validation do not provide an independent-overlap null (§§6-7 pp.6-9).

Criteria Map 4 Evidence Log

Sakai, Song and Larson, When Does Delegation Beat Majority? A Delegation-Based Aggregator for Multi-Sample LLM Inference, arXiv:2606.08098v3 Sources: https://arxiv.org/pdf/2606.08098v3

• §§4-8: MMLU-Pro, 128 samples from one solver, majority/score voting and confidence-routing ablations, answer accuracy, geometric coherence, minority reversals and McNemar significance. Two-block delegation supplies an exact flip condition; Markov/absorbing-matrix analysis supplies stability. A coherent synthetic block example is algebraic illustration, not empirical known-dependence validation. Code and current venue status could not be verified; primary PDF is available. Relevance: closely related content-geometry aggregation, not a null-calibrated step-independence count.

Evidence notes | MinSentinel

Criteria Map 1 Batch 4 Evidence Log

Identity confirmed: Chuan He, Zebin Chen, Zhengyi Yang, Shaobo Qiao, Mingchen Ju, Jiate Liu, Dong Wen, Guanfeng Liu, Minority Sentinel: When to Overturn Majority Voting in Multi-Agent LLM Debates, AgentSearch Workshop at SIGIR 2026, arXiv:2606.29270v1. Full text: https://arxiv.org/html/2606.29270 .

• **§3.2, §4.1:** three heterogeneous closed-API agents (GPT-4o-mini, Gemini-2.0-Flash, Claude Haiku 4.5), each with a different persona. **Round 0:** blind initial answers + reasoning; **rounds 1-2:** each receives all peers' previous responses and tracks stance changes. Fixed 2 debate rounds before majority decision (rows **1-4,17-19,34,37,39-41,45-46,49,51-52,58,64**, X). Identical sampling temperature 0.7 across agents is **not** manipulated as a diversity source (row **38**, -). Its agents are from **different providers**, not one model with several roles (row **32**, -).

• **§1, §3.1, §4.1:** 686 of 1,754 items have 2:1 post-debate disagreement; the minority is correct in 175 (25.5%) of these. Explicitly studies incorrect majority pressure, conformity, false consensus, and correct-minority suppression (rows **9-13,16,25**, X). Shared pretraining/correlated errors are considered (row **48**, X), but the paper does not estimate a pairwise probabilistic error-correlation matrix (row **26**, -).

• **§3.3, Table 1:** a **22-dimensional debate fingerprint** comprises 10 debate-dynamics features, 4 voting metadata features, and 8 **GPT-4o semantic-audit** features (e.g. majority echo chamber, new evidence, blind following). This is a behavioral/semantic audit, not extraction of independently sourced, null-calibrated reasoning **steps** (rows **27-28,44**, X; row **62**, -: no structured evidence is passed between debaters as a discrete packet).

• **§3.4, §4.1: trained LightGBM** classifies when to flip an existing 2:1 majority result; a learned post-debate flip threshold is **not** a debate trigger, a stop condition, or a confidence-based message filter (rows **19,27**, X; **65-70**, -). No model-weight training of the debating LLMs is claimed, but the meta-classifier is trained; do **not** label the entire system training-free.

• **§4.2-4.3, §5.5:** +1.71 percentage-point overall Net Gain; 81.2% flip precision; 20-seed validation; LLM-as-Judge baseline performs worse. No communication-cost metric/optimization, despite acknowledging inference cost of GPT-4o audit (rows **5,9**, X; **6,71**, -).

• **§6.2-6.3:** process reward models for step-level verification are described as **future work**, not as a component implemented here. The paper does not establish a new shared benchmark/resource (row **30**, -): collected debate logs are experimental inputs rather than a separately introduced benchmark.

Criteria Maps 2 3 Batch 4 Evidence Log

Identity checked from the PDF: Chuan He, Zebin Chen, Zhengyi Yang, Shaobo Qiao, Mingchen Ju, Jiate Liu, Dong Wen and Guanfeng Liu, Minority Sentinel: When to Overturn Majority Voting in Multi-Agent LLM Debates, AgentSearch@SIGIR 2026; arXiv:2606.29270v1, June 28, 2026. Source: primary PDF, 10 pages; §§3-6 and Appendix A. The abstract lookup failed; the full PDF was accessible.

• M2 1-2/6-8/14/16-20/22-25/28/31/35-36/39-42: three heterogeneous agents answer independently at R0, then exchange prior-round reasoning for two rounds (§3.2 p.3). The final vote is retained for non-divergent cases; a trained LightGBM adjudicator flips 2:1 cases when its minority-correct score exceeds a threshold (§3.4 pp.4-5). Implemented baselines include MV, single-feature thresholding, logistic regression and a GPT-4o final judge (§4.1 p.5). Tagged answer/reasoning/stance fields support structured exact parsing (A.2 pp.9-10). The continuous minority-correct score is produced before thresholding; it is not a calibrated distribution over answer clusters (10/15/21/32: -).

• M2 3-5/9/11-13/26-27/29-30/33-34/37-38/43-44: -; no final numerical confidence/trust/accuracy vote weights, fitted joint dependence law, delegation, separately defined trajectory score, step/token fusion, confidence/accuracy prior, task-relevance/format-quality weighting or online update. Learned classifier parameters support 42; out-of-fold threshold choice is not held-out vote-weight calibration (43). The complete trajectory is used via stance-change, persistence, novelty and endorsement features; whole-rationale semantic scores support path scope, without atomic step fusion (Table 1 p.4).

• M2 45: X, limited to the paper's qualitative wrong-flip analysis (§5.6 p.7): incorrect minorities display high reasoning confidence and seemingly independent arguments, misleading the classifier. This is not a numerical confidence calibration experiment and verbal confidence is not a fingerprint feature (29: -).

• M3 1/4-5/12-15/19-21/24/28-29/31/34-35: plain MV ignores dependence; the proposed override heuristically suppresses an echo-chamber majority using content/dynamics features (Table 1 p.4). This is dependence mitigation, not

fitted error correlation or explicit per-agent discount factors (3/6/8-10: -). Endorsements count support; echo-chamber and blind-following features treat non-substantive repetition/switching as harmful. blind_follower_count explicitly counts agents switching without substantive reasoning, supporting 31. No proof of copying provenance or honest-shared-versus-copied separation (32: -). Frozen GPT-4o compares whole arguments; no embeddings or structural derivation matcher (16-18/33/36: -).

• M3 7: X for the explicitly stated R0 independent-answer baseline used to measure subsequent shifts (§3.2 p.3; answers_changed and stance fields in Table 1 p.4). This same-panel pre/post control is not a chance-null overlap calibration (22: -). Heterogeneous vendors/personas are diversity, not private information asymmetry (11: -). Dependence is acted on through audit and override, rather than mentioned only in motivation (2: -).

• M3 23/25-27/30: -; fingerprint extraction yields counts, flags and whole-rationale quality scores, not atomic steps, reasoning clusters or reasoning trees/DAGs/graphs. The discussion of AgentAuditor's reasoning tree is related work (§2 p.3). PRM step scoring and RAG/code applications are future directions (§6.3 p.8), not implemented checks (47-49: -).

• M3 37/39-40/44-46/50/54: gold-labeled divergent debates train LightGBM and alternative classifiers, with stratified five-fold out-of-fold predictions; per-dataset thresholds are selected from these predictions (§§3.4/4.1 pp.4-5). Small labeled fold-based threshold calibration supports 39; there is no independent nested threshold-test split, and §6.2 p.8 acknowledges threshold-selection overfitting risk. Frozen GPT-4o supplies extra logic-audit calls and detects logical/arithmetic errors in whole reasoning (§3.3 p.4, §4.2 p.5, A.3 p.10), supporting 46/50 without step extraction. The LLM is not fine-tuned; the classifier is trained. No trained audit LLM, graph encoder or EM model (41-43: -). M3 38 is X only for implemented frozen MV/LLM-judge baselines; main Sentinel is not training-free.

• M3 51-53/55: -; extra audit inference cost is acknowledged (§6.2 p.8), but no token-versus-baseline or explicit token/latency efficiency result is reported. Evaluation fixes three agents/two rounds; scaling to five-seven agents is future work (§§3.2/6.3 pp.3/8).

Criteria Map 4 Evidence Log

He et al., Minority Sentinel: When to Overturn Majority Voting in Multi-Agent LLM Debates, AgentSearch Workshop/SIGIR 2026; arXiv:2606.29270v1 Sources: https://arxiv.org/pdf/2606.29270v1

• §§3-5: six QA/math benchmarks, majority and LLM-as-Judge comparisons, AUROC, flip precision/net gain and 20-seed cross-validation/feature ablations. No synthetic known-copying/independence ground truth; semantic audits classify debate behavior rather than extracted proof dependence. Workshop SIGIR 2026; feature/prompt details published, code release unclear. Relevance: aggregation-stage majority override/behavioral credibility is close enough to require novelty comparison, but trains a meta-classifier.

Evidence notes | OW-ISP

Criteria Map 1 Batch 4 Evidence Log

Identity confirmed: Rui Ai, Yuqi Pan, David Simchi-Levi, Milind Tambe, Haifeng Xu, Beyond Majority Voting: LLM Aggregation by Leveraging Higher-Order Information, ICML 2026, PMLR 306:1311-1340. arXiv:2510.01499v2. Sources: https://proceedings.mlr.press/v306/ai26d.html and https://arxiv.org/html/2510.01499 .

• **§2, §3:** multiple zero-shot LLM agents answer independently; the paper aggregates **discrete option labels**, not debate messages or reasoning traces (rows **1,3,19,49,59,63**, X; **2,4,50-58,60-62,64-70**, -). Conditional independence of agents' answers given truth is an **explicit** theoretical assumption; the authors also study violations and all-eight-model correlated ensembles (rows **24-26,48**, X).

• **§3, Algorithms 1, §4.1-4.3:** OW uses first-order model accuracies for Bayesian-optimal vote weights; ISP estimates second-order inter-agent **conditional answer distributions**. OW-L fits unsupervised accuracies via empirical conditional distributions; OW-I builds pseudo-label accuracies from ISP. **No null-corrected reasoning-step overlap**, and practical unsupervised parameters do not imply a trained debating model (rows **27-29**, X).

• **§5.1-5.4:** simulation with deliberately unequal agent accuracies; eight actual LLMs in four families **GPT, Qwen, Llama, Phi**, each family with strong and weak variants. Comparisons vary balanced vs mixed capabilities and all-eight correlation cases (rows **33-37,40,41,44,47-48**, X). Both open-weight and closed-API model families participate; one-model multi-sample self-consistency is background, not the method (row **31**, -).

• **§5.3-6:** accuracy improvements on UltraFeedback, MMLU, ARMMAN, and special MMLU-hard items with incorrect-answer concentration; robustness and correlation effects are evaluated (rows **5,9-10**, X). The work does **not** label minority recovery as a separate target or measure conformity under discussion (rows **11,16**, -).

• **§3-4, Appendix C:** formal optimality and extensions beyond conditional independence constitute theory, **but the paper is a method plus experiments**, not diagnosis/theory **only** (row **8**, -; row **29**, X). No goal of probabilistic calibration or token/communication savings is established (rows **6-7,71**, -).

Criteria Maps 2 3 Batch 4 Evidence Log

Identity: Rui Ai, Yuqi Pan, David Simchi-Levi, Milind Tambe and Haifeng Xu, Beyond Majority Voting: LLM Aggregation by Leveraging Higher-Order Information, ICML 2026, PMLR 306:1311-1340; primary PDF arXiv:2510.01499v2, May 19, 2026. Sources: proceedings record, primary PDF, 30 pages. Method §§2-5 pp.3-8; beyond-independence extension C pp.15 onward; estimation/prompts F.2-F.3 pp.22-25.

• M2 1/5/9/14/16/18-19/22-24/28/31/34-35/41-42: OW uses inverse-logistic expected-accuracy weights; OW-L estimates them by least-squares fitting the parametric pairwise conditional-response law, and OW-I estimates accuracy from agreement with ISP pseudo-labels (§5.2 p.7, F.2 pp.22-23). This is learned global reliability/accuracy weighting and fitted probabilistic answer-model aggregation, not verbal/token confidence weighting (3-4: -). ISP combines vote counts with empirical conditional-answer probabilities (§4 pp.4-6). MV/SP and clairvoyant best-agent/OPT references are actually evaluated (§5 pp.6-8). Weights remain fixed after dataset-level estimation; no held-out or online weight update (43-44: -). 31 applies to the empirical agreement/response-frequency signal, not a semantic-entropy calculation.

• M2 18: X because nonuniform OW weights can outweigh the raw majority, and the implemented SP/ISP rules can also depart from it (§§3-5 pp.3-8). These are final answer overrides, without an intervening debate. Algorithms 1-2 emit a mapped answer label, not calibrated confidence or a belief distribution (20-21: -). Output space Δ(S) in the formal problem does not mean the implemented argmax rules emit a posterior.

• M2 2/6-8/10-13/15/17/25-27/29-30/32-33/36-40/45: -; no terminal debate consensus, final LLM judge/trained auditor/meta-classifier, calibrated semantic-cluster posterior, delegation, trajectory/step rule, semantic answer clustering, reasoning-path fusion, confidence/trust/relevance/format-quality/evidence rubric or confidence-unreliability result. Models return only discrete choices without explanations (F.3 pp.23-25). Option-order shuffling and parsing are structured label handling, not content/step extraction.

• M3 1/3-4/9/12-14/19-21/28-29/35: empirical pairwise P(Ai|Aj) estimates model answer dependence, and the theory includes a question-difficulty-induced correlation extension (C pp.15-16). MV ignores it. SP baseline discounts expected popularity by subtracting predicted frequency; 4/29 are scoped to that implemented answer-level popularity correction, not a claim that OW/ISP count independent derivations. Main ISP uses predictions under counterfactual alternative answer labels (§4.2 p.5), which is not a no-communication control (7: -). Accuracy/conditional probabilities concern model/agent/dataset scope and decisions are per question.

• M3 2/5-8/10-11/15-18/22-27/30-34/36: -; no separately fitted coupling coefficients, reasoning-content dependence/embeddings/messages/steps, chance-adjusted error overlap, per-agent redundancy factor, asymmetric private evidence, step extraction or reasoning structures/clusters/copying detector. Random label shuffling, the uniform N/K vote reference and expected-popularity subtraction do not estimate independent chance-expected reasoning overlap (22). Conditional answer association supports 9, rather than a content-level overlap metric. Pairwise conditional probabilities are not an Ising-style coupling parameter (6).

• M3 37-38/53-55: frozen-model one-round label aggregation, no fine-tuning, with finite-sample/optimization settings; main fits only unlabeled answer statistics (§§4.3/5.2 pp.6-7, F.2 p.23). It needs no further LLM calls once the ensemble's fixed predictions exist (45: -). §5.4 p.8 states aggregation takes several CPU seconds, supporting the bounded latency-overhead claim (53), not a measured speedup over MV. Eight-model ensembles and two-family strong/weak correlation stress tests are actually evaluated (§6 p.9, F.4 pp.28-29), supporting 55.

• M3 39-43/46-52: -; no small labeled calibration split, trained classifier/auditor/graph encoder, EM fit, dedicated verification judge, executable/PRM/tool/intermediate-step checker, token-cost comparison or token efficiency claim. OW-L's least-squares parameter estimation is not EM or LLM training. Responders select preferences/labels, without a separate correctness verifier. M3 44: X only for the explicitly clairvoyant Single Best/OPT reference; the paper notes selecting the best model requires ground truth (§5.4 p.8). Practical OW-L/OW-I/ISP are label-free. Ground truth for ordinary accuracy evaluation does not by itself support 44.

Criteria Map 4 Evidence Log

Identity: Rui Ai, Yuqi Pan, David Simchi-Levi, Milind Tambe and Haifeng Xu, Beyond Majority Voting: LLM Aggregation by Leveraging Higher-Order Information, ICML 2026, PMLR 306:1311-1340; primary PDF arXiv:2510.01499v2, May 19, 2026. Sources: https://proceedings.mlr.press/v306/ai26d.html, https://arxiv.org/pdf/2510.01499v2

• §§3-6/C/F: MMLU/UltraFeedback/ARMMAN; uniform and reliability-weighted aggregation, answer accuracy and second-order conditional answer information. Simulations define the CI mechanism and vary accuracies; correlated eight-model extensions are also checked, so controlled generative-dependence validation includes the known independence case (18). Own Bayes-optimal weights/conditions and estimation analysis. Published ICML 2026; eight actual models, not ten. Code release unclear; prompts given. Relevant aggregation/theory baseline, without reasoning chains.

Evidence notes | DivEvid

Criteria Map 1 Batch 4 Evidence Log

Identity matched provisionally (Paper Key not provided): Yuante Li, Yicheng Tao, Kate Zhang, Taozhi Wang, Gefei Gu, Yaxin Zhou, Diverse Evidence, Better Forecasts: Multi-Agent Deliberation Under Information Asymmetry, arXiv:2607.01661v1 (July 2026), method named **InfoDelphi**. Primary: https://arxiv.org/html/2607.01661 . The abbreviation “DivEvid” uniquely and directly matches **Diverse Evidence**, and method/topic match this comparison map; still request the Paper Key entry if strict key-level identity certification is necessary.

• **§3.1-3.3:** BM25 ranks each question's retrieved evidence. Highest-ranked documents form a **shared public subset**, remaining documents are round-robin allocated to **disjoint agent-private subsets**. This is designed input information asymmetry and enforced initial diversity-not a dynamically selected agent communication topology (rows **17,25,41-42,45**, X; **53-55**, -). Receives structured `evidence_used` identifiers and emits confidence + label + rationale (rows **60-62**, X).

• **§3.2, Prop. 3.1 & Cor. 3.1.1:** models the predicted cross-agent error covariance as a function of public evidence ratio under an explicit independent private-error assumption. Both challenges and mathematically models error dependence (rows **24-26,29**, X). This is **assumption-driven decorrelation through evidence routing**, not a question-level, null-corrected derivation-independence metric.

• **§3.3, Fig. 2, §4, Appendix G: three same-model agent instances** are independently prompted with their own evidence and then share previous-round **probability, label and rationale excerpts**. Default R=2, with one-round no-deliberation and 3-round ablations; all peers' round-1 summaries are passed (rows **1-4,18-19,31,49-52,60-64**, X). Their roles are not different personas. No question-dependent topology, dynamic sparse peer selection, or early-stopping/trigger rule (rows **39,53-57,65-70**, -). Relevance is used to select source evidence, not to filter peer debate messages (row **69**, -).

• **§4-5, Tables 1-3:** reports Brier score and accuracy on the **new PolyGym** benchmark (375 binary Polymarket items with fixed temporally filtered evidence), and ablates private/public split, information routing, rationale sharing, confidence-weighting, and debate rounds. Primary GPT-5.4-mini API backend plus tests with DeepSeek-V3.2 and Llama-4-Scout; each experimental panel is homogeneous even though multiple model families are tested (rows **5,7,9,30,34,36-37,44,47**, X; **40**, -). Information asymmetry reduces shared erroneous interpretation/herding; after 3 rounds results degrade (rows **10-13,21-22**, X).

• **§5.4:** explicitly measures inter-agent prediction variance before/after debate to quantify diversity retention and conformity (rows **11,44**, X). **Appendix E.4, Table 7** reports API dollar cost and call counts, including 3,375 API calls/run: communication/inference cost reported, without cost saving being the method's main goal (row **71**, X; **6**, -).

• **Limits:** no provenance step extraction, no independence-weighted evidence-chain counting, no shared-pretraining-provider ablation; no message filtering by diversity/trust/confidence, and no debate stopping threshold.

Criteria Maps 2 3 Batch 3 Evidence Log

Identity checked from the primary PDF: Yuante Li, Yicheng Tao, Kate Zhang, Taozhi Wang, Gefei Gu and Yaxin Zhou, Diverse Evidence, Better Forecasts: Multi-Agent Deliberation Under Information Asymmetry, arXiv:2607.01661v1, July 2, 2026; method InfoDelphi. This retains the previously provisionally mapped short key because the Paper Key is absent. Source: primary PDF, 17 pages; §§3-5, Appendices B-H. The abstract-page lookup failed, but the full PDF was accessible and reviewed.

• M2 2-3/6/14/16/18/20-24/29/31/41: InfoDelphi combines terminal probability forecasts using weights |logit(p)| in logit space (§3.3 p.5). It emits a probability/derived binary label, including a Bernoulli belief distribution, not an answer-only or clustered-answer posterior (19/10/15: -). A sufficiently extreme minority forecast can outweigh weak majority-label forecasts: 18 is a mathematical implication of Eq.7. Implemented baselines include uniform means, a median, sampled self-consistency with averaged probabilities, and MoA/AIA LLM aggregation (E.1 pp.13-14). Neither SC nor Standard Debate uses a plain label-majority vote here (1/28: -). No trained adjudicator, fitted dependence-law final rule, delegation, trajectory score or step/DAG weights (4-5/7-13: -).

• M2 23/32-33/35/37/39: BM25 evidence relevance, explicit source-credibility priority and rationale/forecast integration are used (§3.3 p.4, G p.15). Prompt-calibration checks apply to the implemented Halawi/Superforecaster baselines (E.1 pp.13-14), not a fitted InfoDelphi confidence calibrator. The implemented AIA supervisor identifies reasoning disagreements before deciding its forecast. These are prompt-level rubric/signals, not independently verified evidence-quality scores. No answer-diversity statistic selects the initial solver answers (36: -); information diversity is

imposed through inputs.

• M2 17/25-27/30/34/38/40/42-44: -; rationale excerpts provide context without scored path/step/token aggregation, and only previous-round summaries are exchanged. No token likelihood, historical agent-accuracy prior, parse-quality weighting, learned/held-out aggregation fit or iterative numerical weight update. Confidence weights are computed once from the terminal forecasts. M2 45: X; H pp.16-17 reports confidently wrong forecasts, including a correlated-evidence echo chamber reaching 0.996 despite a NO outcome. This is qualitative empirical evidence, not a population calibration-error estimate.

• M3 1/3/7/9/11/13-15/19-21/24/28-29: the error-covariance model separates common/public from independent/private errors (§3.2 pp.3-4, C-D pp.12-13). Disjoint private documents implement information asymmetry; matched R=1/R=2 configurations and before/after forecast variance examine communication effects (§5.4 p.8, E.2 p.14). Ordinary homogeneous-input pooling baselines ignore dependence. Confidence pooling itself has no dependence discount (4/10: -). Evidence routing reduces correlation by design, not by measuring reasoning overlap or counting independent derivations.

• M3 2/5-6/8/12/16-18/22-23/25-27/30-33/36: -; no fitted coupling/content-overlap estimator, excess-error chance statistic, null calibration, atomic step extraction, reasoning graph/clustering or copying/honest-sharing discriminator is implemented. Structured evidence_used IDs and free-text rationales do not constitute extracted steps. BM25 provides lexical query-document similarity (35), and AIA uses LLM reasoning-disagreement comparison (34); both have that restricted scope.

• M3 37-38/45/49/54-55: prompt-only inference with tunable J/R/public-ratio/routing settings; extra deliberation and baseline supervisor calls. Tavily retrieval is actually used in benchmark construction (B p.12), while inference receives fixed pre-retrieved evidence; 49 is precomputed source grounding, not live retrieval verification. Main InfoDelphi has three agents; the paper implements baselines up to ten agents (E.1 p.14), accounting for 55. No calibration split, trained classifier/auditor/encoder/EM or required gold training labels (39-44: -); resolved outcomes are evaluation-only. Evidence verification is future work (A p.11), not a dedicated judge/step/tool-execution/PRM checker (46-48/50: -).

• M3 51-53: -; E.4/Table 7 p.14 reports estimated API dollars/call counts per model, not measured token totals versus an inference baseline, a token-saving claim or a latency-efficiency result.

Criteria Map 4 Evidence Log

Li et al., Diverse Evidence, Better Forecasts: Multi-Agent Deliberation Under Information Asymmetry, arXiv:2607.01661v1 Sources: https://arxiv.org/pdf/2607.01661v1

• §§3-5/C-H: PolyGym resolved-event forecasting, independent pooling/standard debate, ReConcile, MoA/AIA judge baselines; accuracy/Brier/diversity variance and controlled shared-versus-disjoint evidence/round/rationale ablations. Known public/private information sharing supports 18, not labeled copied derivations. Own common/private-error covariance analysis yields a closed variance relation. API dollars/calls are not token-count reporting (15: -). Paper states accompanying code/data are released. Venue lookup failed; PDF remains readable. Motivation/baseline/background.

Evidence notes | R²-MAD

Criteria Map 1 Batch 5 Evidence Log

Identity: Xuanfa Jin, Zhijian Ma, Yongcheng Zeng, Xinyu Cui, Haifeng Zhang, Jun Wang. Remember and Reweight: Enhancing Multi-Agent Debate with Experience Memory and Confidence Estimation, arXiv:2609.03619v1, 3 September 2026. Method name explicitly matches. Primary: https://arxiv.org/html/2609.03619v1 ; https://arxiv.org/pdf/2609.03619 .

• §§3-4,A.3: independent starts, full peer exchange, vote (1-4,18-19,51-52,58 X); retrieval begins after initialization (17 -).

• §4.1,C: assumes conditional independence, models latent peer skew, challenges shared errors (24-26,29 X); no chain-independence estimator/null.

• §4.2,A.3: private memories, consensus/MMR diversity, selected structured cases (42,44-45,61-62 X). Live messages remain unfiltered (68-70 -).

• §4.3: annotates complete responses with reliability (60 X); prior correction is not probability-calibration evaluation (7 -).

• §5,A.3,D pp.22-24: homogeneous persona panels, temperature, analyst summarization (31-32,36-39,41,46 X); no mixed-model/capability-gap tests (33-35,40,47-48 -).

• §5.2,B.4: misconception/minority flips and recovery (9-13,16,21-22,28 X); no seeded hints (43 -).

• A.3: memory construction fixes rounds (64 X); evaluation stops on consensus (65 X); no single-round protocol (63 -).

• B.7: reports tokens/time (71 X), adds cost (6 -).

Criteria Maps 2 3 Batch 3 Evidence Log

Identity: Xuanfa Jin, Zhijian Ma, Yongcheng Zeng, Xinyu Cui, Haifeng Zhang and Jun Wang, Remember and Reweight: Enhancing Multi-Agent Debate with Experience Memory and Confidence Estimation, arXiv:2609.03619v1, September 3, 2026. Source: primary PDF, 24 physical pages; §§3-5, Appendices A-D. The abstract-page lookup failed, but the full PDF was accessible. Page references below were checked using individual PDF pages because chart glyphs introduced spurious form-feed characters in bulk text extraction.

• M2 1-2/14/16/19-20/22/24/28/31/41: final answer is majority over terminal responses; evaluation can stop at consensus (A.3 pp.13-14). CoT/SC/MAD and both MAD-M2 masking variants are actually implemented; SC uses 3/6/9 samples (A.2 p.12, B.2 p.14). Boxed numeric and double-parenthesized option fields support structured exact answer extraction (D pp.22-24). Confidence accompanies peer messages. All final voting remains uniform: internal reliability weighting does not constitute a weighted terminal vote or minority override (3-13/18: -).

• M2 17/23/25/33-37/39-40/44: debate state combines the current task, own previous response, recursive debate summary and consensus ratio; it keys memory retrieval (§4.2 pp.4-5, D p.24). The summary explicitly compares the previous summary with current stances, retaining a compressed trajectory. Historical whole-response cosine matches and correctness labels generate peer reliability (Eq.11 p.5), while MMR mixes relevance, success and redundancy penalties. Confidence/selection are recomputed each round, although the memory bank remains frozen at test time. Aggregation-path scope is whole-response historical matching, not atomic step fusion. Starting-answer consensus already affects first-round retrieval. Historical correctness and subjective-mask checks support quality/reliability inputs.

• M2 30: X only for the implemented objective MAD-M2 perplexity baseline (A.2 pp.12-13, B.1 p.14). Main R²-MAD uses embedding/correctness-based reliability, not token likelihood or LLM self-verbalized confidence (29: -). Confidence tags are externally computed reliability labels. Memory-corrected concept priors are not a demonstrated probability-calibration feature; no fitted confidence calibrator/held-out weight-calibration split or learned aggregation coefficients (32/42-43: -). Similarity retrieval does not cluster equivalent answers or score parse quality (15/38: -). No explicit confidence-unreliability experiment is reported (45: -).

• M3 1/3-4/10-11/13-16/18-21/24/28-29/33-35: the latent-concept/peer-skew theory models repeated majority influence and suppresses it using reliability weights (§§3-4 pp.3-6, C pp.18-21); ordinary voting baselines ignore dependence. Agent-specific memory banks/retrieved contexts provide information asymmetry, without disjoint or statistically independent histories (11). BGE-M3 whole-state/response similarity retrieves historical cases; MMR discounts redundant memory cases, not live derivation overlap (A.3 p.14). Token signal is the tested perplexity baseline. The analyst compares stances/arguments, and exact answer identity defines consensus.

• M3 31: X for the implemented LLM summary's qualitative detection of pressure-driven switching: it distinguishes substantive argument changes from abandoning a unique position without substantive refutation (D p.24). This is a heuristic stance/argument assessment, not proven copying or derivation provenance. It does not separate chance-expected honest reasoning overlap from copied reasoning (32: -).

• M3 2/5-9/12/17/22-23/25-27/30/36: -; no fitted pairwise coupling/correlation model, current-panel content-dependence statistic, excess-error/chance-overlap null, matched no-communication dependence control, atomic step extraction, reasoning clustering/tree/DAG/graph or structural derivation similarity. Extracting round-level memory fields and two abstract summary sentences is not extracting reasoning steps. Conditional independence in the theory is an assumption, not null calibration.

• M3 37-38/44-46/49/54-55: no LLM fine-tuning, but the implemented memory requires outcome-labeled training debates (366/633/727/498 training questions; A.1 p.12) and is fixed during evaluation (Limitations p.10). Gold is reused, not newly annotated; verifier/execution outcomes are offered as alternatives without a tested implementation. No small calibration split, trained classifier/auditor/graph encoder or EM fit (39-43: -). Extra summary calls and internal memory retrieval are real. LLM verification applies to the tested subjective MAD-M2 mask, not a trained auditor. No executable/PRM/intermediate-step checker (47-48/50: -). Sample scaling is the implemented SC 3/6/9 comparison, not increased agent counts for the proposed method.

• M3 51-53: X; B.7/Table 10 pp.16-17 reports tokens and wall-clock time against SC/MAD, then claims bounded constant overhead. It reports higher cost (about 1.5-2× MAD time), not token savings or a latency speedup.

Criteria Map 4 Evidence Log

Identity: Xuanfa Jin, Zhijian Ma, Yongcheng Zeng, Xinyu Cui, Haifeng Zhang and Jun Wang, Remember and Reweight: Enhancing Multi-Agent Debate with Experience Memory and Confidence Estimation, arXiv:2609.03619v1, September 3, 2026. Sources: https://arxiv.org/pdf/2609.03619v1

• §§3-5/A-D: math/knowledge/commonsense tasks; CoT/SC/MAD and masking comparisons, tokens/time, memory/confidence/diversity ablations. Latent-concept priors and peer-skew analysis give formal update/error conditions; aggregate success comparisons do not automatically constitute flip precision or ground-truth dependence. Code and prompts linked; current venue lookup failed. Relevance: misconception/conformity motivation, credibility/memory baseline and latent-prior background; whole-response memories differ from live derivation-chain counting.

Evidence notes | Estornell

Criteria Map 1 Batch 5 Evidence Log

Rows 1-71: ? **(identity unresolved).** The surname does not distinguish Multi-LLM Debate: Framework, Principals, and Interventions (Estornell and Liu, NeurIPS 2024) from ACC-Collab: An Actor-Critic Approach to Multi-Agent LLM Collaboration (Estornell et al., ICLR 2025). Primary records establish both candidates, but no supplied or recovered Paper Key maps this column to either title. Candidate records: https://papers.nips.cc/paper_files/paper/2024/hash/32e07a110c6c6acf1afbf2bf82b614ad-Abstract-Conference.html and https://proceedings.iclr.cc/paper_files/paper/2025/hash/e187897ed7780a579a0d76fd4a35d107-Abstract-Conference.html .

This is missing key-level identity evidence, not a finding that the criteria do not apply. A full title plus arXiv/DOI/venue identifier is needed. These marks must remain untouched in later blank-only sessions unless the user explicitly authorizes their reassessment.

Criteria Maps 2 3 Final Evidence Log

M2 rows 1-45 and M3 rows 1-55: **?**. The intended title/authors/identifier cannot be verified from this alias; the Paper Key remains unavailable. Retained Map 1 logs already document the identity gap. Candidate papers, surname matches and remembered methods were not substituted. These are identity uncertainties, not negative findings about an identified paper. No source can be attributed securely to this column until its Paper Key entry is supplied.

Criteria Map 4 Evidence Log

Identity unresolved; intended Paper Key entry remains unavailable. Sources:

• Rows 1-44: ?. Author surname maps to more than one relevant paper; the intended Paper Key entry is missing.

Evidence notes | Kim

Criteria Map 1 Batch 5 Evidence Log

Rows 1-71: ? **(identity unresolved).** The surname alone cannot establish the intended first author, title, year or identifier. Searches returned multiple Kim-authored works on correlated errors, latent-concept interventions, evaluation and debate. Neither the supplied map/logs nor recovered project context linked this key to a specific primary paper. No candidate's properties were transferred into this column.

This is missing key-level identity evidence, not a finding that the criteria do not apply. A full title plus arXiv/DOI/venue identifier is needed. These marks must remain untouched in later blank-only sessions unless the user explicitly authorizes their reassessment.

Criteria Maps 2 3 Final Evidence Log

M2 rows 1-45 and M3 rows 1-55: **?**. The intended title/authors/identifier cannot be verified from this alias; the Paper Key remains unavailable. Retained Map 1 logs already document the identity gap. Candidate papers, surname matches and remembered methods were not substituted. These are identity uncertainties, not negative findings about an identified paper. No source can be attributed securely to this column until its Paper Key entry is supplied.

Criteria Map 4 Evidence Log

Identity unresolved; intended Paper Key entry remains unavailable. Sources:

• Rows 1-44: ?. No intended full title, author initials or identifier is confirmed; no candidate is substituted.

Evidence notes | Choi

Criteria Map 1 Batch 5 Evidence Log

Identity recovered from the earlier project map and verified: Hyeong Kyu Choi, Xiaojin Zhu, Sharon Li. Debate or Vote: Which Yields Better Decisions in Multi-Agent Large Language Models?, NeurIPS 2025 Spotlight; arXiv:2508.17536v2. Earlier metadata lists Yixuan Li. Key status unavailable. Primary: https://arxiv.org/html/2508.17536v2 ; https://proceedings.neurips.cc/paper_files/paper/2025/hash/934252acd87f254d5d4672fbde283bd2-Abstract-Conference.html .

• §§2-3,5: voting/broadcast/sparse/central protocols, during-debate interventions (1-4,18-19,46,49-53,61 X;17,20 -).

• §4,C-D: independent sampling assumed, belief coupling modeled (24,26 X;25 -); no error-covariance/chain-independence estimator.

• §§3,5: correct responses preserved through oracle locks/practical majority-guided updates (9,11,16,21-23,27 X). No misconception/confidence targeting (10,12-15 -).

• §6,A.1,B.3,G: homogeneous model panels/persona teams (31-32,36-39,41,45 X); no mixed-model/capability-gap study (33-35,40,47-48 -).

• B.1,F: full responses plus label voting (58-59 X); no confidence/structured evidence (60,62 -).

• §3.1,D.3.2,G: fixed/ring/one-round trials (56,63-64 X); no stopping/filtering (65-70 -). Sequential protocols remain future work (57 -).

• A.1,§4: token ceiling is not cost measurement (71 -); semantic overlap only suggested. Oracle validation uses no seeded hints (43 -).

Criteria Maps 2 3 Final Evidence Log

Identity: Choi, Zhu and Li, Debate or Vote: Which Yields Better Decisions in Multi-Agent Large Language Models?, NeurIPS 2025; arXiv:2508.17536v2. Source: https://arxiv.org/pdf/2508.17536v2 .

• M2 1-2/14/16/19/22/24/28/31/41: §2 and B.1/F implement independent voting and terminal label voting after debate, with parsed task-answer fields and equal votes. No learned/weighted final rule, explicit confidence output or extracted-step fusion. Previous-round peer responses are the main input, not a scored whole trajectory (17/25-27: -).

• M3 1/3/7: §§3-4 compare pre-communication voting with debate under the same solver setting; the DCM/martingale model explicitly couples belief updates through observed peer counts. This is analytical dependence modeling, not a fitted dependence-based final decision (M2 9: -). Baseline votes still ignore it. M3 4-6/8-11/22-23/30-33: -; no measured error covariance, chance-corrected overlap, extracted derivations or copying discriminator. §4's semantic agreement extensions are suggestions, not implemented clustering.

• M3 12-15/19-21/24/28-29/35: answer-label dynamics are modeled and measured, with harmful convergence and correct-answer preservation examined (§§3-5); reasoning stays flat. M3 37-38/44-45/54-55: frozen-model rounds/hyperparameters and panel-size experiments; the implemented MAD-oracle uses gold labels to lock correct responses (§5.1), while practical majority-guided variants do not. No trained classifier/encoder/auditor/EM or inference tools. Token ceiling is not a cost comparison (51-53: -).

Criteria Map 4 Evidence Log

Identity: Choi, Zhu and Li, Debate or Vote: Which Yields Better Decisions in Multi-Agent Large Language Models?, NeurIPS 2025; arXiv:2508.17536v2. Source: https://arxiv.org/pdf/2508.17536v2 . Sources: https://arxiv.org/pdf/2508.17536v2

• §§3-6/A-G: seven math/knowledge/commonsense/preference datasets; single CoT/vote, vanilla/topology debates and oracle/majority-preservation interventions. Accuracy/answer corruption and multi-seed/ablation results; own DCM/Bayes martingale and majority success conditions. Oracle locks correctness, not a known-copying mechanism (18: -). Code/prompts released, NeurIPS 2025. Motivation/baseline/theory background, rather than a content-level independence estimator.

Evidence notes | CanDebate

Criteria Map 1 Batch 6 Evidence Log

Rows 1-71: ? **- identity unresolved.** The supplied map contains only this short name. Searches for the exact key did not locate an unambiguous paper, and recovered project context did not supply a title, author list or identifier. A primary candidate, Can LLM Agents Really Debate? A Controlled Study of …, appeared at https://openreview.net/pdf?id=waToHto7G7, but no evidence links it to this column. No candidate properties were assigned. The Paper Key entry is required; this is identity uncertainty, not a negative finding about any criterion.

Criteria Maps 2 3 Final Evidence Log

M2 rows 1-45 and M3 rows 1-55: **?**. The intended title/authors/identifier cannot be verified from this alias; the Paper Key remains unavailable. Retained Map 1 logs already document the identity gap. Candidate papers, surname matches and remembered methods were not substituted. These are identity uncertainties, not negative findings about an identified paper. No source can be attributed securely to this column until its Paper Key entry is supplied.

Criteria Map 4 Evidence Log

Identity unresolved; intended Paper Key entry remains unavailable. Sources:

• Rows 1-44: ?. Exact alias has no securely mapped title/ID; earlier candidate search did not confirm the Paper Key identity.

Evidence notes | Free-MAD

Criteria Map 1 Batch 6 Evidence Log

Identity: Yu Cui, Hang Fu, Haibin Zhang, Licheng Wang, Cong Zuo. Free-MAD: Consensus-Free Multi-Agent Debate, Findings ACL 2026, pp.31977-31997; arXiv:2509.11035. Published primary source: https://aclanthology.org/2026.findings-acl.1600.pdf .

• §§3-4,Algorithms 1-2: independent starts, full broadcast, history-based scoring (1-4,18-19,51-52,58-59 X). No pre-debate/trigger controller (17,20 -).

• §3 Eq.4: explicitly models intrinsic reasoning plus conformity-induced peer dependence (25-26,29 X); no statistical independence assumption, chain extraction, covariance estimator or independent-solver null (24 -).

• §§4.2-4.5,E-F: suppresses conformity/false consensus/minority loss (10-11,16 X); heterogeneous backbones and anti-copy prompts supply diversity (40-41,45 X). Structured self/peer/error/revision sections are transmitted (62 X), not extracted proof chains.

• §5.1,C: mixed Qwen-7B/Qwen-72B and Qwen/DeepSeek teams (33-36,47 X); API access does not establish closed models (37 -); temperature left default (38 -).

• §§5.2,6,D: one- and two-exchange trials, token consumption, 50% blocked receivers (2,6,9,50,63-64,71 X); limited communication refers to the attack experiment.

• A.1 discusses sparse compatibility without evaluating it (53 -). No confidence calibration, confidence messages, adaptive stopping, peer subset filter or seeded-hint intervention (7,43,60-61,65-70 -).

Criteria Maps 2 3 Final Evidence Log

Identity: Cui et al., Free-MAD: Consensus-Free Multi-Agent Debate, Findings ACL 2026; arXiv:2509.11035. Source: https://aclanthology.org/2026.findings-acl.1600.pdf .

• M2 12/14/16-19/24/28/40/41: §4.3 Algorithm 1 scores answer labels across every generation round: initial support, penalties for abandoned answers, rewards for newly adopted/retained answers, and inverse-round decay. Fixed coefficients; updates change scores, not learned weights (42-44: -). Figure 2 explicitly selects a final minority answer. Parsed label transitions are not reasoning-path/step extraction (25-27: -).

• M2 1-3/6/20/29/31: evaluated majority-vote/SC, ReConcile confidence-weighted and Liang judge baselines (§5, Table 1/experimental comparisons) supplement the proposed trajectory rule. Their implementation scope supports confidence sharing and ordinary answer self-consistency. No fitted probabilistic dependence aggregation, calibrated cluster posterior or step/DAG weights (7-11/13: -).

• M3 3-4: §3 Eq.4 analytically separates intrinsic reasoning and peer conformity; §4.3 discounts late, conformity-prone answer contributions. This is not an explicit per-agent dependence factor (10: -), measured content overlap (5: -), or chance null (22: -). M3 7: initialization/voting and debate comparisons (§3.2/Fig.1, §5). M3 23/25-27/30-33: -; structured self/peer/error/revision prose (E-F) does not extract proof units or distinguish honest shared derivations from copying.

• M3 12-15/19-21/24/28-29/35/37-38/45/54-55: flat frozen-model debate, answer-label equality, fixed scoring/round controls, heterogeneous panels and additional peer calls. §5.2/6 reports token consumption and fewer-round acceleration (51-53: X; latency claim is acceleration, not a reported wall-clock benchmark). No training/gold-label fitting. Confidence unreliability is cited in motivation, not established by its own confidence experiment (M2 45: -).

Criteria Map 4 Evidence Log

Identity: Cui et al., Free-MAD: Consensus-Free Multi-Agent Debate, Findings ACL 2026; arXiv:2509.11035. Source: https://aclanthology.org/2026.findings-acl.1600.pdf . Sources: https://aclanthology.org/2026.findings-acl.1600.pdf

• §§3-6/C-F: eight math/logic/knowledge tasks; CoT/SC, standard debate, ReConcile and judge-debate comparisons; accuracy, token consumption, round/prompt/scoring and blocked-receiver ablations. Formal intrinsic/conformity distribution is supplied, without one of the named Bayesian/martingale/Ising theories. Blocking receivers tests communication robustness, not labeled known-copying dependence. Explicit release link and prompts; Findings ACL 2026. Motivation and trajectory-aggregation baseline, without step dependence/null calibration.

Evidence notes | Cortex

Criteria Map 1 Batch 6 Evidence Log

Rows 1-71: ? **- identity unresolved.** Multiple primary works could match this short name: CortexDebate: Debating Sparsely and Equally for Multi-Agent Debate (Yiliu Sun, Zicheng Zhao, Sheng Wan, Chen Gong; Findings ACL 2025; arXiv:2507.03928) and CORTEX: Collaborative LLM Agents for High-Stakes Alert Triage (arXiv:2510.00311). Neither supplied files nor recovered project context links the column to one of them. No candidate properties were assigned. The Paper Key title/identifier is needed. Candidate records: https://aclanthology.org/2025.findings-acl.495.pdf ; https://arxiv.org/abs/2510.00311 .

Criteria Maps 2 3 Final Evidence Log

M2 rows 1-45 and M3 rows 1-55: **?**. The intended title/authors/identifier cannot be verified from this alias; the Paper Key remains unavailable. Retained Map 1 logs already document the identity gap. Candidate papers, surname matches and remembered methods were not substituted. These are identity uncertainties, not negative findings about an identified paper. No source can be attributed securely to this column until its Paper Key entry is supplied.

Criteria Map 4 Evidence Log

Identity unresolved; intended Paper Key entry remains unavailable. Sources:

• Rows 1-44: ?. CortexDebate and another CORTEX paper remain distinct possible matches; intended identity is missing.

Evidence notes | ReConcile

Criteria Map 1 Batch 6 Evidence Log

Identity: Justin Chih-Yao Chen, Swarnadeep Saha, Mohit Bansal. ReConcile: Round-Table Conference Improves Reasoning via Consensus among Diverse LLMs, ACL 2024, pp.7066-7085; arXiv:2309.13007. Primary: https://aclanthology.org/2024.acl-long.381.pdf .

• §4,A.2,Algorithm1: corrective examples enter initial prompts (17 X); full peer discussion/weighted votes (1-4,18-20,51-52,58-62 X). Selected subsets are demonstration examples; live peer explanations are retained.

• B.4,Fig9,Table13: confidence rescaling lowers ECE; weight robustness tested (7,9,12 X). No confidence stopping/filtering (66,70 -).

• §1,6.2,C.4-C.5: shared-pretraining/echo-chamber concerns, confident errors, minority recovery (10-13,16,25,48 X). BERTScore diversity is not chain-dependence estimation/null correction/step extraction (24,26 -;44 X).

• §6.1-6.2,A.1: mixed-family/strong-weak/open/API teams and homogeneous temperature ablations (31,34-38,40-41,45,47 X). Closed-model parameter sizes unavailable (33 ?).

• §4,6.2,B.3: consensus stops or skips discussion (49,65,67 X); actual rounds vary (64 -); one-round outputs analyzed (63 X).

• §6.2,B.3: efficiency uses calls/rounds (6 X), not measured communication-token/message cost (71 -). Cross-agent demonstration sets differ (42 X); these are not current-question seeded hints (43 -).

Criteria Maps 2 3 Batch 5 Evidence Log

Identity: Justin Chih-Yao Chen, Swarnadeep Saha and Mohit Bansal, ReConcile: Round-Table Conference Improves Reasoning via Consensus among Diverse LLMs, ACL 2024, pp.7066-7085, ACL Anthology 2024.acl-long.381; arXiv:2309.13007. Sources: publisher record, published primary PDF, 20 PDF pages. Method §4 pp.4-6, analysis §6.2 pp.8-9, A-C pp.14-20.

• M2 1-3/6/14/16/18-20/22-24/28-29/31-32/35/41/44: each agent produces an answer, full explanation and verbal numerical confidence. Previous-round responses are grouped by their distinct answer labels with explicit group counts (§4 p.5, Fig.5 p.14, Algorithm 1 p.15). Terminal answer maximizes confidence-weighted label support; consensus may end discussion early. SC/unweighted vote and Debate+Judge are implemented baselines (§5 p.6, B.5 p.16). Weighted confidence can outweigh a raw majority (18 is an implication of the rule); changing opinions during debate is not the reason for that mark. Peer records carry answer plus confidence, while the team output is an answer label. The recalibration function is fixed, but per-agent voting weights are recomputed from the new confidence each round (41/44).

• M2 4-5/7-13/15/17/21/25-27/30/33-34/36-38/40/42-43: -; no separate trust/accuracy vote, trained judge/classifier, fitted dependence law, calibrated answer-cluster posterior, delegation, trajectory/step score, semantic answer clustering or whole-history feature. Only previous-round responses are supplied, not a retained full trajectory. Initial model diversity and BERTScore diversity diagnosis are not a runtime starting-answer-diversity score. Raw rationale exchange does not score/fuse extracted paths or steps. No learned recalibrator or held-out aggregation-weight fit is used.

• M2 39: X narrowly for selecting convincing human explanations that rectify an agent's previously wrong answer; four per agent enter other agents' prompts (§4 p.5, A-B pp.14-15). This is demonstration-quality selection, not a current-question evidence-verification rubric. M2 45: X; B.4/Fig.9 pp.15-16/18 reports ECE falling from 0.2565 to 0.1374 after fixed confidence rescaling, and C pp.17-18 illustrates confident wrong answers/echo chambers. The tested weight grids are fixed configurations; the paper explicitly refrains from fitting a learned calibration model (B.4 p.16).

• M3 1/5/12-16/19-21/24/28-29/33/35: ordinary vote/discussion baselines ignore dependence; pairwise BERTScore whole-response embedding similarity diagnoses redundancy/diversity across initial responses and paraphrase controls (§6.2/Table 5 p.8). That is a content-level diversity proxy, not a fitted dependence/error-correlation estimator or an inference discount (3-4/6/8-10: -). Agreement supports voting; repetition/echo chambers are diagnosed as harmful (C.5 pp.17-18). Frozen full explanations remain flat prose, without extracted derivations (17-18/23/25-27/30-32/34/36: -). BERTScore is described here as response-embedding similarity, not a runtime token-confidence feature.

• M3 7/11: X with restricted scopes. Table 8/Fig.4 p.9 compares the same agents' independent round-zero outputs with subsequent discussion, giving a no-communication performance control, not a null-calibrated overlap estimate (22: -). Different agents receive corrective demonstrations selected for the other agents (Fig.3/§4 pp.4-5), creating asymmetry in examples, not disjoint live evidence for the current query. Dependence goes beyond motivation through the similarity

diagnosis (2: -).

• M3 37-38/45-46/50/54-55: prompt-only inference with tunable rounds/temperatures/confidence bins, repeated discussion calls and actually implemented 9-way SC (§5 p.6, A.1 p.14, B.3 pp.15-16). Verification 46 includes the implemented Self-Refine feedback baseline; main ReConcile does not add a dedicated final correctness verifier. Qualitative peer re-reasoning explicitly challenges intermediate chemical/logical claims (Fig.7 p.17), supporting 50 as fallible rationale checking, not a step-indexed validator. No labeled calibration split, trained classifier/auditor/encoder/EM fit, large-scale gold training, executable/PRM or actual tool/retrieval verification (39-44/47-49: -). Small gold explanations are few-shot inputs; references to searching/Wikipedia in generated text do not establish tool use (Table 16 p.20).

• M3 51-53: -; efficiency evidence compares API calls/rounds and faster consensus (§6.2 p.9, B.3 p.15). It does not compare token totals or report/claim wall-clock inference latency. Faster consensus measured in rounds is retained with this limitation.

Criteria Map 4 Evidence Log

Identity: Justin Chih-Yao Chen, Swarnadeep Saha and Mohit Bansal, ReConcile: Round-Table Conference Improves Reasoning via Consensus among Diverse LLMs, ACL 2024, pp.7066-7085, ACL Anthology 2024.acl-long.381; arXiv:2309.13007. Sources: https://aclanthology.org/2024.acl-long.381/, https://aclanthology.org/2024.acl-long.381.pdf

• §§4-6/B-C: seven math/commonsense/logic/NLI datasets; CoT/SC and debate baselines, confidence-weighted versus uniform ablations. Accuracy, ECE improvement, BERTScore diversity and minority/answer-change analyses, mean±SD and confidence/demo/round ablations. No independent-solver overlap null or known-dependence ground truth. Own code/prompts released; ACL 2024. Credibility/debate baseline and motivation, without atomic-step independence.

Evidence notes | DOWN

Criteria Map 1 Batch 7 Evidence Log

Identity: Sugyeong Eo, Hyeonseok Moon, Evelyn Hayoon Zi, Chanjun Park, Heuiseok Lim. Debate Only When Necessary: Adaptive Multiagent Collaboration for Efficient LLM Reasoning, arXiv:2504.05047v2. Primary: https://arxiv.org/html/2504.05047v2 .

• §3.1: independent starts, confidence gate, full peer responses/confidence, majority vote or judge (1-4,17-20,46,49,51-52,58-60,63-64,66-67 X). Single round means the accepted one-generation branch; active debate has two generation rounds. Homogeneous judge variant supplies role differentiation (32 X), not a diversity intervention (39 -).

• §§1,5.2,5.6; Appendix B Table9: unnecessary exchanges propagate errors and overwrite a correct minority (10-11,14,16,22,25 X). No covariance/independence model (24,26 -).

• §3.2: Wilson accuracy bound selects a utility threshold; no confidence calibration, independent-solver overlap null or step extraction (7 -).

• §§4.1-4.2,5.1: homogeneous and mixed-family teams; mixed models intentionally diversify reasoning (31,34,36-37,40-41,45 X). Size-varying Llamas are separate homogeneous panels, not a mixed-size team (33 -); no designated strong/weak pairing or gap study (35,47 -). Temperature zero (38 -).

• §§3.1,4.2,5.4: all peer messages retained; no message filter (61,68-70 -). Cost uses calls/skip rate, not communication tokens/messages (71 -). Controlled ablations test gate/confidence sharing; JSON reasons are not extracted evidence chains (62 -).

Criteria Maps 2 3 Final Evidence Log

Identity: Eo et al., Debate Only When Necessary: Adaptive Multiagent Collaboration for Efficient LLM Reasoning, arXiv:2504.05047v2. Source: https://arxiv.org/pdf/2504.05047v2 .

• M2 1-2/6/14/16/18-20/24/28-31/39/41: §3.1 accepts a confident initial answer or performs two generation rounds with three agents, sharing confidence and reasoning, then uses majority voting or a synthesis judge. Confidence is mean generated-token probability or verbalized confidence for inaccessible logits. It guides entry and peer revision, not a weighted terminal vote (3-5: -). Judge prioritizes supported/coherent arguments and can synthesize beyond the vote. M2 17: -; no all-round trajectory rule is specified.

• M2 45: §§5.2/5.6 and Table 9 show confident wrong revisions/correct-minority losses. M3 1/7/12-15/18-21/24/28-29/35: independent-start versus discussion comparisons and answer-label agreement; no joint-error/content-overlap model, copying provenance or independent-solver overlap null (3-6/8-11/22-23/30-36 except 35: -).

• M3 37-39/44-45/54: §3.2 tunes a utility threshold using validation accuracy, Wilson bounds and skip rates; this needs held-out gold labels but does not calibrate chance overlap or probability confidence. Main solver is training-free, no fitted classifier/auditor/encoder/EM. Confidence is not itself step verification. §4 uses three-agent panels, with no many-agent scaling experiment (55: -). Cost is calls/skip rate, not token totals or an explicit latency result (51-53: -).

Criteria Map 4 Evidence Log

Identity: Eo et al., Debate Only When Necessary: Adaptive Multiagent Collaboration for Efficient LLM Reasoning, arXiv:2504.05047v2. Source: https://arxiv.org/pdf/2504.05047v2 . Sources: https://arxiv.org/pdf/2504.05047v2

• §§3-5/B: MUSR/StrategyQA and other math/knowledge tasks; single/SC/MAD and voting-versus-judge comparisons. Accuracy/skip rate/call cost and answer-corruption analyses, gate/confidence/threshold tests. Wilson accuracy bounds choose a validation threshold; they do not establish ECE/AUROC or a known-dependence experiment. No reported token metric. Prompt-only inference and appendix prompts; code release unclear. arXiv 2025 record has no registered venue. Motivation/selective-debate baseline.

Evidence notes | MARGIN

Criteria Map 1 Batch 7 Evidence Log

Rows 1-71: ? **- identity unresolved.** Exact-name and alternative searches combining MARGIN with LLM, multi-agent reasoning and debate did not find an unambiguous paper named MARGIN. Results used “margin” as an ordinary word. Recovered project context supplied neither a title nor an author/identifier mapping. The Paper Key remains unavailable. No properties were assigned from a candidate title, related paper or memory. This column requires its Paper Key entry before reassessment; these are identity uncertainties, not checked negative findings.

Criteria Maps 2 3 Final Evidence Log

M2 rows 1-45 and M3 rows 1-55: **?**. The intended title/authors/identifier cannot be verified from this alias; the Paper Key remains unavailable. Retained Map 1 logs already document the identity gap. Candidate papers, surname matches and remembered methods were not substituted. These are identity uncertainties, not negative findings about an identified paper. No source can be attributed securely to this column until its Paper Key entry is supplied.

Criteria Map 4 Evidence Log

Identity unresolved; intended Paper Key entry remains unavailable. Sources:

• Rows 1-44: ?. Alias/title/author/identifier mapping is still unavailable.

Evidence notes | ConfLiar

Criteria Map 1 Batch 7 Evidence Log

Identity: Ali Keramati, Justin Cheok, Jacob Horne, Mark Warschauer. The Confident Liar: Diagnosing Multi-Agent Debate with Log-Probabilities and LLM-as-Judge, ACL 2026 Student Research Workshop, pp.1361-1375; arXiv:2606.10296. Primary: https://aclanthology.org/2026.acl-srw.121.pdf .

• §§3.1-3.3, Appendix A: Constructor → Auditor → Synthesizer, one exchange, full structured arguments and evidence (3,46,50,53,57-58,62-63 X;1-2,4,51-52,64 -). Anti-paraphrase/distinct-evidence instructions enforce role diversity (11,39,45 X).

• §§3.4-3.6: confidence captured during generation; separate post-debate judge diagnoses quality/failures (8,12,18-19,27-28 X). Fixed token/percentage windows are not key-step extraction. Cross-signal correlations/within-response variance do not model agent dependence; no independent-solver overlap null (24-26 -).

• §§3.6,4: confidence/quality alignment is correlation and failure AUROC, not calibration or an accuracy-improving intervention (5,7 -).

• §4,§5: only ASAP rubric experiments are reported. GPT-4o-mini fills debating roles; GPT-5-mini judges (32,37 X). Parameter sizes undisclosed (33 ?); other judge families/nonzero-temperature consistency tests are future work (34,38,40 -). No capability-gap or same-provider error-correlation study (35,47-48 -).

• §3,Limitations: fixed protocol; no selective trigger/stopping/filtering or reported communication cost (65-71 -). Proposed math/QA checks do not establish controlled cross-domain validation.

Criteria Maps 2 3 Final Evidence Log

Identity: Keramati et al., The Confident Liar: Diagnosing Multi-Agent Debate with Log-Probabilities and LLM-as-Judge, ACL 2026 SRW; arXiv:2606.10296. Source: https://aclanthology.org/2026.acl-srw.121.pdf .

• M2 6/17-20/24-25/27/30/39/41: §§3.1/3.3 Constructor→Auditor→Synthesizer uses the completed transcript for the final task output; the independent diagnostic judge scores whole arguments. §§3.4-3.5 aggregate token log-probability windows and fixed rubric dimensions, emitting reasoning quality and confidence features. M2 22-23: single-signal summaries and multi-dimension quality aggregation both apply; cross-signal correlation analysis is not a confidence-weighted final vote. M2 45: §4's own experiments diagnose high-confidence low-quality reasoning.

• M2 14-16: -; rubric-score outputs are not grouped into answer-equivalence classes, and explicit argument headings alone do not establish an answer parser.

• M3 1-2: declared independence/distinct-evidence instructions motivate diverse reasoning (§3.3.1), without a fitted or measured inter-agent dependence correction (3-11/22: -). M3 13-15/18-21/24/37-38/45-46/54: token windows and whole-response rubrics, frozen agents and post-debate quality judge. Fixed first/last token windows are not step extraction or derivation units (17/23/25-27/30-36: -).

• §4 reports ASAP rubric scoring only; math step verifier, retrieved QA passages and cross-family repeated-judge consistency are prospective protocol descriptions. Hence executable/retrieval/PRM checks and actual intermediate-step verification are not established (47-50: -), nor token/latency comparison or many-agent scaling (51-53/55: -).

Criteria Map 4 Evidence Log

Identity: Keramati et al., The Confident Liar: Diagnosing Multi-Agent Debate with Log-Probabilities and LLM-as-Judge, ACL 2026 SRW; arXiv:2606.10296. Source: https://aclanthology.org/2026.acl-srw.121.pdf . Sources: https://aclanthology.org/2026.acl-srw.121.pdf

• §§3-4/A: actually reports ASAP rubric scoring with human scores; mathematical/QA/cross-judge protocols are prospective, so benchmark rows 1-2 are -. Reports final scoring accuracy, log-probability/rubric correlations and critical-failure AUROC, window/role feature analyses. Cross-signal correlation is not agent error overlap (16: -), and fixed token windows do not report token cost. No named aggregation baselines or known-copying validation. ACL 2026 SRW; full prompts shared, own code release unclear. Supports confidence-reliability motivation.

Evidence notes | Slim-SC

Criteria Map 1 Batch 7 Evidence Log

Identity: Colin Hong, Xu Guo, Anand Chaanan Singh, Esha Choukse, Dmitrii Ustiugov. Slim-SC: Thought Pruning for Efficient Scaling with Self-Consistency, EMNLP 2025, pp.34500-34517; arXiv:2509.13990. Primary: https://aclanthology.org/2025.emnlp-main.1750.pdf .

• §2.1: explicitly i.i.d. chains and a plurality-probability argument (1,24,29 X). Semantic redundancy does not reject/model statistical independence (25-26 -).

• §§2.4,3.1-3.2: shared wrong logic/correct-chain outvoting motivate diversity-preserving pruning (10,13,16,44-45 X). Delimiter-split thoughts are embedded/compared; this segments thoughts rather than extracting logical proof steps. Delay protects common legitimate openings; thresholds are empirically selected, without an independent-solver overlap null.

• Algorithm1: final answers from surviving chains are voted (3,19,59,63 X). Single round denotes one parallel generation/vote pass. Chains receive no peer messages (2,4,14,17-18,20,50-58,60-62,64-70 -); chain pruning is neither debate topology nor message filtering.

• §§4-5,B.4: two separate open-model configurations (31,36 X;33-35,40,47-48 -); ablations compare global-random/least-similar pruning, delay and thresholds. Tokens/latency/KV cache measured (6,9 X;71 -).

• Row38 ?: temperature unspecified in paper/official README; linked evaluation code could not be read. Repository: https://github.com/hyscale-lab/slimsc .

Criteria Maps 2 3 Final Evidence Log

Identity: Hong et al., Slim-SC: Thought Pruning for Efficient Scaling with Self-Consistency, EMNLP 2025; arXiv:2509.13990. Source: https://aclanthology.org/2025.emnlp-main.1750.pdf .

• M2 1/14/18-19/22-26/28/31/35/41: §2.1/Algorithm 1 vote uniformly over surviving answer labels. §§3.1-3.2 prune chains using inter-chain thought similarity, with random or internal-diversity retention; thus answer, path and thought-unit operations coexist. Pruning changes the effective voting pool and can rescue an original minority (Fig.2). This is not a step-weighted final score (13: -). M2 16: ?; the mathematical answer map g is given, but a concrete structured parser was not verified.

• M3 1/4-5/10/17/23-24/29/33: i.i.d. sampling assumption coexists with semantic redundancy pruning. Stop-word delimiters extract thought segments, embed them with all-mpnet-base-v2, compare cosine similarity and remove one redundant chain (a zero contribution to final voting). Extracted segments are operational thought units, not validated logical premises or proof DAG nodes. No fitted error-dependence model or discrete semantic-clustering algorithm (3/25-27/30: -).

• M3 7-9/22/31-32: -; independent SC is a sampling/pruning comparison, not a matched communication counterfactual. Warm-up delay protects routine shared openings but neither calibrates chance overlap nor detects provenance/copying.

• M3 37-38/44-45/51-55: thresholds, delay and optimal sample count are empirically selected (§4.1/B); task-label accuracy guides optimal N, without an identified held-out calibration split (39: -). Frozen LLMs and embedding model; no classifier/auditor training or tool-based correctness check. §5 compares generated tokens, end-to-end latency and up to 64 samples; measured savings support both efficiency claims. No intermediate-step truth verification (46-50: -).

Criteria Map 4 Evidence Log

Identity: Hong et al., Slim-SC: Thought Pruning for Efficient Scaling with Self-Consistency, EMNLP 2025; arXiv:2509.13990. Source: https://aclanthology.org/2025.emnlp-main.1750.pdf . Sources: https://aclanthology.org/2025.emnlp-main.1750.pdf

• §§2-5/A-B: AIME/AQuA math and GPQA STEM, CoT/SC/pruning comparisons, accuracy, generated tokens, chain/thought cosine similarity and latency, pruning/delay/diversity/sample-count tests. Own plurality discussion states the sufficient probability-advantage condition. Natural semantic clusters do not supply known-dependence labels. Code/model/configuration details released; EMNLP 2025. Thought-segmentation and semantic redundancy pruning are relevant design overlap, while proof provenance and chance-null calibration are absent.

Evidence notes | ConsTrap

Criteria Map 1 Batch 8 Evidence Log

Identity: Jiayuan Liu, Shiyi Du, Weihua Du, Mingyu Guo, Vincent Conitzer. The Consensus Trap: Rescuing Multi-Agent LLMs from Adversarial Majorities via Token-Level Collaboration, arXiv:2604.17139v1. Primary: https://arxiv.org/html/2604.17139v1 .

• §§3.2-3.3,D.1: independent-vote condition and token-level RR/RRMaj studied (1,3,18-19,49 X). RR relays the complete prefix in fixed cyclic order (50,53,56-59 X); no independent-start hybrid or response-level multi-round debate (2,4 -). Chunk turns/M trajectories are not fixed debate rounds (63-64 -).

• §§1,4,A.2,B: coordinated false beliefs, sycophancy and minority loss targeted (10-14,16,25-26,29 X). Dependence is operator/state coupling, not covariance estimation, key-step extraction or independent-solver overlap calibration (7,24,62 -). Traditional debate benefits are discussed, not directly tested (21-23 -).

• §6,D.3: homogeneous, mixed-size/family and strong-weak ensembles; model/context diversity and private distractor injections (31,33-36,40-43,45,47 X). Symmetric continuation roles (32,39,46 -); no shared-pretraining correlation analysis (48 -).

• §§3.3,5-6,D: decoding-budget argument, corruption-strength/chunk-size/capability ablations; no measured communication cost or confidence stopping/filtering (65-71 -). Temperature not specified (38 ?).

Criteria Maps 2 3 Batch 5 Evidence Log

Identity checked from the PDF: Jiayuan Liu, Shiyi Du, Weihua Du, Mingyu Guo and Vincent Conitzer, The Consensus Trap: Rescuing Multi-Agent LLMs from Adversarial Majorities via Token-Level Collaboration, arXiv:2604.17139v1, April 18, 2026; preprint under review. Source: primary PDF, 21 pages; §§3-6 pp.3-9, Appendix D pp.16-21. Abstract-page lookup failed, but the full PDF was accessible.

• M2 1/14/17/19/22/24/27-28/31/40-41: MAJ takes the exact discrete-answer mode; RRMaj votes uniformly over M completed collaborative trajectories (§3.2-3.3 p.4). Within each trajectory agents cyclically generate K tokens after reading the entire accumulated prefix, supporting token aggregation and use of the full history. RRMaj adds answer-level self-consistency across trajectories. No distinct trajectory score, extracted-step weighting, confidence/trust/reliability vote or learned/online weight exists (3-13/26/29-30/32-39/42-45: -).

• M2 2/15/18/20-21/23/25: -; no response-level multi-round consensus, semantic answer clustering, confidence/posterior output, explicit multiple feature fusion or scored reasoning-path pooling. Recovering from a majority of corrupted agents changes generated trajectories, rather than selecting against RRMaj's terminal vote majority (18). Fixed round-robin generation is not vote delegation (11). M2 16: ?; terminal labels are specified mathematically, but the paper gives no structured output schema or answer-extraction/parser procedure; structured parsing cannot be confirmed.

• M3 1/3/7/11-15/18-21/24/28-29/35: the model represents dependence through shared autoregressive state and honest/corrupted transition operators (§4 pp.5-7); it is not a fitted covariance model. MAJ is the no-communication comparison with the same task/team corruption configuration (§§3/5 pp.4/7-9). D.1 p.16 defines the single-draw anchor and matched compute scaling. Privately injected false instructions create asymmetric agent inputs (§3.1 p.4); token chunks/messages share a flat prefix. Repeated final labels support voting; repeated corrupted answers are diagnosed as harmful coordinated error, not measured content-copy redundancy. Heterogeneous families/sizes and model/context conditions are actually tested (§6 p.9, D.3 pp.17-20).

• M3 2/4-6/8-10/16-17/22-23/25-27/30-34/36: -; no pairwise coupling coefficients, statistical answer-correlation measure, chance-adjusted error overlap/null, explicit agent dependence discount, content-similarity measurement, step extraction, reasoning graph/clustering, copying detector or honest-shared-versus-copied separation. Latent projection onto a theoretical truth direction is an analysis abstraction, not an implemented whole-reasoning embedding similarity signal. Token chunks are not extracted derivation steps.

• M3 37-38/45/54-55: frozen models and K/N/M hyperparameters, repeated continuation calls/prefills, and actual trajectory scaling up to M=40 (§6 p.9, D.1 pp.16-17). No labeled calibration split, classifier/auditor/graph encoder/EM training or required gold training at scale (39-44: -). Benchmark gold is used for correctness and adversarial evaluation, not fitting the runtime layer. No dedicated LLM/executable/PRM/tool/step verifier is implemented (46-50: -); implicit continuation-based correction and objectively checkable tasks do not constitute a specified verification module.

• M3 51-53: X with an analytical scope. §3.3 pp.4-5 explicitly compares N×L decode-token steps for MAJ with M×L for RRMaj at M=N, then claims token-cost parity and practically equivalent latency. This is a nominal decoding-budget calculation, not measured total tokens, prefill cost, communication cost or wall-clock timings. The claim of lower cost

than traditional deliberation is in §2 p.3.

Criteria Map 4 Evidence Log

Liu et al., The Consensus Trap: Rescuing Multi-Agent LLMs from Adversarial Majorities via Token-Level Collaboration, arXiv:2604.17139v1 Sources: https://arxiv.org/pdf/2604.17139v1

• §§3-6/A-D: math/spatial/logical tasks, single trajectory and independent majority versus token RR/RRMaj. Deliberately coordinated wrong-belief injections and corruption-strength/mixed-capability experiments provide controlled known-dependence validation. Accuracy/recovery, analytic decoding budget, significance and ablations are reported. Own impossibility/Condorcet and Lyapunov-stability conditions; traditional multi-round debate is discussed rather than directly benchmarked. Prompt/belief perturbations given, code release unclear; arXiv 2026. Motivation, token-collaboration baseline and stability background.

Evidence notes | ARGORA

Criteria Map 1 Batch 8 Evidence Log

Identity: Youngjin Jin, Hanna Kim, Kwanwoo Kim, Chanhee Lee, Seungwon Shin. ARGORA: Orchestrated Argumentation for Causally Grounded LLM Reasoning and Decision Making, arXiv:2601.21533v1. Primary: https://arxiv.org/html/2601.21533v1 .

• D.1-D.9,I: independent role-conditioned claims, then peer review/rebuttal; parsed statements/provenance/polarities passed selectively (1,4,17-19,32,39,41-42,46,58-59,61-62 X). Main claims reach every expert (51-52 X); later selective review coexists with the fixed orchestrator star (50,53,56 X). Argument trees are not dynamic communication topology (54-55 -).

• §4.2,D.4: embedding pruning measures/enforces novelty (14,44-45,68 X); no independent-solver null. QBAF/SCM models argument-score influence, not agent independence (24,26 -). D.1 explicitly addresses conformity/anchoring (11,25 X).

• D.8: prompt-based score calibration only (7 X), without probabilistic calibration validation. §§5-6,G.4-G.5: confident errors/shared misconceptions, capability-threshold analysis, beneficial/harmful flips and voting exceptions (10,12-13,21-23,47 X).

• §5.3,6.1,G.4: homogeneous open/API role teams; temperature zero (31,36-37 X;33-35,38,40 -).

• H.1-H.3: pruning/base-score/round ablations; R=1/2/3 reported (2,63-64 X). §4.4,G.6 confidence gate controls post-discussion override, not stopping/messages (65-67,69-70 -). Runtime/argument counts are not communication-token/message cost (71 -).

Criteria Maps 2 3 Batch 5 Evidence Log

Identity checked from the PDF: Youngjin Jin, Hanna Kim, Kwanwoo Kim, Chanhee Lee and Seungwon Shin, ARGORA: Orchestrated Argumentation for Causally Grounded LLM Reasoning and Decision Making, arXiv:2601.21533v1, 2026 preprint. Source: primary PDF, 58 pages. Abstract-page lookup failed, but the complete PDF was accessible. Earlier reading is reused; targeted checks cover §4 pp.3-5, D pp.21-31, F p.36, and H pp.48-51.

• M2 1-2/6/13-14/16/18-28 except 27/29/31/35/37-39/41: expert main claims, supporting arguments, peer critiques and rebuttals form scored rooted argument trees. Orchestrator scores task relevance, evidence support and logical soundness; deterministic QBAF semantics propagate node strengths, then a confidence-gated intervention may align the winner with a separate LLM judge (§4 pp.3-5, D.3-D.9 pp.23-30). This is graph/step/path scoring, not a confidence-weighted label vote (3-5: -). Implemented independent MV3/CoT baselines account for plain voting, self-consistency and answer-only/flat output scopes (§5.3 p.6). Exact main-statement matching merges authors; JSON schemas and explicit label mapping support parsing (D.3 p.24, F p.36, I.2 p.53). Invalid option formats can lower the task-relevance assessment (D.8 p.30), supporting 38.

• M2 18: X because the maximum-strength main argument and observational override are not constrained to the most numerous answer label. M2 20-21: the emitted technical report includes node confidence/strengths, normalized pQBAF and pobs distributions, winner margins and overrides (F/Table 5 p.36); these are argument consensus distributions, not calibrated semantic answer-cluster probabilities (10/15: -).

• M2 17/40/44: all generation/review/rebuttal stages enter the current argument structures and compressed transcript; implemented multi-round ablations retain each expert's full conversation history and recompute argument graphs/base scores in subsequent rounds (D.10 p.31, H.3/Table 14 p.51). Online-weight scope is those recomputed round-specific node scores, not learned agent reliability. Main experiments use one round. No chronological trajectory-score final rule, fitted dependence law, trained adjudicator/meta-classifier, delegation or token-level fusion (7-12/27: -).

• M2 32/45: explicit prompt-based score calibration counters observed extreme/inflated LLM scores, centers ordinary quality around 0.5 and restricts scores to (0,1) (D.8 pp.29-30, I.1.5 pp.52-53). This supports 32 as heuristic calibration of judge priors, not a gold-fitted probability calibrator or an ECE result. Confidence-gate/override failure analyses are actually tested (G.5-G.6 pp.42-48). No token likelihood, separate trust/accuracy prior, initial-answer-diversity statistic, learned coefficients or held-out weight calibration (30/33-34/36/42-43: -).

• M3 1/4-5/13-15/17/19-21/23/25/27/29/33/35: statement-level all-MiniLM-L6-v2 cosine prunes parent restatements and redundant sibling arguments (Alg.2/D.4 pp.24-25), providing content-level redundancy discounting. Parsed atomic claims carry parent IDs, support/attack polarity and expert provenance (D.3-D.7 pp.23-29). This is argument atomization, not post-hoc CoT sentence splitting. Trees are explicitly represented as bipolar reasoning graphs; no converging multi-parent DAG representation is built (26: -). Main-claim merging is exact-string, and novelty filtering is

not a semantic reasoning-cluster algorithm (30: -). MV3/CoT baselines support 1/24/28; main method discounts redundant argument nodes rather than relying on repeated votes.

• M3 2-3/6-12/16/18/22/31-32/36: -; the deterministic SCM models influence between argument scores, not a fitted agent-error dependence law. No pairwise coupling/error-overlap/second-order answer-correlation estimator, chance-expected overlap correction, per-agent discount factor, copying detector or honest-shared-versus-copied separation. The same-sized independent MV3 baseline uses separate single-model prompts, not a matched no-communication intervention on the same expert traces (7). Role expertise is different emphasis on shared task/key elements, not private evidence (11). Claim embeddings are step-level, not whole-reasoning embeddings (16). Structural interventions are not cross-structure similarity matching (36).

• M3 34/37-38/45-46/50/54-55: frozen LLMs semantically assess support/attack and logical soundness, with extra orchestration/review/judge calls; no fine-tuning. Node-level soundness assessment supports fallible intermediate-claim verification, not execution checking. Counterfactual graph re-evaluation verifies internal numerical influence, not factual argument truth (§4.4 p.4); no executable/symbolic, PRM or actual tool/retrieval verifier (47-49: -). Inference web search is explicitly disabled (I p.54). Thresholds/semantics/rounds/expert counts are tunable; actual expert-count screening ranges from two to seven (H.4 p.51).

• M3 39-43: -; no calibration split, trained classifier/auditor/graph encoder or EM fit. M3 44: X only for the reported per-benchmark best-semantics oracle, which is selected using gold (§5.4 p.7); default DF-QuAD and runtime scoring are label-free. M3 51: -; runtime/argument-count comparisons are reported without token totals. M3 52-53: X; H.1/Fig.8 p.49 explicitly motivates controlling token usage/context length and measures runtime versus pruning threshold; H.3/Table 14 p.51 compares wall-clock runtimes across rounds.

Criteria Map 4 Evidence Log

Jin et al., ARGORA: Orchestrated Argumentation for Causally Grounded LLM Reasoning and Decision Making, arXiv:2601.21533v1 Sources: https://arxiv.org/pdf/2601.21533v1

• §§4-6/G-I: MMLU-Pro/TruthfulQA/GPQA/HLE and cyber-triage case; single/SC/vote/standard debate/ReConcile/MoA/judge comparisons. Explicit positive/negative reversal/net-reversal metrics and semantics/pruning/base-score/round tests. Synthetic incident controls how evidence is supplied, supporting 18 in the shared-information sense, not a copying-ground-truth evaluation. Own modular-strength/causal-influence conditions support 27-28. Runtime/argument counts are not token totals. Code/full prompts shared; arXiv 2026. Argument trees and evidence strength are substantial structural overlap, without null-calibrated solver chains.

Evidence notes | Li-Sparse

Criteria Map 1 Batch 8 Evidence Log

Identity: Yunxuan Li, Yibing Du, Jiageng Zhang, Le Hou, Peter Grabowski, Yeqing Li, Eugene Ie. Improving Multi-Agent Debate with Sparse Communication Topology, Findings EMNLP 2024, pp.7281-7294; arXiv:2406.11776. Primary: https://aclanthology.org/2024.findings-emnlp.427.pdf .

• §§3-4,A: independent starts, five generation rounds, full reasoning from selected neighbors and final label vote (1-4,17-19,58-59,61,64 X). Graph density addresses all-to-all edge growth (15 X), not semantic message deduplication (14 -).

• §5.4,G: controlled wrong references mislead agents; sparsity delays premature consensus (10-11,22,25,43-45 X). Q(n,p) models correctness conditional on reference count/quality (26 X), not joint error covariance, independent-chain counts, step extraction or overlap-null calibration (7,24,62 -). Minority preservation is not separately targeted (16 -).

• §5,Tables4-5: some configurations match voting (23 X), while others improve it (21 X). Scope is configuration/task dependent.

• §6: mixed GPT/Mistral strong-weak teams and centrality ablations (34-37,40,47 X); no mixed sizes within one family (33 -). §3.2,4.5,C: random decoding/temperature explicitly studied (31,38,41 X). Behaviors in G are emergent, not assigned roles (32,39,46 -).

• §§3,6,D-E: dense/fixed sparse, rings/stars and tested probabilistic topology (50-54,56 X;55 -). Consensus timing is diagnostic (65 -); input-token savings measured (71 X).

Criteria Maps 2 3 Final Evidence Log

Identity: Li et al., Improving Multi-Agent Debate with Sparse Communication Topology, Findings EMNLP 2024; arXiv:2406.11776. Source: https://aclanthology.org/2024.findings-emnlp.427.pdf .

• M2 1-2/14/19/22/24/28/31/41: §§3.2/4.3 independently initialize, revise from selected prior-round neighbors, then vote over final answers; SC baseline also votes. Equal votes, answer-level fusion only; no confidence/trajectory/step-weighted final rule or majority override. M2 16: ?; final label/numeric extraction is necessary, but an explicit structured-parsing procedure was not verified.

• M3 1/3/7: main vote ignores dependence; §5.4's Q(n,p) models correctness as a function of reference number/quality, not a fitted joint-error law. §4.5 fixes initial responses across topology comparisons, with independent SC included (§4.3). This is communication control, not a chance-expected content-overlap null (22: -). Sparse edges reduce exposure but do not estimate redundancy discounts (4-6/8-11: -).

• M3 12-15/19-21/24/28-29/35: model/agent/answer/message analyses and equality-based convergence; Appendix G shows harmful peer influence/premature consensus. Communication graphs are not reasoning graphs (25-27: -), and selected full rationales are not extracted steps (17/23/30-36 except 35: -).

• M3 37-38/45/51-52/54-55: prompt-only topology/round/density controls; compares 4/6-agent settings and input inference tokens (§4.4/§5/D). No small labeled calibration set, training, gold-label fitting, verifier/PRM/tools or measured latency gain (39-44/46-50/53: -).

Criteria Map 4 Evidence Log

Identity: Li et al., Improving Multi-Agent Debate with Sparse Communication Topology, Findings EMNLP 2024; arXiv:2406.11776. Source: https://aclanthology.org/2024.findings-emnlp.427.pdf . Sources: https://aclanthology.org/2024.findings-emnlp.427.pdf

• §§3-6/A-G: MATH/GSM8K/MathVista/HH preference labeling, CoT/SC/dense versus sparse debate, accuracy and input-token costs. Fixes initial responses and injects controlled incorrect references (§5.4/G), providing known shared-misinformation tests. Formal Q(n,p) reference-quality conditions and edge-count complexity; topology/model/temperature controls. Findings EMNLP 2024; prompts/settings shared, own code release unclear. Motivation, sparse-debate baseline and reference-quality background.

Evidence notes | MoA

Criteria Map 1 Batch 8 Evidence Log

Identity: Junlin Wang, Jue Wang, Ben Athiwaratkun, Ce Zhang, James Zou. Mixture-of-Agents Enhances Large Language Model Capabilities, ICLR 2025; arXiv:2406.04692. Published primary: https://proceedings.iclr.cc/paper_files/paper/2025/file/5434be94e82c54327bb9dcaf7fca52b6-Paper-Conference.pdf .

• §§2.1-2.2,3.1: independent proposers, successive synthesis and final aggregator (1,3-4,17-19,46,58 X). Complete connections/broadcast between adjacent layers (51-52 X), two-layer final star and layer-wise relay (56-57 X); not a globally complete graph across layer-specific nodes.

• §3.3,Table5: paper implements adapted MAD comparisons (2 X); first aggregation helps, second can reduce MAD quality (21-22 X). One/two aggregation-round variants reported (63-64 X).

• §§3.1,3.3,A,I: repeated-model, same-model functional-role, mixed-size/family/open/closed/strong-weak configurations (31-38,40-41,47 X). Role functions are not persona-diversity interventions (39 -).

• §3.3,B,G: whole-output BLEU/TF-IDF/Levenshtein overlap and role-quality correlations (44 X); no agent independence model, extracted proof steps or independent-solver null (24-26,62 -). Model selection deliberately seeks diversity (45 X).

• F: tested BFGS architecture/model-population search (55 X); fixed after validation, no dynamic sparse routing (53-54 -).

• §3.4,Table5: token-priced inference budgets reported (71 X), including layered context rather than an isolated communication-only total. All generated peer responses retained; no confidence/subset message filter or adaptive stopping (60-61,65-70 -).

Criteria Maps 2 3 Final Evidence Log

Identity: Wang et al., Mixture-of-Agents Enhances Large Language Model Capabilities, ICLR 2025; arXiv:2406.04692. Source: https://proceedings.iclr.cc/paper_files/paper/2025/file/5434be94e82c54327bb9dcaf7fca52b6-Paper-Conference.pdf .

• M2 6/17-19/22-24/34/36/41: §2 uses whole-output synthesis by frozen aggregators, recursively passing previous synthesized outputs. It can prefer/combine minority input content; there is no explicit vote, normalized belief distribution, confidence output or scored proof paths. §2 model population selection uses win-rate and output diversity. These are configuration signals, not learned final-response weights (42-44: -). Adapted MAD/ReConcile comparisons remove JSON and confidence and use chat synthesis (§3.3), so original confidence-weighted voting cannot be inherited from their cited methods.

• M3 1/5/7/12-15/19-21/24/28-29/35: §3.3/B/G compare output diversity, proposers/aggregator lexical overlap (BLEU/TF-IDF/Levenshtein), and single-model versus peer-conditioned generation. These content influence/diversity analyses do not fit joint error dependence or calculate chance overlap (3-4/6/8-11/22: -). They compare whole responses, without extracted derivations, a reasoning graph, semantic clusters or copying provenance (16-17/23/25-27/30-34/36: -).

• M3 37-39/45/54-55: frozen prompting, tested depth/width/model populations; Appendix F tunes architecture by BFGS over 100 validation prompts, judged by benchmark evaluators. This is a small configuration-validation split, not classifier/encoder/auditor training or gold-label fitting (40-44: -). §3.4 compares token-priced budgets and uses TFLOPs as an explicit latency proxy (51-53: X; proxy rather than wall-clock timing). No step verifier, PRM, executable check or retrieval (46-50: -).

Criteria Map 4 Evidence Log

Identity: Wang et al., Mixture-of-Agents Enhances Large Language Model Capabilities, ICLR 2025; arXiv:2406.04692. Source: https://proceedings.iclr.cc/paper_files/paper/2025/file/5434be94e82c54327bb9dcaf7fca52b6-Paper-Conference.pdf . Sources: https://proceedings.iclr.cc/paper_files/paper/2025/file/5434be94e82c54327bb9dcaf7fca52b6-Paper-Conference.pdf

• §§2-3/B/F/I: AlpacaEval/Arena-Hard/MT-Bench/FLASK response evaluation; individual models and adapted MAD/ReConcile. Quality win rates, token-priced budget, proposer/aggregator BLEU/TF-IDF/Levenshtein overlap, model/depth/width/architecture studies and human-rated MT-Bench criteria. More than ten distinct model variants appear across primary experiments/appendices. No own formal probability/stability theorem or known-copying

validation. Code released; ICLR 2025. Synthesis/model-diversity baseline and motivation, not a derivation-independence method.

Evidence notes | CoT

Criteria Map 1 Batch 9 Evidence Log

Identity: Wei et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS 2022, arXiv:2201.11903. Sources: conference paper; supplement.

• Rows 1-4, 24-26, 31, 38, 49, 63: §3.1 uses one greedy generation per problem, without peer communication or answer aggregation. Five seeds permute demonstrations; they are separate robustness runs. Sampling/voting is attributed to subsequent SC work.

• Rows 9, 28, 47: §§3.2-3.4, 5; appendices A.1-A.2 and D test model scale, prompt variants, length generalization and categorize reasoning errors. This is capability analysis, without a mixed agent team (33-35: -).

• Row 30: §5 and supplemental checklist release the synthetic last-letter and coin-flip tasks.

• Rows 36-37: §3.1/Table 1 evaluates UL2-20B and GPT-3/Codex; appendix E.1-E.2 identifies GPT-3 API access. UL2's released checkpoints are independently confirmed by Google's model documentation.

• Rows 17-23, 50-62, 64-71: no debate pipeline, peer messages, filtering or communication-cost evaluation. Producing a rationale is not extracting and transmitting evidence steps; appendix B's external calculator processes equations, not agent evidence chains.

Counts: 10 X; 61 -; 0 ?.

Criteria Maps 2 3 Final Evidence Log

Identity: Wei et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS 2022; arXiv:2201.11903. Sources: https://papers.nips.cc/paper_files/paper/2022/file/9d5609613524ecf4f15af0f7b31abca4-Paper-Conference.pdf and https://proceedings.neurips.cc/paper_files/paper/2022/file/9d5609613524ecf4f15af0f7b31abca4-Supplemental-Conference.pdf .

• M2 19: §3.1 uses greedy single-model reasoning and an answer, without ensemble aggregation, final voting/judging, belief/confidence output or fusion weights/signals (all other M2 rows: -).

• M3 12/14/17/19/21/24: model-, answer- and intermediate-reasoning analyses across questions/datasets; flat generated reasoning (§§3-5, A/D). No multi-agent dependence criterion applies (1-11: -). Generating a stepwise rationale is not extracting units from existing traces or a logical graph (23/25-27: -).

• M3 37-38/45/54: few-shot prompting of frozen models and tested prompt configurations; the source describes one greedy generation, without training/gold labels at scale. The existing row-45 X is disputed below and retained under the blank-only rule. No many-agent/sample-count scaling or token/latency efficiency claim (51-53/55: -).

• M3 47/49/50: supplemental B applies Python eval to all generated arithmetic equations and propagates calculator corrections through later equations. This checks intermediate computations, not the logical validity of a proof or an LLM/PRM verifier (46/48: -). No cross-trace similarity/clustering/null calibration or copying provenance (22/28-36: -).

Criteria Map 4 Evidence Log

Identity: Wei et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS 2022; arXiv:2201.11903. Sources: https://papers.nips.cc/paper_files/paper/2022/file/9d5609613524ecf4f15af0f7b31abca4-Paper-Conference.pdf and https://proceedings.neurips.cc/paper_files/paper/2022/file/9d5609613524ecf4f15af0f7b31abca4-Supplemental-Conference.pdf . Sources: https://papers.nips.cc/paper_files/paper/2022/file/9d5609613524ecf4f15af0f7b31abca4-Paper-Conference.pdf, https://proceedings.neurips.cc/paper_files/paper/2022/file/9d5609613524ecf4f15af0f7b31abca4-Supplemental-Conference.pdf

• §§3-5 and supplement A-E: math, commonsense and symbolic tasks; standard prompting versus single CoT, accuracy, multiple exemplar seeds, prompt/model-scale ablations and hand inspection of erroneous GSM8K traces (19). Distinct LaMDA/GPT-3/PaLM/UL2/Codex variants exceed ten models; variants, not sampling repetitions, are counted. No debate/SC baseline implemented here: SC is later work. Main/supplement share prompts/data; code release and citation standing unverified. NeurIPS 2022. Foundational single-solver baseline; rationale elicitation alone is not independence estimation.

Evidence notes | SC

Criteria Map 1 Batch 9 Evidence Log

Identity: Wang et al., Self-Consistency Improves Chain of Thought Reasoning in Language Models, ICLR 2023, arXiv:2203.11171. Source: published PDF; standard OpenReview/arXiv downloads failed.

• Rows 1, 3, 19, 24, 31, 49, 59, 63: §§2, 3.2 independently sample paths, then vote over parsed answers. Row 19 means standalone aggregation; conditional sampling independence is not error-independence modeling (25-26: -).

• Rows 7, 44: §3.5/Fig. 5 measures agreement versus accuracy; §5 claims calibration benefits. No dependence correction/null calibration.

• Rows 9, 28, 38, 41: §§3.3-3.5 test temperatures, sample counts, corrupt demonstrations and OOD tasks.

• Rows 34-35, 40, 47: A.1.3/Table 10 combines LaMDA/PaLM/GPT-3; weaker models hurt voting. Row 33: separate size trials only.

• Rows 36-37: §3.1, open UL2/Codex API.

• Row 42: §3.4/A.1.4, ensemble members get different demonstrations/permutations.

• Row 60: Table 1 tests likelihood-weighted answers.

• Rows 45, 58, 62, 65-71: no uniqueness constraint, peer rationale transfer, evidence extraction, adaptive stopping, message filtering or communication-cost measurement.

Counts: 24 X; 47 -; 0 ?.

Criteria Maps 2 3 Final Evidence Log

Identity: Wang et al., Self-Consistency Improves Chain of Thought Reasoning in Language Models, ICLR 2023; arXiv:2203.11171. Source: https://openreview.net/references/pdf?id=xIxIkdndPl .

• M2 1/3/14/16/18-20/22-25/27-28/30-31/41: §2/Table 1 implements ordinary answer voting and likelihood-weighted sum/average variants. The likelihood variant may choose differently from the raw majority, using whole-path token-probability products (Eq.1), with task-specific numeric/string parsers (footnote 1 p.3). §3.5/Fig.5 uses consistency fraction as an answer-confidence indicator. No semantic classes, learned/held-out weights, step/DAG aggregation or trajectory score (2/4-13/15/17/21/26/32-40/42-44: -).

• M2 45: §2 footnote 2 p.3 shows correct/wrong solutions receive similar normalized likelihoods, and weighted averages can hurt (Table 1). This is an empirical confidence limitation, not just a citation.

• M3 1/12-14/18-21/24/28/35: independent sampled flat paths; exact answer support and token likelihoods. Answer agreement is treated as support, without a dependence model, content-overlap discount or chance null (2-11/16-17/22: -). Alternative paths are generated rather than extracted, clustered or verified (23/25-27/29-34/36/46-50: -).

• M3 37-38/45/54-55: frozen unsupervised decoding, sampling hyperparameters, additional sampled-generation calls and sample-count curves (§§2-3, A). No labeled fitting, trained classifier/auditor/graph encoder/EM. Ordinary evaluation labels do not constitute required fitting data (39-44: -). Sampling adds computation; no reported token/latency comparison or efficiency claim (51-53: -).

Criteria Map 4 Evidence Log

Identity: Wang et al., Self-Consistency Improves Chain of Thought Reasoning in Language Models, ICLR 2023; arXiv:2203.11171. Source: https://openreview.net/references/pdf?id=xIxIkdndPl . Sources: https://openreview.net/references/pdf?id=xIxIkdndPl

• §§2-3/A: math/commonsense benchmarks, greedy single CoT, unweighted/likelihood-weighted voting, beam and sample/rank comparisons; accuracy, consistency-versus-correctness uncertainty, repeated-sample SD and sampling/prompt/scale/ensemble tests. Calibration benefit is discussed, with empirical reliability/uncertainty characterization (14); no dependence-overlap or known-copying test. Four main solver families/limited variants do not establish ten models. Published ICLR 2023, prompts shared; code/citation/full-reading status unverified. Motivation and essential voting baseline.

Evidence notes | Du

Criteria Map 1 Batch 9 Evidence Log

Identity: Du et al., Improving Factuality and Reasoning in Language Models through Multiagent Debate, ICML 2024, PMLR 235:11733-11763; arXiv:2305.14325. Source: publisher record, published PDF, downloaded after browser retrieval failed. Conference version, not the older arXiv draft.

• Rows 1-4, 18-19, 24, 49, 51-52, 58-59, 63-65: §§2-3/A.3 independently initialize, broadcast rationales for fixed rounds, then majority-vote unresolved answers; independent-voting and answer-only ablations are implemented. Consensus does not trigger stopping.

• Rows 10-12, 22, 25, 44: §§2.2, 3.2-3.3/A.1 study agreeableness, confident errors and consensus rates. Fig. A12 loses a correct answer. Dependence is empirical conformity, without a statistical model/null correction (26: -).

• Rows 6, 50: §3.3/A.1 tests context compression by summarization; no selected-message/evidence-step extraction (61-62: -).

• Rows 9, 28, 30: A.1 robustness checks; §3.2 introduces biography data.

• Rows 32, 34-37, 39-40, 47: §3.3 uses personas and Bard/ChatGPT; the latter score 11/20 versus 14/20 before joint debate. A.1 separately tests GPT-4/Llama.

• Row 46: A.1/Table A6 tests an affirmative/negative/moderator baseline.

• Row 38 (?): sampling temperature is unreported.

• Row 71: A.4/Table A9 reports generated tokens across agents/rounds; excludes broadcast-input accounting.

Counts: 40 X; 30 -; 1 ?.

Criteria Maps 2 3 Batch 6 Evidence Log

Identity: Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum and Igor Mordatch, Improving Factuality and Reasoning in Language Models through Multiagent Debate, ICML 2024, PMLR 235:11733-11763; arXiv:2305.14325. Sources: publisher record, published PDF. The saved complete 31-page published-version text is readable and reused; fresh PDF extraction yielded blank later pages, so evidence comes from that retained source text, not memory.

• M2 1-2/14/16/19/22/24/28/31/41: independently initialize, exchange previous-round rationales, then take the terminal majority if disagreement remains (§2 pp.3-4). Independent voting/answer-only/SC comparisons are implemented. Boxed numeric and option/coordinate fields are parsed (A.3 pp.13-14, prompts p.15). Votes are uniform. M2 6/17 apply to the implemented affirmative/negative/moderator benchmark (A.1/Table A6 p.14), whose judge uses debate history; main Du reads prior-round peer outputs. No weighted vote, trained adjudicator, fitted dependence law, cluster posterior, delegation, trajectory score, step weights or majority override is established (3-5/7-13/15/18: -).

• M2 33/35: prompt variants explicitly modulate self-versus-peer trust (§2.2/Fig.3 pp.3-4), and biography generation revises/omits disputed facts (§3.2/Figs.8/12 pp.6/8). These are prompt/content signals, not numerical trust voting. M2 45: X; direct confidence elicitation yields high confidence on inconsistent answers (§3.2 p.6). Per-token perplexity is plotted as a diagnostic (§3.3 p.6, A.1 p.12), without using it as an aggregation-confidence input (29-30/32/34/36-40/42-44: -). No emitted numeric-confidence/belief output or scored reasoning-path/step/token fusion (20-21/23/25-27: -).

• M3 1/7/12-15/18-21/24/28-29/34-35: majority/discussion does not correct dependence. Independent responses versus post-debate answers provide same-task/panel communication comparisons; error convergence and correct-answer loss are examined (§§2-3, A.1). Token scope is diagnostic perplexity, not vote weights. Shared answers support consensus; harmful convergence is diagnosed. A.3/Fig.A5 pp.13-15 uses a frozen LLM to judge generated biographies against reference facts, supporting semantic comparison (34). No fitted dependence/content-overlap statistic, chance null, agent discount, private evidence, step extraction, reasoning graph/clusters or copying provenance (2-6/8-11/16-17/22-23/25-27/30-33/36: -).

• M3 37-38/45-47/50/54-55: prompt-only inference, extra critique/summary/verification calls, real intermediate arithmetic rechecks (Fig.A6 p.17), biography verification by LLM, and executable chess evaluation with Stockfish (§3.1 p.4, A.3 pp.13-14). Ten-agent debate and fifty-sample independent voting are tested (A.1/Table A4 p.13). No fitted calibration/classifier/auditor/encoder/EM, gold training at scale, PRM or actual retrieval checker (39-44/48-49: -).

• M3 51-52: X; Table A9 p.15 reports generated tokens summed across rounds versus baselines; summary compression is motivated to control context growth (§3.3/Fig.11 p.7). This is not uniformly cheaper generation, and broadcast-input

token totals are excluded. M3 53: -; faster consensus is discussed in rounds, without a wall-clock latency claim/comparison.

Criteria Map 4 Evidence Log

Identity: Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum and Igor Mordatch, Improving Factuality and Reasoning in Language Models through Multiagent Debate, ICML 2024, PMLR 235:11733-11763; arXiv:2305.14325.
Sources: https://proceedings.mlr.press/v235/du24e.html, https://raw.githubusercontent.com/mlresearch/v235/main/assets/du24e/du24e.pdf

• §§2-3/A: arithmetic/math/commonsense/chess/biography tasks; single CoT/SC/vote, standard debate and moderator comparisons. Accuracy, generated tokens, answer corruption/consensus/capability tests, and qualitative human checking of biographies; prompt and model/round/summarization ablations. No trained dependence estimator or known-copying ground truth. ICML 2024, prompts/data supplied; a project page is given but an explicit code release was not verified in this rapid pass. Motivation and principal vanilla-debate baseline.

Evidence notes | Liang

Criteria Map 1 Batch 9 Evidence Log

Identity: Liang et al., Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate, EMNLP 2024:17889-17904; arXiv:2305.19118. Source: published PDF.

• Rows 2-4, 18-20, 51-52, 57-58: §2, sequential debaters/full shared history/judge; no independent initialization phase (4: -).

• Rows 1, 31, 49, 59: §3.2, implemented independent baselines.

• Rows 6, 22, 63-66: §4.3/Fig. 5, adaptive/fixed-round comparison; extra rounds can hurt. Judge stops on a solution, not consensus/confidence threshold (appendix F).

• Rows 10-13, 25, 39, 44-45: §§1, 4.1/4.3/D, misconceptions, opposition directives, disagreement/Self-BLEU. No dependence model/null calibration (26: -).

• Rows 30, 43: A.2, new CIAR; explanations withheld.

• Rows 32, 34-37, 46-48: §§3.2/4.2, Vicuna/GPT mixtures, capability gaps, same-backbone judge bias; no error-correlation estimator.

• Row 33 (?): mixed GPT parameter sizes undisclosed; Vicuna sizes tested separately.

• Row 38: §3.2, temperature zero.

• Rows 61-62, 67-70: no evidence extraction, selective trigger or pruning.

• Row 71: E/Table 12, generated-token budget; input-message cost excluded.

Counts: 41 X; 29 -; 1 ?.

Criteria Maps 2 3 Batch 6 Evidence Log

Identity: Tian Liang et al., Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate, EMNLP 2024:17889-17904; arXiv:2305.19118. Source: published PDF, 16 pages; saved full source accessible.

• M2 1/6-7/14/16-19/22/24/28/31/39-41: sequential debaters and a discriminative/extractive LLM judge use the whole history, selecting a correct solution rather than a terminal vote (§2 pp.2-3). More-debater variants can select against raw answer frequency (18 is a rule-level implication; Table 7 p.7). SC is an implemented uniform-majority baseline; the Rerank baseline selects among four translations using the pretrained wmt21-comet-qe-da quality estimator (§3.2 p.4), supporting trained adjudication (7), not new training by this paper. Structured judge output and arithmetic-answer extraction are specified in prompts/examples (Appendix F pp.15-16). The quality estimator/argument judge assesses final-answer quality, not extracted-path weights. No final consensus vote, confidence/trust/accuracy weights, fitted dependence law, trajectory score, clustered posterior or explicit multi-metric fusion (2-5/8-13/15/20-21/23/25-27/29-30/32-38/42-44: -).

• M2 45: X for the actual confident-but-wrong self-reflection example (Fig.2 p.3); this is qualitative evidence, without numerical confidence calibration.

• M3 1/5/12-15/19-21/24/28-29/35: Self-BLEU measures lexical overlap between initial/current translations as rigidity/diversity (Table 4/§4.1 p.6). Repeated wording is diagnosed as unproductive redundancy; SC agreement supplies vote support. Mixed debater/judge backbones are tested (§4.2 pp.6-7). No fitted dependence/coupling/correlation measure, agent discount, matched no-communication intervention, private evidence, chance null, step extraction, embedding/structural similarity, reasoning graph/clustering or copying discriminator (2-4/6-11/16-18/22-23/25-27/30-34/36: -). Final adjudication is not a dedicated semantic-equivalence similarity judge (34).

• M3 37-38/41/45-46/50/54-55: prompt-only MAD with stopping/round/opposition settings; extra judge/self-reflection calls; peer/judge arithmetic reasoning checks in Fig.2 and F.2 p.16. The pretrained QE scorer accounts for 41; no new classifier/auditor fit, labeled calibration, graph encoder, EM or gold training at scale (39-40/42-44: -). No executable, PRM or retrieval verifier (47-49: -). Two/three/four debaters are actually compared, with decreasing quality (Table 7 p.7); 55 denotes tested scaling, not successful improvement.

• M3 51-52: X; Appendix E/Table 12 p.14 reports 2.46× CoT generated tokens and describes the overhead as acceptable. This is a bounded-cost claim, not token savings. No wall-clock latency claim/comparison (53: -).

Criteria Map 4 Evidence Log

Identity: Tian Liang et al., Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate, EMNLP 2024:17889-17904; arXiv:2305.19118. Source: published PDF, 16 pages; saved full source accessible. Sources: https://aclanthology.org/2024.emnlp-main.992.pdf

• §§3-4/D-F: GSM8K/CIAR/counterintuitive translations and role/judge comparisons, single CoT/SC/vote/debate. Accuracy, token budget, Self-BLEU divergence and correct/wrong changes, capability/round/role/stop ablations and variance plots. CIAR hides explanations to test misconceptions, not solver-copying ground truth. Code/data/prompts released, EMNLP 2024. Supports dissent/persuasion motivation and debate/judge baselines, without independent evidence chains.

Evidence notes | ChatEval

Criteria Map 1 Batch 10 Evidence Log

Identity: Chan et al., ChatEval: Towards Better LLM-based Evaluators through Multi-Agent Debate, ICLR 2024; arXiv:2308.07201. Source: published PDF.

• Rows 1-4, 18-19, 49, 51-52, 57-59, 63-65: §§2-4/B-C implement independent ensembles, sequential discussion and simultaneous broadcast; fixed rounds end in label voting/score averaging. No consensus stopping.

• Rows 6-7, 50, 56, 71: §3.2 uses position calibration; C adds a central summarizer; E/Table 6 reports debate API expense/time. Position calibration is not probability/null calibration.

• Rows 21-22: §§3.3-3.4 show overall improvements; §4.3 reports stagnation/degradation with extra turns. Main findings favor debate over ensembles (23: -).

• Rows 31-32, 36-37, 39, 41, 45-47: §4.1/A assign distinct personas including a critic; C adds an asymmetric summarizer; D studies smaller open models. B explicitly uses homogeneous teams; model sizes/families are separate experiments (33-35, 40: -).

• Rows 24-26, 44, 48: no error-independence model or inter-agent diversity/correlation measurement. §3.3 kappa compares evaluations with human annotations.

• Rows 61-62, 66-70: summaries rewrite all responses; no extracted evidence chains, selective trigger or message pruning.

Counts: 33 X; 38 -; 0 ?.

Criteria Maps 2 3 Final Evidence Log

Identity: Chan et al., ChatEval: Towards Better LLM-based Evaluators through Multi-Agent Debate, ICLR 2024; arXiv:2308.07201. Source: https://proceedings.iclr.cc/paper_files/paper/2024/file/25cc3adf8c85f7c70989cb8a97a691a7-Paper-Conference.pdf .

• M2 1-2/6/14/16-17/19/22-24/28/37/39/41: §§2-3/B-C frozen referee agents judge text quality using role/rubric evidence, maintain full or summarized chat histories, then majority-vote parsed comparison labels or uniformly average evaluation scores. Single-agent and independent-ensemble evaluator baselines are implemented. Quality ratings are not answer-confidence probabilities (20-21: -), and averaging is not reliability weighting. No trajectory score, proof-step weights or fitted dependence rule. The final panel vote does not override its own majority (18: -).

• M3 1/7/12-15/19-21/24/28/34-35: compares independently role-prompted evaluations with debate; agents discuss whole candidate texts and comparison labels. No reasoning-overlap/dependence estimator, extracted derivations, copying detector or chance-calibrated solver null. §3.3 Cohen's kappa corrects agreement with human evaluation labels, not overlap between debating solvers, so 8/22 remain -.

• M3 37-38/45-46/54-55: prompt-only evaluation and additional debate/summary calls, multiple agent-count/turn settings; LLM referee checks candidate-response quality/groundedness. No trained classifier/auditor/encoder/EM, gold labels required at scale or intermediate-step verification (39-44/47-50: -). Appendix E/Table 6 compares time/API expense against human annotators; latency efficiency applies to that comparator (53: X). It does not report a token baseline or token-saving claim (51-52: -).

Criteria Map 4 Evidence Log

Identity: Chan et al., ChatEval: Towards Better LLM-based Evaluators through Multi-Agent Debate, ICLR 2024; arXiv:2308.07201. Source: https://proceedings.iclr.cc/paper_files/paper/2024/file/25cc3adf8c85f7c70989cb8a97a691a7-Paper-Conference.pdf . Sources: https://proceedings.iclr.cc/paper_files/paper/2024/file/25cc3adf8c85f7c70989cb8a97a691a7-Paper-Conference.pdf

• §§3-4/B-E: FairEval/Topical-Chat human-rated response evaluations, single LLM evaluators/role ensembles versus debate, accuracy, rank correlations and chance-adjusted kappa against human labels. This overlap metric is judge-human annotation agreement, not debater error dependence. Human baselines/qualitative analysis, role/communication/agent-count/turn ablations; no known-copying experiment. Costs are dollars/time, not token totals. ICLR 2024; prompts shared; code release unclear. Evaluation/debate baseline and role-diversity motivation.

Evidence notes | Smit

Criteria Map 1 Batch 10 Evidence Log

Identity: Smit, Grinsztajn, Duckworth, Barrett and Pretorius, Should we be going MAD? A Look at Multi-Agent Debate Strategies for LLMs, ICML 2024; PMLR:smit24a. Source: published PDF.

• Rows 1-4, 18-20, 49-52, 56-59, 63-66: §2/A.2/A.5-A.6 implement SC, ensemble refinement, SoM, ChatEval and Multi-Persona. Judge preference controls stopping; neither consensus nor confidence thresholds do.

• Rows 5-6, 9, 21-23, 27-28, 30, 71: §§3-4 benchmark accuracy, cost, time and held-out hyperparameter transfer; release a protocol/evaluation toolkit. Original MAD often trails voting, whereas tuned agreement improves results.

• Rows 10-11, 16, 24-26, 41, 44: §3/Figs. 4-7/A.3-A.4 measure initial diversity, consensus, answer changes, correct-answer loss and peer influence. SC samples independently; no statistical error-dependence model/null calibration.

• Rows 31-32, 36-39, 45-47: §3/A.2 tests personas, opposition, sampling temperatures and transfer across GPT/Mixtral backbones; models are separate runs, not mixed teams (33-35, 40: -).

• Row 61: A.4 records messages removed at context limits; not evidence-chain selection.

• Rows 43, 62, 67-70: no current-question seeded hints, extracted step graphs, selective trigger or relevance/confidence/diversity pruning.

Counts: 44 X; 27 -; 0 ?.

Criteria Maps 2 3 Batch 6 Evidence Log

Identity: Andries Smit, Nathan Grinsztajn, Paul Duckworth, Thomas D. Barrett and Arnu Pretorius, Should we be going MAD? A Look at Multi-Agent Debate Strategies for LLMs, ICML 2024, PMLR:smit24a. Source: published PDF, 23 pages; saved full source accessible.

• M2 1-2/6/14/16-19/22/24/28/31/35/39-41: actually implements SC/Medprompt voting, SoM/ChatEval terminal pooling, Multi-Persona judge and ensemble-refinement synthesis (§2 p.2, A.5-A.6 pp.20-23). Complete-history judge/one-on-one protocols support 17/40. The ensemble-refinement aggregator can choose a minority candidate's correct arguments rather than the most frequent starting answer (18 scope: candidate-pool synthesis, not an override of a later uniform vote). Judge JSON and final capital option fields support exact structured parsing. Agreement-intensity prompts are actual inference controls (§3 pp.6-7), not numerical trust weights. ER's correctness/argument-quality instructions support the heuristic rubric signal (39).

• M2 3-5/7-13/15/20-21/23/25-27/29-30/32-34/36-38/42-45: -; no confidence/trust/accuracy weighted vote, trained adjudicator, fitted dependence law, calibrated posterior, delegation/trajectory/step scoring, explicit multi-metric fusion or learned/held-out/online vote coefficients. Initial diversity, parse-quality and transition counts are diagnostics, not weighting inputs (A.3-A.4 p.19). Full reasoning is context, without explicit scored path/step fusion. A limitations warning about undue confidence (p.9) is not an empirical confidence-reliability experiment (45).

• M3 1/7/12-15/18-21/24/28-29/35: labels, agents/models, whole messages, tokens/cost and initial-versus-final transitions are recorded (Figs.4-7 pp.6-8, A.3-A.4 p.19). Before/after comparisons on the same task/agents support 7 as a communication performance control, not chance-overlap calibration. Agreement can improve performance; wrong-answer reinforcement/correct-answer loss is also diagnosed. No fitted dependence/content-overlap model, coupling/chance-error/correlation metric, agent redundancy discount, private information, step extraction, reasoning structures/clustering, embedding/structural similarity or copying provenance (2-6/8-11/16-17/22-23/25-27/30-34/36: -). Relied-on-other/bullied metrics track answer transitions, not switching without an independent derivation.

• M3 37-38/44-46/54-55: frozen prompt-only systems, extra synthesis/summary/judge calls, and agent/sample/round/temperature/aggregation-count sweeps (§§2-3, Table 3 pp.16-18). Gold is used for held-out-dataset hyperparameter selection (§3 p.5), supporting 44 for the tuning experiment; untuned protocols need no gold. No labeled confidence-calibration split, trained classifier/auditor/encoder/EM or actual executable/PRM/retrieval checker (39-43/47-49: -). Medprompt kNN retrieval is expressly omitted (§2 p.2). M3 50: ?; SPP prompt examples contain intermediate arithmetic rechecks (A.6 pp.21-23), but the paper does not provide generated verification traces sufficient to establish actual intermediate-step checking in the evaluated runs.

• M3 51-53: X; Appendix A.1 pp.12-15 compares token consumption and seconds per question against baselines; A.4 p.19 distinguishes prompt/response totals. §3 p.4 discusses accuracy-versus-resource trade-offs and lower-cost alternatives. These are measured efficiency comparisons, not a claim that every debate system saves tokens/time.

Criteria Map 4 Evidence Log

Identity: Andries Smit, Nathan Grinsztajn, Paul Duckworth, Thomas D. Barrett and Arnu Pretorius, Should we be going MAD? A Look at Multi-Agent Debate Strategies for LLMs, ICML 2024, PMLR:smit24a. Source: published PDF, 23 pages; saved full source accessible. Sources: https://raw.githubusercontent.com/mlresearch/v235/main/assets/smit24a/smit24a.pdf

• §§2-4/A: math/commonsense tasks, SC/ensemble refinement/SoM/ChatEval/persona comparisons. Accuracy/tokens/API cost/time, initial answer diversity, correct-answer losses/changes, confidence intervals and hyperparameter/role/temperature/capability/held-out-tuning controls. No controlled known-copying mechanism. Code and toolkit/prompts released, ICML 2024. Broad baseline selection and negative-result motivation; no formal dependence/chain estimator.

Evidence notes | Huang

Criteria Map 1 Batch 10 Evidence Log

The map supplies only the surname “Huang”; the separate Paper Key could not be recovered. Prior-context retrieval did not identify its intended title, author initials or paper ID. Two relevant but different primary records exist:

• Jie Huang et al., Large Language Models Cannot Self-Correct Reasoning Yet, ICLR 2024; arXiv:2310.01798.

• Jiaxin Huang et al., Large Language Models Can Self-Improve, EMNLP 2023.

Neither can be assigned to this column from the available keyless map. All rows 1-71 are ?, rather than attributing one candidate's method to the user's unspecified paper. This is an identity limitation, not a claim that either paper lacks readable sources. No title-based judgments were made. Counts: 0 X; 0 -; 71 ?.

Criteria Maps 2 3 Final Evidence Log

M2 rows 1-45 and M3 rows 1-55: **?**. The intended title/authors/identifier cannot be verified from this alias; the Paper Key remains unavailable. Retained Map 1 logs already document the identity gap. Candidate papers, surname matches and remembered methods were not substituted. These are identity uncertainties, not negative findings about an identified paper. No source can be attributed securely to this column until its Paper Key entry is supplied.

Criteria Map 4 Evidence Log

Identity unresolved; intended Paper Key entry remains unavailable. Sources:

• Rows 1-44: ?. Several Huang-authored reasoning papers fit the surname; intended citation remains unconfirmed.

Evidence notes | Xiong

Criteria Map 1 Batch 10 Evidence Log

Identity: Xiong, Ding, Cao, Liu and Qin, Examining Inter-Consistency of Large Language Models Collaboration: An In-depth Analysis via Debate, Findings EMNLP 2023:7572-7590; arXiv:2305.11595 (earlier title differs). Source: published PDF.

• Rows 1-4, 17-20, 24, 41, 45, 49, 63-67: §3 independently obtains stances/arguments, debates only inconsistent questions and stops on consensus/max rounds. Fixed caps are not fixed execution (64: -); gating enforces initial disagreement.

• Rows 10-12, 16, 22, 25, 35, 47: §§5-6 measure domination; weaker stubborn peers mislead stronger models, especially in roundtables.

• Rows 9, 28: §7.2 reverses speaking order.

• Rows 34, 36-37, 40, 42, 48: §§2.2/4/A compare same-base variants and cross-family teams; chat models get zero-shot prompts, text models get demonstrations.

• Row 33 (?): mixed GPT sizes undisclosed.

• Rows 26, 44: INCON/dominance measure disagreement/compromise, not error-independence or null-calibrated overlap.

• Rows 46, 50-52, 57-59: §3/B/D-F share complete arguments in sequential/roundtable histories, withhold explicit stance labels, then use stance voting or GPT-4 adjudication.

• Rows 38, 43, 61-62, 68-71: A specifies temperature zero; no current-answer hints, step extraction, message pruning or measured communication costs.

Counts: 42 X; 28 -; 1 ?.

Criteria Maps 2 3 Final Evidence Log

Identity: Xiong et al., Examining Inter-Consistency of Large Language Models Collaboration: An In-depth Analysis via Debate, Findings EMNLP 2023; arXiv:2305.11595. Source: https://aclanthology.org/2023.findings-emnlp.508.pdf .

• M2 1-2/6/14/16-19/22-24/28/31/35/41: §3/Fig.2 obtains independent parsed stances, gates on disagreement, alternates arguments until agreement/cap, then summarizes full debate with GPT-4 adjudication; roundtables also use stance voting. A judge can resolve the tie/minority rather than copy the raw vote. INCON equality/disagreement drives selection; confidence/trust/accuracy weighting is absent. Whole history is used, without a numerical trajectory score (12: -).

• M3 1/7/12-15/19-21/24/28-29/35: §§2-7 compare no-discussion predictions and post-debate results, track label INCON and domination, and identify strong/weak-model influence. INCON is disagreement, not chance-corrected error overlap, second-order correlation, a dependence model or an independent-chain estimator (3-11 except 7/22: -).

• M3 17/23/25-27/30-34/36: -; full natural-language arguments and summarization do not extract logical units, embed reasoning or detect switches without an independent derivation. Qualitative compromise is not provenance/copying detection (31-32: -).

• M3 37-38/45/54-55: frozen prompt-only protocol with caps, judge calls, and two/three-model roundtables. No training/calibration/gold-label fitting or actual dedicated step/tool/PRM verification. No communication-token or latency benchmark (39-44/46-53: -).

Criteria Map 4 Evidence Log

Identity: Xiong et al., Examining Inter-Consistency of Large Language Models Collaboration: An In-depth Analysis via Debate, Findings EMNLP 2023; arXiv:2305.11595. Source: https://aclanthology.org/2023.findings-emnlp.508.pdf . Sources: https://aclanthology.org/2023.findings-emnlp.508.pdf

• §§2-7/A-F: seven commonsense datasets; independent soft/hard collaboration, FORD debate and GPT-4 judge. Accuracy, INCON answer disagreement and dominance/change analyses, speaker-order/team-capability controls; naturally different abilities are not known-dependence ground truth. Code/data/prompts released; Findings EMNLP 2023. Motivation and mixed-capability debate baseline, without formal derivation independence.

Evidence notes | Kuhn

Criteria Map 1 Batch 11 Evidence Log

Identity: Kuhn, Gal and Farquhar, Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation, ICLR 2023; arXiv:2302.09664v3. Source: published-version PDF.

• Rows 1, 3, 19, 24, 31, 49, 59-60, 63: §§4/6/B.5 independently sample answers; aggregate semantic-class likelihoods for uncertainty; implement a p(True) judge baseline receiving sampled answers. Row 19 is standalone post-generation aggregation, not debate. Confidence denotes model likelihoods.

• Rows 5-7, 9: §§6/6.2/B.3 predict correctness and study sample efficiency/metric robustness. Goal 5 concerns correctness discrimination, not improved generator accuracy. Calibration (7: -): §6 explicitly favors AUROC over calibration measures.

• Rows 27-28, 36, 38, 41, 44, 47: §§4/6/B.4 use open OPT, temperature-diverse samples, lexical/semantic diversity measures and model-scale ablations.

• Rows 25-26, 48: equivalence clusters are not an error-dependence model, independent-chain count or null-calibrated overlap.

• Rows 29-30, 33-35, 46: mathematical entropy construction, without a proved theory contribution/new benchmark; OPT sizes are separate runs. DeBERTa is an auxiliary classifier, not a heterogeneous solver team.

• Rows 58, 61-62, 64-71: answer-level clustering, no reasoning-step extraction, debate topology/stopping/message filtering or communication-cost report.

Counts: 19 X; 52 -; 0 ?.

Criteria Maps 2 3 Batch 5 Evidence Log

Identity: Lorenz Kuhn, Yarin Gal and Sebastian Farquhar, Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation, ICLR 2023; arXiv:2302.09664v3, April 15, 2023. Sources: primary record, published-version PDF, 19 pages. Method §4 pp.4-6, evaluation §6 pp.7-9, implementation/ablations B pp.16-19.

• M2 14-16/19-20/22-24/30-31/35-36/39/41: semantic entropy clusters independently sampled answer texts by bidirectional entailment and sums their token-derived sequence likelihoods (§4 pp.4-6). Predictive/normalized entropy, lexical similarity and distinct-answer counts are implemented alternatives (§6/Table 2 pp.7-8). Semantic answer variability is an uncertainty/answer-quality proxy, not evidence-chain verification (39 scope: paraphrase/meaning variation). The scored answer is separately generated with five-beam sampling; entropy uses ten-twenty other sampled answers (B p.16). No vote selects the answer. Exact-match grading is tested; QA-pattern stop markers trim generated continuations (B pp.16-18), supporting 14/16 with that extraction/evaluation scope. Outputs are generated answers and associated uncertainty scores; full semantic likelihood distributions are mathematical intermediates, not a reported posterior output (21: -).

• M2 1-13/17-18/25-29/32-34/37-38/40/42-44: -; no debate/final voting, judge-based answer selection, majority override, fitted dependence law, calibrated answer-cluster posterior, trajectory/step/token fusion, verbal numerical confidence, trust/accuracy prior, task/evidence/format weighting or fitted/held-out/online aggregation coefficients. Token likelihood is a confidence signal (30), not token-level fusion across agents (27). Semantic cluster likelihoods are not calibrated correctness probabilities; §6 p.7 explicitly favors AUROC over calibration metrics. Model parameters and the NLI classifier are pretrained; no new aggregation-weight fit is performed.

• M2 45: X; own comparisons show p(True) and unstructured sequence/margin likelihoods can discriminate correctness poorly, while semantic entropy improves AUROC (§6 pp.7-8, B.6 p.19). B.5 p.18 explicitly defines p(True) as the likelihood of the next token being True, not a verbalized confidence number.

• M3 1/12/14-15/18-19/21/28-29/35: semantic agreement changes uncertainty in the answer distribution, without a model of inter-solver error dependence. Samples sharing a meaning contribute combined probability mass; surface paraphrase variation is treated as irrelevant outcome multiplicity (28/29 scopes), not discounted copied evidence. Signals are model/answer/token probabilities and raw answer text supplied to NLI/self-verification; there are no agents with individual histories (13/20: -). Lexical ROUGE-L similarity is an actual baseline (§6 p.8), not merely related work.

• M3 2-11/16-17/22-27/30-34/36: -; no content-level dependence measure, pairwise coupling, no-communication intervention, chance-adjusted error overlap, second-order answer-correlation estimator, agent discount/private evidence, key-step extraction or reasoning structure/clustering. Clusters are answer meanings, not reasoning chains. The bidirectional DeBERTa NLI classifier uses contextual semantic entailment, not an explicit embedding-distance

similarity score or generative LLM similarity judge (33/34). Related-work embedding/paraphrase methods on p.6 are not implemented. Semantic entropy and length normalization do not correct overlap expected under an independent-solver null (22).

• M3 37-38/40/45-46/54-55: frozen OPT models and sampling/temperature/ROUGE thresholds; ten-twenty samples are actually studied (§§4/6 pp.5/7-9, B p.16). M3 40 is X specifically for the required pretrained MNLI-fine-tuned DeBERTa NLI classifier (§4 p.5); the paper uses it off the shelf and does not train a new uncertainty/correctness classifier or fine-tune OPT. This component dependency coexists with training-free method operation (38/54). Extra generation and p(True) self-verification calls support 45/46; verification concerns final-answer correctness, not intermediate steps. No calibration split, trained auditor/graph encoder/EM, target-task gold training at scale, executable/PRM/tool or step verifier (39/41-44/47-50: -). The 300 manually labeled equivalence-validation samples per dataset validate clustering, without training it (B.2 p.17).

• M3 51-52: -; sample-efficiency curves and computational-overhead discussion do not report token totals versus a baseline or a generated-token saving claim. M3 53: X for the explicit latency/compute bound in §4 p.5: each smaller-NLI-model comparison is claimed much faster than generating one main-model token, with transitivity reducing comparisons. This is an analytical relative-cost claim, not a measured end-to-end latency benchmark.

Criteria Map 4 Evidence Log

Identity: Lorenz Kuhn, Yarin Gal and Sebastian Farquhar, Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation, ICLR 2023; arXiv:2302.09664v3, April 15, 2023. Sources: https://arxiv.org/abs/2302.09664v3, https://arxiv.org/pdf/2302.09664v3

• §§4/6/B: TriviaQA/CoQA uncertainty evaluation, predictive/semantic entropy, lexical/likelihood baselines and p(True) judge given samples. AUROC and accuracy/temperature/sample/NLI-choice tests; explicitly prefers discrimination to calibration metrics. No controlled-copying/null-overlap experiment or newly proved Bayesian/latent-factor theorem. Code/data released, ICLR 2023. Semantic-answer uncertainty/aggregation baseline and motivation; not a reasoning-chain dependence estimator.

Evidence notes | Lightman

Criteria Map 1 Batch 11 Evidence Log

Identity: Lightman et al., Let's Verify Step by Step, ICLR 2024; arXiv:2305.20050. Source: published PDF.

• Rows 1, 3, 19, 24, 31, 49, 58-60, 63: §§2-3 use sampled solutions, trained verifiers and best-of-N selection; compare label voting/RM-weighted voting. No debate. Conditional sample independence is not error-independence.

• Rows 5, 7, 9, 16: §5 tests OOD exams; §6.3/I recover rare correct solutions; H adjusts a negative-label threshold for teacher miscalibration. Annotation efficiency is not token/communication efficiency (6: -).

• Rows 27-28, 30, 38: trained PRM/ORM comparison, PRM800K release and E's temperature-1 sampling.

• Rows 35, 46-48: §§2/4 use generator/verifier roles and a stronger teacher for smaller-model supervision; §6.3 considers shared pretraining contamination, not agent error covariance.

• Rows 33/37 (?): small-model parameter sizes/family details and API access for proprietary research models are unspecified.

• Rows 41, 43, 61-62: §§2.3-2.4/B/D generate newline-delimited steps; phase-1 labelers may supply replacement steps; active learning selects annotation candidates. These are training/data-collection properties.

• Rows 25-26, 64-71: no dependence estimator/null calibration, debate stopping, peer-message filtering or communication-cost measurement.

Counts: 26 X; 43 -; 2 ?.

Criteria Maps 2 3 Batch 6 Evidence Log

Identity: Hunter Lightman et al., Let's Verify Step by Step, ICLR 2024; arXiv:2305.20050. Source: published PDF, 24 physical pages; saved full source accessible.

• M2 1/3/7/12-14/16/18-20/22-26/28/30-31/39/41-42: best-of-N ranks whole solutions by a trained ORM or by the product of PRM step-correctness probabilities; minimum/product variants, plain majority and RM-weighted voting are actually tested (§§2.1/2.5/2.6/3 pp.2/4-5, F.2 p.15). Newline-separated steps and final answers are parsed. A high-scored minority answer can win best-of-N. Signal 30 is the probability of classification-label tokens, not verbal self-confidence. Step-quality labels supply a process rubric. Fixed reductions/uniform vote coexist with learned verifier scoring. No debate-history, dependence-law, semantic answer clustering, calibrated cluster posterior or online inference-weight update is used (2/4-6/8-11/15/17/21/27/29/32-38/40/43-44: -).

• M2 45: X; the large teacher is observed to favor positive labels, prompting an arbitrary 20% negative-label cutoff for synthetic supervision (H p.17). This is a stated miscalibration observation, not a calibrated probability feature or held-out calibration fit.

• M3 1/12/14-15/17-19/21/23-24/28/35/37/40-41/44-45/48/50/55: trained autoregressive PRM/ORM classifiers assess solutions; process labels are attached to individual newline-delimited steps and reward models are fine-tuned (§2 pp.2-4, D-F pp.14-15). PRM800K supplies large-scale gold process supervision, and the large PRM actually generates synthetic labels for smaller models (§§3-4 pp.5-7, H p.17). Up to 1,860 solutions are searched (Fig.3 p.5). No reasoning tree/DAG, semantic overlap/copying analysis or chance-null dependence correction is implemented (2-11/13/16/20/22/25-27/29-34/36/39/42-43: -). Majority repetition is support; the PRM product does not fit step-error dependence.

• M3 38/54: X only for the implemented uniform-majority aggregation baseline over fixed generated solutions; main PRM/ORM training is not inference-only. M3 46/49/51-53: -; the verifier is a trained reward model, without a separate prompted LLM judge, retrieval/tool checker or token/latency efficiency comparison. Data-label efficiency is not token efficiency. M3 47: ?; automatic final-answer grading is specified (§2.1/2.5 pp.2/4), but its checker implementation is not described sufficiently to confirm executable/symbolic verification beyond answer matching.

Criteria Map 4 Evidence Log

Identity: Hunter Lightman et al., Let's Verify Step by Step, ICLR 2024; arXiv:2305.20050. Source: published PDF, 24 physical pages; saved full source accessible. Sources: https://proceedings.iclr.cc/paper_files/paper/2024/file/aca97732e30bcf1303bc22ac3924fd16-Paper-Conference.pdf

• §§2-6/B-I: MATH/OOD math exams, majority/RM-weighted voting and outcome versus process reward models. Best-of-N accuracy, active-learning/data/model/sample/step-score ablations and trained human-labeled PRM800K. Human labels are central (19); class labels/steps are not known solver-dependence ground truth. ICLR 2024, PRM800K data/scripts shared; PRM800K scripts count as a code release; the complete training implementation is not required by row 31. Process verification/voting baseline and evidence-over-votes motivation.

Evidence notes | Cemri

Criteria Map 1 Batch 11 Evidence Log

Identity: Cemri et al., Why Do Multi-Agent LLM Systems Fail?, NeurIPS 2025 Datasets and Benchmarks; arXiv:2503.13657. Uses the published 1,642-trace version. Source: published PDF.

• Rows 2-5, 17-21, 39, 41, 45-46, 58-60: H/L/M implement role-separated solvers, discussion, verifier decisions and workflow interventions. L instructs initial solutions independently; this does not establish statistical independence (24: -). N.5/N.11 show numeric replies and confidence-rated critiques.

• Rows 8-9, 12, 14, 16, 27-28, 30: §§3-5/A/N diagnose failures; N.11 ignores a correct peer answer while giving a wrong critique maximum confidence.

• Rows 31-32, 36-37, 47: Table 1/H/I use homogeneous role teams with open/closed backbones; compare capabilities across runs. Rows 33-35/40: -; separate backbones are not mixed solver teams.

• Rows 42, 50, 53, 56-57: B describes private service information, paired conversations, supervisor stars and assembly-line workflows.

• Rows 38/51-52 (?): sampling temperature and full-message visibility/connectivity are insufficiently specified.

• Rows 7, 25-26, 44, 48, 61-62, 64-71: -. Few-shot annotator refinement is not probability calibration; E correlates failure labels, not agents' errors. No reasoning-chain extraction/null calibration. G's confidence protocols are proposals. K reports annotation API costs, not debate communication costs. H stopping uses verifier approval/caps, not consensus or confidence thresholds. Row 63: §3.3 uses one-pass trace annotation.

Counts: 35 X; 33 -; 3 ?.

Criteria Maps 2 3 Final Evidence Log

Identity: Cemri et al., Why Do Multi-Agent LLM Systems Fail?, NeurIPS 2025 Datasets and Benchmarks; arXiv:2503.13657. Source: https://proceedings.neurips.cc/paper_files/paper/2025/file/b1041e52d3be19f0a9bc491657488e4a-Paper-Datasets_and_Benchmarks_Track.pdf .

• M2 6/14/16-17/19-20/22-25/29/37-39/41/45: §3.3/A/N uses a prompt-only judge to annotate whole execution traces against failure-mode criteria; H/L's implemented solver/coder/verifier uses history, parsed boxed answers and structured code. N.11 records a maximally confident incorrect critique. Rubric/task/format failures are diagnostic inputs; these are not fitted vote weights. No majority, dependence-based belief aggregation, trajectory score or extracted-step weighting (1-5/7-13/18/21/26-28/30-36/40/42-44: -).

• M3 1/11-15/17/19-21/24/28: studied workflows include private service information (B), whole messages and per-action/step failures. Error-dependence is not statistically fitted or content-overlap-calibrated (2-10/22: -). Failure-label co-occurrence (E) is not agent error covariance or copying. Collected system traces are not an extracted reasoning tree/DAG, and workflow DAGs are communication/workflow structures (23/25-27/29-36: -).

• M3 37-39/45-47/49-50/54: §3.3 calibrates few-shot annotation against a small human-labeled set without training model parameters. H/L actually runs Python/SymPy solvers and LLM verifier reviewing intermediate solutions; L.6 contains successful execution and review. Tools/search appear in evaluated system traces (B/N). No trained classifier/auditor/encoder/EM, process reward model or required gold labels at scale (40-44/48: -). H's step-verification intervention is actually tested, not merely G's suggested confidence protocols.

• K reports annotation API dollars, not debate-token totals; no token/latency efficiency comparison or agent-count scaling experiment (51-53/55: -).

Criteria Map 4 Evidence Log

Identity: Cemri et al., Why Do Multi-Agent LLM Systems Fail?, NeurIPS 2025 Datasets and Benchmarks; arXiv:2503.13657. Source: https://proceedings.neurips.cc/paper_files/paper/2025/file/b1041e52d3be19f0a9bc491657488e4a-Paper-Datasets_and_Benchmarks_Track.pdf . Sources: https://proceedings.neurips.cc/paper_files/paper/2025/file/b1041e52d3be19f0a9bc491657488e4a-Paper-Datasets_and_Benchmarks_Track.pdf

• §§3-5/H-N: 1,642 traces across QA/math/software/general-agent tasks; dedicated expert annotation, LLM annotator and workflow intervention comparisons. Accuracy/failure classification and kappa against experts, Wilcoxon/repetitions and role/prompt/topology tests. No named single-CoT/SC/MoA aggregation baseline or known-copying ground truth is established. Annotation dollars are not debate-token cost; failure-label co-occurrence is not agent error overlap.

Dataset/taxonomy/annotator code/prompts released, NeurIPS Datasets and Benchmarks 2025. Strong diagnostic motivation.

Evidence notes | Ladha

Criteria Map 1 Batch 11 Evidence Log

The separate Paper Key remains unavailable. The surname alone does not identify the intended paper, and recovered prior context supplied no exact citation. Two distinct candidates are:

• Krishna K. Ladha (1992), The Condorcet Jury Theorem, Free Speech, and Correlated Votes, American Journal of Political Science 36(3), 617-634; DOI 10.2307/2111584. Publisher record exposed a title but no readable paper text.

• Krishna K. Ladha (1995), Information pooling through majority-rule voting: Condorcet's jury theorem with correlated votes, Journal of Economic Behavior & Organization 26(3), 353-372. Publisher record returned an access error.

Rows 1-71: ?. No candidate was silently selected or classified from its title. The intended title/ID is needed before substantive assessment. These ? marks are now processed cells and must remain unchanged under the current preservation rule unless the user explicitly authorizes revisiting them. Counts: 0 X; 0 -; 71 ?.

Criteria Maps 2 3 Final Evidence Log

M2 rows 1-45 and M3 rows 1-55: **?**. The intended title/authors/identifier cannot be verified from this alias; the Paper Key remains unavailable. Retained Map 1 logs already document the identity gap. Candidate papers, surname matches and remembered methods were not substituted. These are identity uncertainties, not negative findings about an identified paper. No source can be attributed securely to this column until its Paper Key entry is supplied.

Criteria Map 4 Evidence Log

Identity unresolved; intended Paper Key entry remains unavailable. Sources:

• Rows 1-44: ?. Several jury-theory citations share the surname; intended Paper Key title/ID is unavailable.

Evidence notes | Kuncheva

Criteria Map 1 Batch 12 Evidence Log

The separate Paper Key could not be recovered. The column supplies only a surname; the author's own publication list contains several directly relevant, distinct papers. Primary identity source: Kuncheva's ensemble-diversity bibliography. Candidates include:

• Kuncheva and Whitaker (2003), Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy, Machine Learning 51, 181-207; DOI 10.1023/A:1022859003006.

• Kuncheva, Whitaker, Shipp and Duin (2003), Limits on the majority vote accuracy in classifier fusion, Pattern Analysis and Applications 6, 22-31.

• Kuncheva, Whitaker, Shipp and Duin (2000), Is independence good for combining classifiers?, ICPR.

Rows 1-71: ?. No title was silently chosen, and none of these papers' properties was substituted for the intended column. Exact title/ID is required. Counts: 0 X; 0 -; 71 ?.

Criteria Maps 2 3 Final Evidence Log

M2 rows 1-45 and M3 rows 1-55: **?**. The intended title/authors/identifier cannot be verified from this alias; the Paper Key remains unavailable. Retained Map 1 logs already document the identity gap. Candidate papers, surname matches and remembered methods were not substituted. These are identity uncertainties, not negative findings about an identified paper. No source can be attributed securely to this column until its Paper Key entry is supplied.

Criteria Map 4 Evidence Log

Identity unresolved; intended Paper Key entry remains unavailable. Sources:

• Rows 1-44: ?. No specific ensemble-diversity citation is mapped to this surname.

Evidence notes | Dawid-Skene

Criteria Map 1 Batch 12 Evidence Log

Identity: A. P. Dawid and A. M. Skene, Maximum Likelihood Estimation of Observer Error-Rates Using the EM Algorithm, Applied Statistics 28(1), 20-28 (1979); DOI 10.2307/2346806. Recovered prior citation agrees with the publisher and scanned paper. Sources: publisher record; readable published-paper scan. All journal pages 20-28 read; scan page 1 is a cover sheet.

• Rows 1/3/19/49/59/63: §2, pp.21-24 and §4, pp.25-27 independently collect categorical human ratings and infer posterior consensus. Row 19 denotes offline aggregation; 63 denotes one collection/aggregation stage. EM fitting iterations are not debate rounds.

• Rows 5/8/9/27-28/47: §§1/3/4 estimate heterogeneous observer accuracy, diagnose measurement errors and test estimate stability. No externally verified accuracy gain is claimed.

• Rows 10/24-25: §2, pp.21-22 assumes independence conditional on truth, then explicitly warns that shared patient-level variables can invalidate it. Row 26: -; individual confusion matrices do not fit cross-observer dependence or calibrate reasoning overlap.

• Rows 7/29-30/31-46/48/50-58/60-62/64-71: -. No probability-calibration evaluation, new convergence theorem, benchmark release, LLMs, agent topology, rationale extraction, messaging controller/filter or communication-cost report. EM convergence theory and agreement-diversity methods are cited, not newly contributed.

Counts: 15 X; 56 -; 0 ?.

Criteria Maps 2 3 Final Evidence Log

Identity: A. P. Dawid and A. M. Skene, Maximum Likelihood Estimation of Observer Error-Rates Using the EM Algorithm, Applied Statistics 28(1):20-28 (1979); DOI 10.2307/2346806. Sources: https://rss.onlinelibrary.wiley.com/doi/abs/10.2307/2346806 and https://dl.icdst.org/pdfs/files4/72000def5032e547726faf948cd958e8.pdf . Scan page 1 is a cover; references below use printed journal pages.

• M2 5/14/18-21/23-24/28/34/42: §2 pp.21-24 infers posterior categorical consensus using learned observer confusion matrices and class priors; unequal observer reliability can supersede raw majority. Table 4 p.27 emits posterior class probabilities and consensus classifications. These are reliability-conditioned probabilistic contributions, not confidence reports from observers or learned cross-observer dependence. Simple majority/weighted-consensus discussion on p.21 is motivation, and majority initialization is not the fitted model's final rule (1: -). No calibrated semantic answer-cluster rule (10: -); categories are given, not semantically clustered.

• M3 1-2: §2 pp.21-22 explicitly assumes responses independent conditional on latent truth and warns shared patient variables can violate it. Per-observer confusion matrices do not model cross-observer dependence (3-11/22: -). Categorical human judgments contain no reasoning-content units, logical graph, trace embedding or copying provenance (12/15-18/23-34/36: -).

• M3 13-14/19-21/35/37/43/55: observer/category/patient/dataset scope, exact category counts, EM initialization/convergence choices (§§2-4), and five-observer example with incomplete ratings. EM fits without known true labels (44: -), but is parameter fitting rather than a training-free/inference-only frozen-model procedure (38-42/54: -). EM iterations are not online debate-weight updates (M2 44: -). No LLM calls, verification, token or latency claims (45-53: -).

Criteria Map 4 Evidence Log

Identity: A. P. Dawid and A. M. Skene, Maximum Likelihood Estimation of Observer Error-Rates Using the EM Algorithm, Applied Statistics 28(1):20-28 (1979); DOI 10.2307/2346806. Sources: https://rss.onlinelibrary.wiley.com/doi/abs/10.2307/2346806 and https://dl.icdst.org/pdfs/files4/72000def5032e547726faf948cd958e8.pdf . Scan page 1 is a cover; references below use printed journal pages. Sources: https://rss.onlinelibrary.wiley.com/doi/abs/10.2307/2346806, https://dl.icdst.org/pdfs/files4/72000def5032e547726faf948cd958e8.pdf

• Journal §§1-4 pp.20-28: categorical human medical-observer ratings, estimated accuracies/confusion matrices and posterior consensus; reports inter-observer error/variation, not an empirical truth-verified test-accuracy gain. Dedicated five-clinician data are printed. Latent-class/Bayesian inference and EM are its own method; EM convergence is cited, not

a newly proved theorem. No LLM debate, code release, token cost or known-dependence simulation. Published 1979. Conditional independence/latent-observer theory and reliability aggregation background/baseline; explicit previous record says all journal pages were read.

Evidence notes | Prelec

Criteria Map 1 Batch 12 Evidence Log

The intended Paper Key entry was not recovered. The author's own MIT publication list identifies multiple distinct papers relevant to elicitation and aggregation:

• Dražen Prelec (2004), A Bayesian truth serum for subjective data, Science 306, 462-466; DOI 10.1126/science.1102081.

• Dražen Prelec, H. Sebastian Seung and John McCoy (2017), A solution to the single-question crowd wisdom problem, Nature 541, 532-535; DOI 10.1038/nature21054.

Publication identities are verifiable, but neither has been linked to this surname-only column by the missing key. Rows 1-71: ?. No substantive criteria were decided from a title or substituted from an assumed match. Counts: 0 X; 0 -; 71 ?.

Criteria Maps 2 3 Final Evidence Log

M2 rows 1-45 and M3 rows 1-55: **?**. The intended title/authors/identifier cannot be verified from this alias; the Paper Key remains unavailable. Retained Map 1 logs already document the identity gap. Candidate papers, surname matches and remembered methods were not substituted. These are identity uncertainties, not negative findings about an identified paper. No source can be attributed securely to this column until its Paper Key entry is supplied.

Criteria Map 4 Evidence Log

Identity unresolved; intended Paper Key entry remains unavailable. Sources:

• Rows 1-44: ?. Bayesian truth serum and single-question crowd-wisdom papers remain distinct candidates; intended citation unavailable.

Evidence notes | Dong

Criteria Map 1 Batch 12 Evidence Log

Neither author initials nor title/year/ID could be recovered for this column. Searches surfaced several different authors named Dong; in source-dependence research alone, these two published papers have the same author team (Xin Luna Dong, Laure Berti-Equille and Divesh Srivastava):

• Integrating Conflicting Data: The Role of Source Dependence, PVLDB 2(1), 550-561 (2009): publisher PDF.

• Truth Discovery and Copying Detection in a Dynamic World, PVLDB 2(1), 562-573 (2009): publisher PDF.

Both sources are accessible, but there is no evidence that either is the intended key entry. This is an identity failure, not a claim that the candidate papers are unreadable. Rows 1-71: ?. The expected title/ID is required; candidates were not silently selected or classified from their titles. Counts: 0 X; 0 -; 71 ?.

Criteria Maps 2 3 Final Evidence Log

M2 rows 1-45 and M3 rows 1-55: **?**. The intended title/authors/identifier cannot be verified from this alias; the Paper Key remains unavailable. Retained Map 1 logs already document the identity gap. Candidate papers, surname matches and remembered methods were not substituted. These are identity uncertainties, not negative findings about an identified paper. No source can be attributed securely to this column until its Paper Key entry is supplied.

Criteria Map 4 Evidence Log

Identity unresolved; intended Paper Key entry remains unavailable. Sources:

• Rows 1-44: ?. Multiple source-dependence/copying papers share the same author surname/team; intended Paper Key entry is missing.

16 | Glossary and claim checklist

Term	Plain meaning
Ablation	Remove one component to identify its contribution.
Accuracy	Fraction of answers correct under the chosen task labels.
Agent	A solver instance with an identity, role, prompt and information access.
Answer cluster	Outputs treated as equivalent final answers; not a cluster of reasoning.
AUROC	Discrimination of a score across positive/negative cases; not probability calibration.
Calibration	Agreement between predicted probabilities and observed frequencies.
Candidate coverage	Whether the pool contains a correct supported candidate.
Chain	Linked support units that justify an answer under an explicit definition.
Conditional independence	Independence after conditioning on specified variables; assumptions matter.
Confidence	Belief or score attached to an answer; may be verbal, likelihood-based or fitted.
Counterfactual reference	A specified alternative process such as matched no-communication generation.
Credibility	Trust/reliability/quality assessment; distinct from independence.
DAG	Directed acyclic graph; direction and cycles are properties of the representation.
Dependence proxy	Observable statistic related to dependence, without certifying its hidden cause.
Derivation overlap	Shared propositions/inferences/subpaths; requires a meaningful unit and reference.
Discrimination	A score separates correct from incorrect cases; calibration is a different property.
Effective sample size	Independent-sample-equivalent quantity under a specified mathematical model.
EM	Iterative fitting with latent-variable expectations; not automatically training-free.
ECE	Binned calibration-error summary; sensitive to definitions and binning.
Evidence source	Upstream document, observation, theorem, assumption or tool result.
Excess overlap	Observed overlap above a specified honest-solver reference.
False consensus	Shared agreement on a wrong or unsupported answer.
Flip precision	How often changed decisions are beneficial under a defined criterion.
Generation independence	No current peer-output access during generation; errors can still correlate.
Gold labels	Reference outcomes; distinguish evaluation use from estimator fitting.
Ground truth dependence	Known controlled sharing/provenance mechanism, not merely observed similarity.

Term	Plain meaning
Hallucinated premise	A claimed supporting fact or assumption that lacks adequate grounding.
Hyperparameter	A chosen setting such as threshold, budget, similarity rule or round cap.
Ising model	A joint binary-label model with fields/interactions, under specified assumptions.
Judge	A model or other evaluator making a quality/selection decision.
Latent factor	An unobserved variable influencing observations or errors.
Majority vote	Choose the most supported answer label; plain counts omit dependence.
Martingale	A process whose conditional expected next value equals its present value under a specified model.
No-communication panel	Comparable solvers generate without current peer messages.
Null distribution	Distribution of a statistic under an explicitly defined reference hypothesis.
Operational independence	The specific measurable notion tested by the experiment, not unrestricted absolute independence.
Overlap statistic	Numeric comparison of chosen units; its meaning depends on representation.
Paired replay	Different aggregators receive the same recorded candidate pool.
PRM	Process reward model trained to evaluate intermediate reasoning; resources must be stated.
Provenance	Recorded origin, source and exposure history for a claim or message.
Semantic equivalence	Same relevant meaning despite different text; logical conditions can break it.
Support credit	Contribution assigned by a defined aggregation rule; not automatically a probability.
Training-free	Must identify what is frozen, fitted, labeled and inferred at runtime.
Trajectory	History of answers and interactions, not just terminal output.
Verification	Check correctness/grounding of a claim or inference; independent support can still be wrong.

Before a claim enters the thesis

- Does the source implement this, or merely cite/propose it?
- Does the claim concern the main method, a baseline, an ablation or theory?
- What is the conditioning/reference process for independence?
- Were the outcome labels used only for evaluation or also for fitting/tuning?
- Is the reported improvement paired on the same candidates and matched budget?
- Are candidate-generation coverage and aggregation performance separated?
- Can the dependence mechanism be inspected under a controlled origin test?

- Are uncertainty, extraction errors and unresolved identities visible?

17 | Primary references and source index

The source-identifiable paper directory follows the original map order. URLs link to publisher/arXiv papers or records. Some aliases were matched provisionally by title/method in earlier notes because the separate Paper Key was unavailable; those qualifications remain in the evidence appendix. Venue/year below is the reviewed version, not a new live bibliographic certification.

DySCo

Wanshuang Gou and Zihan Liu, Dynamic Trust-Aware Sparse Communication Topology for LLM-Based Multi-Agent Consensus, arXiv:2606.01828v1 (2026).

https://arxiv.org/pdf/2606.01828v1

https://arxiv.org/html/2606.01828

DarkForest

Yi Li et al., DarkForest: Less Talk, Higher Accuracy for Multi-Agent LLMs, arXiv:2605.25188v1 (2026).

https://arxiv.org/pdf/2605.25188v1

https://arxiv.org/html/2605.25188

AgentAuditor

Wei Yang, Shixuan Li, Gengshuo Liu, Heng Ping, Peiyu Zhang, Paul Bogdan and Jesse Thomason, Auditing Multi-Agent LLM Reasoning Trees Outperforms Majority Vote and LLM-as-Judge, arXiv:2602.09341v2 (2026). The PDF author line includes Gengshuo Liu, whom the abstract-page metadata omits.

https://arxiv.org/pdf/2602.09341v2

https://arxiv.org/html/2602.09341

CAGE-CAL

Jiatan Huang, Mingchen Li, Ziming Li, Sunjae Kwon, Hong Yu and Chuxu Zhang, Counterfactual Graph for Multi-Agent LLM Calibration, arXiv:2605.30653v1 (2026).

https://arxiv.org/pdf/2605.30653v1

https://arxiv.org/html/2605.30653

Zhu

Xiaochen Zhu, Caiqi Zhang, Yizhou Chi, Tom Stafford, Nigel Collier and Andreas Vlachos, Demystifying Multi-Agent Debate: The Role of Confidence and Diversity, Findings ACL 2026, Anthology ID 2026.findings-acl.1694.

https://aclanthology.org/2026.findings-acl.1694.pdf

Ising

Balasubramanian, Podkopaev and Kasiviswanathan, Dependence-Aware Label Aggregation for LLM-as-a-Judge via Ising Models, ICML/PMLR 306 (2026). The author-hosted January 29 preprint prints the expanded author names Krishnakumar Balasubramanian and Shiva Prasad Kasiviswanathan.

https://proceedings.mlr.press/v306/balasubramanian26a.html

https://www.shivakasiviswanathan.com/LLMaaJ.pdf

SELENE

Akshay Verma, Swapnil Gupta, Deepak Gupta, Prateek Sircar and Siddharth Pillai, SELENE: Selective and Evidence-Weighted LLM Debating for Efficient and Reliable Reasoning, EACL Industry 2026, Anthology ID 2026.eacl-industry.7.

https://aclanthology.org/2026.eacl-industry.7.pdf

DAR

Manh Nguyen, Anh Nguyen, Dung Nguyen, Svetha Venkatesh and Hung Le, Hear Both Sides: Efficient Multi-Agent Debate via Diversity-Aware Message Retention, arXiv:2603.20640v2, April 14, 2026.

https://arxiv.org/pdf/2603.20640v2

https://arxiv.org/pdf/2603.20640

Goel

Shashwat Goel et al., Great Models Think Alike and this Undermines AI Oversight, ICML 2025, PMLR 267:19621-19678; arXiv:2502.04313v2, June 12, 2025.

https://proceedings.mlr.press/v267/goel25b.html

https://arxiv.org/pdf/2502.04313v2

Wynn

Andrea Wynn, Harsh Satija and Gillian K. Hadfield, Talk Isn't Always Cheap: Understanding Failure Modes in Multi-Agent Debate, ICML MAS Workshop 2025; arXiv:2509.05396v2, October 13, 2025.

https://arxiv.org/pdf/2509.05396v2

https://arxiv.org/pdf/2509.05396

EoT

Zhangyue Yin, Qiushi Sun, Cheng Chang, Qipeng Guo, Junqi Dai, Xuanjing Huang and Xipeng Qiu, Exchange-of-Thought: Enhancing Large Language Model Capabilities through Cross-Model Communication, EMNLP 2023, ACL Anthology 2023.emnlp-main.936.

https://aclanthology.org/2023.emnlp-main.936/

https://aclanthology.org/2023.emnlp-main.936.pdf

PPV

** Yasushi Sakai, Allen Song, Kent Larson, When Does Delegation Beat Majority? A Delegation-Based Aggregator for Multi-Sample LLM Inference, arXiv:2606.08098v3 (July 2026). Full text: https://arxiv.org/html/2606.08098 . “PPV” names the Propagational Proxy Voting mechanism applied to multi-sample LLM inference here; it is not a literal multi-agent discussion protocol. Separate Paper Key absent, so identity remains subject to Paper Key cross-check.

https://arxiv.org/html/2606.08098

https://arxiv.org/pdf/2606.08098v3

MinSentinel

** Chuan He, Zebin Chen, Zhengyi Yang, Shaobo Qiao, Mingchen Ju, Jiate Liu, Dong Wen, Guanfeng Liu, Minority Sentinel: When to Overturn Majority Voting in Multi-Agent LLM Debates, AgentSearch Workshop at SIGIR 2026, arXiv:2606.29270v1. Full text: https://arxiv.org/html/2606.29270 .

https://arxiv.org/html/2606.29270

https://arxiv.org/pdf/2606.29270v1

OW-ISP

Rui Ai, Yuqi Pan, David Simchi-Levi, Milind Tambe and Haifeng Xu, Beyond Majority Voting: LLM Aggregation by Leveraging Higher-Order Information, ICML 2026, PMLR 306:1311-1340; primary PDF arXiv:2510.01499v2, May 19, 2026.

https://proceedings.mlr.press/v306/ai26d.html

https://arxiv.org/pdf/2510.01499v2

DivEvid

** Yuante Li, Yicheng Tao, Kate Zhang, Taozhi Wang, Gefei Gu, Yaxin Zhou, Diverse Evidence, Better Forecasts: Multi-Agent Deliberation Under Information Asymmetry, arXiv:2607.01661v1 (July 2026), method named **InfoDelphi**. Primary: https://arxiv.org/html/2607.01661 . The abbreviation “DivEvid” uniquely and directly matches **Diverse Evidence**, and method/topic match this comparison map; still request the Paper Key entry if strict key-level identity certification is necessary.

https://arxiv.org/html/2607.01661

https://arxiv.org/pdf/2607.01661v1

R²-MAD

Xuanfa Jin, Zhijian Ma, Yongcheng Zeng, Xinyu Cui, Haifeng Zhang and Jun Wang, Remember and Reweight: Enhancing Multi-Agent Debate with Experience Memory and Confidence Estimation, arXiv:2609.03619v1, September 3, 2026.

https://arxiv.org/pdf/2609.03619v1

https://arxiv.org/html/2609.03619v1

Choi

Choi, Zhu and Li, Debate or Vote: Which Yields Better Decisions in Multi-Agent Large Language Models?, NeurIPS 2025; arXiv:2508.17536v2.

https://arxiv.org/pdf/2508.17536v2

https://arxiv.org/html/2508.17536v2

Free-MAD

Cui et al., Free-MAD: Consensus-Free Multi-Agent Debate, Findings ACL 2026; arXiv:2509.11035.

https://aclanthology.org/2026.findings-acl.1600.pdf

ReConcile

Justin Chih-Yao Chen, Swarnadeep Saha and Mohit Bansal, ReConcile: Round-Table Conference Improves Reasoning via Consensus among Diverse LLMs, ACL 2024, pp.7066-7085, ACL Anthology 2024.acl-long.381; arXiv:2309.13007.

https://aclanthology.org/2024.acl-long.381/

https://aclanthology.org/2024.acl-long.381.pdf

DOWN

Eo et al., Debate Only When Necessary: Adaptive Multiagent Collaboration for Efficient LLM Reasoning, arXiv:2504.05047v2.

https://arxiv.org/pdf/2504.05047v2

https://arxiv.org/html/2504.05047v2

ConfLiar

Keramati et al., The Confident Liar: Diagnosing Multi-Agent Debate with Log-Probabilities and LLM-as-Judge, ACL 2026 SRW; arXiv:2606.10296.

https://aclanthology.org/2026.acl-srw.121.pdf

Slim-SC

Hong et al., Slim-SC: Thought Pruning for Efficient Scaling with Self-Consistency, EMNLP 2025; arXiv:2509.13990.

https://aclanthology.org/2025.emnlp-main.1750.pdf

https://github.com/hyscale-lab/slimsc

ConsTrap

Jiayuan Liu, Shiyi Du, Weihua Du, Mingyu Guo, Vincent Conitzer. The Consensus Trap: Rescuing Multi-Agent LLMs from Adversarial Majorities via Token-Level Collaboration, arXiv:2604.17139v1. Primary: https://arxiv.org/html/2604.17139v1 .

https://arxiv.org/html/2604.17139v1

https://arxiv.org/pdf/2604.17139v1

ARGORA

Youngjin Jin, Hanna Kim, Kwanwoo Kim, Chanhee Lee, Seungwon Shin. ARGORA: Orchestrated Argumentation for Causally Grounded LLM Reasoning and Decision Making, arXiv:2601.21533v1. Primary: https://arxiv.org/html/2601.21533v1 .

https://arxiv.org/html/2601.21533v1

https://arxiv.org/pdf/2601.21533v1

Li-Sparse

Li et al., Improving Multi-Agent Debate with Sparse Communication Topology, Findings EMNLP 2024; arXiv:2406.11776.

https://aclanthology.org/2024.findings-emnlp.427.pdf

MoA

Wang et al., Mixture-of-Agents Enhances Large Language Model Capabilities, ICLR 2025; arXiv:2406.04692.

https://proceedings.iclr.cc/paper_files/paper/2025/file/5434be94e82c54327bb9dcaf7fca52b6-Paper-Conference.pdf

CoT

Wei et al., Chain-of-Thought Prompting Elicits Reasoning in Large Language Models, NeurIPS 2022; arXiv:2201.11903.

https://papers.nips.cc/paper_files/paper/2022/file/9d5609613524ecf4f15af0f7b31abca4-Paper-Conference.pdf

https://proceedings.neurips.cc/paper_files/paper/2022/file/9d5609613524ecf4f15af0f7b31abca4-Supplemental-Conference.pdf

SC

Wang et al., Self-Consistency Improves Chain of Thought Reasoning in Language Models, ICLR 2023; arXiv:2203.11171.

https://openreview.net/references/pdf?id=xIxIkdndPl

Du

Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum and Igor Mordatch, Improving Factuality and Reasoning in Language Models through Multiagent Debate, ICML 2024, PMLR 235:11733-11763; arXiv:2305.14325.

https://proceedings.mlr.press/v235/du24e.html

https://raw.githubusercontent.com/mlresearch/v235/main/assets/du24e/du24e.pdf

Liang

Tian Liang et al., Encouraging Divergent Thinking in Large Language Models through Multi-Agent Debate, EMNLP 2024:17889-17904; arXiv:2305.19118.

https://aclanthology.org/2024.emnlp-main.992.pdf

ChatEval

Chan et al., ChatEval: Towards Better LLM-based Evaluators through Multi-Agent Debate, ICLR 2024; arXiv:2308.07201.

https://proceedings.iclr.cc/paper_files/paper/2024/file/25cc3adf8c85f7c70989cb8a97a691a7-Paper-Conference.pdf

Smit

Andries Smit, Nathan Grinsztajn, Paul Duckworth, Thomas D. Barrett and Arnu Pretorius, Should we be going MAD? A Look at Multi-Agent Debate Strategies for LLMs, ICML 2024, PMLR:smit24a.

https://raw.githubusercontent.com/mlresearch/v235/main/assets/smit24a/smit24a.pdf

Xiong

Xiong et al., Examining Inter-Consistency of Large Language Models Collaboration: An In-depth Analysis via Debate, Findings EMNLP 2023; arXiv:2305.11595.

https://aclanthology.org/2023.findings-emnlp.508.pdf

Kuhn

Lorenz Kuhn, Yarin Gal and Sebastian Farquhar, Semantic Uncertainty: Linguistic Invariances for Uncertainty Estimation in Natural Language Generation, ICLR 2023; arXiv:2302.09664v3, April 15, 2023.

https://arxiv.org/abs/2302.09664v3

https://arxiv.org/pdf/2302.09664v3

Lightman

Hunter Lightman et al., Let's Verify Step by Step, ICLR 2024; arXiv:2305.20050.

https://proceedings.iclr.cc/paper_files/paper/2024/file/aca97732e30bcf1303bc22ac3924fd16-Paper-Conference.pdf

Cemri

Cemri et al., Why Do Multi-Agent LLM Systems Fail?, NeurIPS 2025 Datasets and Benchmarks; arXiv:2503.13657.

https://proceedings.neurips.cc/paper_files/paper/2025/file/b1041e52d3be19f0a9bc491657488e4a-Paper-Datasets_and_Benchmarks_Track.pdf

Dawid-Skene

A. P. Dawid and A. M. Skene, Maximum Likelihood Estimation of Observer Error-Rates Using the EM Algorithm, Applied Statistics 28(1):20-28 (1979); DOI 10.2307/2346806.

https://rss.onlinelibrary.wiley.com/doi/abs/10.2307/2346806

https://dl.icdst.org/pdfs/files4/72000def5032e547726faf948cd958e8.pdf

Additional methodology: outside the four maps

Faithfulness / Turpin et al.

Miles Turpin, Julian Michael, Ethan Perez and Samuel Bowman. Language Models Don't Always Say What They Think: Unfaithful Explanations in Chain-of-Thought Prompting. NeurIPS 2023.

Primary record

Faithfulness / Lanham et al.

Tamera Lanham and colleagues. Measuring Faithfulness in Chain-of-Thought Reasoning. arXiv:2307.13702, 2023.

Primary record

Phipson-Smyth

Belinda Phipson and Gordon K. Smyth. Permutation p-values should never be zero: calculating exact p-values when permutations are randomly drawn. Statistical Applications in Genetics and Molecular Biology 9(1), Article 39, 2010; author manuscript arXiv:1603.05766.

Primary record

Toolkit / official documentation

vLLM

Hugging Face Transformers

Sentence Transformers

SymPy

The four source maps

Map 1: 71 criteria, 48 columns, no blanks. Source version: Batch 12 / version 8.

Map 2: 45 criteria, 48 columns, no blanks. Source version: Completed / version 7.

Map 3: 55 criteria, 48 columns, no blanks. Source version: Completed / version 7.

Map 4: 44 criteria, 48 columns, no blanks. Source version: Completed / version 1.